vmware by Broadcom

VMware Lab Platform

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VMware Lab Platform Release Notes

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- What's New in 2024
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- What's New in 2021
- What's New in 2020
- What's New in 2019

Introduction

VMware Lab Platform is a multi-tenant VMware Cloud service that lets you create, manage, and deliver rich-featured IT labs for self-paced, on-demand training, instructor-led training, and live lab events.

Check for additions and updates to the release notes.

What's New in 2024

Check regularly for additions and updates made to the VMware® Lab Platform™ (formerly known as VMware® Learning Platform™ release notes.

What's New January 19, 2024

Updated on January 19, 2024

- **Deprecated the old user interface and the instructor interface—**We are replacing the old user interface and the instructor interface with an improved and revamped new user interface and instructor interface.

  We have retained all of the features from the previous user and instructor interfaces while adding many new ones. Moving forward, all the new features and bug fixes will be limited to the new instructor and user interfaces.
- With this, we'd like to announce that **from February 1, 2024** the new user interface and instructor user interface will be enabled for all, by default.

  The old interfaces will be **permanently deprecated and removed on March 1, 2024**.

  If you need support on using the new user interface and the instructor interface, contact the VMware Lab Platform support team.

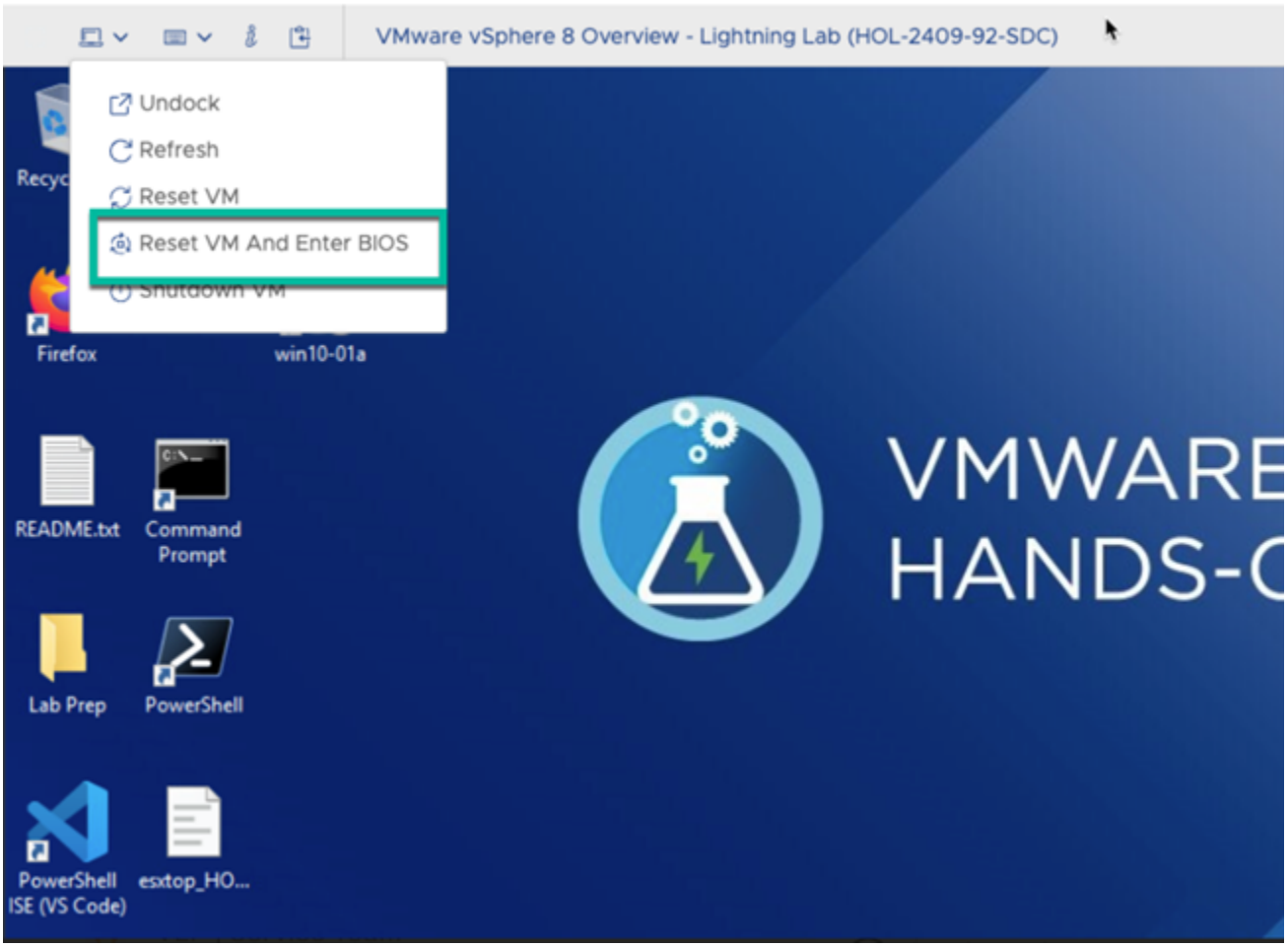
  For more information on how to use the new instructor interface, see the Using and Managing the VMware Lab Platform Instructor Interface guide.

  For more information on the user interface, see the Using and Managing the VMware Lab Platform User Interface guide.

Enhancements in the administrator user interface:

- **Added Reset VM And Enter BIOS option in the Lab Builder and lab console—**With the **Reset VM And Enter BIOS** option, you can easily reset a virtual machine (VM) of a lab, and start the BIOS and make changes to the boot order. To access the lab console from the administrator user interface, go to **Lab Management >Lab Builder**, and

click **Edit**. Open the lab console, and in the console view, click the  **Reset VM And BIOS** icon. This option is also available for end users under the **Console Actions** option in the VMware Lab Platform user interface.

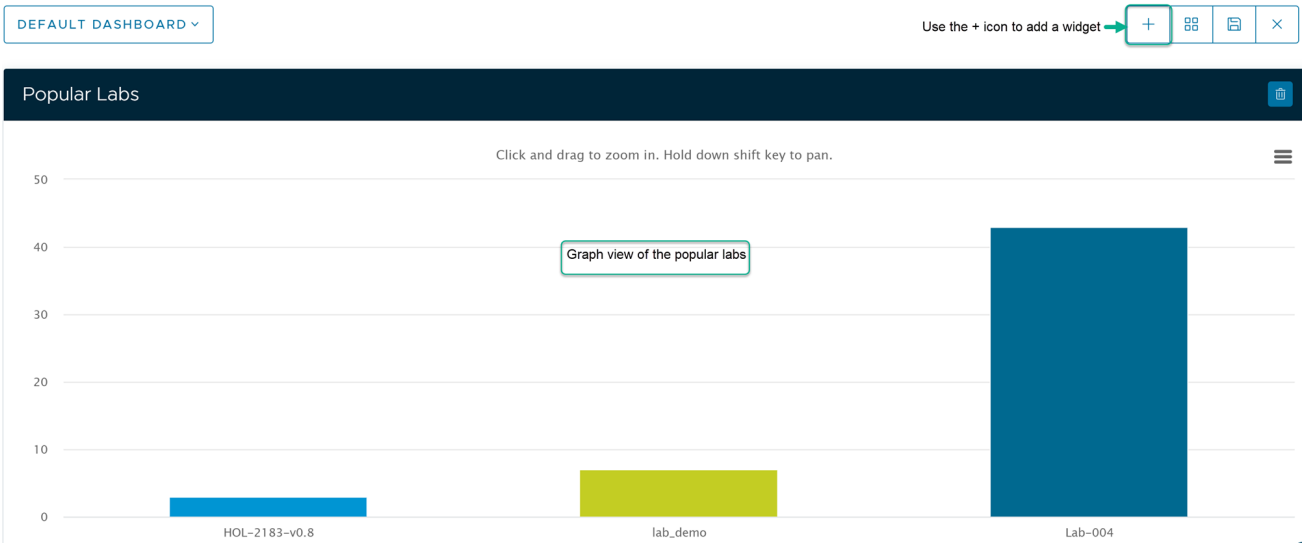


- **Improvements to assign a cloud by region—**To ease the task of assigning a preferred cloud location for a class, we have made improvements to the Classes wizard. Now as an administrator, you don't need to select the **Select a cloud by user's location** setting option to set a "preferred cloud location" for classes. The new improvements will help administrators, who don't want to enable the **Select a cloud by user's location** setting and yet want to define a preferred cloud location for the entire class in the Classes wizard.

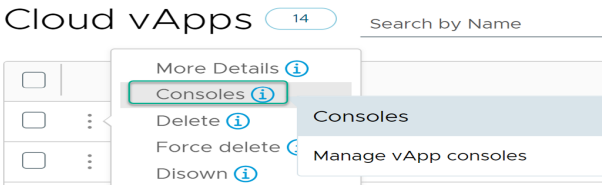
- **View your popular labs from the Dashboard**—Gain lab usage insights from the Dashboard. To see the popular labs for your account, we have added the **Popular Lab** widget in the Dashboard. To add the **Popular Lab** widget in the Dashboard, you need to first activate the **Portal Settings** > **Configuration**> **Show Popular Labs** setting. To set the limit of entries to display in the Popular Labs widget, activate the **Portal Settings** > **Configuration** >**Max Popular Labs Shown** setting option. To view the popular labs from the Dashboard, add the Popular Labs widget,

and view the graph with the results. Like other statistic parameters such as lab usage, cost, and so on, the popular lab data will help you to make informed decisions on your labs.

Figure 1: Popular Labs widget



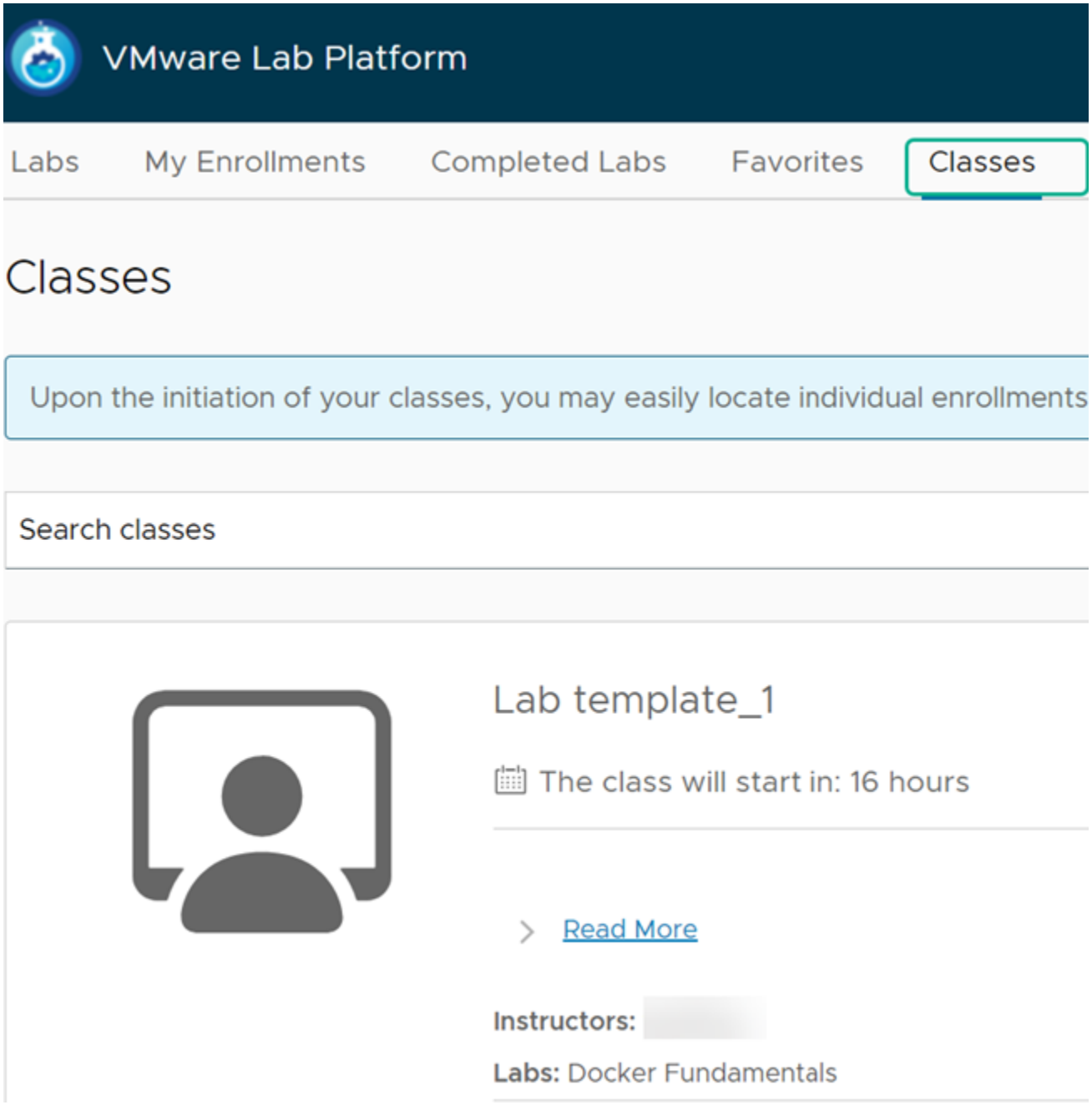
- **Access VM console from the Cloud vApp page—**You can access and open a VM console for your vApp from the Cloud vApps page. To access a VM console, go to **Cloud Management** > **Cloud vApps**, and click the three ellipses for a cloud vApp entry in the table, and click **Consoles**.



- **Access student lab console from the instructor user interface—**We have made it easier for you to access a student console from the new instructor interface. As an instructor, you can access each student's lab console from the instructor user interface. To open each lab console as a new tab with the student name, click the three ellipses on the student tile, and then right-click **Explore console** and select **Open in a new tab**. This feature helps in navigation and visibility in each student's lab console.

Enhancements in the user interface:

- **Self registering for a class—**We have added a new tab called **Classes** in the user interface that allows you or lab users to view their upcoming or active class enrollments, or self register for a class.



The **Classes** tab will be visible only when:

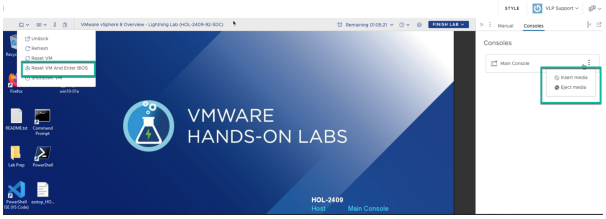
- A student has an upcoming class or has active classes.
- **Note**: You cannot see completed classes that are not yet deleted from the system.
- The class tile in the tab displays the name of the class, a label indicating remaining time for the class, status of the class if the class is active, description, start and end date details, instructor name, and other relevant information.
- The **Classes** tab has an option that will redirect you to the **My Enrollments** page with filtered results. This means that if the class has more than one lab in it, the filtered result will list all the enrollments related to the class for you. The filter will be visible and present in the search bar.
- The title in the **Classes** tab will indicate only the class information, you cannot manage the individual entitlements from the page.

- **Customize the style and themes for your user interface—**As an administrator, you can personalize and tailor the VMware Lab Platform user interface login page according to your brand or organization's identity for your users. You can change the header color, navigation panel, user interface buttons, page banner, add company logo, company icon, and add page background using the **Theme styler** option**.** You also use the custom CSS script to personalize the UI page. The new interface allows you to add your custom image for the login page.

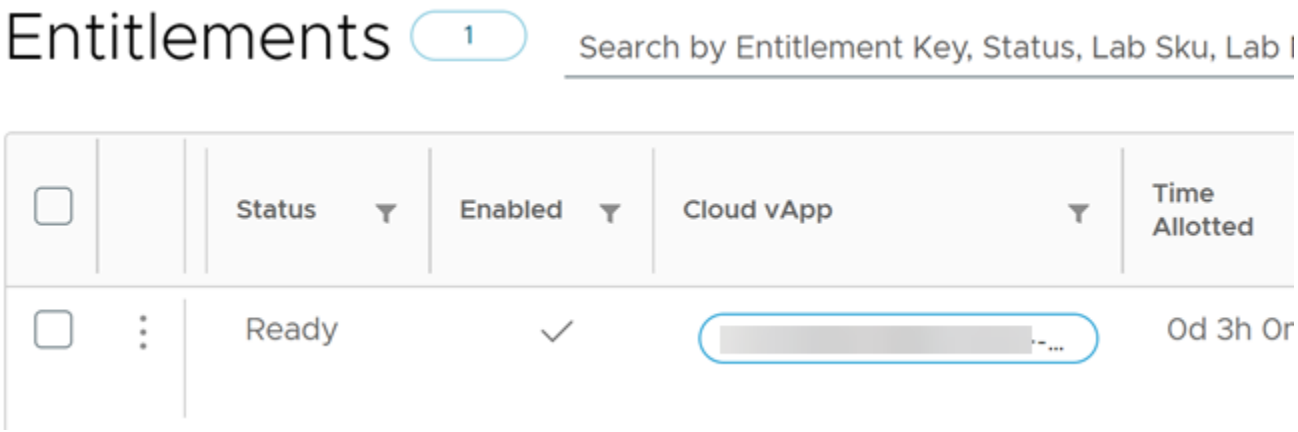
- **Create your own the custom messages for lab actions—**You can use the custom message option to customize the available custom messages when ending, or exiting a lab.

  For more information on how to use a custom message, see the Custom Messages section.

- **Support to mount a CD to a VM—**Now you will be able to attach or eject media to a VM from the user interface. Click the vertical ellipses and use the **Insert media** or **Eject media** options to add or eject media files in the user interface. This will allow you to host labs or classes that require a walk through of an OS installation and other use cases for mounting a CD on a VM.



- **Support to mirror display a lab console—**Use the **Mirror Display** option to view a user/student's lab console without interacting or disturbing the user while the user is taking a lab. To use the Mirror Display option, go to **Entitlements** >**User Entitlements**> **Mirror Display** link. This option lets proctors and the support team to view the lab console as and when a student is taking the lab and troubleshoot any issues when required.



What's New in 2023

Check regularly for additions and updates made to the VMware® Lab Platform™ (formerly known as VMware® Learning Platform™ release notes.

What's New November 21, 2023

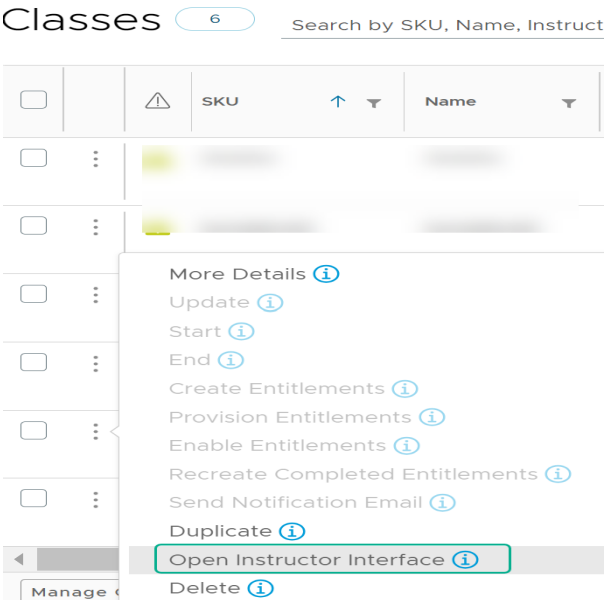
Updated on November 21, 2023

Enhancements in the administrator user interface:

- We have added a new instructor interface with significant feature enhancements:
  - **New instructor interface for a seamless user experience—**In our continuous effort to provide a better user experience, along with the new user interface, we have now added a new instructor interface. To use the new instructor interface, contact the VMware Lab Platform support team, or use the chat option in the application. For more information on how to use the VMware Lab Platform instructor interface, see Using and Managing the VMware Lab Platform Instructor Interface.

    NOTE
    You can use the new instructor interface only with the new user interface.

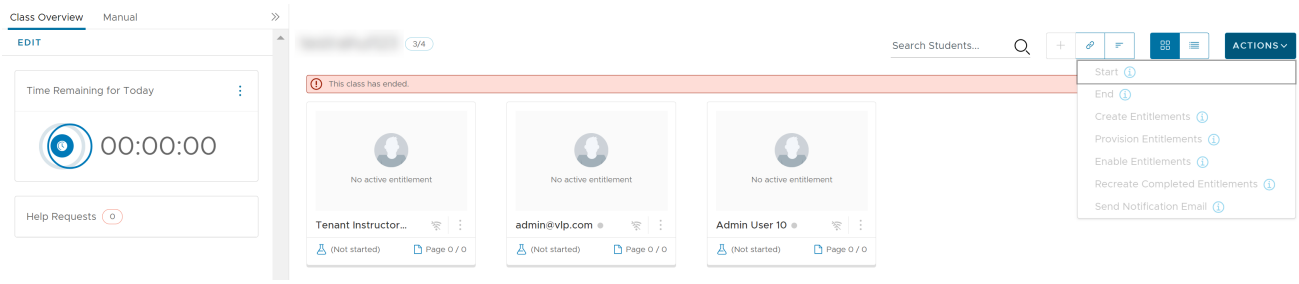
  - **Access the instructor interface from the administrator interface—**We have embedded the instructor interface in the administrator interface so that you can access the instructor interface from the administrator user interface itself. Go to **Class Management** > **Classes**, and click the three ellipses for a class entry, and then click **Open Instructor Interface**, or use the URL link under the **Instructor Launch URL** column for your class.



  - **A modern design**—The adoption of the VMware Clarity Design system for the updated instructor interface signifies a commitment to modern design principles and a more cohesive user experience within the VMware ecosystem. The interface not only appears visually appealing, but also ensures a consistent look and feel across various elements of the platform.

- **Access all class related and student related operations from one page—**As an instructor, you can perform all class related actions such as extend a class, start a class, end a class, create entitlements for students, add students to a class etc from one page. Monitor the class progress, and view help requests from students in the same interface.

Figure 2: The new instructor interface view



- **New billing dashboard for comprehensive view of total cost and usage**

  VMware Lab Platform combines rapid reporting with granular usage control to give you the tools to analyze resource usage and manage costs. The new Billing dashboard provides a variety of reporting parameters and filters that lets you create precise and customized billing reports. You can use the dashboard page to gain insights of your resource usage. To view a billing report, from the navigation panel, select **Reports** > **Billing**. To view details of a billing entry, click on the **Details** page.

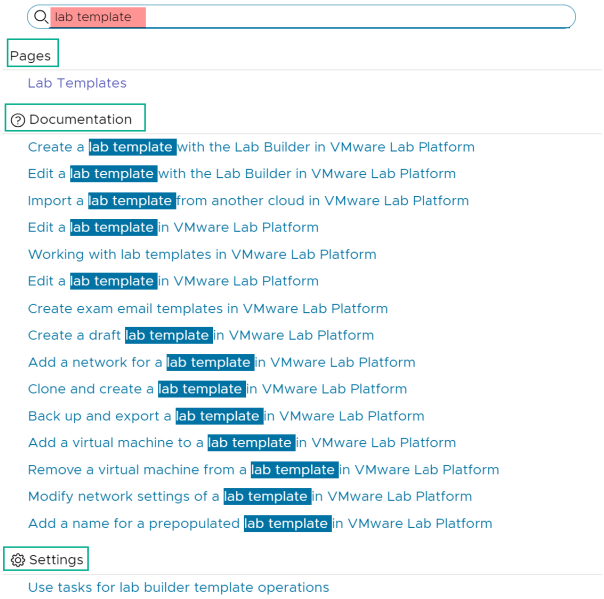


- **Added global search option to improve site navigation**

  Now you can easily find any VMware Lab Platform page, portal setting, or documentation link with the global search bar option in the application. The search bar option is located on top of all the pages in the administrator user interface. Users with the role of administrators, content creators, instructors, and super users who have access to the VMware Lab Platform administrator interface can use the search option.

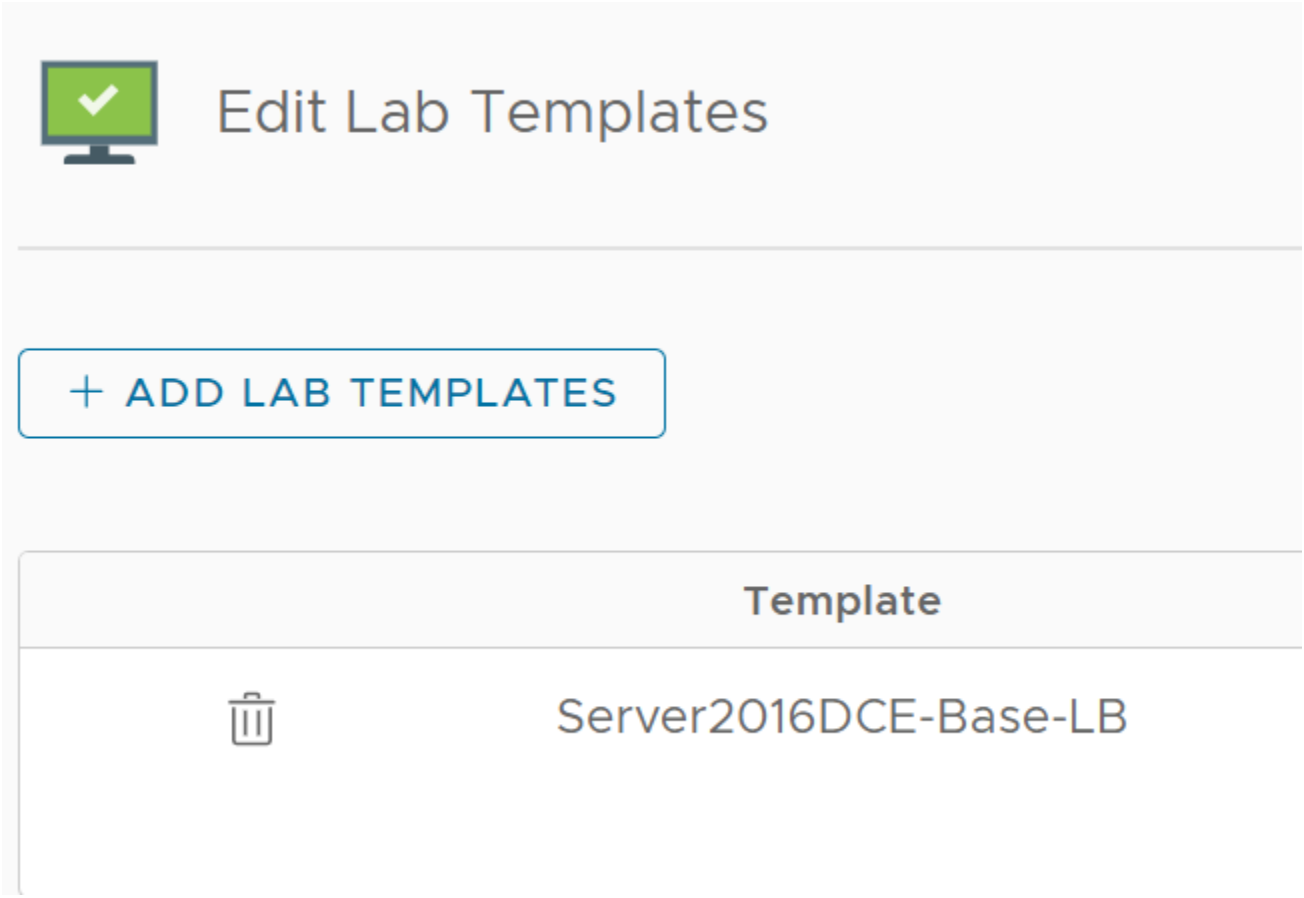


However, the user experience of the search results is based on the permissions that a user has. For example, if you log in as an instructor and search for any page, the search results will show only the pages that an instructor has permissions to. The search option is useful as it helps you to easily navigate and search any page, documentation link, or portal setting in the system.



- **Cloud Catalog column added in the Lab Templates page under Deployment Pools**

  You can view the cloud catalog information from the **Cloud Management**> **Deployment Pools**> **Edit Lab Templates** page. A cloud catalog typically stores lab templates, which are preconfigured and include virtual machines, operating systems, and associated configurations. Storing lab templates in catalogs allow you to easily reuse lab templates for any deployment.



You can view and search for cloud catalogs in the **Lab Builder** page, using the search option.

- **Added the Hide forgot password button for local authentication provider**

  You can hide the **Forgot Password** button for end users with local authentication provider. To hide the **Forget Password** button, activate the **Portal Setting** > **Hide forgot password button** setting.

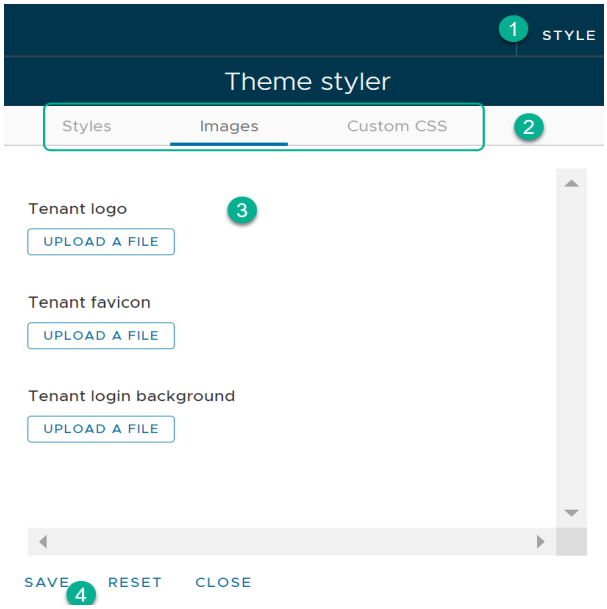
  NOTE
  You can hide the **Forgot Password** button only for the local authentication provider.

Enhancements in the new end-user interface:

- **Customized branding of the login page**

  As an administrator, you can personalize and tailor the VMware Lab Platform user interface login page according to your brand or organization's identity for your users. You can change the header color, page banner, add company logo, company icon, and add page background using the **Theme styler** option.

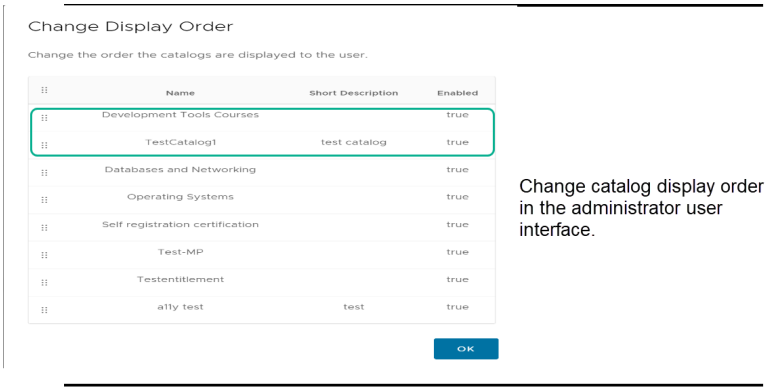
  To customize your login page, click **Style** and use the options under **Theme styler**. Ensure to save your changes.



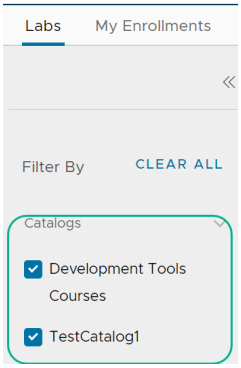
- **Use customized icons for lab resources**

  Like the customized user interface login page, you can add icons for lab resources in the user interface. Use the **Lab Resources** page in the administrator interface and add image for the thumbnail section. In the user interface, the images added for the thumbnail section are used as icons for the **Lab Resources** list .
- **Change the lab order in a lab catalog**

  Define and manage the lab order in the user interface, navigation pane. Now, you can display the same lab order that you define in the administrator interface. To change the lab order, in the administrator user interface, click **Lab Management** > **Lab Catalogs**, then click the three ellipses, and **Change Display Order**. Drag and move the labs in the order list. Save your changes.



Change catalog display order in the administrator user interface.



Catalog display order in the user interface.

- **Random lab order in a lab catalog**

  If you don't want to set the sequence of labs that are displayed in a catalog, you can activate the **Portal Setting > Randomize lab order in lab catalog** setting. When this option is activated, labs are displayed in a random order.

- **Support for lab SKU in URL**

  The on-demand labs can now be accessed and shared using lab SKU in the URL. To use this feature, activate the **Portal Setting** > **Configuration** > **Use SKU in Quick Launch URL** setting.

  Lab URL with lab SKU: `https://www.labplatform.vmware.com/<org_name>/catalogs/lab/<lab_sku>`

  For example, if the lab SKU is Test-Lab-1

  Then the URL will be: `https://www.labplatform.vmware.com/<org_name>/catalogs/lab/Test-Lab-1`

What's New September 25, 2023

Updated on September 25, 2023

- **Support for preferred cloud location added to the bulk upload feature**

  As a user with an administrator role, you can easily add the preferred cloud location to deploy the lab resources for your users. Use the **Classes** wizard, to add bulk users along with the preferred cloud location in the same CSV file (`preferredCloudLocation`), or you can use the API endpoints to add this information for bulk users.

For more information on how to add bulk users, see Upload multiple users.

For information on the API calls for the bulk upload feature, on the VMware Lab Platform administrator interface, click **Support** >**REST API Guide**.

- **Create user accounts with your preferred authentication provider**

  As a user with the administrator role, you can create user accounts with a specific authentication provider of your choice. You need to first create an authentication provider in the system and add it to user accounts. Use the **Users & Groups**>**User Accounts**>**Create New** wizard and add the authentication provider for bulk users using the **Bulk Upload** feature. As an end user, when you log in to the user interface, you can select the authentication provider through the drop-down menu. This feature is useful when you want to register user accounts to log in using a single sign-on (SSO) authentication provider.

  For more information on how to create an authentication provider in the VMware Lab Platform, see Working with an authentication provider, and for information on how to create user accounts, see Create a user.

- **Better support for account migration from one authentication provider to another authentication provider**

  Effortlessly migrate bulk accounts from one authentication provider to another provider. To migrate the authentication provider for bulk accounts, on the VMware Lab Platform administrator interface, go to **Users & Groups**> **User Accounts**. On the User Accounts page, select the users, and click the three ellipses for batch action, and select **Migrate to a different provider**. In the **Migrate to a different provider** form, from the drop-down menu select the preferred authentication provider.

  You can use either the VMware Lab Platform administrator interface or the API endpoints to migrate specific accounts.

  For example; https://login.labplatform.vmware.com/api/account/migrate?providerName=targetProvider&usernames=username1@example.com,username2@test.com

  For more information on how to migrate users from one authentication provider to another, see Edit a user account.
- **Lab Builder improvements**

  During a cloud synchronization operation, to provide stability around the Lab Builder operations and template names, the Lab Builder actions such as renaming, updating, or deleting a lab template, will not be available until the synchronization is completed.
- **New VMware Lab Platform user interface**

  We are excited to announce the launch of the new VMware Lab Platform user interface for lab users and students.

  Here are some of the high-level changes that you may notice while using the new VMware Lab Platform user interface:

  - **Filter and sort labs**—The enhanced navigation and improved discoverability features in the new interface can significantly streamline your experience. You can easily sort labs by name or age, and filter them by catalogs, products, tags, or badges. You can quickly narrow down your search and find the specific lab that meets your needs. Whether you are looking for a particular topic, product, or type of activity, these sorting and filtering options can help you locate the lab more efficiently than ever before.
  - **A modern design**—The adoption of the VMware Clarity Design system for the updated user interface signifies a commitment to modern design principles and a more cohesive user experience within the VMware ecosystem. The interface not only appears visually appealing but also ensures a consistent look and feel across various elements of the platform.
  - **Favorites feature**—By popular demand, you can now mark labs as 'Favorites,' making it quick and easy to return to labs you find particularly useful. This feature not only saves time by enabling quick access to preferred

labs but also facilitates a more efficient learning experience, allowing you to focus on the specific content that matters the most to you.

- **Improved search and indexing**—Our search algorithms have been optimized to ensure that you find what you're looking for faster and more accurately. This means that when you search for labs, the search results will be more relevant and precisely match your query.
- **Introducing dark theme**—You can now switch between dark and light themes based on your user preference. The dark theme can be particularly beneficial for reducing eye strain in low-light environments and creating a more comfortable viewing experience for those who prefer darker interfaces.
- **Accessibility support**—The modern user interface brings improved accessibility features, ensuring an inclusive learning environment for all.

Why the change?

The changes were implemented to improve the overall user experience, with a focus on enhancing the learning process. By making the interface more intuitive and interactive, the goal is to facilitate seamless navigation through various labs and resources. The new design is intended to help you navigate through different labs efficiently, allowing you to focus more on learning and less on figuring out how to get things to work.

How to enable it?

The new interface is available for all users through the following link: https://www.labplatform.vmware.com/<org_name>. Here the "`org_name`" represents the distinct name of your VMware Lab Platform organization (account) name. For example, if your VMware Lab Platform organization name is HOL, the link to the new user interface will be https://www.labplatform.vmware.com/HOL.

To make the new interface your default interface, contact the VMware Lab Platform support team. For more details on how to get started with the new user interface, see the VMware Lab Platform User Interface Guide.

What's New July 17, 2023

Updated on July 17, 2023

- **Administrators can create, enable, and update cloud system credentials**

  As an administrator for a cloud system with bring your cloud, you can now update cloud system credentials. VMware Lab Platform uses the system credentials for shadow creation in a multi Organization's Virtual Data Center (OvDC) architecture of vCD-based clouds. Earlier, if a credential expired, it would auto-disable after a few unsuccessful logins, and the administrator did not have access to enable it again. Now administrators can update the password and enable it.

- **View-column icon is removed from all listing pages in the administrator user interface**

  The view-column icon is removed from all the listing pages. We have added a more intuitive name, **Manage Column**, for your convenience. You use the **Manage Column** button to view all the available parameters on the listed page.

- **Improved load balancing between the clouds when using the APIs**

  In the case of a multiple Organization Virtual Data Center (OvDC) with different locations, VMware Lab Platform avoids lab assignment to a cloud that is out of capacity or at the threshold. This is applicable only when the lab deployments or prepopulated labs are created with the `/api/pool/prepops` API call.
- **Delete option added for email templates**

The option to delete email template is added in the **Portal Settings>Email Templates** setting. To view the delete option for an email template, click the ellipses for an email template.

- **Periodic vulnerabilities fixes**

  We run security scans and vulnerability fixes periodically on our administrator interface, user interface, and clouds to ensure that your data and transactions are safe and less vulnerable to malicious attacks. We have addressed vulnerabilities that affected the VMware Lab Platform user interfaces and other services that were observed over the last few weeks. We value your privacy and security and constantly strive to improve our services.

- **Accessibility improvements for the administrator user interface**:
  - Added user interface labels to the Classes Resource management fields.
  - Support to add dates in all date forms using the keyboard.
  - Aligned check boxes on the Deployment Pool page.

What's New May 18, 2023

Updated on May 18, 2023

Administrator user interface enhancements:

- **Aggregated stats**

  We have added a new dashboard where you can see the total number of users, labs delivered, and lab minutes for a duration of time period. Use the **Aggregated Statistics** widget in the Dashboard to populate the required data. This data can help you recognize the trends for labs, and lab deployments, and you can make informed decision based on the data.

- **Support for Docebo LMS**

  We are thrilled to announce that VMware Lab Platform now supports Docebo Learning Management System (LMS) integration. This means that you can easily create and manage virtual labs for your online courses and users using the Docebo's powerful LMS.

- **Support for tracking ID for Classes**

  Now you can track entitlements for classes using the tracking ID feature in the Classes wizard. For large events, or multiple classes with same entitlements, you can track the entitlements based on the tracking ID associated with the class. To use the tracking ID feature, activate the **Portal Settings>Configuration> Enable Tracking Code for Classes** setting. Use the Classes wizard to add a tracking code for a class.

- **Improvements are made to the custom email templates**

  We have modified the custom email template body text to support HTML type email rendition. To view the custom email templates, click **Portal Settings>Configuration>Email**.

User accessibility improvements:

- VMware is committed to provide a good user experience to all its users. To support this, we have made some site accessibility improvements to make your browsing experience more convenient and user friendly.

We have made the following improvements:

- Changed colors in all the interfaces to meet the recommended color contrast ratio
- Provided better focus management for drop-down list items
- Added notification for deployment pool grids
- Added labels to all form elements
- Improved the grid layout for better alignment and spacing

We have also added warning and error messages for invalid inputs, and keyboard shortcuts for common actions. These changes will help users with different needs and preferences to access our site more easily and efficiently.

What's New April 13, 2023

Updated on April 13, 2023

- **Support for rate limit and captcha on login is added**

  Two new options, **Automatic Account Lockout** and **Login Rate Limiting**, are added to prevent a malicious user from trying many different passwords to login to an account.

  The **Automatic Account Lockout** option locks an account if too many failed login attempts occur. The lockout can be temporary or permanent. The **Login Rate Limiting** option is similar to the **Automatic Account Lockout** and can be used along with the **Automatic Account Lockout** option. The **Login Rate Limiting** option is used to throttle too many login attempts that occur with in a specified time period.

  For more information on how to enable these options, see Modify Password Policy for a Local User.

- **Broadcast user message**

  In the end-user interface, the VMware Lab Platform administrators can broadcast or schedule to send a custom message to all or selected logged-in users. This is great for sending important announcements, reminders, tips, or feedback. The custom message appears as a pop-up notification on the user's screen, and they have to acknowledge the message notification on the system.

  For more information on how to send custom user message, see Create Custom Notification Message.

- **User accessibility improvements**

  We have made some site accessibility improvements to make your browsing experience more convenient and user friendly. We have fixed some issues with focus management, added labels to all form elements, and improved the grid layout for better alignment and spacing. We have also added warning and error messages for invalid inputs, and keyboard shortcuts for common actions. These changes help you with different needs and preferences to access our site more easily and efficiently.

- **Improved class experience**

  - **Better Zoom integration in Classes**

    The Zoom integration in the **Classes** wizard is improved. The **Classes** feature supports Zoom-encoded passwords, so users can click the Zoom link associated with the class.

  - **Get entitlement clone notifications**

Now you get a notification when an entitlement clone operation is completed. You can easily track the progress of your cloning requests and know when it is completed. The **Entitlement clone** operation is used to clone an entitlement with another student, when the entitlement is in bad state or is lagging behind in the class.

- **Support for SSO login for self registered class**

  Now, the self registration feature for a class supports the SSO login for students. Students can use the self registration link and sign up or login to the class with the configured SSO authentication method. If the students login with a different provider, they will be redirected back to the original login URL they were on, which in this case would be the self-register page.

- **Provision of prepops with multiple labs in a class is disabled**

  If you select multiple labs in a class, the **Pre-Provision Labs** count option is grayed out and you won't be able to pre-provision them.

  For more information on how to create a class and provision it, see Create a Class.

- **Warning notifications for classes with missing labs or deployment pools**

  For more information, see Create a Class.

- **Addressed vulnerabilities in the administrator user interface and end-user interface**

  We run security scans and vulnerability fixes periodically on our administrator interface, user interface, and clouds to ensure that your data and transactions are safe and less vulnerable to malicious attacks. We value your privacy and security and constantly strive to improve our services.

What's New February 9, 2023

Updated on February 9, 2023

- **Support for multiple labs in a single class**

  This feature allows you to have multiple labs for a class. All students of a class can have multiple entitlements based on the number of labs in a single class.

  For more information on how to create a class for multiple labs, see Create a Class.

- **Granular permissions are live for everyone**

  Now, you can create user accounts and provide granular access to users based on your organization's requirements. You can either use the in-built user roles or create custom user roles and define the access permissions for the role.

  For more information on how to create custom roles and permissions, see Working with Roles and Permission.

- **Lab Builder improvements**:

  - Role parameter in AWS S3 import and AWS S3 export is optional

The **AWS Role ARN** parameter is optional, instead you can use the **Session Token** parameter to import or export files for the AWS S3 instance.

For more information on how to import or export files to AWS S3, see Export Files to an AWS Instance.

- Import to S3 option moved to Lab Builder page

  The **Import to S3** menu option is moved from the **Lab Templates** page to the **Lab Builder** page. To use the **Import to S3** option, navigate to **Lab Management> › Lab Builder** and click the three ellipses on the top-right of the page, and select **Import from S3**.

- Performance improvement in deleting lab template drafts and clones

  We have enhanced the back-end system to provide a better user experience while deleting a lab template draft or a clone.

- **Simplified the Create User Entitlement user interface page**

  The **Create User Entitlement** page is revamped and the process of creating a user entitlement is simplified. Now the user interface is more intuitive and provides a seamless experience to users.

  For more information on user entitlement, see Create an Entitlement for a New User.

- **Accessibility improvements for the administrator user interface**

  To provide equal access and equal opportunity to use the VMware Lab Platform interfaces, we have made improvements to the administrator user interface such as alternative (ALT) text for all icons and images. Support to add user inputs using a keyboard is added in the administrator user interface.

- **Blur student console when blank**

  Now instructors can blank the screen to make changes in a student console during a live class. The student screen is blurred during the interaction so that students cannot see the changes that an instructor makes in the console. This feature is helpful especially when an instructor assigns tests, grades, or exercises to the students.

What's New in 2022

What's New December 15, 2022

Updated on December 15, 2022

- **Lab Builder enhancements**
  - We have made significant enhancements in the Lab Builder to improve the user experience for creating and removing VMs while editing a lab template. For more information on how to create or remove a VM, see the Add a Virtual Machine for a Lab Template topic.
  - The remove draft labs option in the Lab Builder Drafts page is enhanced to improve the removing draft labs operation. For more information on how to remove draft labs, see the Create a Draft Lab topic.
  - Improvements are made to the Add VM page in the Lab Builder section. When there are no VMs to add in the lab template, the **Add VM** button is grayed out. When there are multiple VMs listed, you can easily remove them using the **Remove** option on the page. For more information on new enhancements, see the Remove a Virtual Machine from a Lab Template topic.
- **Assign classes based on cloud location**

For accounts with lab templates deployed from multiple cloud locations, you can set the preferred cloud location to deploy the lab templates in the Classes wizard. The **Show preferred cloud location dropdown in the classes wizard** setting option is added to allow administrators to set the cloud location for a lab deployment. For more information on the cloud location option in the Classes wizard, see the Create a Class topic.

- **Added Filter option to the Labs Catalogs page**

  Similar to all the other listing pages in VMware Lab Platform, the filter option is added to the Labs Catalogs page. Click the filter icon on the top of the Labs Catalogs page to set your filter criteria for search.

- **Added permissions for Deployment Pool**

  Deployment Pool permission setting is added under the Roles & Permissions page. Administrators can enable and customize Deployment Pool permissions for custom roles. To view the permission categories, navigate to **Users & Groups>Roles & Permissions>Create New>Deployment Pools**.

- **Feedback icon**

  The  feedback icon is now available on the top header of the application. You can submit feature requests or provide feedback about the application using the feedback form. For more information on the feedback forms, see the Working with Feedback Forms section.

- **Help Center panel is revamped**

  The Help Center panel in the administrator user interface is revamped for ease of use and is more intuitive. Click the  question icon to open the Help Center panel. You can view resources such as product tutorials, knowledge-base articles, documentation, and product videos. For more information on Help Center panel, see the Working with Help Center topic.

What's New November 14, 2022

The following updates are included in this release:

- **Improved granular permission**

  We are thrilled to announce that we have made enhancements to user account permissions. Now, you can create user accounts and provide granular access to users based on your organization's requirements. You can either use the in-built user roles or create custom user roles and define the access permission for the role. To enable the granular permission setting, contact the VMware Lab Platform support team.

  For more information on granular permissions, see the Working with Roles and Permissions topic.

- **New side menu navigation panel**

  The option to switch to the old side menu navigation panel is removed in the administrator user interface. By default, all VMware Lab Platform (tenant) accounts will have the new side menu navigation panel. The new side menu navigation panel is better organized, and intuitive, and provides a better user experience.

- **Full-screen scrollable forms**

  In the administrator user interface, all forms and pages, such as create a lab, create a class, and so on open in full-screen page view, and the pages are scrollable. The full screen design provides the same user experience on both the desktop and mobile device, and across different browsers.

- **Auto synchronization of lab templates**

  If a draft template or a clone lab template is deleted from the VMware Cloud Director, the system performs auto synchronization and deletes such lab templates from the VMware Lab Platform.

- **Improved error handling**

  - We have improved the error handling for uploads, lab deployments, cloud tasks and so on.
  - Improvements are made to the email and in-app notification messages.

- **New features in lab manuals**

  If you create lab manuals in ScreenSteps and import the manuals to the VMware Lab Platform, we have added lots of new formatting features around it. Now, the VMware Lab Platform supports the following content formats of ScreenSteps:

  - Collapsible sections
  - Checklists
  - Block styles and display
  - Alt image as captions

  For more information on ScreenSteps in VMware Lab Platform, see the VMware Lab Platform Content Engine guide.

- **Support for help requests on proctor application**

  You can configure the Slack messaging application with the VMware Proctor application to receive help requests from users. For more information, see the VMware Lab Platform Event Management guide.

- **Fix non-ANSI US keyboard layouts**

  The non-ANSI US keyboard layout feature remaps the keys pressed on your keyboard sends the US ANSI key code equivalents to the console. The feature was broken due to a change, and is now restored.

- **Default setting changes**

  By default, the following menu options are deactivated (hidden from the new side menu navigation panel). You can enable these options from the **Portal Settings>Configuration>Side Menu** option.

  - **Analytics**
  - **Help Desk Integrations**
  - **API Domain Allowlist**
  - **Deployment Pools** (for organizations with single cloud)

    For more information on the Portal Settings, see the Working with Settings topic.

- **Wifi icon in the Lab Builder to indicate the internet status for a VM**

  The **Portal Settings>Configuration>Lab Builder>Show a WiFi icon** is enabled by default. When you create a lab template in the Lab Builder, a wifi icon is displayed on the VM to inform you that the VM has internet access.

What's New September 21, 2022

The following updates are included in this release:

- **Single Sign-On (SSO) login option for the end-user interface**

  Similar to the administrator user interface, support for multiple authentication providers is added to the end-user interface. The **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options are added in the administrator user interface to provide different authentication providers to users.

  - **Allow Student SSO Login**: This option lets administrators to provide a non-local authentication provider to users to log in to the end-user interface.
  - **Disable Student Local Provider Login**: This option lets administrators to deactivate a non-local authentication provider option from the end-user interface login page.

  With the **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options, you will have the following four scenarios for the authentication provider setting in the end-user interface:

  - VMware Lab Platform login only: Deactivate both the **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options.
  - SSO only login: Activate both the **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options.
  - SSO and VMware Lab Platform login: Activate the **Allow Student SSO Login** option, and deactivate the **Disable Student Local Provider Login** setting option.
  - No direct login: Deactivate the **Allow Student SSO Login** option, and activate the **Disable Student Local Provider Login** setting option.

  For more information on how to add and configure an authentication provider for the end-user interface, see Working with Authentication Provider.
- **Improved granular permissions for user role**

  Set, create, and edit user roles. With the new granular permission feature setting, administrators can create custom roles and customize the permissions for the custom roles based on their organization requirements. This feature lets administrators to mix and match permissions from in-built roles and create custom roles within their organization.

  We invite you to try this compelling feature and give us your valuable feedback. To activate this feature, please contact the VMware Lab Platform support team. For more information on granular permissions, see Working with Roles and Permissions.
- **Detailed entitlement event history information**

  The Entitlement Event History page for an entitlement provides a detailed event history for each stage of an entitlement, including time stamps. This provides administrators a more comprehensive view when troubleshooting issues that affect multiple students, or when administrators need specific information for a support request. To view the Entitlement Event History option, navigate to **Entitlements>User Entitlements**, click on the three ellipses for an entitlement, and click **Event Entitlement History** from the list of options.
- **Enhancements in the Lab Builder:**

  The following enhancements are made in the Lab Builder for better user experience:

  - **Ability to upgrade VM hardware version from the Lab Builder**: Administrators can easily upgrade the VM hardware version from the Lab Builder menu. To upgrade the hardware version for a VM, open the VM console in the Lab Builder, and click on the **Upgrade VM hardware version** icon.

- **Access cloud network information from the Lab Builder**: You can use the **Cloud Network** tab added under the **Lab Builder>VM Networking** menu to open the Cloud Network page, and easily apply or reset tags associated with the cloud.
- Improved Hard Disk support: All bus types for a hard disk are supported. Now, you can select and delete multiple hard disks from the Hard Disk list with a single batch action. To remove multiple hard disks from a lab template, select the hard disks and click the **Delete** icon.
- Back-end improvements are made to the process of uploading a VM or ISO file using the **Upload VMs/ISOs** option in the Lab Builder.

What's New August 17, 2022

The following updates are included in this release:

- **Option to never expire a password**

  For the password expiry policy feature, we received several feedback, to never expire password for certain accounts which are used for automation. To address this requirement, we have added the **Never Expire Password** feature in the **Users & Groups** menu. To configure the Never Expire Password feature for a user account, or multiple users, navigate to **Users & Groups>User Accounts**, click the three ellipses and select the Never Expire Password option for a user. You can also apply this option for bulk users from the **Batch Action** menu option. Alternatively, you can also use the Accounts API to set the **Never Expire Password** for user accounts. To download the latest version of the VMware Lab Platform API Guide, on the VMware Lab Platform Administrator page, select **Support>Rest API Guide**.

  Points to consider before you change a password expiry policy:

  - If a password for an account is changed, the **Never Expire Password** action must be performed again for that account.
  - For user accounts with the Local Provider authentication, you can set the **Password Expiration** option to 0 days in the **Authentication Providers** page. This deactivates the expiration check during login and allows users to login even with an expired password.
  - The default password expiration value is increased from 180 to 720 days in the tool.
- **Enhancements in Lab Builder**

  Following enhancements are made in the Lab Builder for better user experience:

  - Now you can view the internet access and availability for individual VMs for your labs from Lab Builder. WiFi visual icon is added to lab templates when you edit a lab template. To add the Wifi icon for your lab templates, activate the **Portal Settings>Configuration>Lab Builder>Show a WiFi icon in Lab Builder to indicate VM Internet Status** option.
  - You can power on and power off individual VMs in a lab.
  - Various bug fixes and error handling improvements are made in listing, uploading, and networking tasks.
- **Import and export lab templates to Amazon Web Services (AWS)**

  If you have your own S3 instance and want to back up your lab template, or import an existing lab template to VMware Lab Platform, you can do that with the AWS S3 integration with VMware Lab Platform.

  To import your S3 instance:

  1. Navigate to **Portal Settings>Configurations**, and select **AWS S3 Integration**
  2. Add the required details.
  3. Activate the **awsBucketEnabled** option.
  4. To view the Export to S3 option for a lab, navigate to **Lab Management>Lab Templates** and, click on the three ellipses next to the lab.

  Note:

Only the OVF file format is supported, and the OVA file format is not supported. If you have an OVA file, you must first extract it in the AWS Bucket. The export and import action can also be performed using the VMware Lab Platform APIs. To download the latest version of the VMware Lab Platform API Guide, on the VMware Lab Platform Administrator page, select **Support>Rest API Guide**.

- **Improvements in user interface**

  We have made several improvements in the VMware Lab Platform user interface for better error handling, logging and audit history, upload process, and sync process.

What's New June 23, 2022

The following new improvements are included in this update:

- **Support for German keyboard characters**

  We have added support for German keyboard character inputs to student VM console. Keys with AltGR modifier are now supported for console input.
- **Drag and drop ISO files in Lab Builder**

  We have added a feature to drag and drop VMs/ISOs to Lab Builder.

  To do this:

  1. Navigate to **Lab Builder** and select **Upload VM/ISO's**.
  2. In the Uploads section, you can drag and drop a VM, ISO, OVA, and OVF files.

- **Support for vCloud Director version 10.4**

  VMware Lab Platform now supports vCloud Director version 10.4 (with vCloud API version 34).
- **SaaS Console option added in the new Lab wizard**

  The SaaS Console option is added in the new Lab wizard. To create or add a SaaS console to VMware Lab Platform, navigate to **Lab Management>SaaS Consoles>Create New**. SaaS Consoles provide a simulated lab experience to users in the user interface. You can use third party applications such as Walnut to create your simulated labs and add them to VMware Lab Platform.
- **Show hidden VMs to instructors**

  We have added a feature to hide one or more VMs from students and make them visible only to instructors.

  To use this feature:

  1. Enable the **Portal>Configuration>Show all VMs in instructor interface** setting to view all the VMs.
  2. Navigate to **Deployment Pool>Update>Edit VM's** and choose the VM that needs to be visible/invisible to students. This setting is also available in **Labs>Update>VM Configuration**.

- **Updated Tutorial section**

  The following new tutorials are added under the **Help Center>Tutorials** section:

  - Working with User Accounts
  - Working with Email Templates
  - Working with Content Engine
  - Uploading a Template or a Media
  - Working with Classes
- **Lab Tags in transcripts records**

Tags associated with labs are now included in transcripts records. You can use this feature to manage and differentiate user data for better querying through the user interface and API calls.

- **Find and delete non-VMware Lab Platform cloud vApps in a cloud**

  When you perform the Sync Cloud vApps operation on the Cloud page in the Administrator user interface, a list of non-VMware Lab Platform vApps is displayed. You can select all vApps or multiple vApps, and delete it.
- **CC multiple support email address**

  This feature enables you to CC all the class related email notifications to multiple email addresses. Administrators can add multiple email addresses to get all the class related notifications from **Portal Settings>Configuration>Add support email in cc in class emails**.

What's New May 17, 2022

The following new features and improvements are included in this update:

- **Rebranded product name**

  VMware Learning Platform is rebranded as **VMware Lab Platform**. All VMware Lab Platform interfaces, such as administrator user interface, user interface, instructor interface, Content Engine interface, and product pages, product documentation, logo, ticketing tools, and all product communication channels are renamed to VMware Lab Platform.
- **Introducing all new Event Management user interface**

  The Event Management user interface is now redesigned and enhanced for better user experience and the interface is consolidated in the VMware Lab Platform administrator user interface. The Event Management user interface allows administrators, proctors, and technical support staff to manage lab station registration, seating maps, and help requests for virtual events and physical events. Administrators can define and schedule multiple events per VMware Lab Platform account. The Event Management feature is widely used by VMware for VMware Explore (formerly known as VMworld) and other fortune IT companies for their world events.

  See *VMware Lab Administrator Guide*.
- **Show and hide console connection signal strength in user interface**

  Now administrators can hide the console connection strength bars in the user interface, and monitor the connection strength from the turned on console in the administrator user interface.

  To use this feature, in the administrator user interface, enable the **Show console connection strength** Portal setting.
- **Insert and eject media**

  Insert and eject media are now two separate actions in the Lab Builder console. This avoids confusion and provides better user experience for lab creators.
- **Updated Zoom integration**

  The built-in Zoom application in the **Classes** menu is updated to latest version.
- **New filters in transcripts record**

  New filters such as class name, instructor assigned, and company name are added in the transcript records for better sorting and filtering data.

What's New April 20, 2022

The following new improvements are included in this update:

- **Ability to port user authentication from one provider to another**

VMware Lab Platform provides support for multiple authentication providers for users, and enables users to login through different authentication providers. Administrators can change the user authentication provider from one provider to another with just a click from the administrator user interface. To change the authentication provide for a user, click **Users & Groups>User Accounts**.On the User Account page, click on the three ellipses next to the username, and click **Migrate to a different provider**. Enter the authentication provider details and update the page.

- **New password policy for local users**

  Passwords are one of the key essentials to protect a user account and user details. To ensure that user accounts are secure, VMware Lab Platform is enabling administrators to have more control over the password policy for its local users. Navigate to **Users & Groups>Authentication Providers** and click on **Local Authentication Provider**. In the Authentication Provider page, you can make the changes as per your requirements.

  - Strong passwords (passwords must have at least 1 uppercase alphabet, 1 numeric value, and 1 special symbol. This is not enabled by default).
  - Password length (minimum password length must be 8 characters).
  - Password expiration (number of days before a password expires and a password change is required. Default value is 180 days).
  - Previous password (number of previous passwords that users cannot use as their new password. Default value is 0).

    The default values are implemented for all the current users and will be effective once the user changes the password, or when the user's password expires after 180 days. This change will not affect the currently logged in users.

    Note: The changes are not applicable to VMware Cloud service logins or any other authentication provider logins.

- **Support for custom permissions in CSV user uploads**

  We had customized permissions in our previous release, we now support CSV upload for user creation with custom permissions. Contact the VMware Lab Platform support team to enable granular or custom permissions for your account.

- **Enhancements in Lab Builder**

  - Administrators can now edit or create new NAT rules in Lab Builder while editing a template. This is useful when you need a special port to be enabled. This is applicable for both NSX-T and NSX-V based clouds.

    1. To do this, navigate to **Lab Management>Lab Builder**.
    2. Select an unpublished lab template. and from the Actions menu, click **Edit**.
    3. On the Edit Lab Template page, click **Lab Networking** tab**.**
    4. On the Lab Network page the network diagram is displayed. Click on the lab network in the network diagram.
    5. In the right pane, network details are listed. Click on the three ellipses, and click configure services.
    6. Click on the **NAT** tab and add a rule based on your requirements.

  - Fixed “Unknown” import status message while importing a lab template.
  - Fixed virtual machine (VM) console access issues.
  - Improved services like template replication, network tagging, and user experience.

- **Warning for insufficient cloud resources in class creation**

  Warning messages for insufficient resources for a class are added. The warning forecast helps administrators to allocate resources or modify the class settings. After creating a class, you can see the forecast of resources for a class based on number of students added to the class. If there is shortage of resources for the class, a warning message is auto populated. The warning message is added in the end of class creation wizard and the Classes list page.

- **Access Content Engine user interface from Admin user interface**

  The Content Engine user interface, and content manual design and building process is now merged within the Admin user interface. Previously, administrators and content creators had to navigate to separate links to access the Content Engine user interface. Also, the error handling of corrupted file upload for manuals is improved for a seamless user experience.

What's New February 10, 2022

The following new improvements are included in this update:

- **Granular permissions for roles (Beta release)**

  Set, create, and edit user roles. With the new granular permission feature setting, administrators can create custom roles and customize the permissions for the custom roles based on their organization requirements. This will enable administrators to mix and match permissions from in-built roles and create custom roles within their organization.

  We invite you to try this compelling feature and give us your valuable feedback. To try this beta feature, please contact the VMware Lab Platform support team.
- **Monitoring Lab Builder template replication from API**

  APIs for Lab Builder template replication task are updated. The new API's include; listing of template replication tasks, creating a new replication, tracking progress of template replication, and stopping template replication.

  For information about the APIs, refer the VMware Lab Platform REST API Guide from the Help Center panel.
- **Improved user experience for VMware Login**

  When you have an active login session of a VMware Lab Platform tenant (organization) and attempt to login to another org account, instead of throwing an error, the system displays a message to log out from the current org and login to the other org. You cannot login to two orgs simultaneously in the same login session.
- **Always detect VM power state**

  In certain use cases, when users power off VMs within a guest OS or execute the power off command in the user console, VMware Lab Platform detects the state of the VM and provides an option to power on the VM.
- **Get alerts for nearly exhausted cloud networks IP pools**

  Network IP pool shortage issues are always frustrating, especially when you want to add more labs for deployment. To address this issue, VMware Lab Platform now monitors network IP pools and alerts you when it is about to exhaust. Notifications are presented in the notification bar and can be subscribed as an email alert with **Category: Cloud and Severity: Medium**. Administrators can view and receive real-time notification alerts from the system. The alerts are categorized based on the severity and priority of the issue as High, Medium, and Low. To view all notification alerts for an account, click the bell icon on the top of the page header. You can view the list of subscribed notifications from the Notifications menu option. You can create subscriptions for notifications and customize them as per your requirements.

  For information about notifications, see the *VMware Lab Platform Administrator Guide*.
- **Enhancement to In-product Help Center**

  The VMware Lab Platform Help Centre panel provides in-product support experience. Our support experience is constantly evolving with new features being rolled out for our services. Currently, you might view some or all the following features in the Help Center panel.

  - New videos
  - Tutorials – Our tutorials are designed to assist new users, or current users who need help, with completing tasks in VMware Lab Platform.
  - FAQ’s
  - Status page,

  - REST API guides etc.
  - Feedback page
    We will keep adding more content here to improve the service.

What's New in 2021

What's New December 16, 2021

The following new improvements are included in this update:

- **Password reset option**

  Administrators, instructors and content creators can now reset their passwords using the "Forgot Password" option from the Administrator login page. Currently only local user accounts password reset is supported. The VMware Cloud Services Provider and SSO login password reset options are not supported. By default, the feature is enabled for all VMware Lab Platform organization accounts that don't have an alternative authentication provider.
- **Create anonymous users for one-time lab use**

  Administrators can create temporary, anonymous user entitlements for one time lab use, or for a class. Once the entitlement is completed, the user account is deleted from the system. This feature is useful for administrators who create content for quick demos or live events to demonstrate their products to users who might not use the system regularly. Anonymous users can be created using the **Users & Groups>User Accounts>Create New> Bulk Upload** option or with the Account API call. Administrators can share the quick launch URL for the lab with the user.

  See Create an Anonymous User.
- **Create Account API**

  Added the `anonymouseUser` and `providerName` API parameters to create anonymous user and create a user with existing authentication provider.
- **Add custom email templates for a class**

  To create and add custom email templates for a class, administrators can use the Email Templates option and add the custom emails from the Create a Class page. To see the menus in the class creation wizard, you need to enable the **Settings> Class and Instructor>Send student class assigned email from email template** and **Send instructor class assigned email from email template** options.

  See Create a Class.
- **Add PDF manuals from Content Engine**

  Support to add PDF manuals for a lab from Content Engine user interface is added. Earlier, instructors and content creators had to add PDF manuals as a lab resource in a lab. Now, you can directly add PDF manuals to a lab from the Manuals drop-down menu.

  For more information, see the *VMware Lab Platform Content Engine Guide*.
- **Help Center panel**

  With VMware Lab Platform in-product **Help Center** support experience, you can view help content, tutorial guides to help you perform your tasks, search for answers to your questions, and for those times when you want to contact a member of our customer support team. Click the help icon to open the **Help Center** panel and view the Help Center panel options; Getting Started guide, tutorials, videos, what's new in VMware Lab Platform, FAQs, and the Help and Support feature.

  See Working with Help Center.
- **Add personalized support site**

Now **Support>Get Support** menu for instructors can be configured with a custom site or email address. To enable it, navigate to **Portal Settings>Configuration>Get Support (Instructor)** and add a site or email address in the **Get Support (Instructor)** field. If left empty, all support requests will redirect to the default VMware support site.

You can also modify the support link for your VMware Lab Platform organization administrators by contacting the VMware Lab Platform customer support team.

What's New November 17, 2021

The following new improvements are included in this update:

New side menu navigation

The side menu navigation panel is restructured to provide better user experience. The menu options are grouped in logical order based on the different user roles for VMware Lab Platform. Menu labels/names are changed to make it more relevant and intuitive to the user.

To enable the new side menu navigation, contact the support team.

Full page wizard

All the menu pages such as Create a lab, Create a class, now provide a full page wizard view. This helps the users to easily navigate across different work flows/tasks. Users can easily exit a task mid way or resume tasks by selecting the options from the full page wizard.

Bulk Upload, Remove, and Download options

In the User Account page, the **Bulk Upload** option is moved under the **Create New** drop-down list. Access the **Bulk Remove** and **Bulk Download** options from the three ellipses drop-down list available on the top of the page.

Multiple connections to VMs from a single lab

If a lab has multiple VMs associated with it, end-users can open the VMs in separate tabs to access each of the VM consoles in the end-user interface.

Create a lab template from an entitlement

You can create a lab template from an entitlement by using the **Capture Template** action menu. This captures the contents of the lab template and creates a new lab template in the Lab Builder page. This feature is useful in scenarios where administrators and content creators want their own version of the lab templates for the same lab.

View the cloud org virtual data center associated for a lab, from the Class tab

Administrators can view the cloud org virtual data center of a user from the **Class >Student** setting. Use the view-columns menu on the page to add the **Cloud Org Vdc** option on the listing page.

Create and enable verified accounts

Administrators can now verify an unverified account from the Administrator user interface. Navigate to **Users & Groups>User Accounts** listing page, click on the username and mark the account as verified in the **User Access** tab.

Recreate entitlements from a class

Added support to recreate entitlements from the **Classes** menu. If a student ends an entitlement before the class ends and wants to rejoin the class or there are updates, changes made to the class, administrators can recreate the entitlement for the student without creating a new entitlement for the student. To recreate an entitlement for a user, on the Classes page, select the student and click the three ellipses next to student name and select the option **Recreate Completed Entitlements**.

Note: You can recreate entitlements only for the completed/ended entitlements.

CPU/Memory/Storage columns added in download entitlement statistics

The “Download Entitlement Statistics” report from the user dashboard now provides the CPU, memory and storage utilization for all entitlements.

Lab Builder improvements

- Improved error handling mechanism is added when a lab template is "busy" in vCD and users try to create a lab replica of the lab template in VMware Lab platform. Now, you cannot create the replica and a system message is displayed.
- All the menu options and listing in **Lab Builder** and **Edit Lab Template** pages are moved to the left-side of the page. With this change, users will have more workspace area in the center and a better user experience.
- You can view and edit multiple VMs parallelly in **Lab Builder** without impacting the performance or powering off the VMs. Now, you can undock the VMs from the main page and open the VMs in separate tab.
- **Power on and force customization** option is added on top of VM consoles in the **Edit Lab Template** page, which will force and apply the changes enabled in **Guest Customization** page.
- The template network connection type is enhanced to provide options to connect the network over the internet from VMware Lab Platform. The following options are available for a network connection:
  - none
  - isolated (no internet)
  - default (with internet)
  - mixed

What's New October 14, 2021

The following new improvements are included in this update:

- **Network diagram for lab templates**

  You can view the network configuration associated with a lab template and project it in the form of a diagram. This feature is useful for lab creators to identify the network complexities, metadata, and configurations associated with a lab template without deploying a template instance.

  See Lab Builder Action Operations.
- **Hide time remain when lab is part of a class**
  - We have added an option to make time allotted for an entitlement equal to the class duration and hide the remaining lab time. This will avoid confusion with lab timer and unnecessary lab extension requests for instructors.
  - To use this feature, enable the setting **Make Time Allotted for entitlement equal to class duration and hide remaining lab time**.

    See Working with Portal Settings.
- **Default columns**

  In all the menu listing pages, the column display is updated to show the most frequently used and relevant columns by default. To activate or deactivate a a column, click the View Column icon, and use the checkboxes to activate or deactivate a column. To see the default list, select **Restore Defaults**. See Customizing the Administrator User Interface.
- **Filter entitlements by a class**

  Entitlements can now be filtered by class names. This feature helps to differentiate and list entitlements belonging to different classes.

- **Copy URL in a class**

  In the Classes page, the **Instructor Launch URL** and **Registration URL** columns have the options to “copy to clipboard” and “open in a new tab”.
- **Change vApp template to lab template**

  As we work on simplification of the Administrator user interface, the vApp templates are now renamed as lab templates.

What's New September 20, 2021

The following new improvements are included in this update:

- **SSO authentication**

  VMware Lab Platform also supports user authentication from various third-party authentication providers. You can enable this for VMware Lab Platform user interface and administrator interface login. Some of the supported authentication providers are—Facebook, Google, VMware Workspace One and every OpenId provider.

  See Working with Authentication Provider.
- **Custom user messages**

  Administrators can push custom notification to all, or selected users logged in the end-user interface.

  Administrators can create a “user message” which will be pushed to all the logged in users for a specified time. You can broadcast the message to All Accounts/Logged in users/ Accounts in the lab console. To setup a user message, navigate to **Portal Settings>User Messages** and create a new user message.

  See Create Custom Notification Messages.
- **Variable tags in lab manual**

  The lab manual content creators can display some of the important information in lab manuals with help of variable tags. Lab manual content creators can create and inject certain data in lab manuals when a user takes a lab. The variables are injected when the step is first rendered and do not update dynamically if they are changed. Ensure that the variable tags are within double curly brackets {{}}.

  Some of the supported variable tags are:

  - lastName: **lastName**
  - firstName: **firstName**
  - labSku: **sku**
  - classId: **{{classId}}**
  - className: **{{className}}**
  - entitlementId: **entitlement-id**

    For more information, see Inserting Variables into a Manual.

- **Administrator account creation without VMware ID**

  Previously, a new administrator account was created only from the VMware Cloud services console. Now, new administrator accounts can be created from VMware Lab Platform.

  See Create a User Account.

What's New August 17, 2021

The following new improvements are included in this update:

- **Class**

  - Improved the class video conferencing feature to enable seamless integration of Zoom with a class. To enable this feature, contact the VMware Lab Platform support team.

    See Create a Class.
  - Administrators can now edit the start time of a started class. The new start time for a class will be effective from the following day.

    See Modify a Class.
- **Notification alerts for cloud resource usage threshold**

  Added notification alerts when cloud resource usage is at 90%. This will help administrators to avoid a production issue and will provide enough time for administrators to contact cloud administrators to increase or clean up the resources.

  See Create Notification Subscription.
- **Added Notification menu in the side navigation panel**

  We have added a new menu for notifications. On this page administrators can see all the notifications. You can perform various actions such as acknowledge, flag, and subscribe to notifications.

  See Working with Notification.
- **Parallelly power on VMs in Lab Builder**

  Previously, when you had multiple VM's in the Lab Builder editor, the VM power-on operation was sequential and time consuming. Now you can power on all the VM's parallelly, which will save time and you can work on one of the powered-on VM and reboot another VM at the same time.
- **User accounts**
  - Ability to modify usernames:

    The username field is an email address that cannot be modified. Now, modifications and corrections can be made in usernames.
  - Administrators can merge user accounts:

    We have added a feature to merge user accounts. This is useful when a user has changed their email address and wants their transcripts to be transferred to a new account.

    See Edit User Accounts.
- **Administrator user interface**

  Improved administrator user interface listing pages by providing a uniform view of titles, search bar, filters, sorting icons, bulk action options for all the listing pages.

What's New July 19, 2021

The following new improvements are included in this update:

- **Exam module**

  With the exam feature, administrators, instructors, proctors can grade entitlements (users) and provide certification on passing exams. You can define the exam in VMware Lab Platform and add grading options in the system. Administrators can customize email templates to send the exam report to the users.

  See Working with Exam Module.
- **Power on vApps from the entitlement list**

Support to power on a vApp of an entitlement from the VMware Lab Platform administrator interface is added. To power on a vApp assigned to an entitlement, click **Entitlements>User Entitlements**. Click on the three ellipses of an entitlement and select **Power on Cloud vApp**.

See Managing Lab Entitlements.

- **Simplified lab creation wizard**

  Create new labs with a new interface for simplified lab creation. A simple and easy-to-use lab creation form that lists only the required fields for a lab. This option is useful when administrators want to create a quick lab.

  See Working with Portal Settings.

What's New June 15, 2021

The following new improvements are included in this update:

- **Network visualization diagram**

  With the network diagram feature, administrators can create and modify lab templates with suitable network configurations across virtual machines. A network diagram provides a graphical view of virtual machines and networks in a vApp.

  See Edit a Lab Template with the Lab Builder.
- **Import template from any cloud**

  Lab creators can import a lab template from a cloud (vCloud Director) that is not registered in VMware Lab Platform.

  See Import Lab Template from Another Cloud.
- **Improved backup monitoring**

  Enhancement are made to improve the backup monitoring process with appropriate system messages.
  Several useful visualizations, alerts, messages, and tooltips during lab builder operations are added to enhance the user experience of VMware Lab Platform.
- **CC a support email address**

  This feature enables you to CC all the class related email notifications to a particular email address. Administrators can add an email address to get all the class related notifications from **Portal Settings>Configuration>Add support email in cc in class emails**.

  See Working with Settings.
- **Skip provisioning**

  We have added an option to skip provisioning of resources for classes. This option can be used when administrators want to save the resources and not assign it to students unless the lab is launched.
- **Improved notification experience**

  For better customer experience, the user interface elements of notification alerts are enhanced. The new **View all Notification** list page can be used to see all the alerts and perform actions like **Acknowledge**, **Acknowledge All**, **Mark as read**, **Flag** etc.

  See Working with Notification Alerts.
- **Power off vApps from the Entitlement list**

  Support to power off a vApp of an entitlement from the VMware Lab Platform admin interface is added. To power off a vApp assigned to an entitlement, click **Entitlements>user entitlements.** Click on three ellipses of an entitlement and select **Power off Cloud vApp**.

  See Managing Lab Entitlements.

- **Power reset**

  The power reset option resets the power cycles for all the VMs in a vApp, except for entitlements that have **Awaiting Grade** status.

  See Managing Lab Entitlements.
- **Quick load of listing in deployment pool**

  Added improvements to the **Deployment pool** list page, which helps in loading and editing of deployments quickly. The template selection list for a deployment pool is also improved with appropriate warning and listing.

For detailed information about the features, see the *VMware Lab Platform Administrators Guide*.

What's New May 13, 2021

The following new improvements are included in this update:

- **In-App notification for admin interface**:
  - Get easy access to all alerts and errors with the in-app notification feature in the admin interface. Real-time pop-ups of alerts can help you to fix the issues on the go.
  - Option to subscribe for alerts based on various categories and priority for effective monitoring of the platform.
- **File uploader**

  Upload a file (of any type) while you build or modify a template in Lab Builder. The uploaded files generate a temporary link that can be used to download the file within a VM to help you build a lab quickly.
- **Warning for powered on labs**

  While creating a class, if resource-saving option for a class is not enabled, the system displays a warning alert before you can publish the class. To save resources, by default the resource management setting is set to suspend the vApps of a class after 30 minutes, at the end of each day.
- **Duplicate a running class**

  Now you can duplicate a running class. This feature is useful when you have to duplicate an existing class for recurring instances. Administrators and instructors can duplicate a class from the **Class** list menu option.
- **Clone entitlements**

  Support to clone entitlements from an existing student's entitlement for another student is added. Suppose a student's lab is corrupted or the student is running behind the class schedule. Instead of starting an afresh lab, students' entitlement can be cloned with one of the existing student's entitlement. Administrators can clone entitlements from the **Entitlements>User Entitlements** list menu option.
- **Log in without a password**

  Administrators can enable pre-registered students or end-users to log in to VMware Lab Platform student interface without providing a password. This is useful for on-demand lab access during a large event, such as conferences. Administrators can enable this feature from **Portal Settings>Configuration>Events>Allow users to login without providing a password**.
- **Compact view for end-user interface**

  You can modify the student interface to be compact and unobtrusive and provide maximum screen space to work on the virtual machine. Administrators can enable this feature from **Portal Settings>Configuration>Console Interface>Compact Panel** option.
- **Customize the side menus**

  You can optimize the admin interface by hiding the less frequently used side navigation menu items. Administrators can show or hide the required menu items from the **Portal Settings>Configuration>Side menu** option.

What's New March 28, 2021

The following new features and improvements are included in this update:

- **Resource use and cost estimation**:
  - **Widget to view cost per lab per user**: The **Download Entitlement Statistics** widget in the **Dashboard** provides cost (in USD), compute, and storage utilization data for each entitlement.
  - **Entitlements**: Get resource usage and cost estimates of an entitlement in real-time. The **More Details** option in the **Entitlements** page provides the storage, compute units, and cost (in USD) data for an entitlement. You can also use the API to monitor the resource usage of an entitlement in real-time.
  - **Transcripts**: The **More Details** option in the **Transcripts** page displays the storage, compute units, and cost (in USD) inputs associated with a transcript.
  - **Class**:The**More Details** option in the **Class** page provides the storage, compute units, and cost (in USD) inputs associated with a class.

    This feature is currently available to all commercial customers with VMware Lab Platform owned Cloud.

    For more information or support, contact the VMware Lab Platform Support team.
- **Create lab from the Create Class menu**

  Now you can create a lab for a class from the **Create Class** menu. The **Create New Lab** is accessible from the **Create Class** menu. You can choose an existing lab from the list view, or from the **Recently created labs** view, or create a new lab from the **Create Class** menu.
- **Send email to students before or after starting a class**

  Instructors can send emails to students with a link to join a class before or after the class has started. To use this feature, on the **Class** page, click on the three ellipses next to a student's name and select **Send Notification Email**.
- **Quick launch emails**

  For existing entitlements or enrolments, administrators can send an email to user or student with a quick launch URL in one click.
- **Send keyboard shortcuts to console**

  In the end-user interface, you can send a prompt with several new frequently used keyboard shortcuts to the console from a student's keyboard. The new shortcut user interface replaces the **Ctrl+Alt+Del** button that is above the console. Enable this feature using the **Show Shortcut Keys** button in the **Portal Settings>Configuration>Lab Console** page.
- **Delete vApps in an error state**

  In a scenario, when a running vApp or a prepop fails to deploy because of an issue in the cloud, you can delete such vApps from the **Cloud Vapps** page. Enable the **Delete prepops in error state after specified hours** and **Delete vapps for entitlements in error state after specified hours** options in the **Portal Settings>Configuration** page. All vApps in error state are not billed.

  A running vApp is your lab instance deployed for a user (entitlement) or a prepop by VMware Lab Platform deployment pool using a lab template in a cloud.
- **Improved administrator user interface**

  Improved administrator interface for better user experience. With this enhancement, overlapping of tooltips issue is resolved and it provides for better scrollbar experience.
- **API guide under the Support menu**

  Administrators can access the VMware Lab Platform API guide from the **Support** menu.

- **New notification subscription page**: The severity levels in the email **Notification Subscription List** are represented as **Low**, **Medium**, and **High**. To subscribe for notifications, use the **Create Notification Subscription** option.

What's New February 19, 2021

The following new features and improvements are included in this update:

- **Backup lab template**

  With the **Backup** feature, an administrator can easily create a backup for a lab template. The **Backup** option displays the template file name, creation date, and the validity period for the backup. Click **Lab Builder**,and select the template for which you want to create a backup and click on the plus sign to add a backup. Use the **Content Migration** widget to view the status of the backup.
- **Import and upload virtual machines, ISOs or vAPPS**

  Import virtual machines (VMs) and media (ISOs) to your VMware vCloud Director catalog through the VMware Lab Platform **Admin** UI. OVF, OVA, VMKS, VMDK, MF, NVRAM, and ISO file types are supported. The import process is a two-phase process, where you first upload the file and then import it in the template. Click **Lab Builder**, and click the ellipsis next to the **New Lab Template** button. In the Upload VMs/ISOs window, click **Attach Files**, and upload the required files. The upload progress information is displayed. After the file is successfully uploaded, you can import it for a lab.
- **Copy and move templates**

  Copy and move lab templates from one catalog to another catalog in using the **Lab Builder** menu.
- **Compute and storage units information**

  Administrators can now view compute and storage unit information from the **Transcripts** menu for a lab or a user. Navigate to **Entitlements> Transcripts** and click on the ellipsis next to the transcript file to view the compute and storage details.
- **Enable management of user roles from VMware Lab Platform**

  VMware Cloud services users of VMware Lab Platform can use the Admin interface to manage all roles, except for the administrator role.
- **Support menu**

  For better support accessibility, the **Admin** interface is updated with hyperlinks for the Support site, Documentation, YouTube and Compatibility test results.

What's New in 2020

What's New October 14, 2020

The following new features and improvements are included in this update:

- **Introducing the billing dashboard**

  Keep track of current usage with the Billing Dashboard (**Reports>Billing**). Storage and compute utilization for labs are presented in an easy-to-read graphical representation.
- **Audit log**

  We are excited to announce a new log that keeps an audit trail for every administrator including Create, Update, and Delete actions. (**Reports>Audit History)**
- **Class resource capacity forecast**

Streamline resource planning with the **Upcoming Classes Resource Usage** widget. The widget provides a timeline graph of estimated resource usage for upcoming classes.

- **New email setting for class-related notification**

  Send customized VMware Lab Platform alerts to students and instructors when a new class is created.
- **Faster lab provision on reset of an existing entitlement**

  We have improved the algorithms to quickly provision a new vApp when an existing entitlement is reset.
- **Deployment-based on geo location (API)**

  Control where entitlements are deployed when you have multiple clouds.
- **Compact manual and console tab**

  Lab users have more options to choose on how the full-screen mode should be handled.
- **Intercept paste**

  Copy text from the clipboard and paste it directly to a VM console.
- **Fix non-ANSI US keyboard layouts**

  Remap the keys pressed on your keyboard to send US ANSI key code equivalents to the console.

What's New August 26, 2020

The following new features and improvements are included in this update:

- **Detailed entitlement history**

  More information is always better than not enough. Administrators can now see a detailed event history for each stage of an entitlement, including timestamps. This gives admins a more comprehensive view when troubleshooting issues that are affecting multiple students, or when providing more specific information for a request for support.
- **Lab Builder - running workload identification**

  After stability, cost-consciousness is at the forefront of every cloud user's mind. Keeping this in mind, users can quickly and easily identify which templates in the Lab Builder have a VM or draft powered on. If the dot next to the template name is green, it is running. If it’s gray, it’s powered off.
- **Improved LTI interface**

  LTI v1.1 and v1.3 tool settings have been consolidated into a single interface for easier navigation and integration between VMware Lab Platform and your LMS of choice.
- **Deployment delay notification**

  Know if your students are having deployment issues as soon as they do. Template deployment delay alerts are now included in the Notifications widget. You can read more in the Working with Widgets documentation.

What's New June 09, 2020

The following new features and improvements are included in this update:

- **Bitnami**

  A catalog of possibilities at your fingertips. Bitnami provides no-cost, pre-packaged templates for standard applications such as WordPress, Redmine, GitLab, MariaDB, and over 100 more. These templates can be used directly from the VMware Lab Platform to build, deploy and launch new labs quickly. Bitnami applications are always secure, up-to-date, and packaged according to industry best practices. Check out what we have to offer at the VMware Cloud Marketplace.
- **Suspend vApp**

If you do not use it, snooze it. The Suspend feature allows the administrator to hibernate a vApp upon exit, keeping its existing state in memory and ready to go after a lunch break or the next day of class while lowering compute costs for the time that it is not used. Less waiting and lower charges, the best of both worlds.

- **Console connection monitor**

  A number of improvements have been made to the Console Connection monitoring with respect to sensitivity, and the amount of information available to the instructor about the state of a student's connection to the console. If the connection is determined to be unstable from the student's side, this information will be conveyed to the instructor in the class overview page, or when viewing an individual student's console.
- **Class duplication**

  Saving time is always in everyone's best interest. That is why we have introduced the functionality to duplicate an existing class, targeted towards recurring classes. Navigate to the Classes section in the Admin UI. Choose a class from the list and click on the ellipsis on the left. Select Duplicate from the sub-menu and confirm. A window for class creation will appear and all fields should have the same values as the original class (except date and time zone fields).
- **Speed test improvements**

  Performance and latency issues are always aggravating, especially when it is difficult to determine where in the network path the issue resides. The VMware Lab Platform Speed Test will help students in determining where the issue is by looking at the connection to both the VMware Lab Platform servers and the Cloud where the lab is being hosted, which in turn will assist administrators and instructors to mitigate the issue as quickly and effectively as possible. The UI is improved making it easier to understand where the connectivity issue might lie, as you can run the test manually as opposed to only during the compatibility test before launching the lab. Lastly, the administrator can also specify which cloud to run the test against in a multi-cloud environment. For further information, see VMware Lab Platform Speed Test.
- **Instructor vApp cloning**

  The ability for an instructor to clone a vApp can now be restricted via a newly added **Configuration** setting.

What's New April 22, 2020

The following new features and improvements are included in this update:

- **Active lab template update**

  Lab (vApp) Templates can now be easily updated in-flight without any disruption to existing students (active entitlements) via Lab Builder. This functionality is very convenient in situations where an improvement is identified and introduced to a Lab template, but existing students are still active on the previous iteration of the lab. Once the lab template is updated and distributed, and new lab entitlements will be off and running on the newly released iteration.
- **Media management**

  Media uploaded through Lab Builder can be managed through a new menu in the UI called **Cloud Media** under **Cloud Management**, removing the need to manage it through vCloud Director.
- **Cloud catalog management**

  Cloud OvDC catalog administration can now be performed directly from the Admin UI. This provides the capability to create, update or delete catalogs, removing the need to manage it through vCloud Director. Navigate to **Cloud Management** > **Cloud OvDC Catalogs** and click on the ellipsis next to the catalog name for menu options.
- **Deactivate console controls**

  Administrators or class instructors can activate or deactivate controls visible in a student lab on a per lab basis based on case by case scenario. Navigate to **Portal Settings>Configuration** and search for **console header buttons**.
- **Student interface quick launch**

If you want to see how the changes made by you will impact the end user console, we have the quick launch capability in the Admin UI which leverages your existing credentials to save time while testing. The link is located at the very bottom of the main menu.

- **Class start time auto-calculation**

  When creating a class, the administrator or instructor can choose to set the start and end time in either their local time, or the time where the class will be taking place. Updating one automatically updates the other, so no more referring to a time zone calculator.
- **Deactivate on-screen manual**

  For those courses that do not require any on-screen instructions or additional materials for students to follow, administrators can now choose to deactivate the instructions panel to avoid confusion for the students and free up screen real estate. Navigate to **Portal Settings>Configuration** and search for **Hide Instructions panel when no manual.**
- **vCloud Director vApp identification**

  An administrator can easily identify a template that was created in vCloud Director as opposed to VMware Lab Platform Lab Builder. Development of this functionality is part of an ongoing process to consolidate management interfaces into a single pane of glass. In addition to identification, vCloud Director vApps can also be deleted from the platform, putting more aspects of capacity management at your fingertips. Navigate to **Cloud Management** > **Clouds** and select **Sync cloud vApps** from the ellipsis menu next to the relevant org.
- **API additions**

  Instructors can now be added, removed or changed in the same way as students via API.

What's New March 26, 2020

The following new features and improvements are included in this update:

- **PDF lab manuals**

  VMware Lab Platform now supports the PDF document format for student lab manuals. This feature enables content creators to use external authoring tools to develop lab manuals for use in VMware Lab Platform. In addition, it provides customers who have preexisting content an additional import option. Navigate to **Lab Management > Lab Resources > Create Lab Resource**.
- **Detailed billing**

  The new billing dashboard provides granular usage metering of on-demand service usage. This dashboard enables administrators to optimize usage of the service to help control cost. The new billing dashboard is available in the **Tenant Management module > Billing**.
- **Lab replication**

  With replicas, an administrator can easily and reliably replicate lab templates within the same region or to other geographical regions throughout the cloud. This feature helps enable several key use cases including distributing templates for high availability, remote backups, and improving performance for students by providing labs in-region.
- **Create user during entitlement creation**

  This functionality enables you to create a user during the entitlement creation process, without the need for the user to register for or create an account in VMware Lab Platform. This feature can be useful in one-time lab use situations, or to provide a quick hands-on demo without having the user go through the standard registration process.
- **Class self-registration**

  Instructors now have the capability to enable a class for student self-registration. This feature is advantageous in scenarios where an instructor may not have the student roster well in advance of a scheduled class. During class creation the instructor simply configures the number of seats the class is intended to support, and then provides their students with a self-registration URL. No account creation is required beforehand.

What's New January 23, 2020

The following new features and improvements are included in this update:

A new option called **Send Mail to User on Entitlement Creation** is included with this update. An administrator can use the option to send an email that notifies a user that their entitlement has been created. Parameter values for the option are set from the **Portal Settings>Configuration** page on the admin user interface.

What's New January 9, 2020

The following new features and improvements are included in this update:

You can now import Microsoft Word manuals directly into VMware Lab Platform, by using a process similar to the one used for ScreenSteps.

For more information, see the Upload and convert Word documents topic in the *VMware Lab Platform Content Engine* guide .

Administrators can now include a quick launch link when sending email to a user from the manual entitlement option.

What's New in 2019

What's New December 2, 2019

The following new features and improvements are included in this update:

Correction to the feature announcement for Quick Launch URL: administrators do not generate a quick launch URL. Instead, they access the Quick Launch URL that is automatically generated when a lab is assigned.

What's New November 21, 2019

The following new features and improvements are included in this update:

- Additional pages of the Admin UI are available in an updated UI style, including Survey Configurations, Entitlement Callbacks, Tags, Products, Network Topology, and Badges.
- A new interface for options controlled by administrators is available along with several new options. See **Portal Settings>Configuration** in the *VMWare Lab Platform Administrators* guide.
- Administrators can access a Quick Launch URL containing an access token directly from the Entitlements page on the Admin UI. This is in addition to using the API to generate a token.
- You can create an entitlement for a user from the Admin UI at **Entitlements>User Entitlements>Create Entitlement**.
- You can install VM tools from Lab Builder.
- The start and stop order and boot delay of VMs can now be modified in Lab Builder.
- You can resize the console window in Lab Builder.
- The Configuration Settings page replaces the Tenant Options form. You can display settings organized by function or with detailed information and enable or deactivate settings from either view. Searching on settings and filtering on fields is supported.
- You can bulk upload users by class from the Admin UI at **Security>User Control>Account Bulk Upload.**
- The tenant user interface includes an enhancement to notify users when prepops are not available.
- VMware Lab Platform supports LTI 1.3 integration.

What's New July 17, 2019

The following new features and improvements are included in this update:

- Customize completion criteria for labs. Additional features for setting completion criteria for labs are included with the latest update.

For more information, see: Setting completion criteria for a lab in the *VMware Lab Platform Administrator Guide*. As an administrator, you can set values for the following settings.

  - Set a minimum time a student must spend on a lab before it can be considered completed.
  - Set the percentage of steps in a lab that a student must complete before the course can be considered completed.
  - Set tenancy-wide behavior to link **End** button availability to lab completion criteria.
- Pre-provisioning ‘prepop’ values are now configured on the Update vApps Template screen.

Getting Started with VMware Lab Platform

VMware® Lab Platform™, formerly known as VMware® Learning Platform™, is a VMware Cloud service application that lets you deliver hands-on virtual IT training labs at cloud scale.

The following interfaces let you create and deliver a lab, manage documentation for a lab, monitor an instructor-led class, provide user access for class registration, provide access for lab interaction, and manage and track labs at events.

- Administrator user interface: The administrator user interface lets users with the administrator role to perform all administrator tasks such as create and deliver a lab. For more information on the administrator user interface see, Using and Managing the VMware Lab Platform Administrator User Interface.
- Content engine interface: The content engine user interface lets you to author, import, or manage and associate lab manuals with a lab. VMware Lab Platform supports multi-language lab manual projects and allows you to import Microsoft Word, or PDF files as lab manuals. For more information on the content engine user interface see,VMware Lab Platform Content Engine Guide.
- Instructor user interface: The instructor console allows users with the instructor role to add or remove students from a class, and monitor class activity. For more information on the instructor console see,Using and Managing the VMware Lab Platform Instructor Interface.
- User interface: The VMware Lab Platform user interface allows end users to take on-demand labs or instructor-led labs. For more information on the user interface see, Using and Managing the VMware Lab Platform User Interface.
- Event management interface: The event management user interface allows administrators, proctors, and instructors to manage lab offerings at large-scale events such as conferences. For more information on the event management user interface see, Using and Managing the VMware Lab Platform Event Management Interface.

Complete product documentation is available at VMware Lab Platform Documentation.

Intended audience

The primary audience for this guide is VMware Lab Platform administrators and lab authors, who create lab templates, labs, and lab manuals for on-demand learning labs and classes.

About Getting Started Guide with VMware Lab Platform

The *Getting Started with VMware Lab Platform* guide takes you through the tasks required to set up and run a lab using the VMware Lab PlatformLab Builder wizard.

It describes creating resources for your lab environment, the steps required to incorporate a lab manual with your lab course, and the steps required to deliver a lab.

Terminology

Before you start to create a lab, it is important to be familiar with the terms used in this guide.

Lab Template

A vApp that exists in a catalog and is not powered on. vApp templates are called lab templates in the Lab Builder.

Lab Template Working Copy

A lab template that can be actively worked on for the purpose of copying iterations into the catalog.

Deployment Pool

Defines the lab template deployment parameters for a lab. A deployment pool is used to manage where lab templates are deployed, how many lab templates can be deployed, whether a certain number of lab templates can be prepopulated (that is, deployed and powered-on before a user needs it), and so on. The Lab Builder automatically creates a deployment pool for your lab. Users with the administrator role can also create deployment pools.

Prepopulated

A deployment pool capability used to pre-provision (deploy and power-on lab template) ahead of time. Prepopulated lab templates minimize the time required between an end user request for a lab and when it has been delivered. This is especially useful for more complex labs that can require time to power on the lab template's underlying virtual machines, run any user-defined preparation scripts, and so on.

Lab Manual

A manual written in a specific language for a lab course.

Lab Manual Project

A Lab Manual Project is a collection of one or more lab manuals.

Before you begin using VMware Lab Platform

Before you begin using VMware Lab Platform, review the system and the account requirements to create a VMware Lab Platform lab.

NOTE
This guide shows you how to create a lab template using the resources in the VMware Cloud. You can also use virtual machines (VMs) or lab templates developed on-premises, but you must upload them through the VMware Lab Platform administrator user interface. See Upload VMs or ISOs.

Credentials

- Interface URLs: You must have the URL for both your VMware Lab Platform user interface and the administrator user interface.

  Administrator URL: https://admin.labplatform.vmware.com/

  End-user interface URL: https://www.labplatform.vmware.com/<account_name>. For example, the user URL for the vlpdemo2 account, the URL will be: https://www.labplatform.vmware.com/vlpdemo2
- Administrator credentials: You must have administrator credentials for VMware Lab Platform.
- Organization name: You must have a registered account name of your organization with VMware Lab Platform.

Browsers

Browser settings must accept cookies, run JavaScript, and support web sockets. The following browsers, and browser versions are supported with VMware Lab Platform:

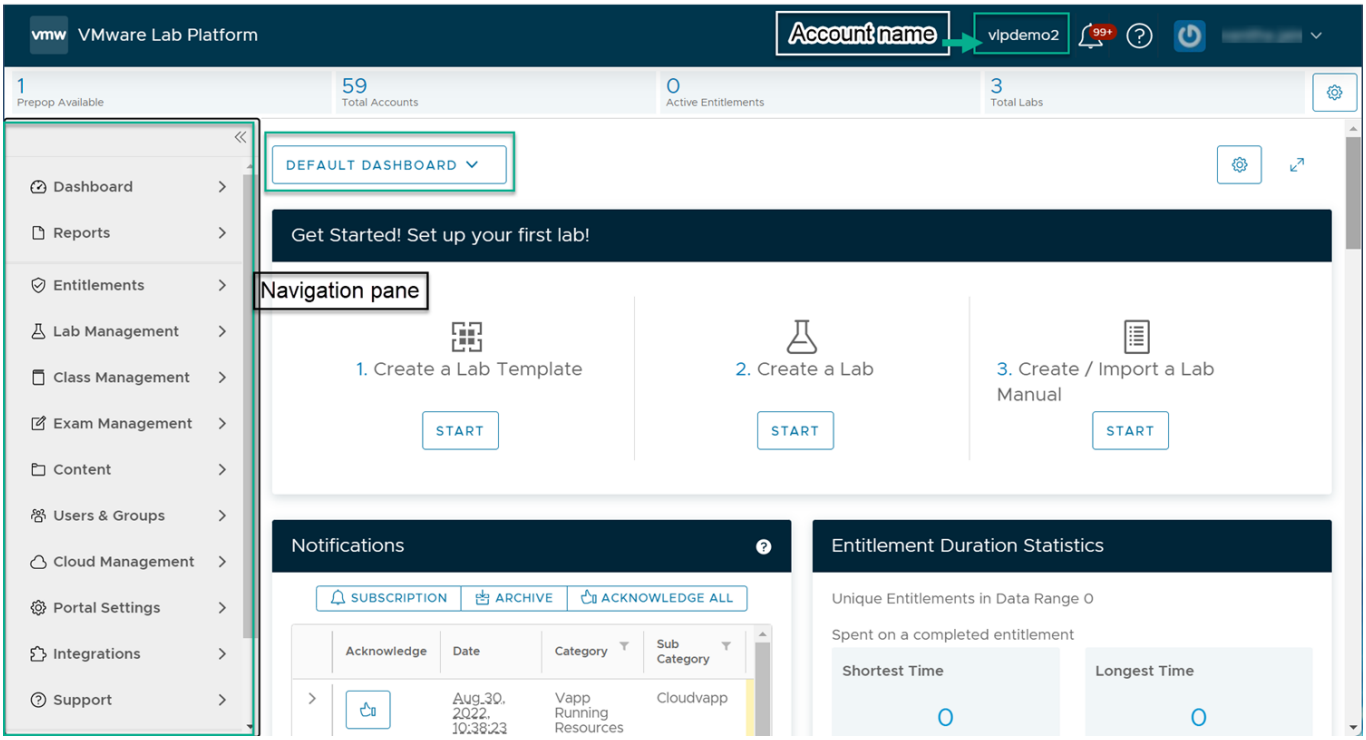
- Internet Explorer version 10 or later
- Firefox version 13 or later
- Chrome version 18 or later
- Safari version 6.0 or later

Overview of the VMware Lab Platform administrator user interface

The first screen you see when you open the VMware Lab Platform administrator user interface is the default dashboard. From the dashboard, you can track and view all aspects of building and running labs.

The pane on the left is the administrator navigation menu. Click the double arrow in the lower right corner to open or close the menu.

Figure 3: The default dashboard of the administrator user interface



VMware Lab Platform includes both a basic administrator interface and an advanced administrator interface view of the navigation menu.

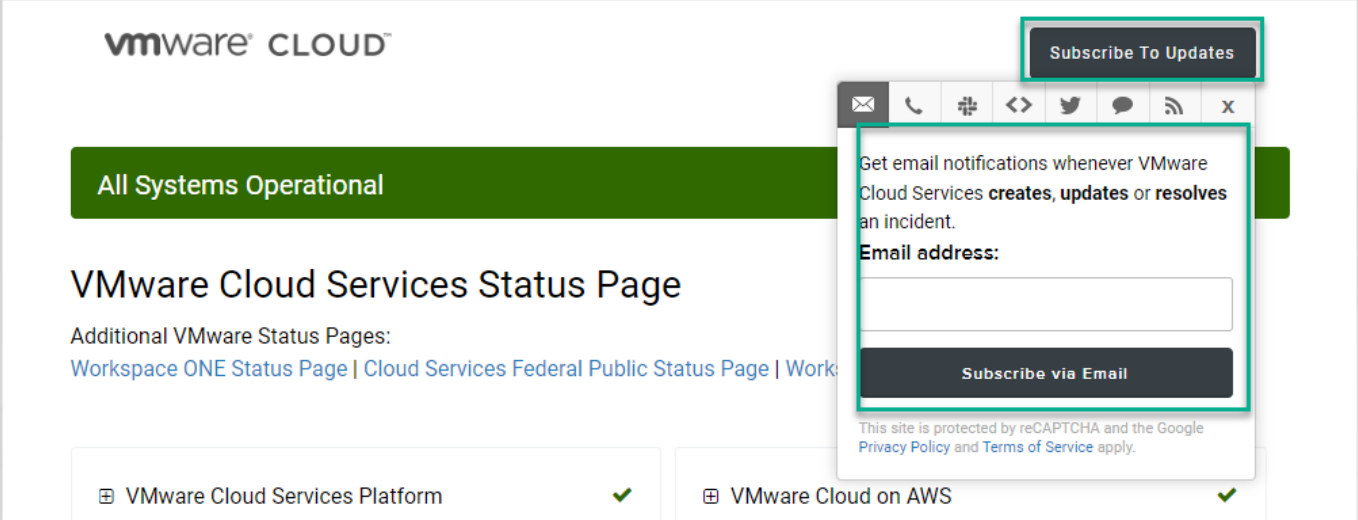
- The basic administrator UI is essentially a subset of the advanced administrator UI. It includes only those options that are required to deliver a lab, providing an efficient and simple presentation of the menu options.
- The advanced administrator UI provides additional options to deliver a lab with advanced features.

Subscribe to status updates for VMware Lab Platform

Subscribe to the VMware Cloud Services status updates for VMware Lab Platform to receive email notifications about incidents such as scheduled maintenance or outages.

When you subscribe to the VMware Lab Platform notifications, you are also notified of the cloud service provider-related incidents.

1. Open https://status.vmware-services.io/ in a browser.
2. Click **Subscribe to update**, and enter your email address.
3. Click **Subscribe**.



4. From the subscription page, select VMware Lab Platform and save the preferences.
5. Check your email inbox and confirm your subscription.

Contact VMware Lab Platform support

To contact the VMware Lab Platform support team for any help or to resolve any issue, you can use the **VMware Cloud Services** console.

To manage your support requests or open a new support request, access the **Support Center** from the VMware Cloud Services Console. To access the **Support Center**, you must have a support role. To get the required permissions to open and manage support tickets for your organization, contact your organization owner.
When you work with support requests, you do so within the context of your current active organization. If you belong to several organizations, make sure that the organization for which you want to manage the request is your active organization. For more information on how to switch to another organization, see how to switch to another organization.

1. Log in to the VMware Cloud Services Console.
2. Click **Support Center**.
3. Click **Create Support Request**.
4. Complete the **Create Support Request** form, including the appropriate category and severity, and click **Submit**.

Create a lab template in VMware Lab Platform

Lab templates provide the basis for your lab's software environment. VMware Lab Platform uses templates that you build-based on a lab template virtual appliance.

A lab template comprises of one or more virtual machines, metadata, a power-on and shutdown sequence, and a networking configuration. VMware Cloud services include everything you require to create a lab template.

Guidelines for lab templates in VMware Lab Platform

As a best practice, use the following guidelines when you set up lab templates for VMware Lab Platform.

Right size the workload

To maximize the cost efficiency while still providing seamless experience to lab users, size the virtual machines efficiently.

- To function optimally, use as few virtual CPUs as possible for the application.

- Avoid configuring virtual machines with more than 4 virtual CPUs.
- **NOTE**
  Virtual machine with more than 8 virtual CPUs is not supported.
- Do not under provision memory as it leads to paging or swapping within the virtual machine and impacts the performance, but also do not over provision memory, which results in excessive waste of the resources.

Managing templates

- To conserve lab resources after a lab template is added to a catalog, delete or power off the working copy. Unlike VMware vSphere, when a lab template is added to a catalog in a virtual cloud director, a copy is made.
- For a successful lab deployment, ensure that the source lab template virtual machine is gracefully shut down (lab template stop command) before you add it to the catalog as a template.

Importing templates

- To import a lab template in VMware Lab Platform, use the process described in Upload a VM or an ISO file in VMware Lab Platform . VMware Lab Platform provides a reliable mechanism to import a virtual machine or lab template exports (OVF/OVA), and ISOs files.
- If the lab template was created and exported from the same vCloud Director environment, it might require some clean-up after the import is finished. Review and reconfigure the network configuration after the import is finished.
- Importing individual virtual machines results in the creation of a single virtual machine lab template. You must then copy or move the virtual machine lab template as required.

Supported outbound ports and protocols for lab template to internet communication

The following ports and protocols are supported for lab templates:

HTTP	TCP 80
HTTPS	TCP 443
DNS	UDP 53
NTP	UDP 123
FTP	FTP 21
SSH	TCP 22

Cloud capacity management for lab templates in VMware Lab Platform

Use the following guidelines when you plan configuration limits for your lab template in VMware Lab Platform.

Calculate the cloud capacity consumption using the allocated resources, and not the utilized resources. For example, if you allocate a 100-Gbyte disk to a virtual machine, but the disk contains 50-Gbyte of data, then the 100-Gbyte of storage is consumed from the storage allocation.

- Avoid over-sizing workloads intended to be used for labs. Idle virtual CPUs, unused memory, and free disk space is consumed from the capacity allocation, incurring extra costs.
- Maintain a good housekeeping of a lab template and templates within the virtual data center.

NOTE
For trial users, the trial capacity is limited to 25 virtual CPUs | 50-Gbyte RAM | 2500-Gbyte storage unless otherwise noted in your on boarding session. This allows you to deploy the configurations described in the following table.

Table 1: Sample configurations

VMs	Virtual CPUs	Memory	Storage
25	1	2 GB	100 GB
12	2	4 GB	200 GB
6	4	8 GB	400 GB

NOTE
A lab template that is powered on, consumes compute (CPU and memory) and storage resources from the cloud allocation, whereas when powered off, a lab template only consumes storage resources.

Authoring lab templates

A lab template is a virtual machine image that is loaded with an operating system, applications, and data. The lab templates ensure that lab environments are configured consistently for all lab participants.

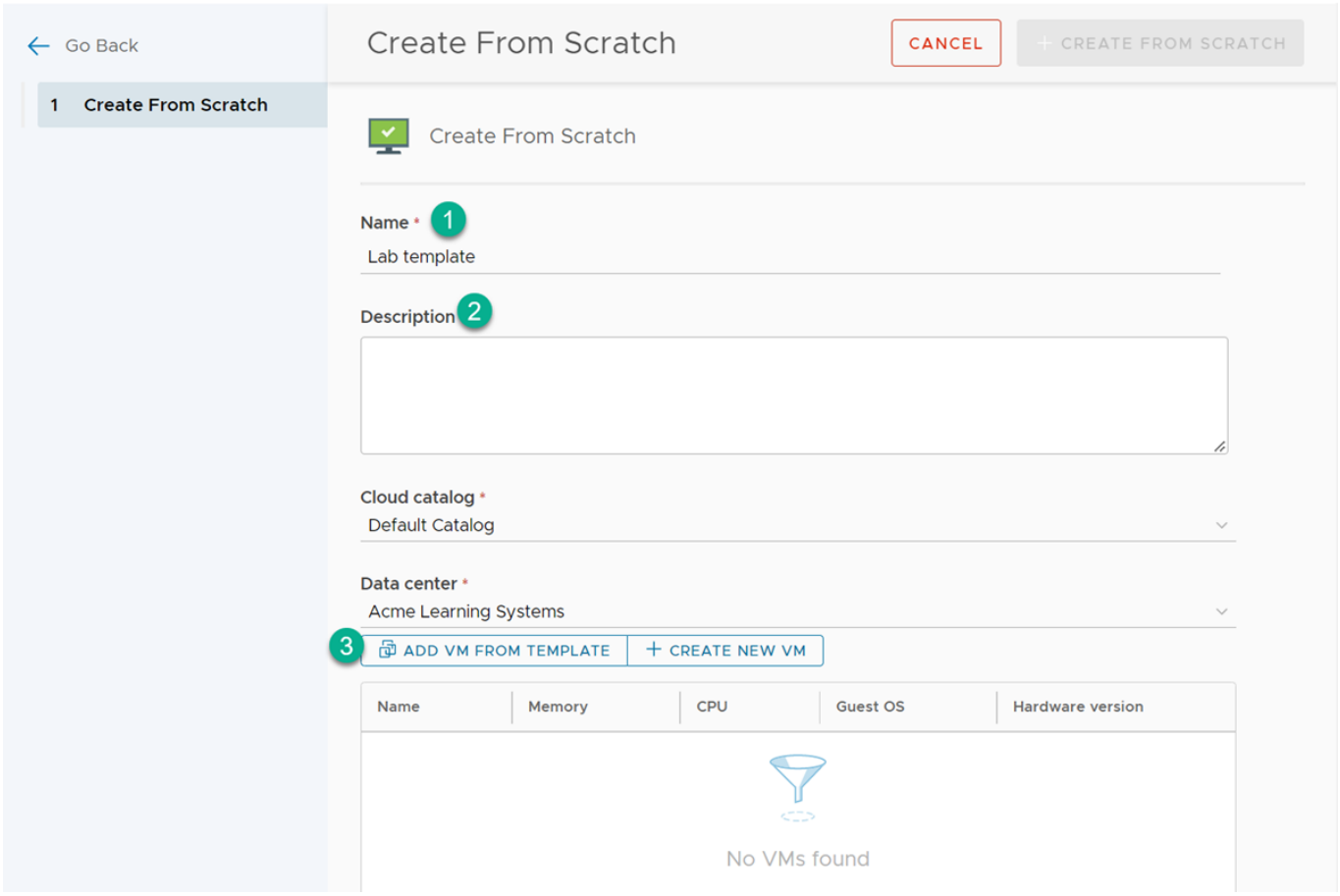
The following exercises show how to deploy a lab template containing a single Linux VM from a base template in the public catalog, how to power on the lab template and log in to the Linux VM, and how to power off the lab template, and create a template by copying it in to your catalog.

Create a lab template in VMware Lab Platform

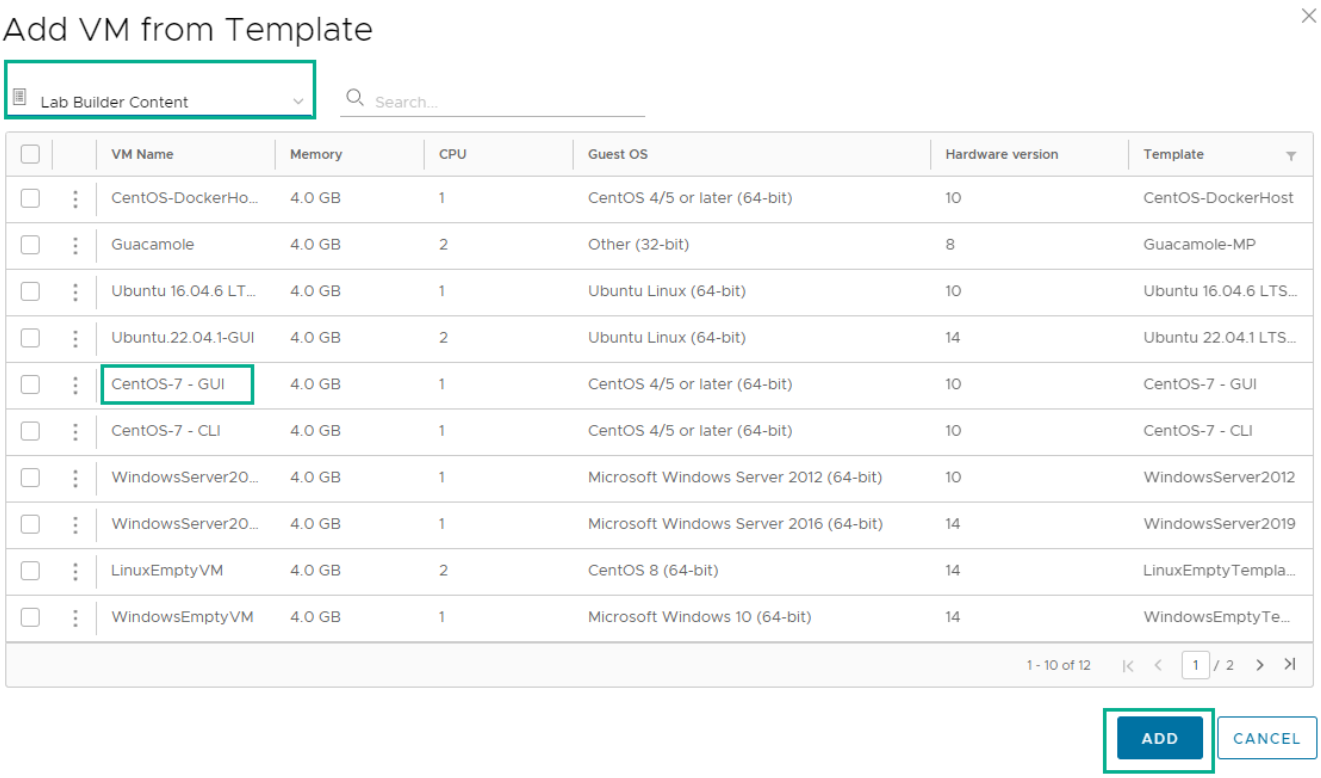
In VMware Lab Platform, you create a lab template by deploying a base template in the public catalog.

You must have administrator role permissions.

1. Enter the **VMware Lab Platform** administrator user interface URL in a browser.
   https://admin.labplatform.vmware.com/
2. From the **Default Dashboard**, open the navigation menu on the left and select **Lab Management › Lab Builder**.
3. Click **New Lab Template**.
4. Update the Name and Description text boxes as needed and click **Add VM**.



5. On the Add VM page, select **Lab Builder Content (Public)** from the drop-down menu to display the information for the VMware Lab Platform Public catalog and select **centos-cli**, CentOS 4/5 or later (64-bit).



6. Click **Add**.
7. Review the lab template details and the VM you just added. If everything is correct, click **Create from Scratch**.

You will be directed to the Lab Builder page, where you can see your newly created lab template.

Customize a lab template in VMware Lab Platform

You can customize a lab template to include information specific to your lab in VMware Lab Platform.

1. To open the Lab Templates page, from the administrator UI, select **Lab Management › Lab Builder**.
2. On the Lab Templates page, locate the lab template you created, and click **Edit** on the **Actions** menu.
3. From the Edit Lab Template page, you can access the console of the VMs within your lab.
   a) To access the console directly, select the CentOS-CLI VM.
   b) Power on the VM.
   c) Log in with the default user name and password:
      - User name – `Root`
      - Password – `VMware1!`



4. Modify the VM as needed with new user accounts, installed software, or application-specific configurations for the lab you are creating.
5. Optionally, you can include multiple virtual machines within your lab.
   a) Click **Add Virtual Machine** for the Edit lab template screen.
   b) Click **Add VM from Template**.
   c) Select the *CentOS-gui* VM.
   d) Click **Add**.

   The New VM is now included within your lab.
6. The VM Details page allows you to view of the name of the virtual machine and the hardware configuration for the CPU and memory.
   a) With the VM powered off, set the memory to 2048 on the Centos-CLI VM.
   b) Click **Update** to save your changes.
7. Complete your form edit, and click **Update and Publish**.

Complete the steps to deliver a lab. See Delivering a lab in VMware Lab Platform.

Uploading VM or ISO file in VMware Lab Platform

To create lab templates for VMs or lab templates that are developed on-premises, upload the VM images into the cloud.

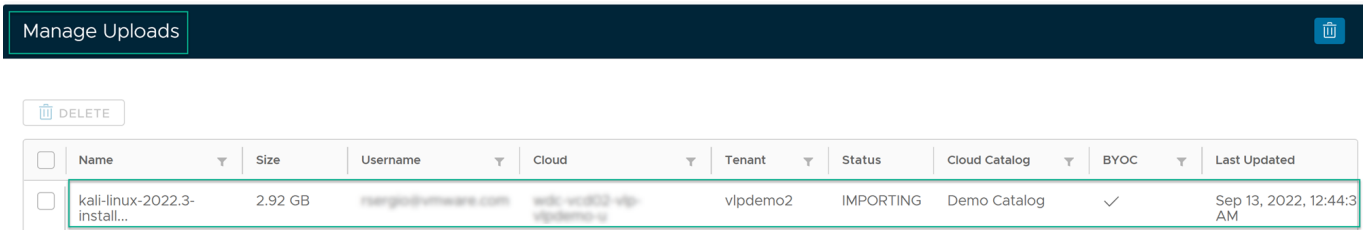
In addition, if a VM is built in the cloud, you might need a unique ISO operating system.

Upload a VM or an ISO file in VMware Lab Platform

If you develop virtual machines (VMs) or applications (VM bundles) on-premises, you must upload them to the cloud to create templates for VMs or lab templates. In addition, when you build a VM in the cloud, you have to install a media file (an ISO) with an OS.

You can import a virtual machine (VM) and a media file (ISO) to your cloud catalog using the administrator user interface. The supported file types are: OVF, OVA, virtual machine disk (VMDK), and ISO.

1. Navigate to **Lab Management › Lab Builder**.
2. Click the three ellipsis next to the **Create New** tab.
3. Select **Upload VMs/ISOs** from the drop-down menu.
4. On the Upload VMs/ISOs form, select a cloud catalog destination.
5. To upload a file, select a file and click **Attach Files**. Or, you can directly drag the file from your local system to the Upload VM/ISOs window.
6. When the file is displayed in the Upload window and ready for upload, click **Upload**.



When an upload is completed, there is a waiting period before you can use the file, because the file is further processed and imported into the cloud. You can view the file upload status in the **Manage Upload** widget in the **Dashboard**. When the file is successfully uploaded, it automatically moves for the import operation. You can see the status of the file import in the Import window page. When the import operation is complete, the template is available in the **Lab Builder** as a published template for use. An automatic job regularly checks for uploaded items,

updates the status, and syncs them with VMware Lab Platform and the **Lab Builder**. You can also start the resynchronize job manually.

7. Click **Close**.

   You can view the ISO (media files) in your cloud catalog from the **Lab Management › Cloud Media** page.

8. Go to the dashboard (**Dashboards › Default Dashboard)** and monitor the progress under the **Content Migration Task** widget. To see the history of tasks, click the arrow next to the task name.
9. Automatically, the resynchronize job is run every hour. To manually synchronize a lab after the upload, go back to **Lab Builder** and click the ellipsis next to the **Create New** tab, and click **Full Resync Lab Templates for current catalog**. To synchronize a lab for the cloud org, select the **Full Sync Cloud Org** option from the drop-down menu. The resynchronize action syncs all the media files and lab templates for the selected cloud.

Synchronize and update template changes with the cloud

It is important to synchronize your lab template with the cloud org after you modify your lab template in VMWare Lab Platform. You synchronize the cloud org with the changed lab templates and then update the corresponding deployment pool.

1. Go to the VMware Lab Platform administrator UI login screen and enter your credentials.

   https://admin.labplatform.vmware.com

2. Navigate to the Cloud Org List, select **Cloud Management › Clouds** from the navigation menu on the left. This option is available when advanced options are displayed.
3. Synchronize any modified templates with your cloud organization to make them generally available.
   - To synchronize an individual cloud org, select it and click the vertical ellipsis icon, then click **Sync Lab Templates**.
   - To synchronize templates for all cloud organizations, click the vertical ellipsis icon next to any cloud org name, and then click **Full Resync Lab Templates**.
4. Navigate to the Deployment Pool List, select **Cloud Management › Deployment Pools**.
5. On the Deployment Pool List, select the deployment pool to which you want to add the changed lab template, and click **Edit Lab Templates**.
6. On the Edit Lab Templates page, click **Add/Remove Lab Templates**.
7. On the Lab Templates list, select the lab template you want to use, then click **Done**.
8. On the **Edit Lab Templates** tab, click **Update**.

If the structure of the lab template you changed is significant enough, your update might fail. In such a case, you must remove all lab templates from the deployment pool, then add the change.

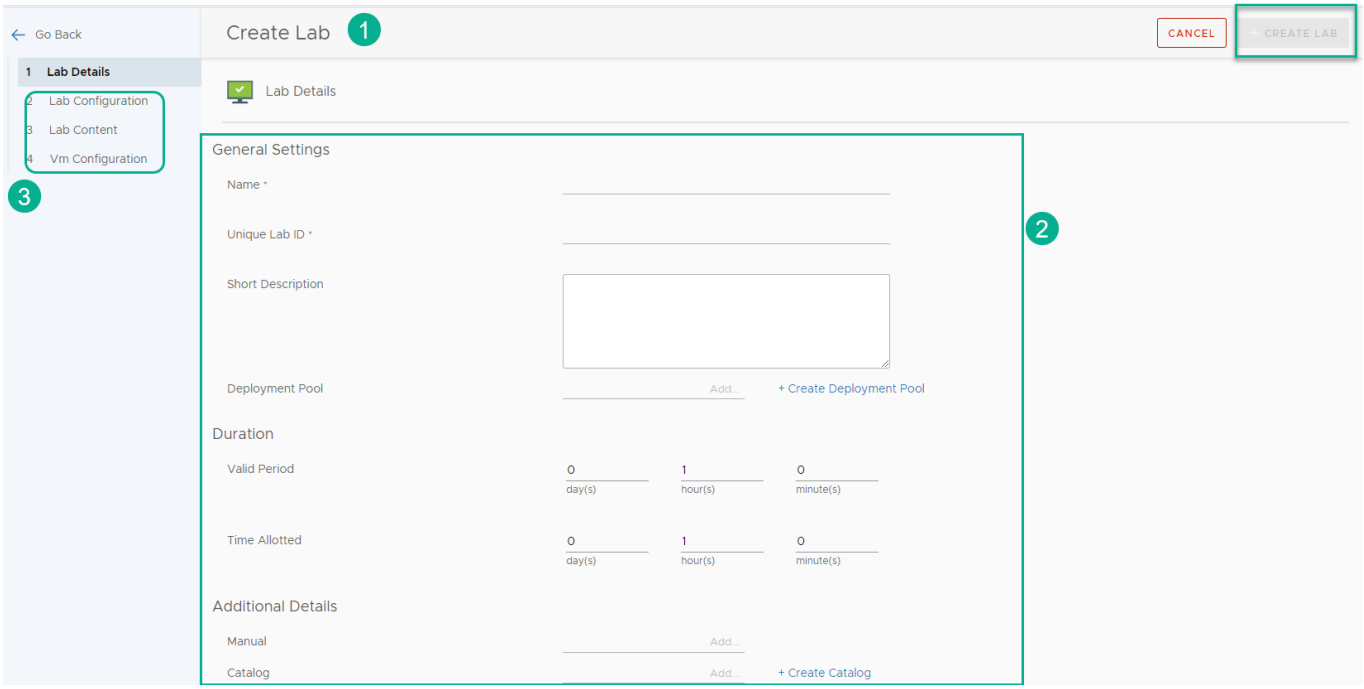
Delivering a lab in VMware Lab Platform

After you create a lab template, you create a VMware Lab Platform lab and set its deployment parameters.

Create a lab in VMware Lab Platform

A lab is a collection of metadata, a single lab template, and an instructional manual in VMware Lab Platform.

1. To open a list of current labs, from the navigation menu panel, select **Lab Management › Labs**.
2. Click **Create Lab**.



3. In the Create Lab window, enter the following information.

Unique Lab ID	A unique identifier for your lab, such as HOL-2085-01-EMT.
Name	A name for the lab you are creating.
Deployment Pool	A collection of deployment parameters for your lab template created by the Lab Builder when you create your lab template. Select a deployment pool from the drop-down menu.
Time Allotted	The time you allot for the student to complete the lab.
Manual	The user manual for the lab. If you want to upload the manual at a later time, select **Manual Empty**.

4. Click **Next**.
5. On the **Lab Configuration** page, accept the defaults and click **Next**.
6. On the **Lab Content** page, accept the defaults and click **Next**.
7. Update the VM settings on the VM Configuration page, and click **Create Lab**.

The lab you created is listed in the table on the Labs page.

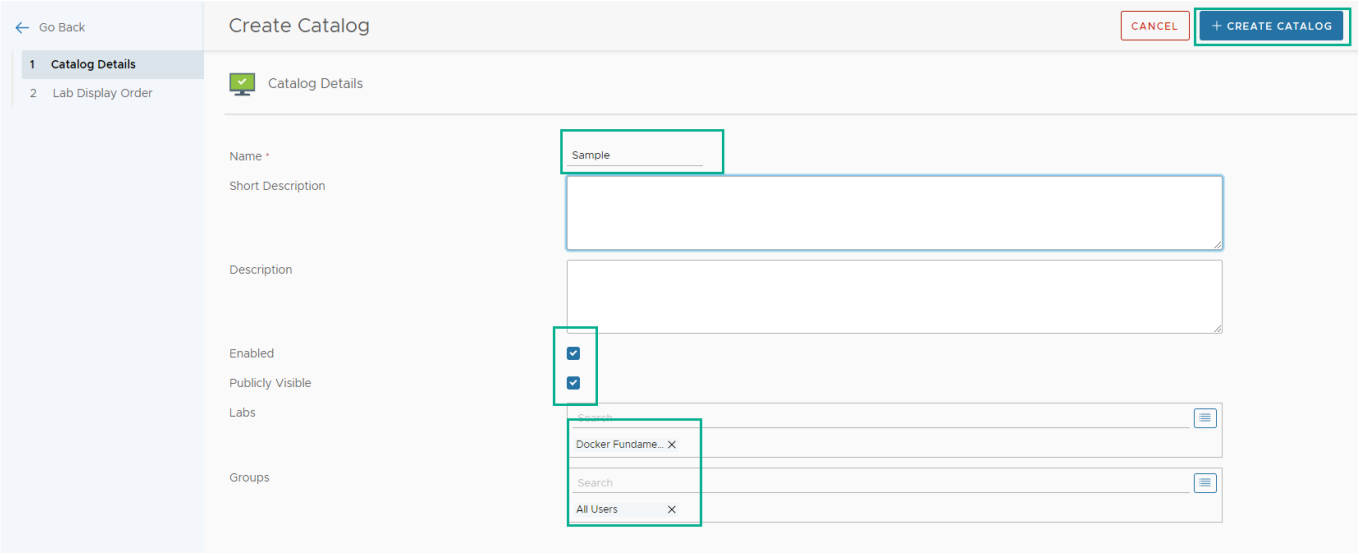
Create a catalog.

Create a catalog in VMware Lab Platform

Create a catalog for your lab. In VMware Lab Platform, a catalog is an organized inventory of labs and their descriptions that users can browse or search in the VMware Lab Platform user interface.

To control who can see the catalog when you create a catalog, use the **Users & Groups › Groups** option. This option allows you to restrict the visibility of labs that must not be available to all users.
Every account includes the All Labs catalog. Labs added to this catalog are visible to all users.

1. From the navigation panel, click **Lab Management › Lab Catalogs**.
2. To create a catalog, click **Create New**. Enter a name for the catalog. Select **Enabled** and **Public Visible**.



3. Click the Labs search box and select the lab you created. Under Groups, select **All Users**.
4. Click **Create Catalog**.
5. You can change the order of labs listed on the Lab Display Order tab.
6. Click **Finish**.

Create end user accounts in VMware Lab Platform

End users are the students who take your lab. Each student requires a user account.

1. From the navigation panel, click **User & Groups › User Accounts**.
2. On the Accounts page, click **Create New**.
3. To create a single user account, click **Single account** from the drop-down menu.
4. Enter the user account details in the Create Account page.
   a) Enter a legitimate email address as the user name.
   b) Complete the other required text boxes.
   c) Select **Generate Password**, **Show Password**, and **Welcome email**.
   d) Click **Create Account**, which creates the account and displays the User Access page for the account.
5. Accept the selected default for **User Role** and click **Update Account**.
   The user account is created, and you can view the account details on the **Users & Groups › User Accounts › Accounts** page.

Developing a lab manual in VMware Lab Platform

Lab manuals provide instructional content for a lab. You can develop manuals with the Content Engine editor in the VMware Lab Platform administrator user interface, or import manuals from Word or applications such as ScreenSteps. PDF documents can be uploaded and used with the VMware Lab Platform PDF viewer.

You associate lab manuals with the lab's manual project. A lab manual project consists of one or more lab manual instances. If you have multiple language versions of an manual, you can make them all available to lab users by adding them as instances to the same lab manual project.

Create a lab manual project in VMware Lab Platform

Create a container project, called a lab manual project, for your lab's instructional content. You upload one or more lab manuals to the lab manual project, including any translations you need for your lab manuals.

Activate the **Portal Settings › Configuration › Show Page Contents page**.

1. From the navigation panel on the administrator user interface, select **Content › Content Engine** to open the Lab Manual Projects list.
2. Click **Add New**.
3. Click **Start a Manual Project**.
4. On the Start a Manual Project form, specify a name for the manual project and then click **Create**.

Add a manual to a lab manual project in VMware Lab Platform

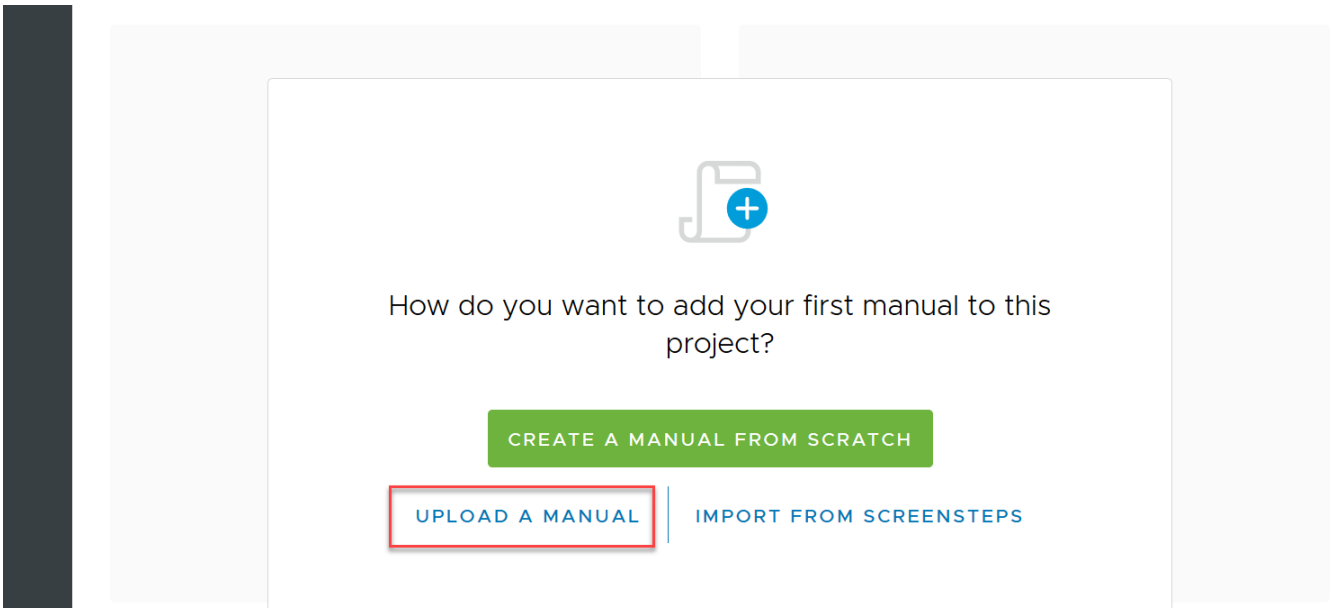
Add a manual to a lab manual project in VMware Lab Platform

Lab manuals are added to a lab manual project. You can import a manual from the ScreenSteps authoring tool, or upload instructional content from your local machine.

Activate the **Portal Settings › Configuration › Show Page Contents page**.

In addition to ScreenSteps, VMware Lab Platform supports Word DOCX and PDF files. ScreenSteps and Word files are changed to the VMware Lab Platform lab manual format. PDF files are displayed in a PDF viewer.
For more information on importing different types of source files for lab manuals, see the *VMware Lab Platform Content Engine Guide*.
The following steps show you how to upload a manual to a lab manual project.

1. From the navigation panel on the administrator user interface, select **Content › Manual Projects** to open the Lab Manual Projects list.
2. Click **Add New**.
3. Click **Upload a Manual** and follow the steps on the Upload Manual page.



Publish a manual with VMware Lab Platform.

Publish a manual with VMware Lab Platform

You must publish the course manual to make it available to an administrator who can then associate the manual with a lab.

1. From the navigation panel, select **Content › Manual Projects › Lab Manual Project List** page.
2. On the Lab Manual Projects list, click a lab manual project link to open it.
3. Click the vertical ellipses and select **Publish**.

The instructional manual can now be associated with a lab. When an end user starts a lab, the console presents the appropriate language version of the manual based on the user's profile.

Add a manual to a lab in VMware Lab Platform.

Add a manual to a lab in VMware Lab Platform

After you have published a lab manual, you can associate it with a lab.

You must have administrator privileges.

1. Return to the VMware Lab Platform admin user interface at https://admin.labplatform.vmware.com.
2. Select **Lab Management › Labs** from the navigation panel.
3. Select the lab you previously created from the list of labs.
4. On the Create Lab form, open the **Update** tab.
5. From the Manual drop-down menu, select the manual project.
6. Click **Update** to associate the manual you created to the lab.

Hide the on-screen instructions panel in VMware Lab Platform

If there is no manual present or required for the lab, you can use a setting to hide the instruction panel on the student interface.

You must have administrator privileges.

Hiding the instruction panel helps students to avoid confusion when the lab has no manual, and provides more space for the Student UI console. This setting applies for all labs in the account.

> **NOTE**
> Because this setting hides the entire instruction panel, which includes the **Resources** menu, lab resources such as diagrams or supplemental documents are not accessible from the student user interface.

1. Log in to the administrator UI.
2. From the navigation panel, **Portal Settings › Configuration**.
3. In the Interface section, enable the **Hide Instructions panel when no manual** setting.

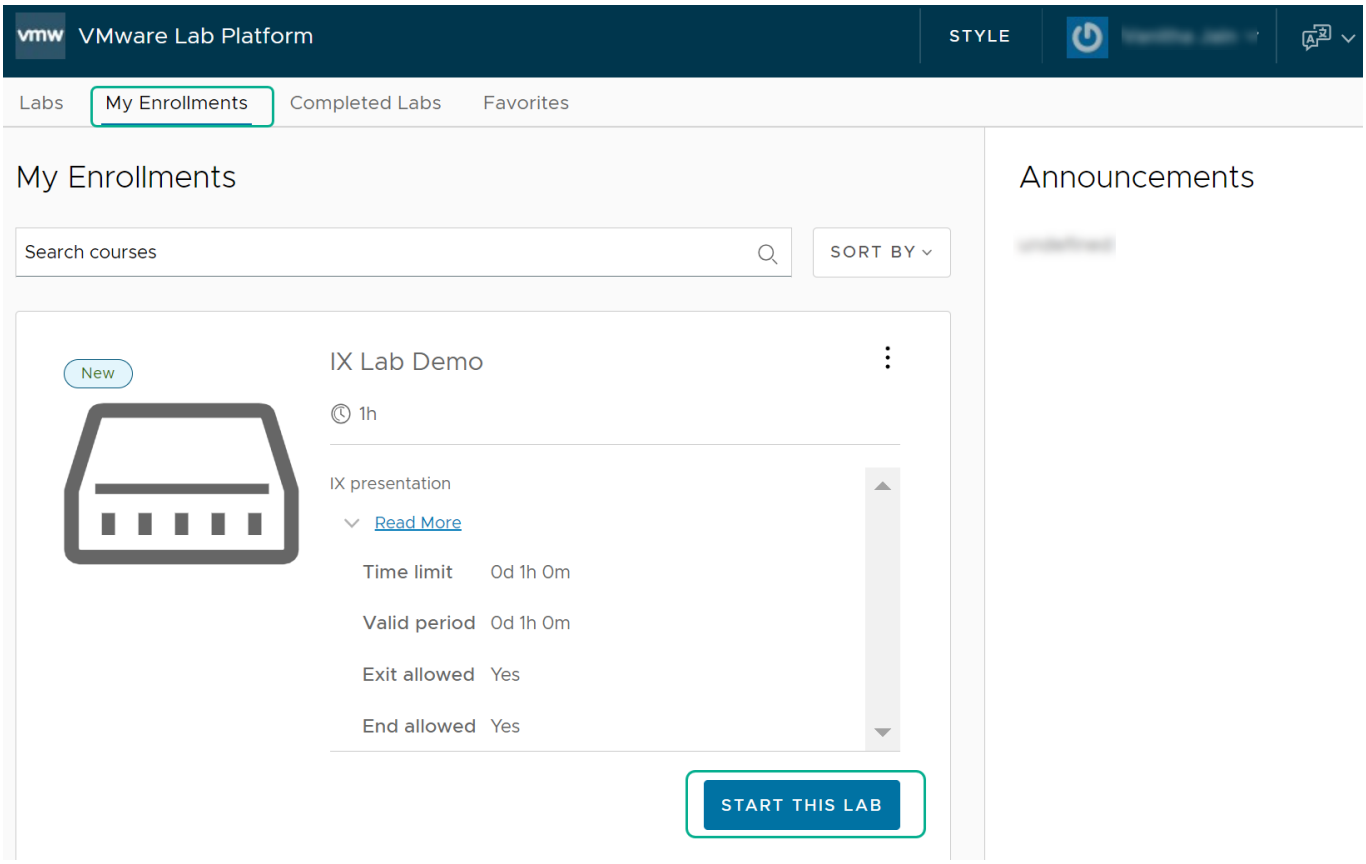
Start your lab in VMware Lab Platform

After you have finished the steps to deliver a lab, your lab is open for enrollment.

Use the following steps to enroll in a lab.

1. Go to the console for the lab you created: https://www.labplatform.vmware.com/<org_name>.
2. Log in to the console.

The enrollment window for your lab is displayed.



3. Click **Enroll** to enroll into your lab.

You have finished the getting started exercise for VMware Lab Platform.

Next steps

This getting started exercise showed you how to create a lab and add it to a catalog. VMware Lab Platform also helps you with course delivery and event management.

For information about course delivery for instructor-led or on-demand training, see the VMware Lab Platform *VMware Lab Platform Instructor Guide*.

For information about event management for administrators, proctors, and help desk staff, see the *Event Management Guide*.

Using and Managing the VMware Lab Platform Administrator User Interface

The *VMware® Lab Platform™ (formerly known as VMware® Learning Platform™) Administrator User Interface Guide* provides information on how to use the VMware Lab Platform administrator user interface. You use the administrator user interface to connect your cloud vApps to your labs and create labs, create and manage user accounts for labs, and perform administrator tasks.

Intended audience

The primary audience for this guide is VMware Lab Platform administrators who want to create on-demand learning labs for their users.

About the VMware Lab Platform administrator user interface guide

The *VMware® Lab Platform™ (formerly known as VMware® Learning Platform™) Administrator User Interface Guide* provides information on how to use the VMware Lab Platform administrator user interface. You use the administrator user interface to connect your cloud vApps to your labs and create labs, create and manage user accounts for labs, and perform administrator tasks.

Intended audience

The primary audience for this guide is VMware Lab Platform administrators who want to create on-demand learning labs for their users.

Before you begin using VMware Lab Platform administrator interface

Before you begin your administrator tasks, it is good to be familiar with the cloud resource and credential requirements to use the VMware Lab Platform.

About customer accounts

Customer accounts in VMware Lab Platform is a company or an organization (or a department within a company or an institution) that has purchased access to VMware Lab Platform. The VMware Lab Platform technical staff provisions your account based on the information provided by your organization. Administrators have administrative rights to one or more customer accounts.

Cloud resources and administrator credentials

You must have procured cloud resources and received administrator credentials from the VMware Lab Platform support team. Typically, the VMware Lab Platform support team sends you an email containing your user name and password soon after you or your organization purchases VMware Lab Platform services.

Supported browsers and environment

You access VMware Lab Platform through a browser. VMware Lab Platform supports the following browsers:

- Internet Explorer version 10 and later
- Firefox version 13 or later
- Chrome version 18 or later
- Safari version 6.0 or later
- VMware Cloud Director Version 5.5 or later
- VMware Cloud Director Version 10.4 is with the VMware Cloud API version 34

NOTE
Your browser settings must accept cookies, run JavaScript, and allow web sockets to run VMware Lab Platform.

Instructional content

Instructional content (manuals) is required for an administrator to deliver a lab. You can include instructional content when you create a lab, or you can import an existing manual later using the VMware Lab Platform Content Engine user interface.

Administrator workflows for VMware Lab Platform

In VMware Lab Platform, the administrator role has permission to use all the VMware Lab Platform user interfaces. Administrators can enable, optional VMware Lab Platform features on per account basis, and assign predefined roles and features of those roles to other VMware Lab Platform users. A user with the VMware Lab Platform administrator role performs these tasks, so that end-users, or students can take on-demand labs and learn about your products, product owners can demo the products to their customers.

Administrator tasks for delivering an on-demand lab

As an administrator, you perform the following set of tasks to prepare an on-demand lab:

1. Use the Lab Builder automation to create a lab. See Create a lab template.
2. Manually create a deployment pool and lab. See Create a deployment pool. See Create a lab.
3. Create a catalog for a lab. See Create a catalog.
4. Create users for a lab. See Create user accounts.
5. Import a lab instructional manual. See VMware Lab Platform Content Engine Guide.

Administrator tasks for enabling a class delivery

You perform the following tasks to enable delivery of an instructor-led lab.

1. For instructor-led classes, assign a VMware Lab Platform user, the role of an**Instructor**.
2. Create user accounts for lab users or let the instructor to bulk upload users.
3. Create a class. A class is made up of selected users who take a specified lab at a specified time and are monitored by a specified instructor.
4. Manage the student roster and entitlements.

Log in to the VMware Lab Platform administrator user interface

VMware Lab Platform provides you a user interface from which you can manage VMware Lab Platform for your organization or a customer account.

To log in to the VMware Lab Platform administrator user interface:

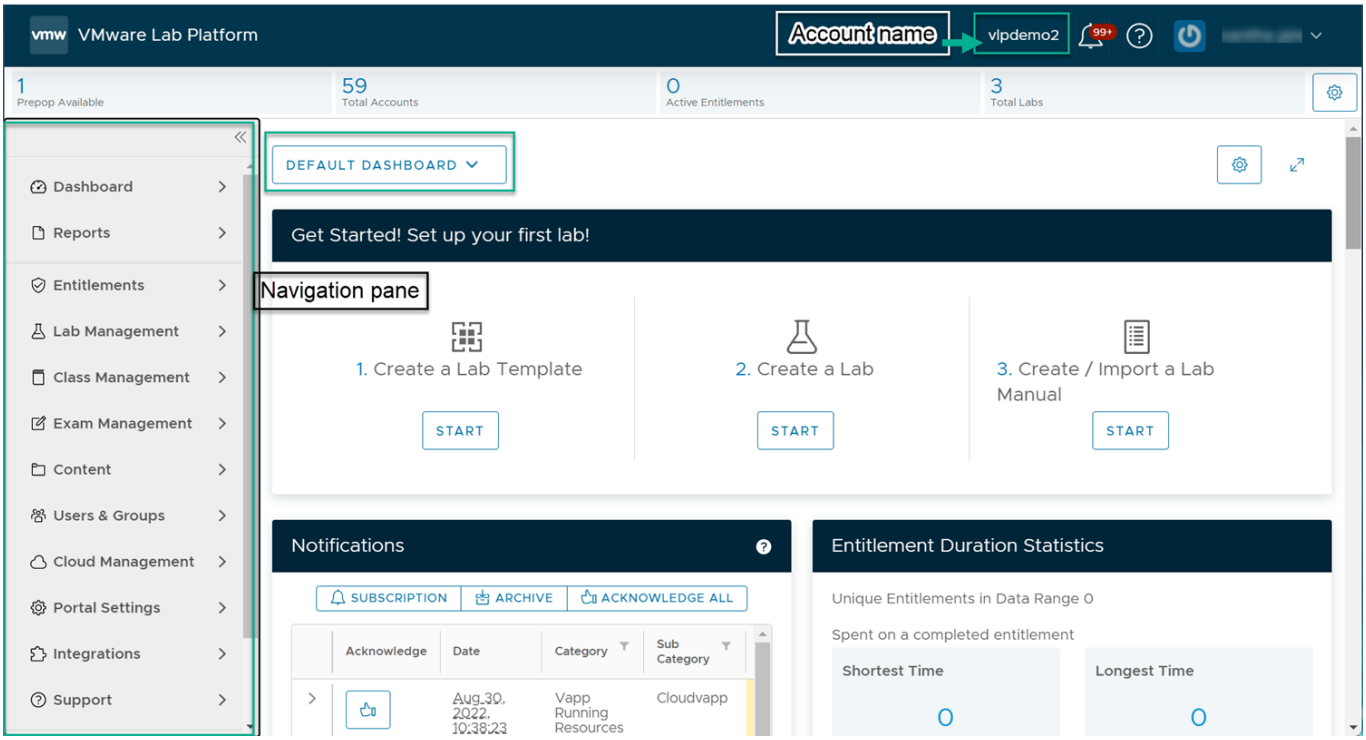
1. Enter the `https://admin.labplatform.vmware.com/` URL in your browser.
2. In the **Organization** field, enter your customer account name.
3. Enter your login credentials.
4. Click **Continue** .

> **NOTE**
> If you forget the password for your account, use the **Forgot Password** option to reset your password. By default, the password reset option is available for a local user account. VMware Cloud Services Provider and SSO login password reset is not supported. If you wish to activate password reset options for your SSO or other authentication provider methods, contact the VMware Lab Platform support team.
>
> For information on how to add authentication provider, see Add an Authentication Provider.

Customize the VMware Lab Platform administrator user interface

You can customize the content that is displayed on the VMware Lab Platform administrator user interface (Admin UI).

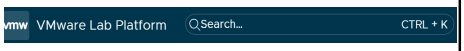
Figure 4: The default administrator user interface view



Manage Columns	Manage Columns	You use the **Manage Columns** button to select or deselect the table column options. You can customize the table headings that are displayed on the administrator user interface. In all the menu listing pages, the column display shows the most frequently used and relevant columns by default.
	Filter icon	Use the filter option to select headings that you want to display in the table list.

Table continued on next page

Continued from previous page

	Search icon	Use the search option to search for a specific entry on the page that you are in. You can reorder the search results by using the drag action.
	Global search icon	Use the global search icon to easily find any VMware Lab Platform page, portal setting, or documentation link. The search bar option is located on top of all the pages in the administrator user interface. Users with the role of administrators, content creators, instructors, and super users who have access to the VMware Lab Platform administrator interface can use the search option. Or, you can use the **CTRL+k** shortcut key.
	Bell icon	You use the bell icon to view all the notifications and alerts associated with your VMware Lab Platform account or instance.
	Help icon	Use the help icon to view all the help and support options. For more details on the Support panel, see Working with Support Options

Warnings and errors column

Lists in VMware Lab Platform include a column for warnings and errors to alert you of an issue that exists with an item in the list. You can click the warning or error icon to get more information about the issue.

Action operations

You can perform single, batch, and page operations for different tasks such as adding users for a class, modifying a lab, and so on.

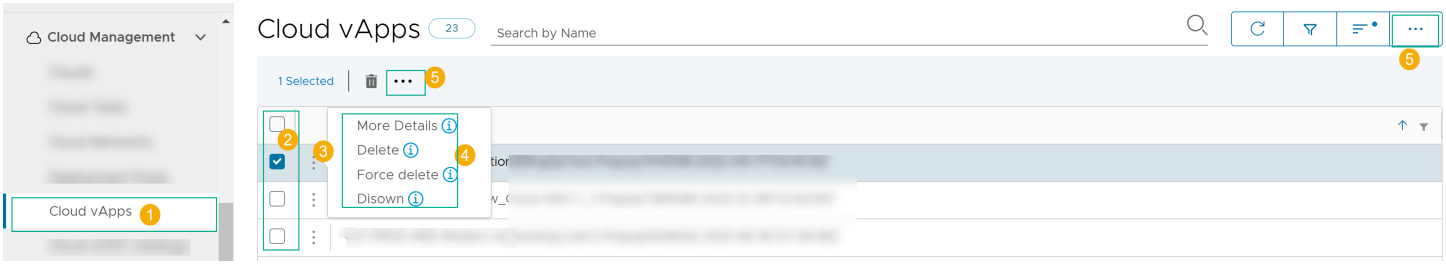
Click the three ellipses on top of the page, and view the list of action options for that page.

Actions	Description
Batch Actions	Use the **Batch Actions** option to perform mass or bulk actions for multiple objects in the page.
Page Actions	Use the **Page Actions** option to perform actions that you want to perform only on the current page. For example, download the page content for the current page that you are in.
Global Actions	Use the **Global Actions** option to perform actions that impacts other parts or features of the VMware Lab Platform and not the current object. For example, synchronize template (new templates are displayed in the Lab Builder and some templates are deleted).

Managing cloud vApps in VMware Lab Platform

You use the **Cloud vApps** page to manage the cloud vApps that VMware Lab Platform need to use to deploy a lab.

To open the **Cloud vApps** page:



1. From the navigation pane, click **Cloud Management › Cloud vApps**.
2. Select the check boxes for all the cloud organizations that you want to update.
3. To open the action drop-down menu, click the three vertical ellipses on the top of the table.

   a. Select an action item from the menu. The following table lists the actions that you can perform on a single cloud vApp.

Options	Description
Consoles	You can access and open a VM console for your vApp from the Cloud vApps page.
Delete	The **Delete** cloud vApp action deletes the selected vApps. This action is useful when you want to delete a corrupted prepopulated vApp. This action removes the vApp from both the VMware Lab Platform and the cloud.
Force delete	The **Force delete** cloud vApp action deletes the selected vApp immediately.
Disown	The **Disown** cloud vApp action is available only for non- Cloud Services Provider customer accounts, and this action permanently removes the selected vApps from VMware Lab Platform, but retains the vApps (unchanged) in the cloud. This action is useful when you think there might be a problem with the vApp functioning in the cloud. In such a case, you can disown the vApp and the problematic vApp will not affect your customer account.

4. To perform bulk actions on cloud vApps, click the ellipses on top of the page and select the action operation.

Synchronize vApps from VMware Cloud Director

You can synchronize VMware Cloud Director lab templates from the VMware Lab Platform administrator user interface. The lab template synchronization is available on per cloud basis as opposed to a full sync of lab template across all clouds. The sync result displays all the vApps of the selected cloud and specifies whether a vApp is a VMware Lab Platform vApp or not.

To synchronize a vApp, perform the following steps:

1. To open the **Clouds** page, from the navigation panel, select **Cloud Management › Clouds**.
2. Select the cloud, and click the vertical ellipsis next to the cloud listing on the page.
3. Select **Sync Cloud vApps** from the drop-down menu.

   The **Sync Cloud vApps** page displays the vApps, including a column designating if it originated from VMware Lab Platform or from VMware Cloud Director. (Optional) You can select a single, or multiple non- VMware Lab Platform cloud vApps in a cloud and delete it.

4. Click **OK**.

Working with cloud networks in VMware Lab Platform

You can define a network change for lab templates that are deployed across different cloud networks or cloud data centers.

You can group networks, by tagging them with the same tag name. For example, in a template metadata, if a network is set with, "vlp" tag name, and a lab template is deployed on any region or data center, VMware Lab Platform looks up for all the networks that are tagged with the “vlp” tag name, ignoring the cloud org network name.

Network prefix ensures that the lab templates are connected to the available organization virtual data center network. For example, if an organization virtual data center network name is *`wdc-vcdxx-vlp-xx`*, you can assign the full name as the organization virtual data center network name, or can make an entry as "*`wdc-`*" in the **Network** page. In this example, XX is a variable.

When a lab is deployed, and there is more than one network available to connect to a lab, the lab is connected to the first network that has IP addresses available.

> **NOTE**
> If both metadata attributes for prefix and tagging are set, tags have the highest priority and are preferred over the prefix. If no match is found by the tag name, then a prefix lookup is performed. If both the criteria are not met, the lab template does not connect to any cloud network.

You can use the **Cloud Management › Cloud Networks** page to:

- Apply a tag name to cloud networks
- Reset a tag
- Sync networks
- View cloud details
- View total IP count available and used IP count

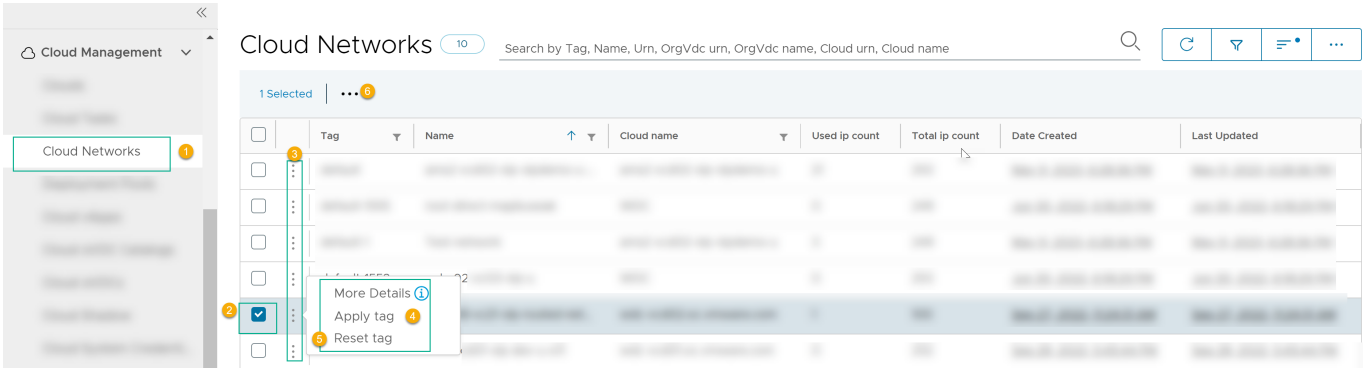
1. To view all the **Cloud Networks** options, click the  icon.

Apply a network tag in VMware Lab Platform

To categorize or reuse templates across different regions and clouds, you can tag networks. By default, all networks are tagged under the "default" tag.

To apply a new network tag or change the default tag name:

1. Click **Cloud Management › Cloud Networks**.



2. Select one or multiple cloud networks from the list.
3. Click the three ellipses next to the cloud for an individual cloud network tagging. For batch action, select multiple cloud networks from the list, and click the three ellipses on top of the table.
4. Click **Apply Tag**, and enter a tag name.

   The network tag is applied for lab templates from the **Lab Builder › Network Editor** option.

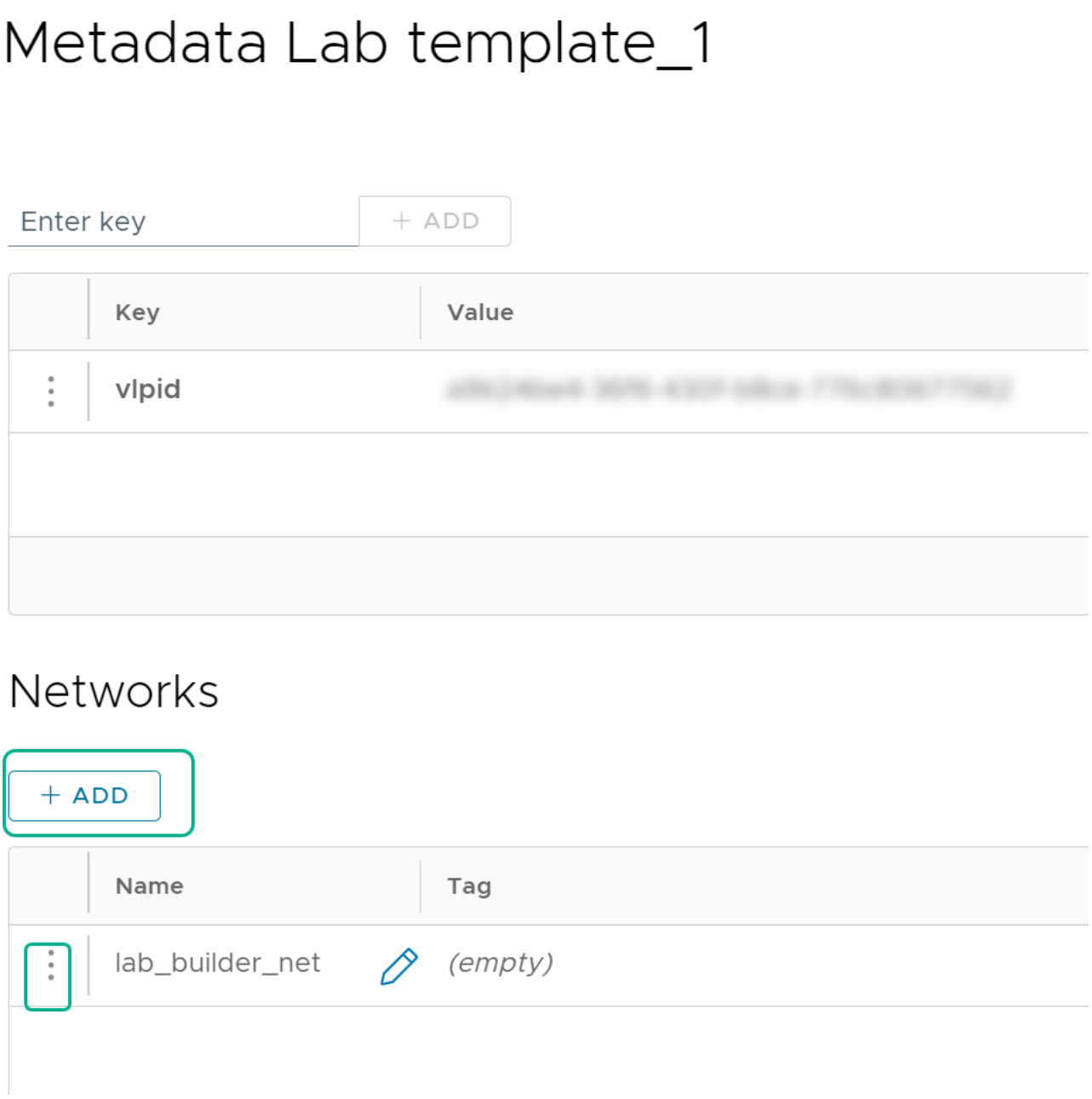
5. To change the tag to default name, click **Reset Tag**.
6. To synchronize the network updates from VMware Cloud Director, from **Global Actions** option, click **Sync Networks**.

Add metadata for a cloud network

The **Cloud Network Details** page provides the lab template metadata information. If a lab template is used for deploying lab templates across different regions and clouds, you must manually edit the template metadata, or clone the template and make the required network changes in VMware Cloud Director. You can group such networks, by tagging them with the same tag name in VMware Lab Platform.

1. Navigate to the **Lab Management › Lab Builder › Lab Templates** page, select a cloud and a lab template from the list.
2. Click **Actions**, and select **Metadata** from the drop-down menu options.
3. Click **Add** under the **Networks** section, and add a network and a network tag for the lab template. Use the edit icon to edit the network name.

   Network tags are created in the **Cloud Management › Cloud Networks** page. For more information on how to apply a network tag, see Apply a Network Tag.



4. Click **Update**.

Add a network for a lab template in VMware Lab Platform

You can add a network to a lab template to make the network available to the virtual machines in the lab template. You can add a lab template network or an organization virtual data center network to a lab template. The network connections can be direct, isolated, or mixed.

To add a network tag to a network, your administrator must have created such a network tag in the **Cloud Management › C loud Networks** page.

To add a network configuration for a lab template:

1. To open the **Lab Templates** page, select **Lab Management › Lab Builder**.
2. On the **Lab Templates** page, open the cloud drop-down menu and select the cloud you want to use for the lab template you are building. (Only available if you have more than one cloud).

   The first cloud in the drop-down menu is the Default cloud.

3. To open the **Edit Lab Template** form, select an unpublished lab template and click **Actions › Edit**.

   On the **Edit Lab Template** form, you can update or modify the lab template. Specify a name, a description, add or remove VMs and change the advanced properties for the lab template.

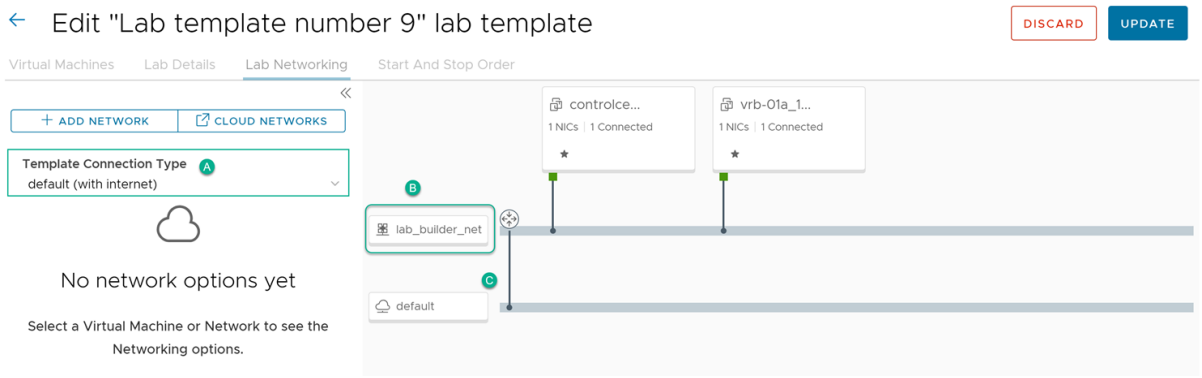
4. Click the **Lab Networking** tab.

   The **Lab Networking** page displays the network diagram of all the network connections for the VMs in the lab template. In the following sample network image, two VMs are connected to two networks, one without the internet and the other network with the internet.

   The networks displayed in the image are:

   - A: A network connection tagged as default and connected to the internet.
   - B: lab_builder_net connection between the VMs is an isolated network connection without the internet. An isolated network is contained within the lab template.
   - C: The default template connection with internet that connects the VMs to external networks.

   Figure 5: Lab Template Network Diagram View



5. To add a new network, click **Add Network**.
6. Enter the network information in the form. Select the connection type from the Connection drop-down menu. The connection list tags are populated based on the tags created in the **Cloud Management › Cloud Networks** page. For more information on how to apply network tags, see Apply a Network Tag.
7. To manually configure the following network information, use the **More Options** slider button. You can define the:
   - Primary DNS
   - Secondary DNS
   - Static IP pool range to use for the network
8. Click **Save**.

   The network diagram reflects the network changes made in the **Add Network** form.

9. To save the changes, click **Update**.

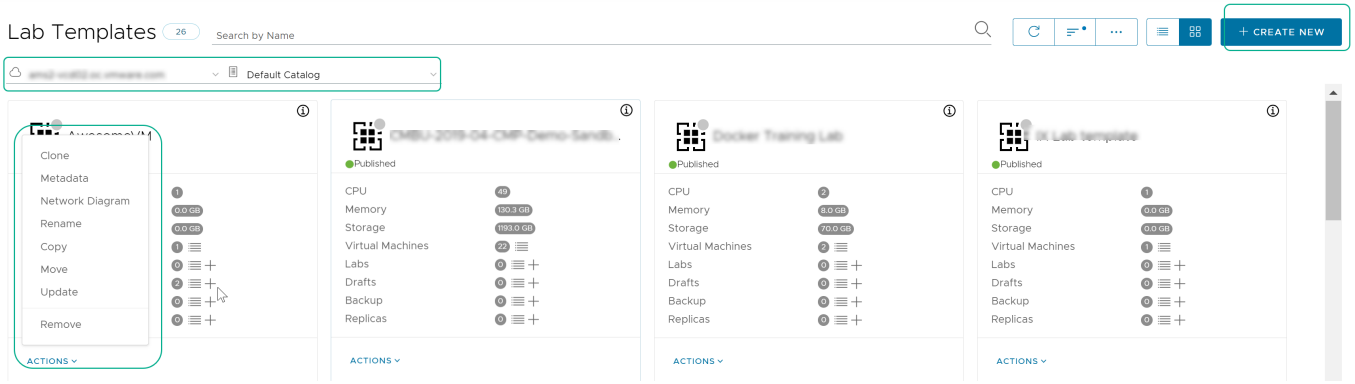
To modify network settings of a lab template, see Modify Network Settings of a Lab Template.

Working with Lab Builder in VMware Lab Platform

A lab template consists of one or more virtual machines and is required to run your labs. Use **Lab Builder** to create VMware Lab Platform lab templates.

To access the **Lab Builder** page, navigate to **Lab Management › Lab Builder**.

Figure 6: Lab Builder home page



You can use Lab Builder to:

- Upload virtual machine files and media.
- Create a lab template from scratch.
- Clone an existing lab template.
- Edit an existing lab template. A lab template must be in a draft state to edit it.
- Publish a lab template. Publishing a lab template sends it to a lab template catalog and makes it available to run your labs. You cannot edit a lab template once you publish it.
- Create a replica (only available if you have more than one cloud assigned to your account).
- Backup a lab template. The purpose of a backup is to create a copy of a lab template, if there is failure or changes made to the deployed lab template.
- Create a draft. This is identical to cloning. Use a draft as a base. When a modified draft is applied to a template, all deployment pools that contain the template are updated.
- Remove a lab template. This deletes a lab template. All drafts and prepopulated resources or labs activated for the lab template are also deleted.
- View visual representation of the network of a lab.
- Upload any file from your local system and download it in your VM while building a lab template in Lab Builder.

Lab Builder provides a card view and a list view of all the lab templates. The card view is the default view, and you can toggle between the views.

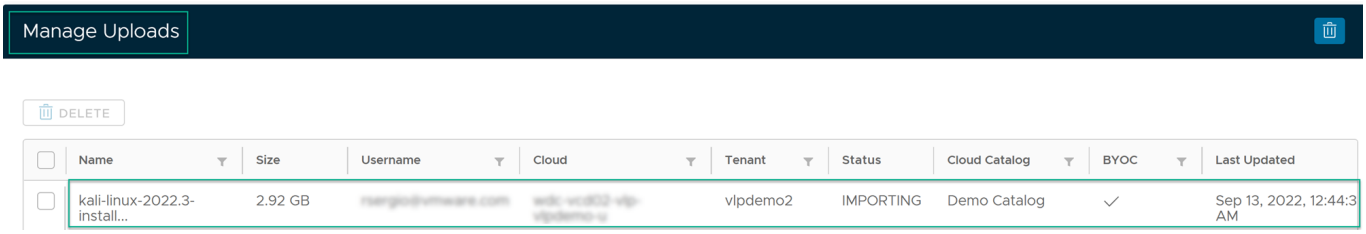
> **NOTE**
> Published lab templates are displayed in green and unpublished lab templates are displayed in yellow.

Upload a VM or an ISO file in VMware Lab Platform

If you develop virtual machines (VMs) or applications (VM bundles) on-premises, you must upload them to the cloud to create templates for VMs or lab templates. In addition, when you build a VM in the cloud, you have to install a media file (an ISO) with an OS.

You can import a virtual machine (VM) and a media file (ISO) to your cloud catalog using the administrator user interface. The supported file types are: OVF, OVA, virtual machine disk (VMDK), and ISO.

1. Navigate to **Lab Management › Lab Builder**.
2. Click the three ellipsis next to the **Create New** tab.
3. Select **Upload VMs/ISOs** from the drop-down menu.
4. On the Upload VMs/ISOs form, select a cloud catalog destination.
5. To upload a file, select a file and click **Attach Files**. Or, you can directly drag the file from your local system to the Upload VM/ISOs window.
6. When the file is displayed in the Upload window and ready for upload, click **Upload**.



When an upload is completed, there is a waiting period before you can use the file, because the file is further processed and imported into the cloud. You can view the file upload status in the **Manage Upload** widget in the **Dashboard**. When the file is successfully uploaded, it automatically moves for the import operation. You can see the status of the file import in the Import window page. When the import operation is complete, the template is available in the **Lab Builder** as a published template for use. An automatic job regularly checks for uploaded items, updates the status, and syncs them with VMware Lab Platform and the **Lab Builder**. You can also start the resynchronize job manually.

7. Click **Close**.

   You can view the ISO (media files) in your cloud catalog from the **Lab Management › Cloud Media** page.

8. Go to the dashboard (**Dashboards › Default Dashboard)** and monitor the progress under the **Content Migration Task** widget. To see the history of tasks, click the arrow next to the task name.
9. Automatically, the resynchronize job is run every hour. To manually synchronize a lab after the upload, go back to **Lab Builder** and click the ellipsis next to the **Create New** tab, and click **Full Resync Lab Templates for current catalog**. To synchronize a lab for the cloud org, select the **Full Sync Cloud Org** option from the drop-down menu. The resynchronize action syncs all the media files and lab templates for the selected cloud.

Create a lab template with the Lab Builder in VMware Lab Platform

You can create a lab template from scratch with Lab Builder.

The prerequisites for creating a lab template with **Lab Builder** are:

- You must have a cloud organization to create a lab template. For more information on cloud and adding a cloud to VMware Lab Platform, see Working with Cloud Organizations.

  NOTE
  Lab Builder displays a message if there are no cloud organizations for the customer account.

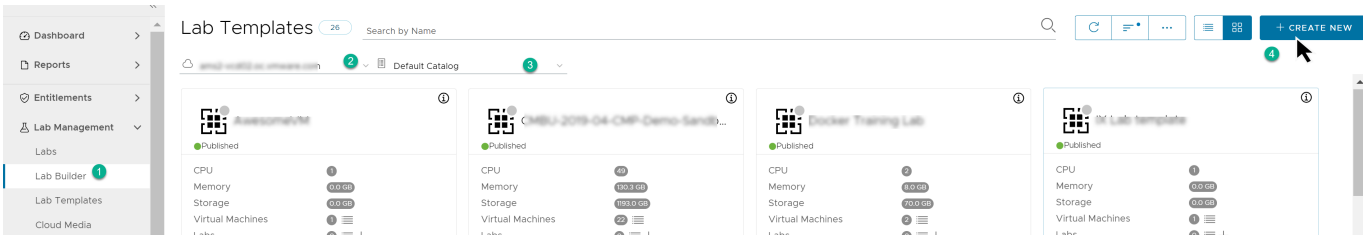
- Your cloud organization must contain at least one cloud catalog.

  NOTE
  Once you publish a lab template, you cannot edit it. You can only clone, delete, backup, or create a draft of the published lab template and modify the template.

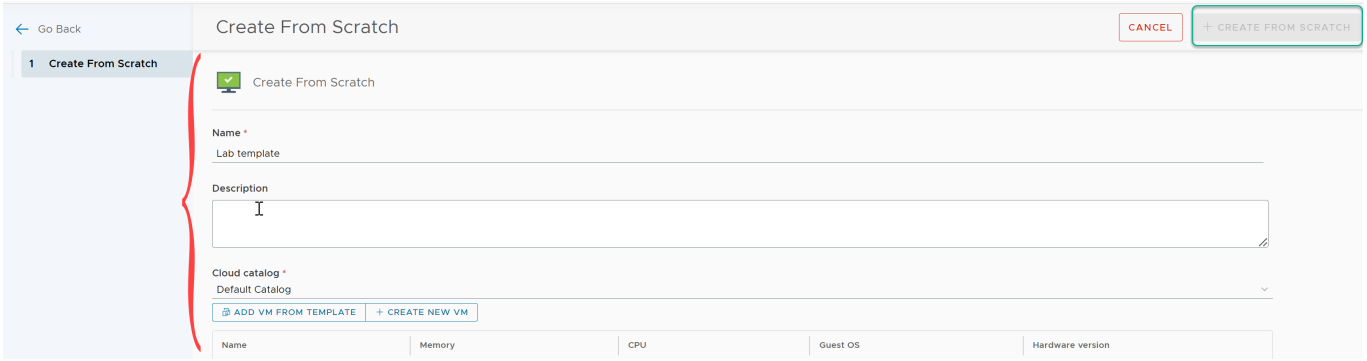
1. To open the **Lab Templates** page, select **Lab Management › Lab Builder** .
2. On the **Lab Templates** page, click the cloud drop-down menu and select the cloud that you want to use for the lab template you are building. (Only available if you have more than one cloud).

   The first cloud in the drop-down menu is the default cloud. You can add a display name for a cloud in the **Cloud Management › Clouds** page.

3. Select a catalog from the catalog drop-down menu.
4. To open the **Create from Scratch** form, click **Create New** .



5. On the **Create from Scratch** form, specify a name and description for the lab, and select a data center. (Only available if you have more than one data center). If you don't select a cloud, the default cloud is selected.
6. On the **Create from Scratch** form, click **Add VM From Template**, and open the Add VM form. Or, to create a new VM, click **Create New VM**.



7. To add virtual machines in your lab template, on the **Add VM from Template** form, select one or more virtual machines you want, and click **Add**. Or, to create a VM, click **Create New VM**.
   The Lab Builder adds the VMs you selected, or created.
8. On the **Create from Scratch** form, you can also edit the VMs you selected. Click the vertical ellipsis icon, and select **Edit VM**.
9. On the **Edit VM** form, you can select the number of CPUs and memory. Click **Save** after you select. The **Create from Scratch** form reopens with the CPUs and memory values you selected.

   If you mark the oVDC to use sizing policy you do not have the option to manually select the number of CPUs and memroy for the VM in Lab Builder. Instead, you must select a sizing policy from the **Sizing policy** drop-down , where the values for the number of CPUs and memory are predefined.

10. Click **Create From Scratch**. The **Lab Builder** adds your lab template, as unpublished, to the **Lab Templates** list. Once you click **Create From Scratch**, Lab Builder displays a progress bar.
11. You can edit, publish, or delete the lab template while it is in an unpublished state. To create the lab template and make it available to run a lab, click **Actions › Publish**.
12. To force a resync, click on the ellipsis next to the **Create New** tab, and click **Full Resync Lab Templates for current catalog**. The resync action will sync all the media and lab templates for the selected cloud.

Edit a lab template with the Lab Builder in VMware Lab Platform

You can edit a lab template while it is in a draft state using the Lab Builder.

> **NOTE**
> Once you publish a lab template, you cannot edit it. You can only clone, delete, rename, or create a draft of the published lab template and modify it.

1. To open the **Lab Templates** page to edit, select **Lab Management › Lab Builder**.
2. On the **Lab Templates** page, open the cloud drop-down menu and select the cloud you want to use for the lab template you are building. (Only available if you have more than one cloud).

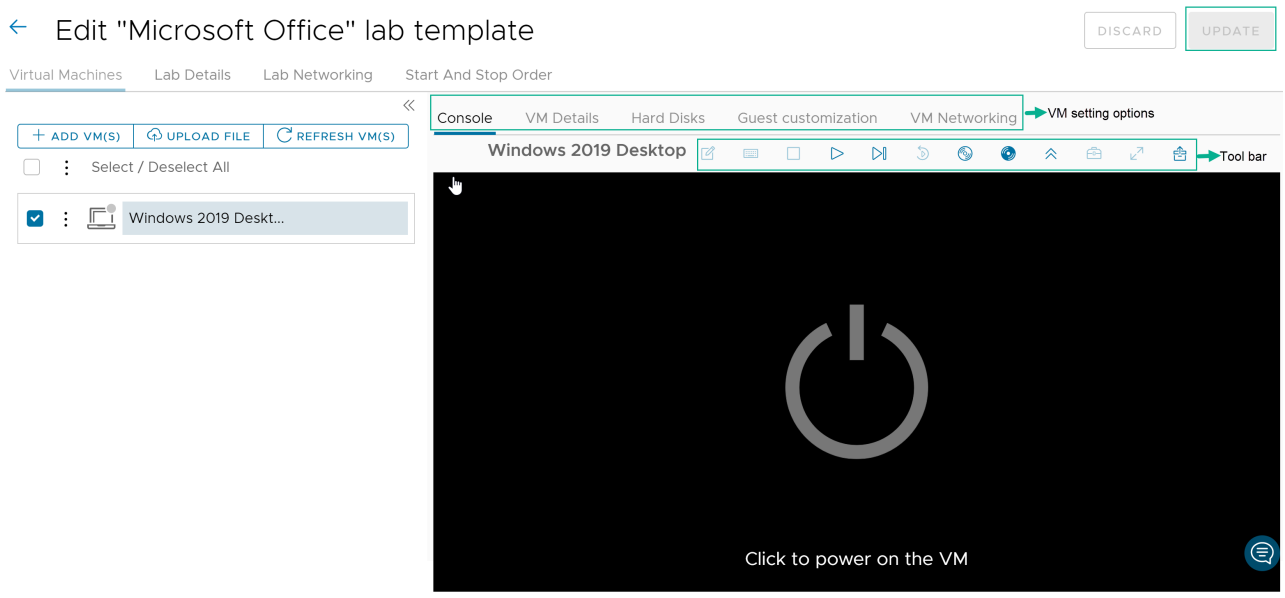
   The first cloud in the drop-down menu is the Default cloud.

3. To open the Edit Lab Template form, select an unpublished lab template and click **Actions › Edit**.

   On the **Edit Lab Template** form, you can update or modify the lab template. Specify a name, a description, add or remove VMs and change the advanced properties for the lab template.

4. To add VMs to the lab template:
   a) Click the **Virtual Machines** tab, and click **Add VM(s)** .
   b) To add a VM from a template, click **Add VM From Template**. Select the VMs from the list and click **Add**.
   c) To create a new VM and add to the template, click **Create New VM**. Fill the details in the page and click **Add**.
5. (Optional) To power on the VMs, click the three ellipses next to the VM, and click **Power On**. Additionally, you can use the **Power Off** option to power off the individual VMs. To power on all the VMs associated with a lab, check the **Select/Deselect All** check box, and click the ellipses to perform the required action.
6. To add media files to the lab, on the Virtual Machines page, click **Upload File** and add the required files for the lab.
7. To access the VM console, and view or modify the VM settings, click on the VM name on the **Edit Lab Template** form. The console view and VM settings options are displayed.

Figure 7: VM Console view



Actions	Description
Console	Use the tool bar on the Console page to perform an action for the console. Hover the mouse on the icons to view the name of the action option on the toolbar. • **Send Text**: Send text or notes. Enter text (copy and paste from your terminal into the console. • **CTRL+ALT+DEL**: Sends the CTRL+ALT+DEL key combination to the virtual machine (VM). The result of sending the key combination is operation-system dependent. • **Power Off**: Power off the VM. • **Power On**: Power on the VM. • **Power on and force customization**: You can include the Guest OS Customization information necessary for the virtual machine to join a Windows domain. • **Reset**: If you reset a lab, all its virtual machines (VMs) are restored to their original state. You are not able to recover any work you have done. • **Reset VM And Enter BIOS**: Reset a virtual machine (VM) of a lab, and start the BIOS and make changes to the boot order. • **Insert CD/DVD from cloud catalog**: Insert an ISO or media file. • **Eject CD/DVD**: Remove a media file. • **Upgrade VM hardware version**: Upgrade the hardware version. • **Install VMware tools**: VMware Tools is a suite of utilities that enhances the performance of the virtual machines guest operating system and improves management of the virtual machine. For more information on VMware Tools, see the knowledge base article. • **Expand**: Expand the console view. • **Undock**: Undock the console and go full screen mode.
VM Details	You can view and edit VM details such as VM name, VM description, modify CPU capacity and memory capacity, change the Guest OS of the VM.
Hard Disks	You can add or remove hard disks from a VM. To remove multiple hard disks from a lab template, select the hard disks from the list, and click the **Delete** icon.

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Actions	Description
Guest Customization	The **Guest Customization** tab includes the information necessary for the virtual machine to join a Windows domain. An organization administrator can specify default values for a domain that Windows guests in that organization can join. Not all Windows virtual machines must join a domain, but in most enterprise installations, a virtual machine that is not a domain member cannot access many of the available network resources. For more information on the Guest Customization settings, see the Change the Guest OS Customization Properties of a Virtual Machine topic in the VMware Cloud Director documentation.
VM Networking	To view or modify VM network settings, click **VM Networking**. You can add, delete, or modify the NIC settings for a VM.

8. To specify a VM name, and add description for the VM, click **Lab Details**.
9. To add or edit the network information for the VM, click the **Lab Networking** tab, and click **Add Network**. Enter the network specifications and click **Save**. By default, lab_builder_net template is selected.
10. If you have more than one VM and want to define the order of booting up your VMs and resources, click the **Start and Stop Order** tab. Set the power on and power off order for your VMs. Make the necessary changes and click **Save**.
11. To view the visual depiction or diagram of network connections for a lab, click the **Diagram** option on the **Lab Networking** page. You can add, modify, or delete NIC cards, network configurations in the **Diagram** view.
12. If you have local files or VMs that you want to add to your lab, click **Upload File**, and click **Attach File** to add resources, add files from your local system to the lab template.

    Once the file is uploaded to a temporary repository, you can download the file from the **Copy Link** option to your lab environment. The file link expires after an hour.

13. To save your changes, click **Update**.
14. To close the editing form and to go back to Lab Builder, click **Discard**.
15. To publish and use your lab template, click **Update and Publish**.

Lab Builder action operations in VMware Lab Platform

In VMware Lab Platform, you can perform the following actions for a lab in the **Lab Builder**.

1. Click **Lab Management › Lab Builder**.
2. On the **Lab Templates** page, select an existing lab and click **Actions**.

Actions	Description
Clone	Clone an existing lab.
Metadata	Add or modify metadata of a lab network.
Network Diagram	Display visual diagram of the lab network.
Rename	Rename the existing lab.

Table continued on next page

Continued from previous page

Actions	Description
Copy	Copy the existing lab across cloud catalog.
Move	Move the existing lab across cloud catalog.
Update	Edit the existing lab details.
Remove	Remove the lab from the Lab Builder. **CAUTION** If you a remove a lab template, its subsequent prepopulated lab templates and drafts are also removed.

Clone and create a lab template in VMware Lab Platform

In the VMware Lab Platform administrator user interface, you can create a lab template by cloning and modifying an existing published lab template.

1. To open the **Lab Templates** list page, select **Lab Management › Lab Builder**.
2. In the **Lab Templates** page, open the cloud drop-down menu, and select the cloud you want to use for the lab template that you are cloning.
3. Select a lab template, and click **Actions › Clone**. The **Clone Lab Template** form opens.
4. In the **Clone Lab Template** form, specify a name and description for the template.
5. In the **Clone Lab Template** form, click **+ Clone <*lab_template*>**. The **Lab Builder** adds your lab template, as a draft, to the **Lab Templates** list. You can edit, publish, or delete the lab template while it is in the draft state.
   Once you click **Clone**, the **Lab Builder** displays a progress bar. After the clone is created, Lab Builder displays a message with a lab template editing link.

Create a draft lab template in VMware Lab Platform

You can create a draft lab template for an existing published lab template.

1. To open the Lab Templates list page, select **Lab Management › Lab Builder** .
2. In the **Lab Templates** page, open the cloud drop-down menu, and select the cloud you want to use for the lab template that you want to create the draft template.
3. Select a lab template, and click the **+** icon listed for the **Drafts** option. The **Create draft** form opens.
4. Click **Create Draft**.
5. To edit, remove, or apply changes to the draft lab template, click the three ellipses icon next to the **Drafts** option.

The **Drafts** option lists the number of drafts created for the lab template.

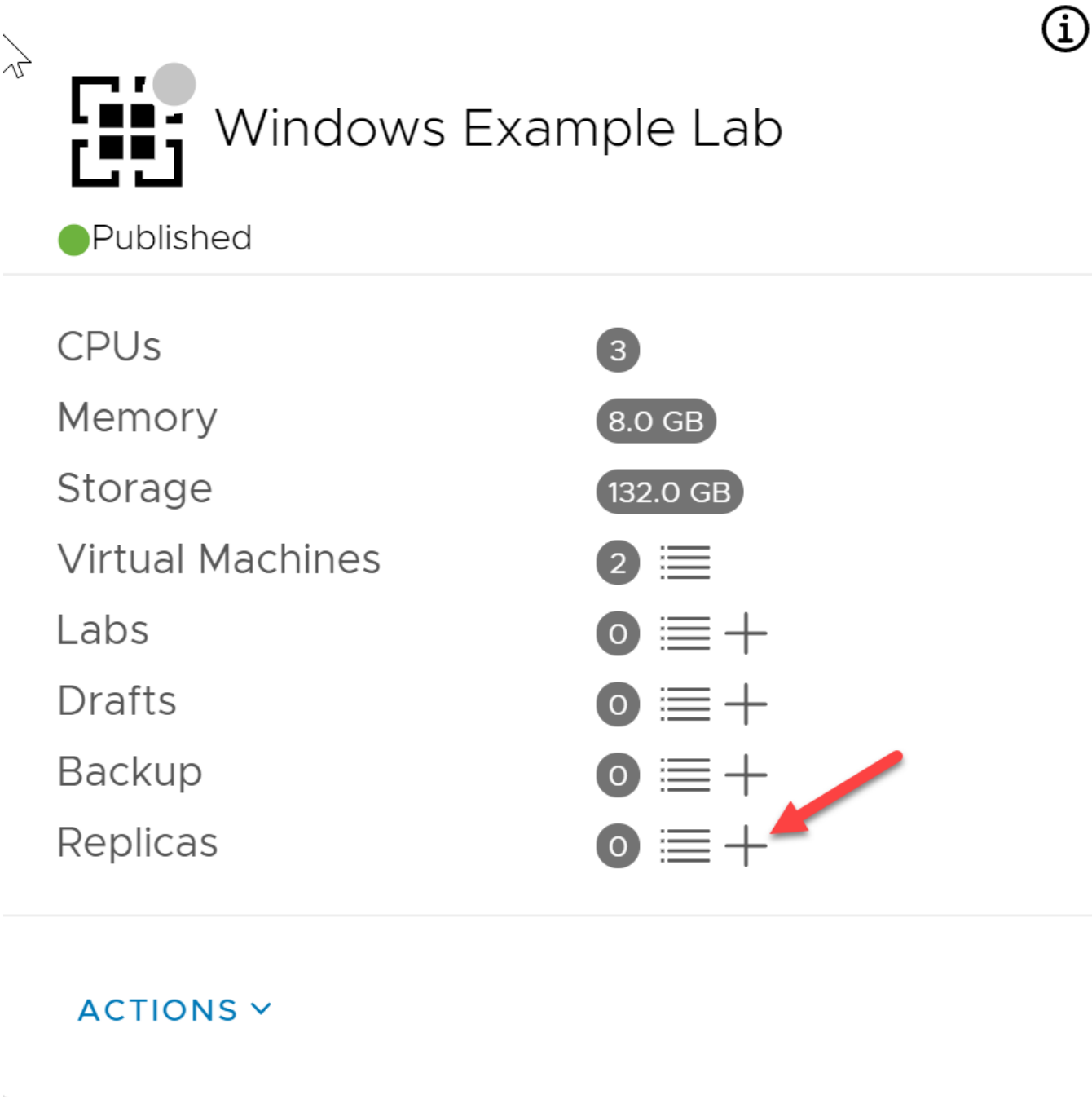
Create a lab replication in VMware Lab Platform

You can create a replica of a lab in one or more regions. Any changes to a lab instance are synchronized to all existing replicas of the lab. This feature is only available if your organization has multiple availability zones.

The goal of this feature is to let you deploy a lab quickly, regardless of the global region the students are in. To use the Lab Replication feature, activate the **Enable template replication** setting. By default, this setting is activated for all users. To deactivate this option, contact the VMware Lab Platform support team.
To create a replica of a lab template:

1. Navigate to **Lab Management › Lab Builder**.

2. Select the lab you want to replicate and click the **+** symbol for the **Replicas** option.

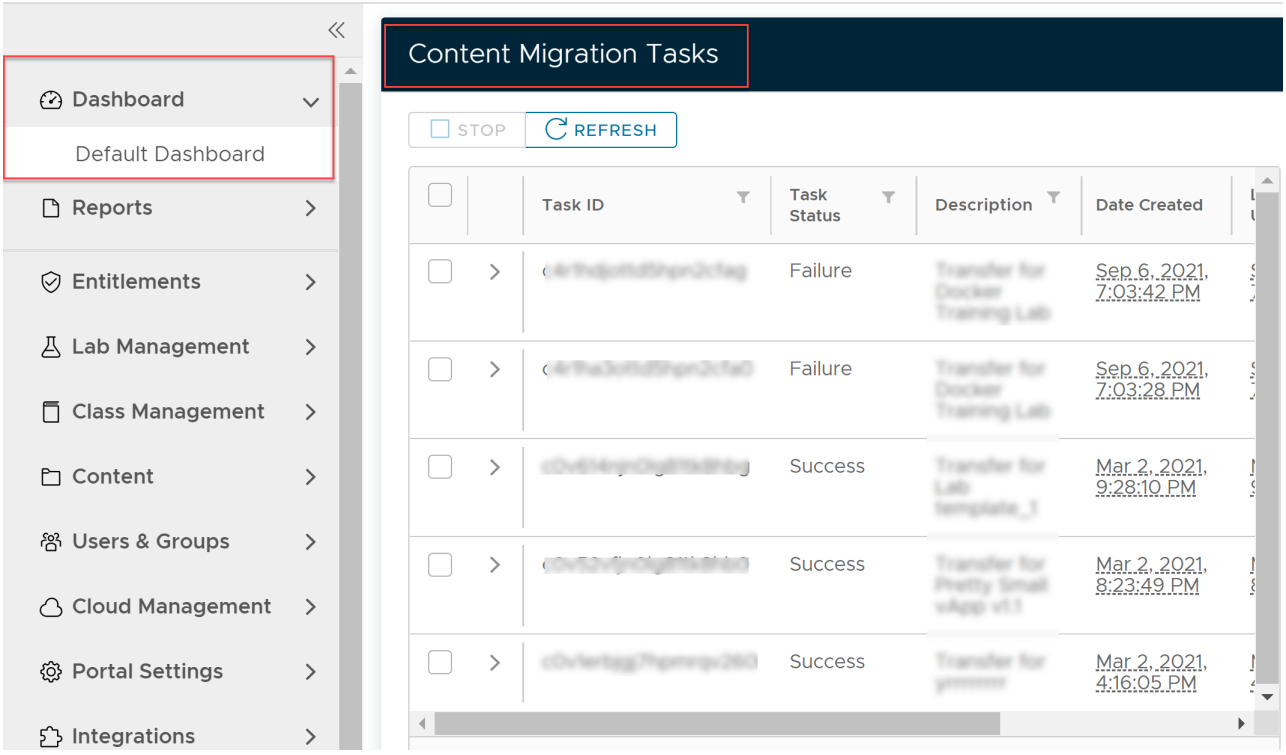


3. On the **Replicate Lab Template** form, select the regions and cloud catalogs you want to replicate to.
4. If you want the lab template to export outside the VMware Lab Platform cloud, select **Backup (Optional)**.

> **NOTE**
> The administrators initiating the replica receives an email with a URL to access the backup. This option is also available from the **Actions** menu.

To check the progress of the task, go to **Dashboards › Default Dashboard**, and see the status of the replication task from the **Content Migration Tasks** widget.

Figure 8: Content Migration Tasks



If the widget is not visible, you can follow the instructions on how to add a widget from Working with widgets.

NOTE
Once a lab is replicated, there is no distinction between the source template and the replicated template.

To delete the replicated template, select any one of the linked labs, and click **Remove** from the **Actions** menu on the lower left.

To delete all linked labs, check the **Delete Replicas** checkbox.

NOTE
If you choose **Backup (Optional)** during the replication process, then you can also **Export** the files from the **Actions** menu. Selecting this option opens a window with a list of individual files for download, and the expiration date and time.

Back up and export a lab template in VMware Lab Platform

You can create a back up of a lab template and export it to one or more regions.

The Backup feature lets you to save and keep a working copy of an active lab template prior to modifying it.

1. Navigate to **Lab Management › Lab Builder**.
2. Select the template you want to back up, and click the **+** icon listed for the **Backup** option.
3. On the **Backup** form, click **Backup**.

To check the progress of the task, you can go to **Dashboards › Default Dashboard** and see the status of the replication task from the **Content Migration Tasks** widget. If the widget is not visible, you can follow the instructions on how to add a widget from Working with widgets.

Once you click **Backup**, Lab Builder displays a progress bar. After the back up is created, Lab Builder displays number of backups for the lab. You can view file name, backup date created and validity of the backup for the template. From the Backup lab template form, select the files you want to export and download.

NOTE
Backup files are saved for seven days. After the validity period ends, the backup files are deleted from the system.

Import a lab template from another cloud in VMware Lab Platform

You can import a lab template from a cloud (VMware Cloud Director) that is not registered in VMware Lab Platform.

1. To open the **Lab Templates** page, select **Lab Management › Lab Builder** .
2. Click the ellipsis next to the **+ Create New** tab, and click **Import External Cloud Template**.
3. Enter the cloud credential details in the **Import External Cloud Template** page.
4. Click **Next**.
5. From the templates list, select a template and a catalog.
6. To import the cloud in VMware Lab Platform, click **Import** .

   If the cloud template is successfully imported, a success message is displayed on the system. To view the status of the import, see the status progress in the **Content Migration Tasks** widget.

Add a virtual machine to a lab template in VMware Lab Platform

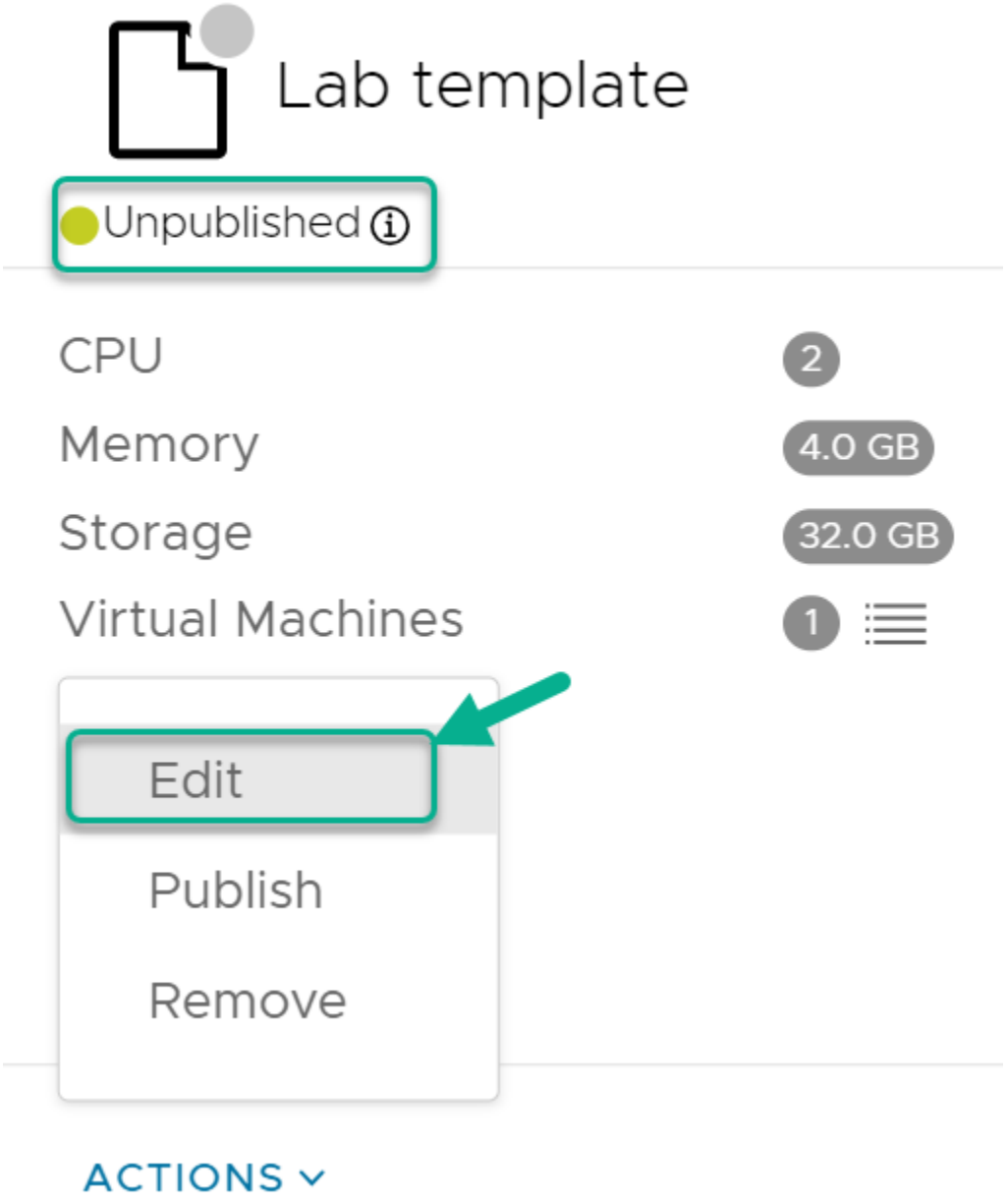
You can create and add a virtual machine (VM) for an unpublished lab template in the Lab Builder.

To create and add a VM for a lab template in the Lab Builder:

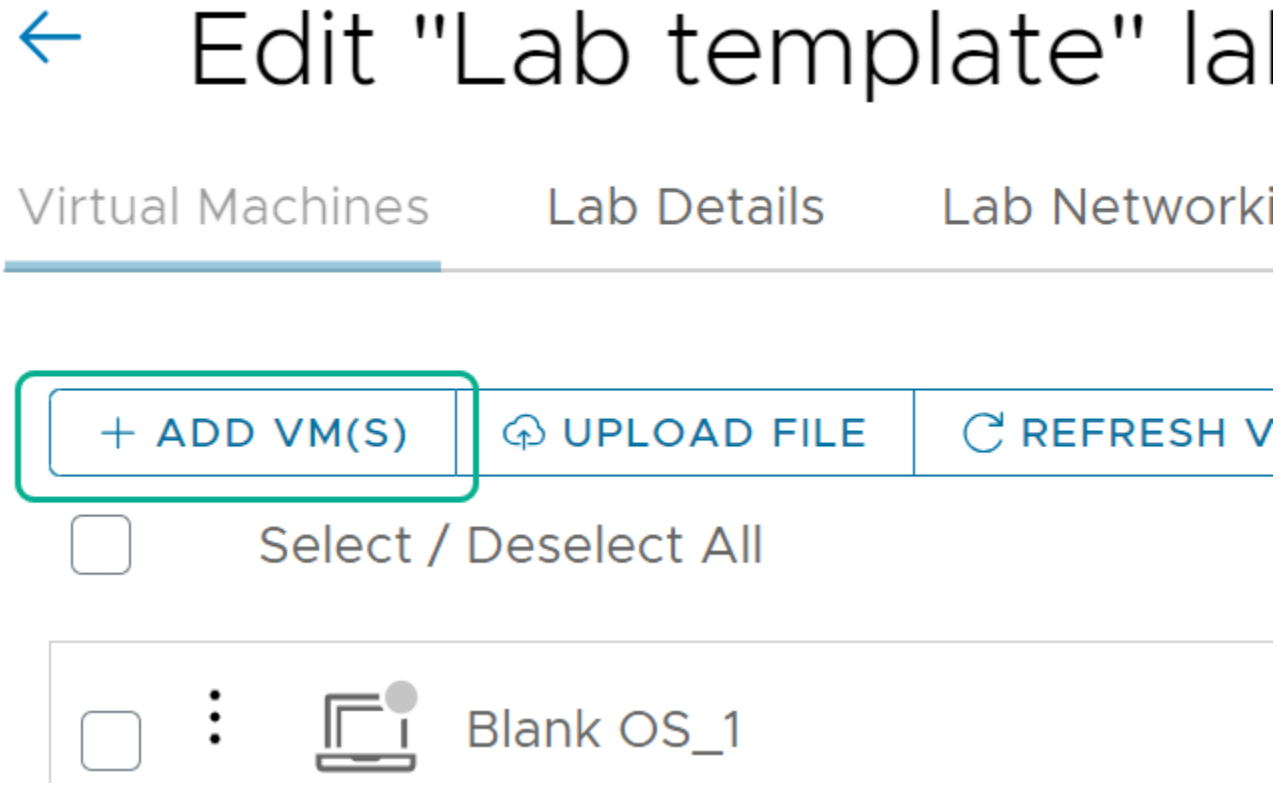
1. Open the **Lab Templates** page to edit, select **Lab Management › Lab Builder**.
2. On the **Lab Templates** page, open the cloud drop-down menu and select the cloud you want to use for the lab template you are building. (Only available if you have more than one cloud).

   The first cloud in the drop-down menu is the Default cloud.

3. To open the **Edit Lab Template** form, select an unpublished lab template and click **Actions › Edit**.



4. Click **Add VMs**.



5. To add a VM from an existing template, click **Add VM from Template** and select a VM from the list and click **Add**.
   To create a new VM from scratch, click **+ Create New VM**, fill the details in the form and click **Add**.
6. Click **Add**.
   The new VM is added to the lab template.

Remove a virtual machine from a lab template in VMware Lab Platform

You can remove a virtual machine (VM) from an unpublished lab template.

To remove a VM from a lab template:

1. Open the **Lab Templates** page, select **Lab Management › Lab Builder**.
2. On the **Lab Templates** page, open the cloud drop-down menu and select the cloud you want to use for the lab template you are building. (Only available if you have more than one cloud).

   The first cloud in the drop-down menu is the Default cloud.

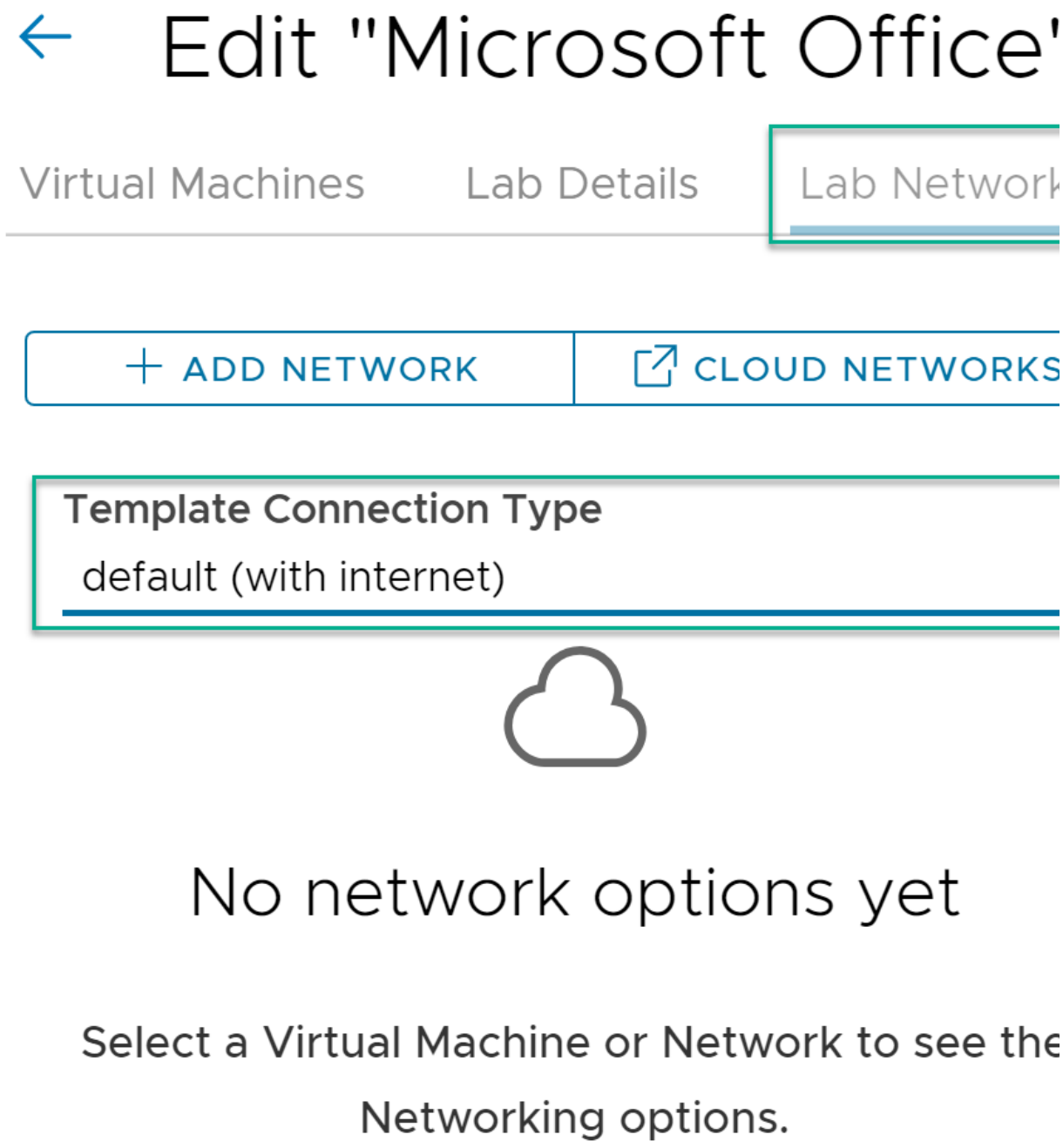
3. To open the **Edit Lab Template** form, select an unpublished lab template and click **Actions › Edit**.
4. Select the VM on the **Edit template** page, and click the three ellipses icon and select **Remove**.

Modify network settings of a lab template in VMware Lab Platform

By default, lab templates are created with a network connection between the virtual machines and the outside world. This topic describes how to manage the network settings in lab templates and check for network problems.

NOTE
You can modify the network settings of a VM only when a lab is in unpublished or in draft state.

1. From the navigation pane, select **Lab Management › Lab Builder**.
2. On the **Lab Builder** page, for labs in draft or unpublished state, click **Actions › Edit Lab Networking**.
3. Click **Lab Networking**.
4. The **Template Connection Type** and a network diagram for the lab template is displayed.



5. Under the **Template Connection Type** field, the following template connection type options are available:
   - **none**: There is no network connection at all. A completely isolated lab template, including the connection between the virtual machines.
   - **isolated (with no internet)**: There is a network connection between the virtual machines but, not to the outside world.
   - **default (with internet)**: There is network connection between the virtual machines and the outside world.

  - **mixed**: A user-defined network with multiple lab templates and networks.
6. To add a new VM or modify the VM network setting, click **Add VM**.

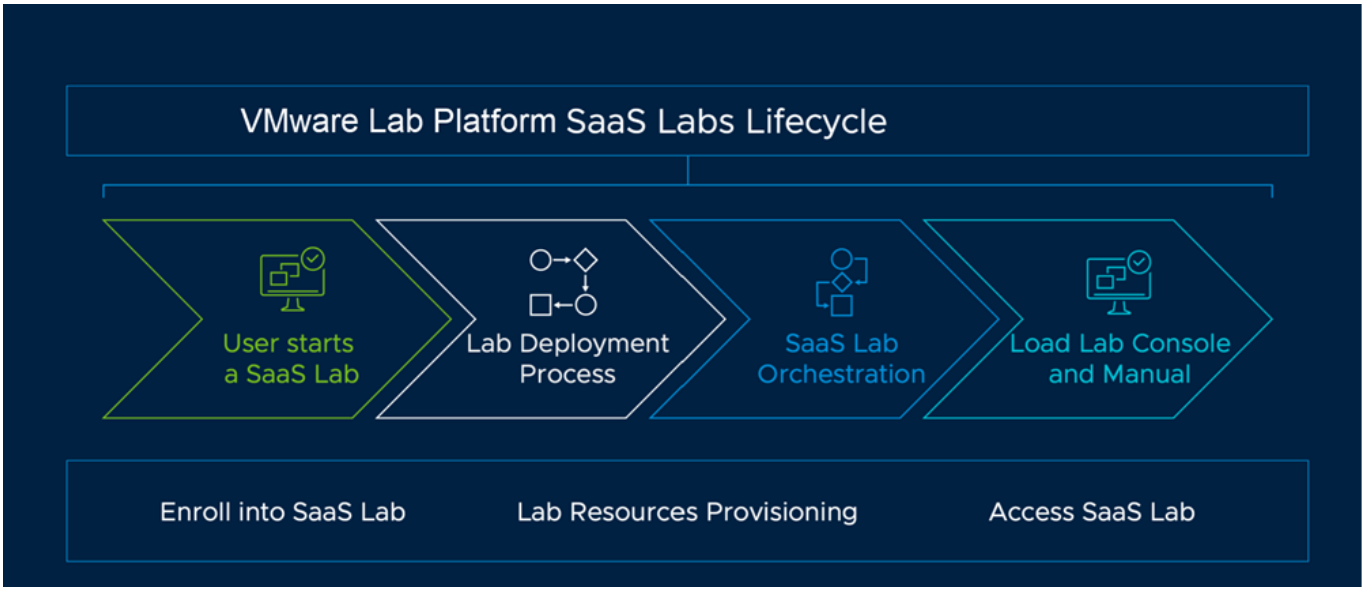
   If you do not provide any network settings, the system takes the default settings of the lab_builder_net_2 network.
7. Click **Save**.
8. Use the network diagram to add, modify, or delete NIC cards, and network configurations for the lab template.

Working with VMware Lab Platform SaaS Consoles

For an on-premises product, the traditional on-demand lab environment with pre-configured VMs and resources as lab templates work best. However for SaaS applications, where feature updates are more frequent, users need labs that provide the same experience as the SaaS application.

With the SaaS Consoles feature, you can provide SaaS applications as labs, simulated labs, and web URLs in the labs. To create simulated labs and add links to them in the VMware Lab Platform, you can use third party applications such as Walnut. A simulated lab provides pre-recorded labs in an interactive mode to users in the VMware Lab Platform user interface.

Figure 9: VMware Lab Platform SaaS labs life cycle



Add SaaS applications in the SaaS Consoles in VMware Lab Platform

As an administrator, you can add SaaS applications as labs and also integrate customized code and provide user login credentials, user instructions, and run the SaaS applications in the VMware Lab Platform user interface. You can add URLs for simulated content in the **SaaS Console** and provide an interactive lab experience to your users. You can add an `iframe`, or an `Isim` type of simulator link for SaaS consoles.

- Enable the **Portal Settings › Configuration › Console Interface › Enable SaaS Consoles** setting. To enable this option, contact the VMwware Lab Platform support team.
- An HTML URL link for your simulated content, or a SaaS application.
- Lab manual for your SaaS application.
- Web application that supports HTTP and HTTPs request from VMwware Lab Platform for service orchestration.

To create a SaaS console for your lab:

1. From the navigation pane, select **Lab Management › SaaS Consoles**.

2. Click **+ Create New**.
3. Enter the basic details for the SaaS console.

Name	Enter a name for the console.
Display Name	Enter a display name for the console. The name can be same as the **Name** field.
Type	To embed the simulated content link in the lab, from the drop-down menu select an option type. • **Iframe** • **iSim** • **Test Console** • **Other**
URL	Add the URL of the content to embed in the lab.
Integration Code	(Optional) You can enter any customization related to the URL code here.
Notes	(Optional) If you have any notes related to the SaaS console, you can enter it in the **Notes** section.

4. Click **+ Create SaaS Console**.
5. Enter the details for the start end point in the **Start Endpoint** page. The "startEndpoint" is called when an entitlement is created for a SaaS lab.

Target URL	Enter the external URL address to be called.
HTTP Method	Enter the HTTP method to be used. • GET • POST • PUT • DELETE • PATCH
Authentication Method	Specify the authentication method be used. • None—Username and password fields are ignored. • Basic—The provided username and password are used to construct the HTTP Basic Authentication header. • Header—Allows you to specify custom header name in the username field and value in the password field. • Bearer—The value provided in the password field is used in Authentication: Bearer <value> header.
Username/header name	Enter the username.
Password	Enter the password.

6. Click **Update SaaS Console**.
7. Enter the details for the stop end point in the **Stop Endpoint** page. The "stopEndpoint" is called when the entitlement is ended, or is canceled to signal to the external system that resources allocated for this entitlement can be released.

Target URL	Enter the external URL address to .
HTTP Method	Enter the HTTP method to be used. • GET • POST • PUT

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	• DELETE • PATCH
Authentication Method	Specify the authentication method be used. • None—Username and password fields are ignored. • Basic—The provided username and password are used to construct the HTTP Basic Authentication header. • Header—Allows you to specify custom header name in the username field and value in the password field. • Bearer—The value provided in the password field is used in Authentication: Bearer <value> header.
Username/header name	Enter the username.
Password	Enter the password.

8. Click **Update SaaS Console**.
9. Enter the SaaS console property details in the **SaaS Console Property Definitions** page.

   A Saas console can have 0 or more configuration properties defined. It describes a property that is expected to be configured or created in the external system and a value for it is provided in VMware Lab Platform. The properties can be used in a manual as placeholders for the actual values. Use double curly brackets for the property name - It {{propertyName}} is replaced with the provided property value.

10. Click **Update SaaS Console**.

The SaaS console link is available for you to add in the **Labs › Update Content** page of your lab. The SaaS lab is provisioned and configured during the lab startup process.

How to deliver an on-demand SaaS lab in VMware Lab Platform

As a product owner, or a product administrator, you can deliver an on-demand lab for your VMware Cloud SaaS product or a service using the VMware Lab Platform SaaS Consoles feature. A SaaS lab provides your users on-demand access to a virtual lab environment.

Using the "How to deliver an on-demand SaaS lab in VMware Lab Platform" document you can learn how to integrate a VMware Cloud Services Provider SaaS service with VMware Lab Platform, and deploy and deliver a SaaS lab using the VMware Lab Platform framework.

Intended audience

The *How to Deliver a VMware Cloud Services Provider SaaS Lab* documentation is intended for VMware Cloud Services Provider cloud architects and administrators who are familiar with and want to use the VMware Lab Platform SaaS Consoles feature to deliver a VMware Cloud Services Provider SaaS service as an on-demand lab.

Overview of the VMware Lab Platform SaaS Consoles

The SaaS Consoles feature allows you to deliver hands-on labs, virtual or in-person training labs, for any VMware Cloud Services Provider SaaS services. The end-to-end SaaS learning environment takes care of the entire technology stack within the SaaS offering including—providing the infrastructure, provisioning the lab, user management, entitlement creation, lab life cycle management, lab manual creation, and lab delivery.

To use the SaaS Consoles feature, all you need is the web URL of the application (VMware Lab Platform) and administrator user credentials to use the application.

Key benefits of an on-demand SaaS lab

Some of the key benefits of offering a SaaS service as an on-demand lab are:

- Easy access: You can access the lab environments from anywhere, anytime on a device that supports the internet, and the required user credentials.
- Same user experience as the actual product: Provides real-time user experience of the actual product.
- Scalability: Based on the lab requirements, you can provision the compute and storage resources. You can easily add, manage or re-allocate resources, and VMs based on the demand for labs.
- Rich learning environment: You can integrate lab manuals, support chat, interactive videos, social media applications, and classroom events in one unified interface.

Life cycle management of a VMware Lab Platform SaaS lab

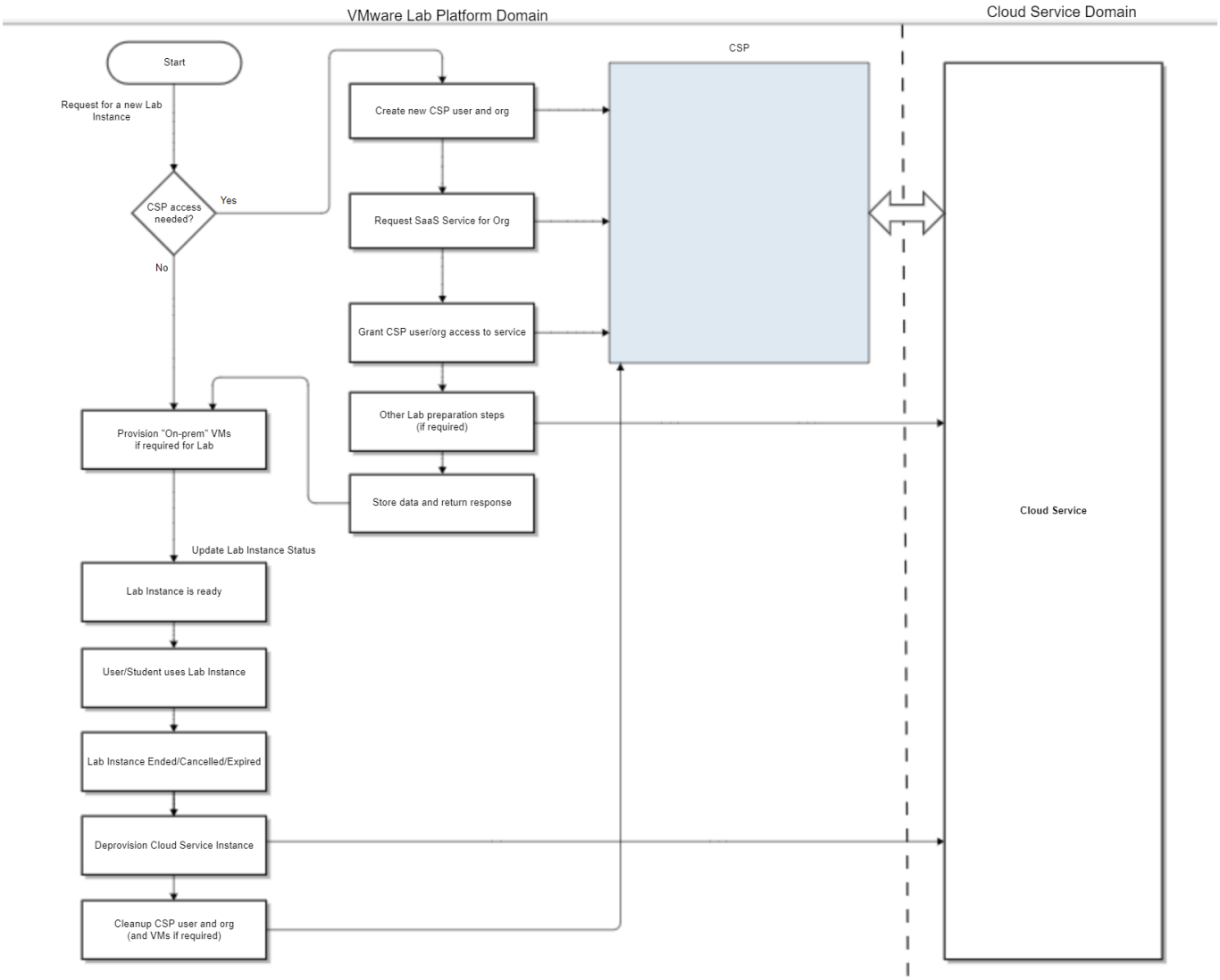
In a traditional on-premises lab, resources are provisioned before a user starts a lab, while in a SaaS lab, an orchestration service is created and lab resources are created and provisioned for the lab.

Before you learn how to deploy and deliver a SaaS lab, it is important to know the life cycle management of a SaaS lab. This section describes the life cycle management of a VMware Lab Platform SaaS lab. How a SaaS lab is configured, deployed, provisioned, and finally deprovisioned for an internal VMware Cloud Services Provider SaaS service.

> **NOTE**
> You need to integrate the VMware Cloud Services Provider service with VMware Lab Platform to deploy and deliver the SaaS service as a lab. For more information on how to integrate VMware Cloud Services Provider service with VMware Lab Platform, see Integrate a VMware Cloud Service.

After a VMware Cloud Services Provider service is integrated with VMware Lab Platform, the SaaS lab includes the following phases:

Figure 10: VMware Lab Platform SaaS lab life cycle



- Launch a SaaS lab using the VMware Lab Platform user interface.

  NOTE
  If a VMware Cloud Services Provider user access is required, an HTTP request is sent to the external service (in this case, a request is sent to VMware Cloud Services Provider) to create a user access for this lab instance. A callback is sent to the system.

  For more information on the VMware Lab Platform user interface, see the *VMware Lab Platform User Guide*.
- VMware Lab Platform provisions the VMs and resources for the lab.
- The entitlement status of the lab user is made **Ready**, and the lab user details are embedded in the lab manual.
- Start the lab and follow the instructions provided in the lab manual and complete the lab.
- After the lab is completed, the entitlement is ended, expired or is canceled. The "`stopEndpoint`" SaaS console signals to the external system to clean up the resources created for this entitlement (for example, VMware Cloud Services Provider org or a VMware Cloud Services Provider user).

NOTE
The external system does not provide a callback for the "`stopEndpoint`" response.

System requirements for a VMware Lab Platform SaaS lab

Before you configure your system to deploy and deliver a SaaS lab, it is important to understand the system requirements for a SaaS lab.

- Customer accounts.
- Cloud resources and administrator credentials.
- Supported browsers and environment.
- SaaS lab connections: Web URL of the SaaS application or service that is accessible over the internet and accepts HTTP and HTTPs request from VMware Lab Platform.
- VMware Cloud Services Provider integration: You need to integrate your service with VMware Lab Platform. For more information on how to integrate a VMware Cloud Services Provider service with VMware Lab Platform, see Integrate a VMware Cloud Service.

Integrate a VMware Cloud Services Provider with VMware Lab Platform

To deliver a VMware Cloud Services Provider as a SaaS lab, you need to first integrate the service with VMware Lab Platform. Contact the VMware Lab Platform support team to integrate the VMware Cloud Services Provider service with VMware Lab Platform.

To integrate a VMware Cloud Services Provider:

- On the administrator user interface, activate the **Portal › Configuration Settings › Enable SaaS Console** portal configuration settings. This allows you to view and access the **SaaS Consoles** menu from the navigation pane. For more information on portal configuration settings, see Working with Settings.
- Activate the **Portal › Configuration Settings › Enable CSP SaaS Labs (Orgs and user creation in CSP)**. This adds two new fields, `createCspOrg` and `cspServiceId`, in the SaaS Console. You can add a VMware Cloud Services Provider service ID in VMware Lab Platform and create a temporary VMware Cloud Services Provider organization environment for lab provisioning.

```
Boolean createCspOrg

String  cspServiceId
```

NOTE
The **Enable CSP SaaS Labs (Orgs and user creation in CSP)** portal setting is applicable only for internal VMware Cloud Services Provider services.

- You import the VMware Lab Platform`OAuth App` API in your service organization and provide the "service enabler" rights to VMware Lab Platform. This lets VMware Lab Platform to activate the service for any new VMware Cloud Services Provider organization that is created.
- Use the `createCSPOrg`, and `cspServiceId` APIs and create the orchestration service that accepts HTTP requests and automates the setup of the external service. The values defined in the orchestration service are used in the SaaS Console configurations for a SaaS lab.
- You can also add custom properties in the SaaS console, such as VMware Cloud Services Provider user name, VMware Cloud Services Provider password, VMware Cloud Services Provider org name and so on. This information is embedded in the lab manual of the lab.
- As a normal clean-up, the system deletes the user account and organization created for a lab. If you need any additional code service clean up after an entitlement is completed, inform the VMware Lab Platform support team.

Deploy and deliver a SaaS lab in VMware Lab Platform

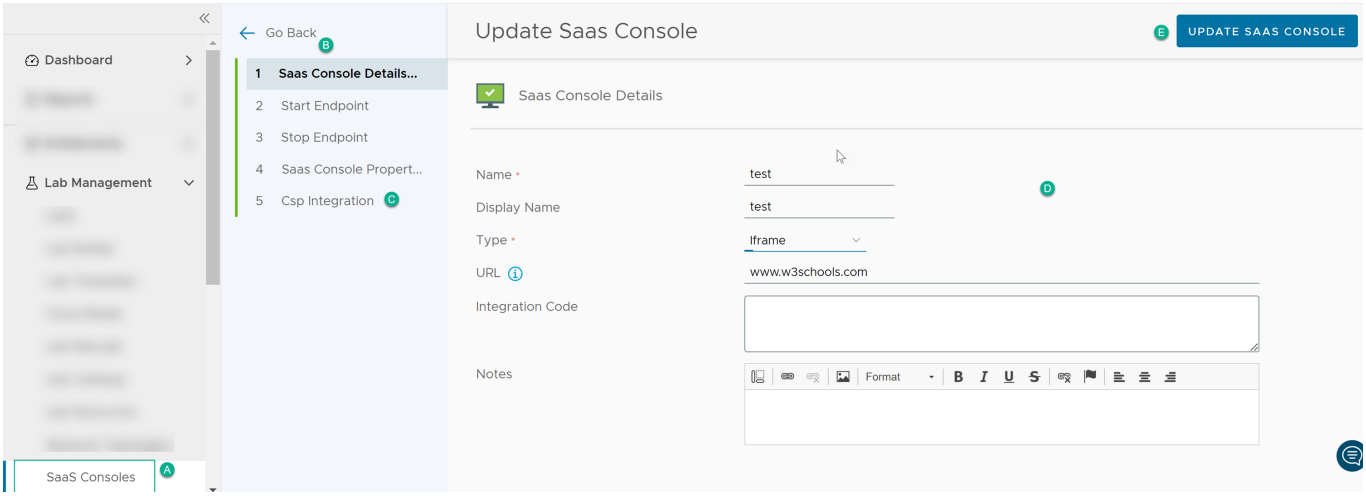
You use the VMware Lab Platform SaaS Consoles feature to deploy and deliver a SaaS lab.

Before you start deploying the lab, you must set up an environment that provides external services as a lab to users.

- System Requirements.
- Integrate a VMware Cloud Services Provider.

To deploy a VMware Cloud Services Provider SaaS lab in VMware Lab Platform:

1. Add the SaaS console configuration properties in **SaaS Consoles**, click **Lab Management › SaaS Consoles › Create New**.
   In this example, sample SaaS details for a `test` SaaS console instance is added.



A	You use the left-navigation pane to select the **SaaS Consoles** menu option.

B	Add details for the SaaS console using the SaaS forms.
	
	You can add the SaaS console configuration properties, such as username, password, or any other custom property that you want to share with the user while using the lab. The output values for the configuration property are displayed in the lab manual.
	
	`For example;`
	
	`cspUsername:saasCspLabEnvironment.orgOwnerUsername,`
	
	`cspPassword:saasCspLabEnvironment.orgOwnerPassword,`

C	In the VMware Cloud Services Provider Integration form, add the VMware Cloud Services Provider ID of the VMware Cloud Services Provider service and enable it.
D	Detail view of the form to enter the required data.
E	To save your changes, click **Update SaaS Console**.

2. Click **Update Lab**.

Your lab is created and listed on the Labs page.

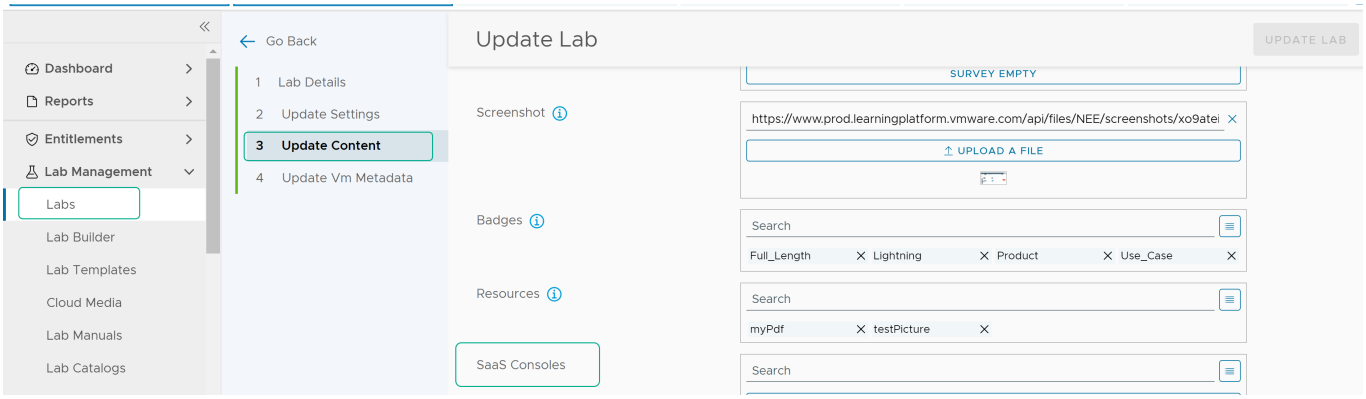
Add a SaaS console to a lab in VMware Lab Platform

As an administrator, you create a lab and add a SaaS console to a lab so that your VMware Cloud Services Provider users can use the lab assigned to them.

Administrator role-based access.

After you create the SaaS console properties, you need to create a lab and add the SaaS console to a lab instance for your users to access and use the SaaS lab.

1. To create a lab and add the new SaaS console to a lab, click **Lab Management › Labs › Create New**. On the **Update Content** page, select the SaaS console from the drop-down list, and add the **SaaS Console** to the lab.



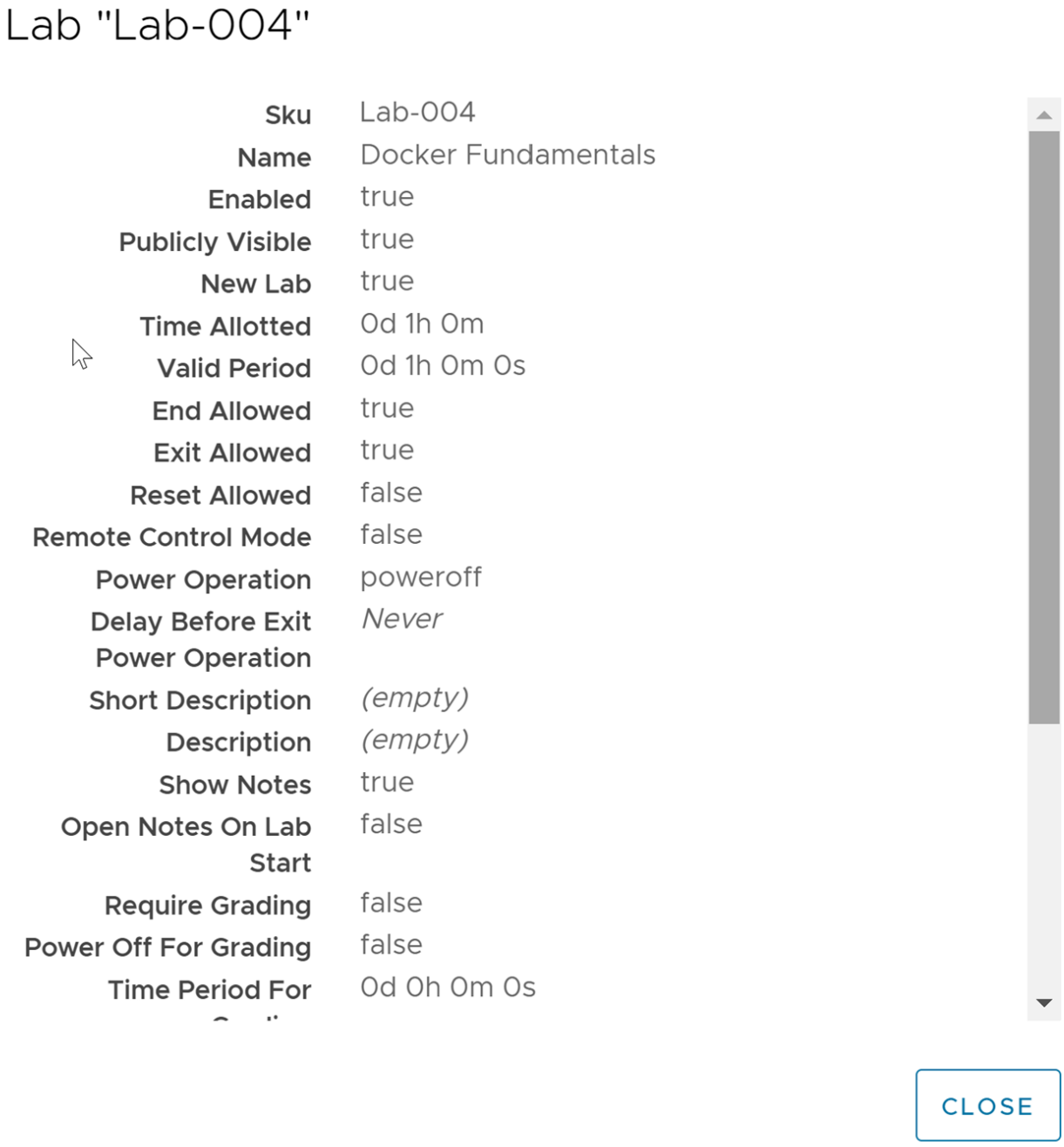
   For detailed information on how to create a lab, and lab template for the lab, see Create a lab.

2. Click **Update Lab**.

The saved lab with SaaS console details is ready and can be assigned to a user account.

Sample lab configuration example

This is a snapshot of a sample lab configuration. You may want to verify your lab details before you assign the lab to a user.



Create a user account and entitlement for the user so that the user can use the lab, see Create an entitlement for a new user.

Launch and start a lab in VMware Lab Platform

You use the end-user interface to start a VMware Lab Platform lab.

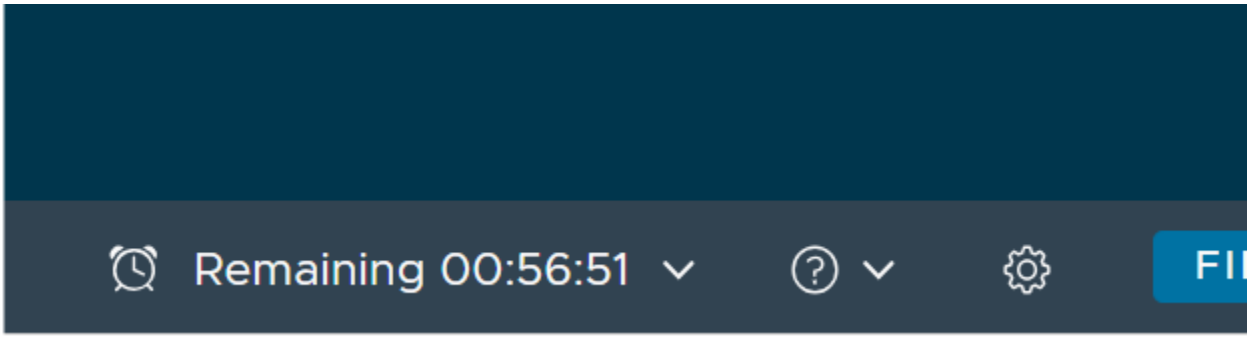
As a lab user or product service owner, you want to view your product or service in the lab environment in the user interface.

1. Use the quick launch URL for your lab and launch the lab. Enter your VMware Lab Platform user credentials in the user interface.

   The authentication provider for login are defined by your organization administrator in the VMware Lab Platform administrator user interface.

   For example, https://www.labplatform.vmware.com/<organization_name>/catalogs/catalog/<lab_name>

2. Select your lab from the lab catalog list.
3. Click the Manual tab to access the lab login details. VMware Cloud Services Provider login details for a SaaS lab are embedded in the lab manual.



4. Enter the login credentials and click **Start Lab**.

   You take the lab and follow the instructions provided in the lab manual to complete the lab. When you complete the lab, click **End**.

After the lab is completed or ended, VMware Lab Platform does a clean up in the back end and deletes the temporary VMware Cloud Services Provider organization, VMware Cloud Services Provider user, and any other resources created for the lab.

To learn more about VMware Lab Platform, see the VMware Lab Platform Release Notes document.

Working with deployment pools in VMware Lab Platform

A deployment pool is a collection of one or more lab templates and is specific to a single account or an organization.

Use the following guidelines regarding the relationship between labs and deployment pools:

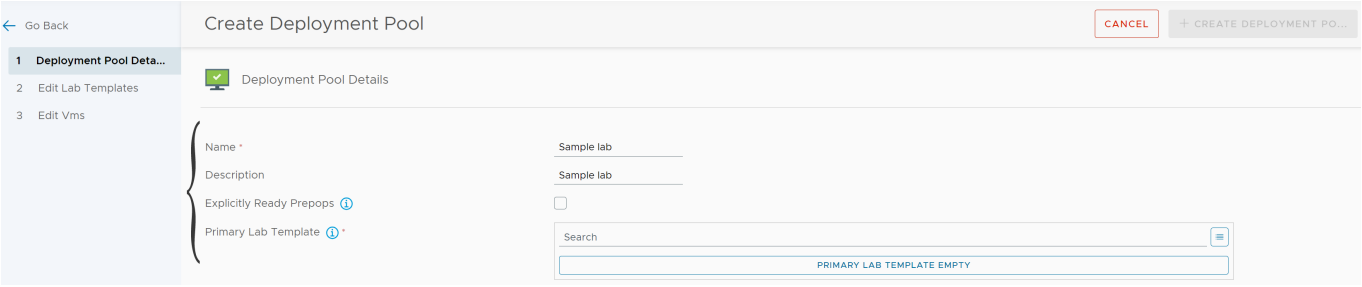
- When you create a lab, you must select an existing deployment pool to associate with that lab.
- A deployment pool includes virtual machine metadata. The metadata primarily includes the username, password, notes and a flag for each virtual machine within the lab template associated with the deployment pool.
- Any lab that does not override the virtual machine metadata values in its associated deployment pool gets those metadata values.
- A deployment pool is automatically created for a lab template built in the **Lab Builder**.

Create a deployment pool in VMware Lab Platform

It is best to configure a deployment pool before creating a lab because of the deployment pool and the lab relationship.

If you create your lab template with the Lab Builder, a deployment pool is automatically created for you. For more information on lab templates and deployment pool creation, see Create a lab template with the Lab Builder.
To create a deployment pool:

1. Navigate to **Cloud Management › Deployment Pools**. The Deployment Pool page opens.
2. Click **+ Create New**.
3. Enter details about the deployment pool that you want to create.



Name	Enter a descriptive name for the deployment pool.
Description	Add a description text about the deployment pool.
Explicitly Ready Prepops	**NOTE** The **Explicitly Ready Prepops** option is available only if the **Portal Settings › Configuration › Deployments › Explicitly ready prepops** setting is activated. If you activate the **Explicitly Ready Prepops** option in the **Deployment Pools** page, you can view the prepopulated lab template status under the **Current values** column on the **Deployment Pools** page. If the prepopulated lab templates are powered on, then you must explicitly mark them as ready to make them available for assignment. When prepopulated lab templates are ready and available, the status for the prepopulated lab templates changes from the blue column to the green column, and the prepopulated lab templates are available for assignment to entitlements.
Primary Lab Template	Select a lab template from the **List View** drop-down menu. The lab template you select is contained in the deployment pool you are creating. **NOTE** When you create a deployment pool, you can only select a single, “primary” lab template. However, later you can add additional lab templates. The role of a primary lab template is to enforce identical structure. That is, all templates in a pool must have the same number of virtual machines, same operating system, same CPU and memory, same virtual machine names, and so on. The system starts with one template (primary) and checks any template being added to the pool to ensure it matches the templates currently in the pool.

4. Click **+ Create Deployment Pool**.
   The system creates your deployment pool and opens the **Edit Lab Templates** form. For more details on the Edit Lab Templates, see Edit Lab Templates.
5. Verify the lab template details on the **Update Deployment Pool** page. Add or modify the prepops value . Click **Update Deployment Pool**.
6. To edit VMs, click **Edit Vms** option from the left-navigation pane. For more details on the **Edit Vms** page, see Edit Virtual Machine Metadata.

7. Click **Update Deployment Pool**.

To view the deployment pool, navigate to **Cloud Management › Deployment Pools**.

Edit a lab template in VMware Lab Platform

To edit a lab template, add a template, or remove a lab template, select the primary lab template and use the **Edit Lab Templates** tab.

Some lab templates are hundreds of gigabytes in the size and it can take time for a lab to deploy. To create a better user experience, VMware Lab Platform uses the Prepop feature. The Prepop feature deploys an instance of a lab ahead of a user demand and leaves it running. As a result, when a user requests an instance of that lab, VMware Lab Platform connects the user to an already running copy in a matter of seconds, and then queues up another deployment of a prepopulated lab template for the future.

As you set up your lab templates, be aware of the following guidelines.

- All lab templates must have an identical structure. As a result, any lab template that you add becomes part of a collection of a lab templates.
- There is no difference between a primary and an additional lab template after the primary lab template enforces an identical structure.
- If you add the lab templates, then the deployment pool aggregates the lab template values such as **Refill At** and **Fill Up**.

  NOTE
  If you include the same lab template in multiple deployment pools and later edit the values of the shared lab template, then each deployment pool that shares the lab template recalculates the aggregate values.

- You cannot remove all lab templates from a deployment pool when there is one or more entitlements for labs that use the pool.
- You cannot change all the templates in a pool with a single edit when there is an entitlement for a lab that uses the pool. That is, you cannot remove all existing templates and then add new ones. You can, however, make two edits: First, add the new templates to the pool, next, remove the old templates (you cannot do this action in a single edit).
- To view the status of cloud tasks, use the **Cloud Management › Cloud Tasks** option. All active tasks, queued tasks, failed tasks, and held tasks are listed on the page. You can purge or remove running tasks from the queue.

Lab Template options

Use the **Cloud Management › Edit Lab Template** form to edit the lab template you chose (the primary lab template) or to add or remove templates.

Option	Description
+ Add Lab Templates	Displays a list of available lab templates. You can add a lab template to the deployment pool in addition to the primary lab template.
Remove All Lab Templates	You can also delete lab templates from the deployment pool.
VM Equality Check	When selected, this option will check the CPU, memory, and disk configurations of the VMs and also check the number of VMs and naming conventions.
Enabled	Activates or deactivates the lab template. For example, if there are two templates and one template is deactivated, then the deactivated template does not get any prepopulated (prepops) lab templates.

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Option	Description
Cloud Org VDC	Displays the status (activated or deactivated) of the Cloud Org or Cloud Org virtual data center (VDC).
Cloud Catalog	Displays the cloud catalog information. A cloud catalog typically stores lab templates, which are preconfigured and include virtual machines, operating systems, and associated configurations. Storing lab templates in catalogs allow you to easily reuse lab templates for any deployment.
Refill At	Sets value for the least number of available prepops. If the number of available prepops goes below the Refill At value, then the system deploys prepops (refills) until the refill value is met.
Fill Up	Sets value to reach (fill up) for the number of prepops to be deployed.
Prepops	Sets the current number of prepops available for the lab template. If you adjust this number, it will add/remove prepops from the lab template. A warning message appears if a deployment pool is deactivated. A deactivated deployment pool cannot deploy prepops.
Available	Displays the available lab templates.
Pools Using	Displays the number of deployment pools using the lab templates.

Show or hide the statistics display

To activate or deactivate the display of lab template-related statistics, use the  gear icon drop-down menu on the **Lab Template** page.

Average Prepop Deploy Time is the average deploy time for prepops, in seconds, based on a seven-day rolling time. A value of 0.0 means that the system does not currently have an average deploy time available.

Add a name for a prepopulated lab template in VMware Lab Platform

The **Deployment Pool List** includes the **Prepop Naming Template** option.

You can specify the pattern the system uses to generate a name for a prepopulated lab template. By default, the system uses a single, static pattern for naming the prepopulated templates.

To view the prepop naming template:

1. Navigate to **Cloud Management › Deployment Pools**.
2. Click the three ellipses on top of the table, and click **Prepop Naming Template**.
3. To change the prepop naming template, click **Prepop Naming Template**.

Add a name for a vApp in VMware Lab Platform

The **Deployment Pool List** includes the **vApp Naming Template** option and this is similar to the Prepop Naming Template option.

The system uses the vApp Naming Template option to generate a name for an assigned vApp (either an assigned prepopulated template or an on-demand deployment). It specifies the fixed text (outside the square brackets) and the

variable names (inside the square brackets) from which the system generates the name by replacing the variables with their current values.

To view the prepop naming template:

1. Navigate to **Cloud Management › Deployment Pools**.
2. Click the three ellipses on top of the table, and click **vApp Naming Template**.
3. To change the prepop naming template, click **vApp Naming Template**.

Edit a virtual machine metadata in VMware Lab Platform

You can configure metadata for the deployment pool on the VM Metadata form.

1. To edit the metadata of the deployment pool, navigate to **Cloud Management › Deployment Pools**, select a deployment pool from the list, and click the three ellipses icon and click **Edit VMs**.

Use Individual VM Power On Settings	If activated, then VMware Lab Platform controls the powering on of VMs according to the setting of the **Power on VM when lab starts** option. If deactivated, then the cloud controls the powering of VMs. Cloud powering settings are determined outside VMware Lab Platform. **NOTE** This option only regards the delegation of VM power on operations either to the cloud or to VMware Lab Platform. Prepop deployment always delegates power on operations to the cloud.
Initial Console	If activated, a lab associated with the deployment pool uses the virtual machine as its initial console.

2. Click **>** to edit metadata.

Visible	
Power On VM When Lab Starts	Specifies that the VM in the lab template must be powered on when the entitlement starts. If activated, then the system can more efficiently determine the power state of the VM and as a result reduce the time it takes to start the lab. Note the following: • If all the VMs in a lab are marked as being powered on, then the system starts the lab template instead of the individual VMs which delegates the powering-on operation to the cloud. This delegation means that the VMs in the lab template are powered on as configured for the template in the cloud. As a result, the system observes any configured power-up order, pauses, or other tasks as configured in the cloud. • If the VM marked as the initial console is not marked to be powered on at the start, and other VMs are marked for power on, then the system displays a warning in the Lab list.
Allow Console Resize	If activated, then a lab user can manually resize the lab console.
Automatic Console Resize	If activated, then the system automatically scales a lab console so that it fits the available space. **NOTE** If both **Allow Console Resize** and **Automatic Console Resize** are activated, then the system initially sizes the lab console but the lab user can still resize it manually.

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Username	You can specify a user name specific to the virtual machine.
Password	You can specify a password specific to the virtual machine.
Notes	Notes an end user might need when interacting with a virtual machine. For example, you might include license keys in a note. Notes supports rich text/ HTML information on the virtual machine for the lab. VMware Lab Platform displays the notes when an end user clicks the **Notes** option on the lab console toolbar.

Export deployment pool configuration information in VMware Lab Platform

The **Deployment Pool List** includes an option that you can use to export full deployment pool configuration information including the relationship between the lab, deployment pool and the lab template. The **Lab Template List** also includes the same option.

1. Navigate to **Cloud Management › Deployment Pools**.
2. Click the three ellipses on top of the table, and select the **Download page** action. A CSV or Excel file with the data is downloaded to your local system.
3. Click the CSV or Excel file.

The system opens the file and provides you a choice to save the file or open the file.

Edit an existing deployment pool in VMware Lab Platform

You can edit a deployment pool from the **Deployment Pool** page.

To perform an action on all deployment pools, use the **Batch Action** menu on the **Deployment Pool** page. The menu includes the **Apply Tags**, **Remove Tags**, and **Clear Errors** options. When a prepopulated lab template fails to deploy, VMware Lab Platform marks it as an error and shows the count of errors. The **Clear Errors** option clears all errors for all the deployment pools listed, and drops the current count of errors to zero.

1. To open the Deployment Pool page, click **Cloud Management › Deployment Pools**.
2. To edit a deployment pool, click the vertical dots next to the deployment pool. A drop-down menu opens with the following action options:

Action	Description
Update	Opens a three-tab editing form. The **Edit Deployment Pool** tab is current.
Edit Lab Templates	Opens a three-tab editing form. The Edit Lab Templates tab is the current tab and described in the Editing Lab Templates section.
Edit VMs	Opens the Edit VMs section.
Delete	Deletes the deployment pool.
Refill At	Value for the least number of available prepopulated lab templates. If the number of available prepopulated lab template goes less than the Refill At value, then the system deploys the prepopulated lab template (refills) until the refill value is met.
Fill Up	Value to equal (fill up), amount of prepopulated lab template that must be deployed.
Drain Prepops	Removes all deployed prepopulated lab template and drops the number of deployed prepopulated lab templates to zero.

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Action	Description
Sync Prepops	Sends a start request for all the prepopulated lab templates currently deployed. The start request does nothing if a prepopulated lab template exists. It creates a prepopulated lab template if one does not exist. **NOTE** Use this option only if you suspect there are prepopulated lab template in an invalid state.
Remove Lab Templates	Removes lab templates from the deployment pool.
Remove Obsolete VMs	Removes obsolete VMs.
Apply Tags	Applies tags to deployment pool.
Remove Tags	Removes tags from a deployment pool.
Clear Errors	When a prepopulated lab template fails to deploy, VMware Lab Platform marks it as an error and shows the count of errors. Clearing the errors marks the database record for the error as cleared and drops the current count of errors to zero.
More details	Opens the **View Deployment Pool** page, which displays details about the selected pool including the following: • **Refill At**: Value for the least number of available prepopulated lab templates. If the number of available prepopulated lab templates goes less than the Refill At value, then the system deploys prepopulated lab templates (refills) until the Refill At value is met. • **Fill Up**: Value to equal (fill up) - amount of prepopulated lab templates that must be deployed. • **Available**: Number of prepopulated lab templates ready to use for taking a lab. • **Provisioning**: Refers to the number of lab templates being booted up but not yet ready to use for taking a lab. • **Error Count**: Number of prepopulated lab templates that failed to deploy. • **Hit**: Number of times users got prepopulated lab templates when requested. • **Miss**: Number of times users got a full deployment of a lab and had to wait. • **Labs**: List of labs included in the deployment pool. • **VMs**: List of virtual machines included in the deployment pool. • **Lab Templates**: List of lab templates included in the deployment.

Working with VMware Lab Platform labs

You create labs, which are a collection of metadata, a single VMware Lab Platform template, and an instructional manual.

When you create a lab, you specify a collection of metadata, a single lab template, and an instructional manual. Specifying a lab template ensures consistent configuration for all lab users. Metadata includes information for using the lab such as username, password, and notes about the course. Metadata also sets parameters for the lab about things such as the expiration date for the lab, how much time is allowed to take the lab, which manual to use and so on. Instructional manuals are typically a series of procedures that guide a student taking the lab.

Create a lab in VMware Lab Platform

You can create labs, which are a collection of metadata, a single VMware Lab Platform template, and an instructional manual.

1. To open the **Labs** page, from the navigation panel, select **Lab Management › Labs**.
2. Click **+Create New**.
   The **Create Lab** form opens with the **Lab Details** tab current.
3. Enter details about the lab you want to create.

Option	Description
Name	A descriptive name for the lab.
Unique Lab ID	An unique ID for the lab. A short string used to identify the lab; it must be unique in the system.
Short Description	A short description about the lab.
Deployment Pool	Click **Add** and select a deployment pool from the list. The lab's virtual machine metadata, which is primarily the user name, password, notes, and a flag for each virtual machine, is based on the lab templates included in the deployment pool you select. To create a deployment pool, click **Create Deployment Pool**. **NOTE** If you have one cloud and the **Enable Optimized Deployment Pool Usage** option is activated, the **Deployment Pool** option is hidden and the system automatically creates deployment pools for the lab template. Instead of **Deployment Pool**, **Cloud Lab Template** option is available in the lab and you can select the lab template from the list menu.
Valid Period	The time period for which a user has an entitlement for a lab. A user can access the lab at any time during this period and use it for the length of the time specified with the **Time Allotted** value.
Time Allotted	Amount of time, in hours, a user gets to use the active console. For example, if you want to give a student three hours to finish a lab, you set **Time allotted** to three hours.
Manual	Click the list view, and select a manual from the **List View** page. If you do not select a manual, then the Manual window in the lab is blank. Selecting a manual ensures that all the language versions of the manual that have been published are available for the lab. **NOTE** You can also specify PDF manuals with the **Resources** setting on the **Update Content** page. **NOTE** You make manuals available to labs by importing them using the Content Engine (**Content › Content Engine**). For more information, see the Add a Lab Manual Project.

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Option	Description
Thumbnail	To display for your lab, click the list view and select a thumbnail image. If you do not select an image, then VMware Lab Platform uses the default lab thumbnail image. **NOTE** To import a thumbnail image, use the **Lab Management › Thumbnails** option.
Catalog	Click the list view, and select one or more catalogs from the List View page for your lab. To create a catalog for the lab, click **Create Catalog**.

4. Click **Create Lab**. The lab is created and a quick launch URL is generated.
5. On the **Lab Configuration** page, configure your lab settings.

General Settings	
Enabled	Activates or deactivates the lab.
Publicly Visible	Controls whether the lab is publicly visible. Labs have a public-visible property as do customer accounts. If you configure an account to allow public browsing and make your lab publicly visible, then an end user can browse the Catalog page on the end-user interface without being logged in.
New Lab	Adds the VMware Lab Platform**New** banner to the lab description. The banner is displayed in the top-right corner of the course description. An end-user can click the banner to display a listing of labs with the same label.
End-User Interface Settings	
End Allowed	If activated, once an end-user ends a lab, then that user cannot return to the lab again. If deselected, the **End** button in the end-user interface is hidden.
Exit Allowed	If activated, an **Exit** button is visible on the end user console. When a user exits a lab, the lab is temporarily halted and the user can return to the lab later. If not selected, the **Exit** button is hidden from the user.
Reset Allowed	If activated, then the lab console includes a **Reset** option. If a lab user resets a lab, then all its virtual machines (VMs) are restored to their original state. The lab user is not able to recover any work done.
Remote Control Mode	Deactivates user control of the manual and requires an admin-accessible API call to advance manual pages.
Show Notes	If activated, then the **Notes** button is included on the lab console. When an end user clicks **Notes**, the console displays metadata such as notes, user name, and password for each VM. Select **Use this lab's value** to display locally specified notes. Select the **Use Value from Tenant Option** check box to use the account-wide default. You (the administrator) can use the **Update VM Metadata** tab to specify such metadata. The Configurations page also includes this option as an account-wide default. You can activate or deactivate the account-wide default using the **Use Value from Tenant Option** check box.
Open notes on lab start	If activated, then the lab console, by default, displays metadata (notes, user name, and password for each VM) for this lab once an end user starts it. The Configuration

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	page also includes this option as an account-wide default. You can activate or deactivate the account-wide default using the **Use Value from Tenant Option** check box.
Chat Settings	
Use Chat From Tenant Option	Use the chat options set by the tenant option. **NOTE** Clear this check box if you want to use a live chat URL.
Grading Settings	
Require Grading	Activates the grading feature. You can use the **Entitlement List** to enter grading information and provide a means for the end user to review that information. If you enable the **Require Grading** setting, then the following options are displayed: • Power Off for Grading • Time Period for Grading
Power Off for Grading	If this setting is not activated, then the lab templates keep running. The **Power Off for Grading** setting saves resources by powering off the lab template when it is not used.
Time Period for Grading	Specifies a time period during which the lab can be graded. During this time period, the system retains the entitlement (and retains the lab template). An end-user can take the lab and the entitlement remains active until either the lab has received a grade, or the grading period has passed. The time period is in Days/Hours/Minutes format. **NOTE** If you set the **Time Period for Grading** to zero (0), then the system retains the entitlement (and lab template) indefinitely.
Time Settings	
Time Allotted Override	Activate it to override the default lab time settings. The override can be done using API. If this is unchecked, the API override will not happen. Time Allotted: Amount of time, in hours, the user gets to work on the lab. This is different from the Valid Period option. For example, if a user is allotted 30 hours to complete a lab within a 30-day period, the time limit is the 30-hours setting. Valid Period: Amount of time the lab stays active. This is different from the Time Allotted option. For example, if a user is allotted 30 hours to complete a lab within a 30-day period, the valid period is the 30-day setting. You can include hours and minutes in this setting.
Callback Settings	
Entitlement Callbacks	Select an entitlement callback. If no entitlement callback exists, then you can add one by clicking **No Selected Entitlement Callbacks**. To create entitlement callbacks, click **Create Entitlement Callbacks**. See the section Working with entitlement callbacks in VMware Lab Platform for more information.

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Callback Metadata	Metadata the system includes in the callback you selected in the Callbacks text box. For example, if you enter the text test callback metadata and the callback you selected has an email return method, then the callback returns an email similar to the following example: `{ "entitlementKey":"NEE-97385361_1_07500" "account": "jsmith@project.com", "sku":"shortlab", "tenant": "NEE", "eventTime":"01-07-2016 18:11:25 UTC", "event": "enroll", "firstName": "John", "lastName": "Smith", "metadata": "test callback metadata" }`
Completion Settings	
Minimum time spent for lab to be completed	Sets the amount of time an end user must work on a lab before it can be marked as completed. If the value is zero, the lab can never be marked as completed. The time period is in Days/Hours/Minutes format.
Percentage of steps visited for lab to be completed	The minimum number of steps the end user must complete before a lab can be marked as completed.
Other Settings	
Prepop Enabled	If activated, the system deploys one or more prepopulated instances of the lab template to the cloud ahead of a user request. When an end user selects the lab, the user is then linked to a prepop. This process reduces the amount of time it takes for VMware Lab Platform to build a lab. However, Prepop uses cloud resources that might incur costs to your organization. **NOTE** The system derives prepop values from the deployment pool you selected for this lab. You can edit those values by editing the deployment pool.
Power Operation	Sets the power operation functionality for a lab template in a lab. • **None**. Lab templates continue to run until the entitlement is not completed or ended. • **Suspend vApp on Exit**. If activated, suspends the lab template upon exit from the lab. Suspend writes memory to disk and the guest OS enters a hibernate/ sleep state. Using this feature saves compute costs. • **Power off vApp on Exit**. If activated, and an end user exits lab, the system powers down the lab template. If not activated, the lab template remains powered on (using cloud resources) after the user exits. This allows for faster resumes when the user returns. • **Shut down and Power off vApp on Exit**. If activated, each virtual machine within the lab template attempts to shut down gracefully before powering off.
Delay before exit power operation	Time in minutes to wait before the exit power operation is started.
Undeploy After	Powers off lab resources for an inactive lab. Enter a value more than 0 to save the lab resources when a lab is inactive. The system waits till the time as set in the **Undeploy After** setting before powering off the resources. A value of 0 means that there is no set limit. **NOTE** Activate the **Use the Show "Undeploy After" option for lab settings** Portal setting to use the **Undeploy After** lab setting.

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Concurrent Usage Limit	Limits the number of concurrent users that can start this lab. The system performs a check on the limit versus the number of users currently running the lab. A value of 0 means that there is no set limit. **NOTE** The system's concurrency check is independent of running lab templates for the lab and does not consider any deployed prepops. Powered-down lab templates due to an exited lab, powered-down labs due to being idle and redeployed lab templates due to reset are all not considered in the system's concurrency check.

6. To open the **Lab Content** page, click **Next** . Configure the following lab settings.

Option	Description
Short Description	Provide a short description of the lab. This short description appears within the Lab List (administrator UI) and also within the lab console's Labs tab.
Description	Enter a complete description of the lab. This description appears within the lab console's Labs tab if the end user clicks **Read more**.
Network Topology	Add an network topology image showing the network topology of the VMs used by the lab. If you do not select an image, VMware Lab Platform hides the **Network Topology** button in the user console. To create the network topology image and add to the system, click **+ Create Network Topology** and upload the image.
Survey	Add survey for a lab from the drop-down menu. Use the **Create Survey** tab, to create a survey configuration and add to a lab. For more information on survey configurations, see Working with Survey Configuration.
Screenshot	Uploads the image you select. It appears in the lab console user interface when the end user clicks **Lab Details**. You can browse and upload any screenshot that meets the display limits. **NOTE** On the Screenshot field of the Edit Content and Metadata tab, you can click the X to delete an existing screenshot.
Badges	Click the list view, and select a badge from the list. Badges are images that VMware Lab Platform displays in the lab console. An end user taking a lab can click a badge to display a list of labs associated with that badge. VMware Lab Platform includes a set of commonly used badges or you can create a badge. You import badges for use with your customer account from **Lab Management › Badges**.
Resources	Click the list view and select a resource from the list. Resource files supply supporting information for the lab, such as additional reading, a syllabus, and so on. To create new lab resources, click **Create Resources**. You can also use this setting to select a lab manual formatted in PDF. PDF manuals appear in a standard PDF viewer in the lab.

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Option	Description
	To use a PDF document as a lab manual, leave the **Manual** text box on the first page of the **Create Lab** form empty. Select the PDF file to use as a lab manual as the first PDF resource here. **NOTE** To use a PDF document as a lab manual, deactivate the option **Hide Instructions panel when no manual** under the **Portal Settings › Configuration** page.
Products	Click the list view and select a product to associate it with the lab. Labs are often categorized by product. For example, a lab might be about the VMware Horizon product. **NOTE** You must create a product entry using the **Lab Management › Products** option.
Tags	Tag a lab with a text string. The text string appears on the lab console. An end user can click the tag to display other labs with the same tag. **NOTE** You can create a tag using the **Lab Management › Tags** option.
Authors	Select an author for the lab. An end user can click an author name to display the author's public profile.
Related Labs	Lets you to relate a lab you are creating to another lab. VMware Lab Platform displays these related labs when an end user clicks **Lab Details** on the lab console user interface. For example, if you are creating an introductory lab about VMware Cloud Director, you can select related labs such as *`Advanced VMware Cloud Director`* or *`Using Log Insight with VMware Cloud Director`*.

7. To save your changes, click **Next**.
8. To configure the VM settings for your lab, click **Vm Configuration**.

Use from Deployment Pool	If activated, the lab defaults to the initial console as defined in the vApp of the deployment pool you selected.
Use Custom Settings	If activated, the lab uses this virtual machine as its initial console instead of the one from the deployment pool.
>	Click the greater than icon (**>**) to edit the metadata of an initial console. **NOTE** You can enable the **Use value from Deployment Pool** setting to use the configuration contained in the deployment pool to which the lab is associated.
Visible	If activated, then the virtual machine is visible in the lab. (If not activated, then VMware Lab Platform does not have a console connection.) For example, consider a lab with 12 virtual machines. You can use the Visible

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	option to give the lab's end users access to only a single virtual machine. **NOTE** If you select only one virtual machine to be visible to the end user, then VMware Lab Platform hides the console chooser that is normally on the left side of the console screen of the lab console.
Power on VM when lab starts	Specifies that the VM in the vApp must be powered on when the entitlement starts. If activated, then the system can more efficiently determine the power state of the VM and then reduce the time it takes to start the lab. Note the following: • If all the VMs in a deployment pool or lab are marked as being powered on, then the system starts the vApp instead of the individual VMs which delegates the powering-on operation to the cloud. This delegation means the VMs in the vApp are powered on as configured for the template in the cloud. So the system observes any configured power on order, pauses, or other tasks as configured in the cloud. • If the VM marked as the initial console is not marked to be powered on at start (and other VMs are marked for power-on), then the system displays a warning in the Lab List.
Allow Console Resize	If activated, then a lab user can manually resize the lab console.
Automatic Console Resize	If activated, then the system automatically scales a lab console so that it fits the available space. **NOTE** If both **Allow ConsoleResize** and **Automatic ConsoleResize** options are activated, then the system initially sizes the lab console, but the lab user can still resize it manually.
Username	You can specify a user name specific to the virtual machine.
Password	You can specify a password specific to the virtual machine.
Notes	If activated, then the lab console, by default, displays metadata (notes, user name, and password for each VM) for this lab once an end user starts it.
Use individual VM power on settings	If activated, then VMware Lab Platform controls the powering on of VMs according to the setting of the **Power On VM When Lab Starts** option. If not activated,

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	then the cloud controls the powering of VMs. Cloud powering settings are determined outside VMware Lab Platform. **NOTE** In **Create a Lab**, the **Use individual VM power on settings** option sends the VM power on operations either to the cloud or to VMware Lab Platform. Prepop deployment option always sends power on operations to the cloud.

9. Click **Update Lab**.
   The lab is created. The **Labs** list displays the lab that you created.

Delete a lab in VMware Lab Platform

To delete a lab, use the **Delete** lab option on the **Lab** list page.

If there is an active prepopulated lab template or an entitlement for a lab, you cannot delete the lab. If you cannot delete a lab, try the following solution:

1. Ensure that the lab is active and prepopulated lab template are active.
2. Go to the **Prepops** page and set the value to 0 (zero).
3. Enable and acknowledge all alerts.

On the **Labs** list page, click the three ellipses next to a lab, and click **Delete**.

Export lab notifications in VMware Lab Platform

You can export notifications from the **Lab** list using the **Download** option.

1. On the **Labs** page, click the three ellipses, and select the **Download** option.
2. You can export all notifications to a CSV, or Excel file to your local system.

Use query strings to modify the quick launch URL in VMware Lab Platform

You can control the user console experience by adding query strings to the quick launch URL for a lab. For example, during an event or a demo, if an instructor wants to bypass the usual "Welcome Message" in a running lab, you can add the `showWelcomeTips=false` string at the end of the quick launch URL. The welcome message is not displayed in the end-user interface.

You can use one or multiple combination of parameters in the query string. Currently the following query strings are supported:

Query String	Description
`checkVerifyByDate=false`	In the user interface, the account verification message popup is not displayed.
`checkCompatibility=false`	Some browser compatibility checks such as cookies, private browsing, local storage are skipped.

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Query String	Description
`checkBrowserSupported=false`	Browser support check is skipped. Users can browse the catalog, but not log in and take a lab, and this option is different from the `checkBrowserRejected` query.
`checkBrowserRejected=false`	An unsupported browser is rejected and does not let you look at the catalog page or do anything.
`checkBrowserCompatibility=false`	This query is equivalent to `checkCompatibility=false`, `checkBrowserSupported=false`, and `checkBrowserRejected=false`. You can use the `checkBrowserCompatibility=false` query to skip the browser supported, browser rejected and browser compatibility checks.
`showWelcomeTips=false`	If activated, the welcome tips does not show on the admin interface and user interface.
`checkEula=false`	Skips if a account or organization's End User License Agreement (EULA) is accepted.

Working with notifications in VMware Lab Platform

You can view and receive real-time notification alerts from the system. The alerts are categorized based on the severity and priority of the issue such as High, Medium, and Low.

To view all notification alerts for an account, click the  icon on the top of the page header. You can view the list of subscribed notifications from the **Notifications** menu option.

Figure 11: Notification quick view



Severity of alerts and the color codes associated with a notification alert are:

- Green (low): Low severity notifications refer to information such as daily summaries.
- Orange (medium): Medium severity notifications mean that a user is unable to use the system (for examples include failure to start labs or errors deploying vApps).

- Red (high): High severity notifications refer to issues that impact multiple users. Typically, a high severity notification stems when a cloud is unavailable.

You can perform the following operations in the **Quick Notifications** pane:

- **All Notifications**: To view all the notifications.
- **Unread**: To view all the unread notification.
- **High Priority**: To view all high priority notifications.
- **Medium Priority**: To view medium priority notifications
- **Low Priority**: To view low priority notifications.
- **Flagged**: Select notifications that you want to flag and take an appropriate action
- **More Details**: Click the arrow to expand and view the details of a notification alert.
- **Settings**: Click the  icon to open the **Settings** option. You can select the alert categories, and specify the number of notification alerts, display time of the alerts and also, subscribe to alerts and receive email notifications.
- **Do Not Disturb**: Move the slider to stop notification alerts.
- **Advance Settings**: To view the **Notification Subscription** page and create notification subscriptions, use the **Advance Settings** option.
- **Notification Subscription**: While notifications appear on the page, you can also enable the system to email the notifications. To open the **Notifications Subscription List**, click **Subscription**. From there, you can add an email address and set criteria for which the system must send out notifications. Once you define a subscription, each time a notification that meets your criteria occurs, the system emails it to everyone who has subscribed, and showing it on the dashboard.
- **Delete**: Deletes a notification from the notification list.
- **Activate**: Activates the subscription for the notification.
- **Deactivate**: Deactivates the subscription for the notification.
- **View All Notifications**: Opens the Notifications page. You can perform single and bulk functions for notification alerts such as:
  - **Acknowledge All**: All notifications, including those marked as read and flagged, are moved to the Notification Archive page.
  - **Read All**: All notifications are marked as read and the notification count under the bell icon is reset.
  - **Flag All**: All notifications are marked as flagged and you can take action for them later. However, the flagged notifications are not marked as read.
- **Archive**: Displays the **Notification Archive** page that lists all archived notifications.

To get the notifications within the app, select the notifications listed under the **In-App Notification** menu. To get the notifications in an email, select the notifications listed under the **Email Notification** option. Use the Advanced Settings option to customize the maximum notification counts and duration for each notification.

> **NOTE**
> The **Notification Archive** page includes an export option. The **Download** option lets you to export all notifications to a CSV, Excel, or PDF.

Notification categories

Notifications are categorized as follows:

- Administrative
- Billing
- Cloud Template
- Cloud vApp

- Cloud
- Generic Task
- Prepop
- System

Setting completion criteria for a lab in VMware Lab Platform

You can specify the requirements that must be met for a lab user to complete a lab.

You can specify the following information:

- The minimum time a student must spend on a lab before it can be considered finished.
- The percentage of steps in a lab that a student must complete before the course can be considered finished.
- Set account-wide behavior to link the **End** button availability to lab completion criteria.

After you set values per lab, you enable tenancy-wide behavior to link **End** button availability to lab completion criteria.

Set criteria for lab completion in VMware Lab Platform

You can set criteria for lab completion from the Update Settings form for labs.

1. Select **Lab Management › Labs**.
2. Click the vertical ellipses next to the name of the lab you are working with and select **Lab Configuration** from the drop-down menu.
3. On the Update Lab form, specify the completion criteria for the lab.
   - **Minimum time spent for lab to be completed**. You can specify days, hours, and minutes.
   - **Percentage of steps visited for the lab to be completed**. The percentage is based on the total number of steps in the lab manual.
4. Click **Update Lab**.

You must activate the **Disable End button until completion criteria (Minimum Time Spent and Percentage Steps Visited) is met** setting.

Deactivate the Lab End button in the in VMware Lab Platform user interface

After you have specified the conditions required to end a lab, you activate the setting to apply these values.

For labs with completion criteria, the lab's **End** button is dimmed and the student sees an informational pop-up presenting the criteria for finishing a lab.

1. Click **Portal Settings › Configuration › Interface**.
2. Select the **Disable End button until Completion Criteria (minimum Time Spent and Percentage Steps Visited) is met** setting.

Working with lab templates in VMware Lab Platform

A lab template is a container that can contain one or more virtual machines. Lab templates ensure that virtual machines are consistently configured across an entire organization.

The **Lab Template List** provides many options to manage your organization's lab templates.

Using Prepopulated Labs

The Prepop feature deploys an instance of a lab ahead of user demand and leaves it running. When a user requests an instance of that lab, VMware Lab Platform connects that user to an already running copy in a matter of seconds, and then queues up another deployment of a prepop for the future deployment.

The Prepop feature is configured when you set up a lab template or create a lab.

Edit a lab template in VMware Lab Platform

The **Lab Management › Lab Templates › Update Lab Template** tab includes options for setting values for lab templates.

NOTE
In addition to editing for individual templates, the **Lab Template** page includes a batch action menu that you can use to perform an action on all lab templates. The menu includes the **Clear Errors** and the **Update network wiring** options. When a prepopulated template fails to deploy, VMware Lab Platform marks it as an error and shows the count of errors. The **Clear Errors** option clears all errors for all lab templates listed and drops the current count of errors to zero.

To edit a lab template:

1. Navigate to **Lab Management › Lab Templates**, click the three vertical ellipses next to the lab template you want to edit, and click **Update**.
2. Open the **Update Lab Template** form for the selected lab template with the **Update Lab Template** tab current.
3. Set values for your lab template.

Enabled	Activate or deactivate the lab template. If not activated, then the lab template cannot deploy prepopulated lab template or an on-demand lab. This text box is useful when you want to deactivate prepopulated deployment for a lab template hosted in a particular cloud. For example, you might have a lab template hosted in a cloud located in the United States and a clone of that lab template hosted in Europe. In such cases, you might want to deploy prepopulated lab templates only in the lab templates hosted in the United States but not in Europe.
Fixed Size	If you activate the **Fixed Size** text box, then the **Refill At** and the **Fill Up** values do not display. Instead the **Pool Size** value displays. The **Pool Size** value, sets the **Refill At** and the **Fill Up** values to the **Pool Size** value. Most importantly, if you activate **Fixed Size**, then the system keeps the number of running lab templates equal to the pool size limit value. As a result, the **Fixed Size** text box allows an administrator to ensure that the number of running lab templates is constant, regardless of the system refilling **prepops**. Example scenario: If an administrator has set the lab template to a pool size of 10, then: • System deploys 10 lab templates and sets **Refill At** and **Fill Up** to 10. • If an end user then starts a lab, the system uses one of the 10 lab templates and assigns the lab template to the end user's entitlement. • Lab template continues with 10 running templates. If all end users continue to start labs and use all the deployed templates from the fixed size pool, then upon an 11th entitlement, the system returns an error.
Refill At	Sets value for the least number of available prepopulated lab templates. If the number of available prepopulated lab templates goes less than the **Refill At** value, then the system deploys prepopulated lab templates (refills) until the **Refill At** value is met.
Fill up	Sets value to reach (fill up) for the number of prepopulated lab templates to be deployed.
Prepops	Sets the current number of prepopulated lab templates available for the lab template. If you adjust this number, it adds or removes prepopulated lab templates from the lab template.

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Concurrent Usage Limit	The maximum number of lab templates that the system allows to run. If a user starts a lab while below the concurrent use limit, then the system allows the user to start the lab. However, if a user starts a lab while the concurrent use limit has reached, then the system does not allow the user to start the lab. The concurrent use limit increases once a student successfully starts a lab but decreases when a student ends or exits a lab.
Placement Algorithm	Specifies the placement algorithm for this lab template and overrides the account-wide placement algorithm as set in the page (within **General Options**). The placement algorithm is described in the **Name of Default Prepop Placement Algorithm** option.
Time To Live (Hours) (0 Implies No Limit)	Specifies the amount of time, in hours, that a prepopulated lab template can exist before the system deletes it. This text box allows you to specify the amount of time a prepopulated lab template must be left waiting in the cloud until either the system assigns it to an entitlement or deletes it. **NOTE** When the system successfully deploys a prepopulated lab template, it adds a time stamp to the database record. By using the time stamp and the value of the template's **Time to Live** text box, the system calculates when it must delete the prepopulated lab templates. When the time to delete the prepopulated lab templates arrives, the prepopulated lab templates becomes eligible for recycling. The recycling does not occur until the next time the job that checks prepop status is run, meaning prepopulated lab templates can continue to live past their end of life time by several minutes.
Auto-Deploy Replacements For Expired Prepops	If activated, then the system deploys a new prepopulated lab template each time a prepopulated lab template is undeployed because it has reached its **Time To Live** limit. **NOTE** For fixed size pools, auto-redeployment occurs whether this field is active or not.
Assignment Algorithm	Specifies the assignment algorithm for this lab template and overrides the account-wide placement algorithm as set in the **Portal Setting** page (within **General Options**). The **Assignment Algorithm** is described in the **Name of Default Prepop Placement Algorithm** option.
Assignment Prepop Preference	Specifies the assignment prepopulated lab template preference for this lab template and overrides the account-wide prepopulated lab template preference as set in the **Portal Setting** page (within **General Options**). The **Prepop Preferences** is described in the **Name of Default Prepop Placement Algorithm** option.
Prepop Template Name Example	Displays an example of a prepopulated lab template name generated with the pattern used by the system. By default, the system uses a single, static pattern for the naming.
Drain Prepops Tab	Opens a tab that lists all the cloud organization VDCs having the template running within them as prepopulated lab templates. This list allows you to select a specific organization VDC to be drained as opposed to draining all prepopulated lab templates.

You can also use the action options from **Lab Management › Lab Templates** page, click the three ellipses next to lab template from the list view and select an action option. To view the entire list of action options, click the  list view icon.

- **Sync Prepops**: Sends a start request for all the prepopulated lab templates currently deployed. If a prepopulated lab template exists, the start request does nothing. It creates a prepopulated lab template if one does not exist.
- **Drain All Prepops**: Undeploys all prepopulated lab templates and drops the number of deployed prepopulated lab templates to zero.
- **Drain Prepops**: Opens a tab that lists all the cloud organization VDCs having the template running within them as prepopulated lab templates. This list allows you to select a specific organization VDC to be drained as opposed to draining all prepopulated lab templates.
- **Clear Errors**: When a prepopulated lab template fails to deploy, VMware Lab Platform marks it as an error and shows the count of errors. Clearing the errors marks the database record for the error as cleared and drops the current count of errors to zero.
- **Set to fill Up**: Sends a fill up equal to the Fill up value (number of prepopulated lab templates that must be deployed).
- **Export to S3**: This option is available only if the **Portal Settings › Configuration › AWS S3 Integration › awsBucketEnabled** setting is activated. The **Export to S3** option lets you to export the lab template to your Amazon Simple Storage Service (S3) instance.
- **Set to Refill at**: Value for the least number of available prepopulated lab templates. If the number of available prepopulated lab templates goes below the Refill At value, then the system deploys prepopulated lab templates (refills) until the Refill At value is met.
- **Reset Average Deploy Time Statistics**: VMware Lab Platform monitors the average lab deployment time and continuously updates the average as end users continue to start (deploy) labs. This is the estimated deployment time an end user sees when launching a lab. **Reset Average Deploy Time Statistics** clears the statistics for the selected template to restart the calculation for the average deploy time. You can enable display of the average deploy time statistics on the Lab Template List. Doing so allows you to easily know which are the slowest lab templates and so which to reset.

  NOTE
  You can click **Reset all average deployment time statistics** on the Lab Template List to reset all the lab templates at once.

- **More Details**: Displays details of the selected lab template.

4. Click **Update Lab Template**.

Export files to an AWS instance in VMware Lab Platform

Amazon Simple Storage Service (AWS S3) is an object storage solution that lets you to import or export VM files in highly available and easy-to-use buckets. You can add and configure your own AWS S3 instance with VMware Lab Platform.

To export files from VMware Lab Platform to an AWS S3 instance:

1. Enable the **Portal Settings › Configuration › awsBucketEnabled** setting.

   When you enable the **awsBucketEnabled** option, the import and export options for AWS are made available in the **Lab Templates** page. Additionally, you can also configure the other AWS configurations either from the **Portal Settings › Configuration › AWS** setting, or on the **Export to S3** form.

   For more information on the AWS setting, see Working with Settings.

2. To export an OVF file from VMware Lab Platform to AWS, click **Lab Management › Lab Templates**.
3. Click the three ellipses for a lab template, and select **Export to S3**.
4. Enter the details in the **Export to S3** form.

   If the AWS details are added in the **Portal Settings › Configuration › AWS** setting, the **Export to S3** form auto populates that information in the form. You can modify the data on the **Export to S3** form.

Bucket Name	The name of the AWS S3 Bucket.
Path within the bucket	Enter the path within the AWS S3 Bucket to export the file.
Access Key ID	Enter the AWS API key.
Access Key Secret	Enter the AWS API secret to be used with the AWS Key.
AWS Role ARN	(Optional) Enter the Amazon Resource Name (ARN) of a role with access to the Bucket.
AWS Region	Enter the Bucket's AWS region.
Session Token	(Optional)For temporary credentials retrieved through AWS Security Token Service (AWS STS).

5. Click **Export to S3**.
   You can see the file transfer progress in the **Content Migration Task** widget on the **Dashboard** widget.

Import files from AWS to VMware Lab Platform

Amazon Simple Storage Service (AWS S3) is an object storage solution that lets you to import or export VM files in highly available and easy-to-use buckets. You can add and configure your own AWS S3 instance with VMware Lab Platform.

To import files from an AWS S3 instance to VMware Lab Platform:

1. Activate the **Portal Settings › Configuration › awsBucketEnabled** setting.

   When you enable the **awsBucketEnabled** option, the import and export options for AWS are made available in the **Lab Templates** page. Additionally, you can also configure the other AWS configurations either from the **Portal Settings › Configuration › AWS** setting, or on the **Import to S3** form.

   For more information on the AWS setting, see Working with Settings.

2. To import an OVF file from AWS, click **Lab Management › Lab Builder**.

   ATTENTION
   Only OVF files are supported for AWS imports. You cannot import OVA files to your AWS instance.

3. Click the three ellipses on the top-right of the page, and select **Import from S3**.
4. Enter the details in the **Import From S3** form.

   If the AWS details are added in the **Portal Settings › Configuration › AWS** setting, the **Import from S3** form auto populates that information in the form. You can modify the data on the **Import from S3** form.

Bucket Name	The name of the AWS S3 Bucket.
Path within the bucket	Enter the path to the OVF files to be imported. This should be a directory containing '`descriptor-with-id.ovf` and `.vmdk` files.
Access Key ID	Enter the AWS API key.
Access Key Secret	Enter the AWS API secret to be used with the AWS Key.
AWS Role ARN	(Optional) Enter the Amazon Resource Name (ARN) of a role with access to the Bucket.
AWS Region	Enter the Bucket's AWS region.

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Session Token	(Optional)For temporary credentials retrieved through AWS Security Token Service (AWS STS).
Target Template Name	Enter the name for the template in the virtual cloud director.
Target Catalog URN	Select the target catalog URN from the drop-down menu.

5. Click **Import From S3**.

   After a file is added, there is a waiting time before the file is available for use. You can see the file transfer progress in the **Content Migration Task** widget on the **Dashboard** widget.

Working with custom lab fields in VMware Lab Platform

You can create custom lab fields and mark your labs.

Custom lab fields allow for the marking of labs with a value from a predefined list or with a free text. After the user completes a lab that has custom lab fields they are passed on to the transcript. By utilizing the custom lab fields, you can filter the transcripts and get a better understanding on the lab usage.

Create, update, or delete custom lab fields

You can create, update, or delete custom lab fields from the custom lab fields list page.

1. Navigate to **Lab Management › Custom Lab Fields** and click **Create New**.
   The Create Custom Lab Field page opens.
2. In the **Name** text box, enter a name for the Custom Lab field.
3. If you want to create a custom lab field with predefined values, from the **Type** drop-down menu, select **Select** and in the **Value** text box you must enter the values separated by commas.
4. If you want to create a custom lab field with a free text value, from the **Type** drop-down menu, select **Text** and in the **Value** text box you can enter a default value.
5. Click **Create Custom Lab Field**.

The custom lab field appears on the custom lab fields scripts page.

After you create a custom lab field, you can assign it to a lab.

You can update or delete a custom lab field by using the **vertical ellipsis** menu next to each custom lab field.

Assign custom lab fields to lab

After you create a custom lab field, you can assign it by using the lab wizard when you create a new lab or when you edit an existing one.

1. Navigate to **Lab Management › Labs**.
2. Select a lab for update or create a new one.
3. In the **Update Lab** wizard, navigate to the **Custom Fields**.
4. In the **Select custom field** drop-down, select a custom field and click **+ Add**.
   The custom lab field is displayed in the table.
5. (Optional) You can add all defined custom lab field by clicking **Add all defined custom fields**.
6. (Optional) You can change the value of the custom lab field by modifying the field value in the table.

When you change the value of the custom lab field in the lab wizard you are modifying only the value of the custom lab field associated with that lab. The values of that field in other labs and in the custom lab fields list page remain unchanged.

7. Click **Save**.

A custom lab field is now successfully assigned to a lab.

You can now filter labs by custom lab fields on the Labs list page. Additionally you can filter transcripts of completed labs by custom lab fields on the Transcripts list page.

Working with VM scripts in VMware Lab Platform

You can create and run VM scripts at different levels such as labs, deployment pools, or in Lab Builder.

VM scripts allow for the automation of time-consuming tasks or setup activities before delivering a lab to the end user. Additionally they can perform post-deployment configurations or operational checks to ensure the functionality of the lab environment. By utilizing the same base lab blueprint and customizing content for different needs, VM scripts save resources. After you create a VM script, you assign it either to a lab or to a deployment pool. You can then, optionally, verify how your VM script runs by using Lab builder. As part of the verification process, you need a custom created VM script agent. To get that agent you must contact VMware Lab Platform support.

Tenant settings used with VM scripts

To turn on the assignment of VM scripts to a VM in deployment pools you must switch on the **Explicitly Ready Prepops** tenant setting.

To turn on the VM script run in Lab Builder you must switch on the **Enable VM script run** tenant setting.

For more information on enabling settings, see Working with settings in VMware Lab Platform.

Create, update, or delete VM scripts

You can create, update, or delete VM scripts from the VM scripts list page.

You can create and store the VM script content externally when you want to modify it outside of VMware Lab Platform, or you can do that in the system.

1. Navigate to **Lab Management › VM Scripts** and click **Create New**.
   The Create VM Script page opens.
2. In the **Name** text box, enter a name for the VM script.

   This name identifies the file that the system creates on the VM before script runs.
3. If you want to create your script and store it in VMware Lab Platform, from the **Script Sourcing** drop-down menu, select **Enter Script content** and in the **Script Content** text box, enter the script.
4. If you want to use the script as a predefined script that you manage externally, from the **Script Sourcing** drop-down menu, select **Load Script from URL** and in the **URL** text box enter a web address.
5. From the **Language** drop-down menu, select the programming language for your script and click **Create VM Script**.

The VM script appears on the VM scripts page.

After you create a VM script, you must assign it either to a lab or to deployment pools.

You can update or delete a VM script by using the **vertical ellipsis** menu next to each VM script.

Assign VM scripts to labs

After you create a VM script, you can assign it by using the lab wizard when you create a new lab or when you edit an existing one.

When you assign VM scripts to a lab they run on entitlement start. You can assign multiple VM scripts to a single VM. However, you can only assign a VM script once per VM.

1. Navigate to **Lab Management › Labs**.
2. Select a lab for update or create a new one.
3. In the **Update Lab** wizard, navigate to the **Update Lab Scripts** page and click **Add Script**.
4. In the **VM Name** drop-down, select a lab VM.
   The **Scripts** page opens.
5. From the VM Name drop-down menu, select the VM to run the script.
6. From the **Script** list view, select the VM script.
7. In the **Timeout (secs)** text box, enter the approximate seconds needed for running the script.

   VMware Lab Platform marks a VM script as failed if it does not return a success code within the specified period.
8. (Optional) Select the **Fail on error** checkbox to mark the script as failed.
9. In the **Retry count** text box, specify the number of retries before the system stops retrying the script and click **Save**.

A VM script is now successfully assigned to a VM in the lab.

Assign VM scripts to a VM in deployment pools

After you create a VM script, you can assign it to a VM in deployment pool by using the deployment pool wizard.

Verify that you enabled the **Explicitly Ready Prepops** tenant setting.

When you assign VM scripts to a deployment pool they run on prepop deployment. You can assign multiple VM scripts to a single VM. However, each individual VM script can only be assigned once per VM.

1. Navigate to **Cloud Management › Deployment Pools**.
2. Select a deployment pool for update or create a new one.
3. Navigate to the section **Edit Scripts** wizard step and click **Add Script**.
4. In the **VM name** drop-down, select a lab VM.
5. In the **Script** drop-down, select a VM script.
6. In the **Timeout (secs)** text box, enter the approximate time needed for running the script.

   VMware Lab Platform marks a VM script as failed if it does not return a success code within the specified period.
7. (Optional) Check the **Fail on error** checkbox to mark the script as failed.
8. In the **Retry count** text box, specify the number of retries and click **Save**.

A VM script is now successfully assigned to a VM in a deployment pool.

Verify VM scripts in Lab Builder

You can verify that a VM script is operational and performs the desired tasks by running it in Lab Builder. To verify the VM script you use a VM that is part of a template. This template must be in an unpublished state and it must have a configured network with access to Internet.

- Verify that you enabled the **Enable VM script run** tenant setting.
- Verify that you have the custom created VM script agent from the VMware Lab Platform support.

1. Navigate to **Lab Management › Lab Builder**.
2. On your unpublished template, click **Actions › Edit**.
3. On the **Virtual Machines** tab, click the vertical ellipsis of the VM where you want to verify the script, click **Enable script run**, and confirm.

   Wait for the process to complete. It might take several minutes.

4. Power on the VM.
5. On the VM, install Java Runtime Environment (JRE) 11 or later.
6. On the VM, download the custom VM script agent.
7. On the VM, run `java -jar egw-agent-<agent-version>.jar --spring.profiles.active=prod` to start the agent.
8. Back on the **Virtual Machines** tab, click the vertical ellipsis of the VM, click **Run script**.
9. On the **Run Script** page, select your VM and script and click **Run**.

The VM script runs and you can verify the result in the VM.

Working with catalogs in VMware Lab Platform

Students sign up for labs through catalog listings. You create or modify catalogs from the **Catalogs** page.

Create a catalog in VMware Lab Platform

You create catalogs to contain labs. Users can sign up for labs through catalogs.

A catalog name must be unique within an account.

1. To open the **Catalogs** page, select **Lab Management › Lab Catalogs**.
2. Click **Create New**.
3. Complete the **Create Catalog** form.

Name	Give the lab catalog a meaningful name. This name appears in the **Catalog List**. Catalog names are case sensitive and should be unique per account organization (tenant).
Short Description	Provide a short description of the catalog. This description appears, if activated, in the **Catalog List**. It appears in the lab console within the **Labs** tab to describe each catalog.
Description	Provide a more detailed description of the catalog. This description appears, if activated, in the **Catalog List** (administrator UI).
Enabled	Select to activate the lab catalog. If not selected, the lab catalog is saved in the system but, not available to any VMware Lab Platform accounts.

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Publicly Visible	Activate or deactivate the public visibility of the catalog. If public visibility is deactivated in the catalog, then any labs in the catalog that have their public visibility activated are not publicly visible.
Labs	Select an available lab. The labs you select become associated with this catalog. A lab must be associated with a catalog in order for it to be listed within that catalog in the end-user interface.).
Groups	Select the groups of users in your account who can see this lab. Do not leave this text box blank. You must select a group or the catalog you are creating will not be visible.

4. Click **Create Catalog**.
   The system creates the catalog, and opens the **Lab Display Order** tab.

You can optionally change the order of the labs included within the catalog. Click the first column's header to sort by that column, then drag a row. The lab order within the **Lab Display Order** list determines the order in which the system displays the labs in the lab console.

After you specify the lab ordering, click **X** to close the catalog form. You can preview the catalog listing in the end user interface.

Change lab display order in a catalog in VMware Lab Platform

You can change the order of the catalogs as they appear in the **Catalog List** and in the lab console.

1. To open the **Catalogs** page, select **Lab Management › Lab Catalogs**.
2. To open the **Change Display Order** page, click the three ellipses on the top of the page, and select **Change Display Order**.
3. To move a lab, click and drag a lab. Click **OK** when you are done reordering the labs.
   The lab console reflects the ordering in the Catalog List.

Cloud Services Platform accounts for VMware Lab Platform users

As a cloud organization owner, you can create VMware Cloud Services Platform accounts, grant access to the VMware Lab Platform service, and assign VMware Lab Platformroles.

The **CSP Permission management** can be accessed through the **Identity & Access Management** link at the top left of the VMware Cloud Services Provider user interface.

As an organization owner, you control user access to your organization and its resources. When you invite users to your organizations, you assign two types of role-based access.

- Role-based access to the organization. An organization owner has full access and an organization member has read-only access.
- Access to one or more of the cloud services of the organization. You grant users access to the service by assigning them one or more of the roles provided by the service.

Assigning access permissions to groups is more efficient than assigning the same permissions to individual users one at a time. As an organization owner, you determine the members that make up your groups and what roles and permissions they are assigned.

Add users to your cloud organization and services in VMware Lab Platform

As a cloud organization owner, you invite users to your organization and give them access to the services associated with it.

You can also track the invitations that you send. Invitations are valid for up to seven days. If you have sent an invitation as an error, you can revoke it. The users you invite can hold several roles:

- A role within the organization - Organization Owner or Organization Member.
- A role within the cloud service to which you are inviting the user. Each cloud service has its own roles. For a description of roles in VMware Lab Platform, see Roles in VMware Lab Platform.

1. On the VMware Cloud Services Provider toolbar, click the VMware Cloud Services icon and select **Identity & Access Management**.
2. Click **Add Users**.
3. On the **Active Users** tab, enter the email address of the user you want to add to your organization. You can add more than one user at a time. Email addresses are delimited with a comma or a new line.
4. In the **Role in the Organization** text box, assign a role to the user.
5. The organization owner role includes access to the Support Center. If you are assigning the role of organization member and you want to give the user access to the Support Center, select the **Support User** check box.
6. Select **VMware Lab Platform** from the Service Role drop-down menu.
7. Assign one or more roles for the account by selecting roles from the drop-down menu.
8. To give the user access to another service, click **Add service access** again, if needed.
9. To send an invitation to the user, Click **Add**.

   The invitations you send are valid for seven days. You can view the status of an invitation on the **Invitations** tab.
10. If you sent an invitation in error, you can revoke it. Select the check box next to the invitation, and click **Revoke Invitations**.

Roles in VMware Lab Platform

When a cloud organization owner creates a services account for VMware Lab Platform, the account owner assigns one or more VMware Lab Platform roles. Each role has a different set of permissions for accessing VMware Lab Platform user interfaces.

All roles, except for the **User Role**, must be set from the VMware Cloud Services Platform user interface by a cloud organization owner. The **User Role** assigned from the VMware Lab Platform administrator UI or from the VMware Cloud Services Provider interface.

Role	Description
User	Entitles user to log into the end-user interface and take a lab. The **User Role** does not include permission to use any other user interface (for example, no permission to the administrator UI or the **Content Engine** user interface).
Admin	You can assign the role of “Admin” to the user you are creating and as a result create another administrator. The **Admin Role** is the most powerful user within an account. It includes permission to all the user interfaces (for example, the administrator UI, the end-user interface, the **Content Engine** user interface, and so on).
Proctor	User you consider to be familiar with the content in the various labs of the account. When end users have a question with a lab, they can use an option on their lab console screen. The option runs an application running on the proctor’s mobile device to notify the proctor that someone needs help. A user having an **Event/ Proctor Role** can log in to the administrator UI.

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Role	Description
Support	Assigns a subset of the administrator permissions. A user having a **Support Role** can log in to the administrator UI. However, the system defaults to the **Entitlement List** and allows only the **Reset/Remove vApp** action.
Instructor	Users assigned this role appears in a list accessible from the **Create Class** form. An instructor can monitor the progress of selected students (a class), taking a specified lab at a specified time (instructor-led lab). A user having an **Instructor Role** can log in to the administrator UI. However, the system defaults to **Class List** and allows the user to modify a class. The system also allows a read-only permission to the **Entitlement List**. **NOTE** The **Portal Settings › Configuration** page includes the setting **Allow Instructors Access To All Classes**. If activated, then an instructor can go to the **Class List** and view all the classes in the account. If deactivated, then the **Class List** lists only those classes to which the instructor is assigned.
Certification Grader	Users assigned this role can view a list of entitlements that must be graded and can assign them a score and a grade string. A user with a **Certification Grader** role user can edit these anytime until clicking **Graded**.
Content Creator	Users assigned this role have permissions only to the **Content Engine**. The **Content Engine** allows a user to create/edit/manage instructional content (lab manuals).
Group	Select a group or groups to associate with the user account.

Managing Cloud Service Provider user accounts

A cloud organization owner can modify a user account or remove a user from an organization.

For more information, see Manage Users in the *VMware Cloud Services* documentation.

Working with roles and permissions in VMware Lab Platform

VMware Lab Platform provides a set of built-in roles that can be assigned to users in an organization. You cannot modify role permissions of the built-in roles, because the built-in roles are predefined. You can create custom roles and customize the permissions for the custom roles based on their organization's requirements.

To view base roles and custom roles for an account or organization, click **Users & Groups › Roles & Permissions**.

VMware Lab Platform has the following built-in roles:

Built-in Role	Permission
Student	Entitles user to log in to the end-user interface and take a lab. The **User Role** does not include permission to use any other user interface (for example, no permission to the administrator UI or Content Engine user interface).
Admin	You can assign the role of “Admin” to a user you are creating and as a result create another administrator. The **Admin Role** is the most powerful user within an account. It includes permission to all the user interfaces (for example, the administrator UI, end-user interface, Content Engine user interface, and so on).

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Built-in Role	Permission
Instructor	An instructor can monitor the progress of selected students (a “class”), taking a specified lab at a specified time (“instructor-led lab”). Users assigned to this role appear in a list accessible from the **Create Class**form. A user having an **Instructor Role** can log in to the administrator UI. However, the system defaults to **Class List** and allows the user to modify a class. The system also allows a read-only permission to the **Entitlement List**. **NOTE** The **Portal Settings › Configuration**page includes the setting **Allow Instructors Access To All Classes**. If activated, then an instructor can go to the **Class List** and view all the classes in the account. If deactivated, then the **Class List** lists only those classes to which the instructor has been assigned to.
Content Creator	Users assigned this role have permissions only to the Content Engine interface. The Content Engine interface allows a user to create, edit, and manage instructional content (lab manuals).
Certification Grader	Users assigned this role can view a list of entitlements that must be graded and can assign them a score and a grade string. A user with the **Certification Grader** role user can edit these anytime until clicking **Graded**.

Create a custom role in VMware Lab Platform

You can create custom roles for your organization accounts.

Guidelines for creating custom roles:

- To control access of members, you can mix and match permissions from the allowed predefined built-in roles and can create custom roles within your organization.
- The custom role name must be unique in the organization.

To create a custom role:

1. Click **Users & Groups › Roles & Permissions › Create New**.
2. Enter a unique name for the custom role.
3. If you want to create the custom role based on a built-in role, then select the role from the **Use Existing Role** drop-down menu. Alternatively, you can also choose an existing custom role and modify the permissions.

   Based on the built-in role selected, the default permissions for the role are auto-selected under the Permissions category.
4. Select the permissions for the custom role from the **Permissions** drop-down menu.
5. Click **Create User Role**.

NOTE
The new custom role is listed on **Roles & Permissions** page under the **User Role** category. Administrators can assign the roles to users using the **Users & Groups › User Accounts › Accounts › Update › User Access** option.

Edit a custom role in VMware Lab Platform

You can edit a custom role and permissions for user accounts.

To edit a custom role:

1. Open the **Roles & Permissions** page, click **Users & Groups › Roles & Permissions › User Roles**.
2. Click the custom role that you want to modify. The **Update Account** form opens.
3. On the Update form, make your edits to the custom role fields, and then click **Update Account**.

Managing VMware Lab Platform user accounts

You can create and manage VMware Lab Platform user accounts.

For information about assigning other roles for the VMware Lab Platform service, such as Instructor or Support User, see Cloud Services Platform accounts for VMware Lab Platform users.

Create a user account in VMware Lab Platform

As an administrator, you create a user account from the administrator user interface. You create user accounts to provide users access to the different VMware Lab Platform interfaces such as the administrator interface, end-user interface to take labs, event management interface to create and manage an event.

> **NOTE**
> You can also create non-cloud services provider administrator accounts. To create non-cloud services provider administrator accounts from VMware Lab Platform, click **Users & Groups › User Accounts › Create New**.

For information about VMware Lab Platform roles, see Roles in VMware Lab Platform.
VMware Cloud services users need to assign administrator roles from the https://console.cloud.vmware.com for their product.

1. To open the **Accounts** page, click **Users & Groups › User Accounts** .
2. To open the **Create Account** form for a single user, click the **Create New** drop-down menu and click **Single Account**. To create user accounts for multiple users, select the **Bulk Upload** option.
3. In each section of the Create Account form, enter the information for the user account that you want to create. Required information is marked with an asterisk.

Setting	Description
Username/Email	Enter an email address for the user you want to create.
First Name	Enter user's first name.
Last Name	Enter user's last name.
Display Name	VMware Lab Platform displays the name you enter on the user's lab console screen beneath the text Welcome, (for example, Welcome, Jane Smith).
Language	Select a language from the language drop-down menu.
Company	Enter user's company or organization name.
Password	Enter the password to use for authentication.
Confirm Password	Re-enter the previous password.
(Optional) **Contact Details**	Enter the contact details of the user.

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Setting	Description
	When you activate the portal setting **Select a cloud by user's location**, system picks a prepop that is closer to the user's location. In a multi-cloud environment, when a deployment pool has templates in clouds located in different regions, users will receive a lab template that is in the same region or that is closer to the user's location. For example; if an organization account has two clouds, in two different regions (one in the USA and another one in EU). Lab (L1) that points to deployment pool 1 (DP1). DP1 has two lab templates —T1 (located in the USA cloud) and T2 (located in the EU cloud). Now when you activate the portal setting **Select a cloud by user's location**, and a user starts a lab, VMware Lab Platform takes the account's location field and looks for a prepop in that location. If the account's location is set to EU, and user starts lab L1, the system checks if template T2 has prepops, assigns one prepop for the entitlement. **NOTE** To check the location of a cloud org, click **Cloud Management › Cloud vApps**, view the details under the cloud org details column.
Authentication Provider	Allows you to select an authentication provider for users using the VMware Lab Platform user interface. You need to first create the authentication provider in the system. To learn more about how to add an authentication provider, see Working with an authentication provider for single sign-on in VMware Lab Platform.
Generate Random Password	Enable the system to generate a user password.
Show/Hide Password	When activated, displays the password for the user account.
Send Welcome/Verify	When activated, sends Welcome to VMware Lab Platform email to the user address. The Welcome to VMware Lab Platform email includes a verification link. The user must use the link to verify the account and configure a password for their account. Once the account is activated by the user, the account is verified in the system. You can check the account status from the **Users & Groups › User Account › Accounts** page. View the **Account Verified** column on the

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Setting	Description
	Accounts page to see the account verification status of the user. If the user does not verify, then VMware Lab Platform provides a one-week grace period. During the grace period, the user has access to VMware Lab Platform, but some features are restricted. Restricted features include social media, the sending of transcripts and sending an invite to friends. If the grace period expires and the user still has not verified the account, then VMware Lab Platform blocks all access. **NOTE** Users can access more than one account. If the user you are creating already has access to another account, then that user does not get a Welcome to VMware Lab Platform email. A user gets the welcome email only for the first customer account to which access is granted.
Contact Details	Enter the contact details of the user.
Tenant Groups	From the drop-down menu list add all the groups that the user needs to be part of.
User Permissions	Select the role for the user from the User Permissions page.
Account Settings	Verify the account settings and select the relevant options in the **Account Settings** page.

4. Click **Create Account**.
5. Click **Finish**. VMware Lab Platform creates the user account and opens a page that displays the information about that user.

View the account details and account status from the **Users & Groups › User Account › Accounts** page. View the **Account Verified** column on the Accounts page to see the account verification status of the user.

Edit a user account in VMware Lab Platform

You can modify a user account and change the user account information, or assign a user a different role.

1. To open the **Accounts** page, click **Users & Groups › User Accounts** from the navigation panel.
2. To modify a user account, click the three ellipses next to the user name account you want to modify. The **Update Account** form opens.
3. On the **Update Account** form, make your edits to the user account fields and then click **Update Account**. Likewise, you can open the User Access form to make changes to the user's role.

View user account options in VMware Lab Platform

The Account page includes a drop-down menu of common account management tasks. You can make changes for an individual user account or bulk accounts.

1. To open the Account page, select **Users & Groups › User Accounts.**
2. On the Account List, select the check box next to a user account, then click the ellipses to open the list of action options.

The following actions are available.

Setting	Description
More Details	Displays the details of the user account.
Update	Edit and update an account.
User Permissions	Edit and update a user's access and roles.
Account Settings	Displays additional details about the user account such as address, phone number, and so on.
Login As Account	Use the **Login As Account** option to log into the lab console as the selected user. If an end user reports a lab console problem, you can use the **Login As Account** option to see the problem or issue that the end user has reported. A login link to the user account's console is generated.
Impersonate Account	Use the **Impersonate Account** option to log in as an administrator into the VMware Lab Platform administrator UI.
Flush Questions	Removes the security questions associated with the user account.
Migrate to a different provider	Lets administrators to migrate the user account from one authentication provider to another. In the Migrate to a different provider page, select the authentication provider from the drop-down menu.
Send New Token	Sends a verification email to the user when a customer user account is set up. The purpose of the email is to validate that the user's email address is authentic.
Migrate Transcripts	Migrates transcripts from one user account to another. This is useful when a user has changed their email address and wants their transcripts to be transferred to a new account.
Download	Lets administrators to download comma-separated values (CSV) of the users selected in the list.
Expire Password	Expires the current password and forces the user to set a new password.
Never Expire Password	Lets administrators to never expire the password for the user account.
Deactivate	Deactivates the account.
Verify	Verifies the account.
Delete	Deletes the accounts.

1. To update multiple accounts, select the check box next to user accounts that you want to update, and select the **Batch Actions** options listed on the page.

Action	Description
Flush Questions	Removes account's security questions.
Migrate to a different provider	Lets you to migrate the bulk user accounts from one authentication provider to another. In the Migrate to a different provider page, select the authentication provider from the drop-down menu.

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Action	Description
Send new token	Sends a verification email to the user when a user account is set up. The purpose of the email is to validate that the user's email address is authentic.
Download	Allows you (the administrator) to download comma-separated values (CSV) of the users selected in the list.
Unlock Account	Unlocks all accounts in the list of accounts. **NOTE** The system locks an account when an end user, while trying to log into a lab, answers the security questions wrong three times. At the time of locking (third wrong answer), the system sends an email to the email address of the account. The email contains a verification link that unlocks the account.
Expire Password	Expires the current password and forces the users to set a new password.
Enable	Activates the accounts.
Deactivate	Deactivates the accounts.
Verify	Verifies the accounts.
Delete	Deletes the accounts.

Using additional account options

There are several other useful options that you can use from the Account page: Flush Questions, Log in as User, and Print One Time Password.

Flushing user questions for a user account

Flush questions removes a user’s security questions. The option is useful when a user role has set security questions, but cannot remember the answers. Flushing the user’s security questions lets the user to log in with a one-time password (OTP) and then set new security questions.

NOTE
Flush Questions must be activated per-customer account (by VMware support team). If you note that the Flush Questions option is inactive for a user, the likely reason is that the user has a role greater than a “user role” in another customer account.

Log in as a user

If an end user reports a lab console problem or issue, you can use the **Login as User** option to log in to the lab console as the selected user. The**Login as User** option lets you to see exactly what problem/issue the end user has reported. To use this option, proceed as follows:

1. Select a user, click **Login as User**.
2. On the notification that appears, click or enter the Login URL into your browser's address field.

   The system logs you out of your current session and logs you into the lab console as the user.

Print one time password

You can print the selected user's one-time password with the Print One Time Password option.

Add multiple users in VMware Lab Platform

You can upload a list of users with the **Bulk Actions** options.

The **Account Bulk Upload** option supports CSV files that specify user roles (user, admin, proctor, support, instructor, grader, and content creator). Administrators can create anonymous users for a one-time lab or a class using the **Account Bulk** option.
To upload bulk users:

1. On the **Users & Groups › User Accounts › Accounts** page, click the **Create New** drop-down menu and select **Bulk Upload**.
2. To select a CSV file, click **Attach File**.

   NOTE
   You can optionally open the File upload format information page. To download a sample CSV file, select the **Download Sample of CSV file** format option. This option helps in understanding the file format that the system supports.

   To add bulk users and assign to a class, add `classSku` field and add specify the SKU for the class in the CSV form.

3. On the **Upload CSV** tab, you can select any of the following options:
   - **Send verification email**: Controls whether a welcome email is sent to each user in the CSV file.
   - **Download Results**: An email that reports the status of the operation is sent to the administrator. If the operation is a success, the email includes a list of the imported accounts.
   - **Export fields**: Selects the columns to be included in your CSV file.
   - **Cancel**: Cancels the upload CSV function.
4. Click **Next**. The Preview tab opens.
5. After you have verified your bulk upload, click **Finish**. The system creates the users included in your CSV file and adds them to your account. Remember to refresh your Account List.

   To cancel a bulk account upload, remove, or download operation, click **Cancel.**

Remove multiple users in VMware Lab Platform

You can download or remove multiple users using the **Bulk Actions** option.

The following procedure shows how to download, or remove bulk users.

1. On the **Users & Groups › User Accounts › Accounts** page, click the three ellipses and select **Bulk Download** from the Batch Actions drop-down list. To remove users, select the **Bulk Remove** option.
2. Click **Attach File** and browse to select a CSV file.

   NOTE
   You can optionally open the File upload format information page. Select the **Download Sample of CSV file** format option to download a sample CSV file that is useful to gain an understanding of the file format the system expects.

   To cancel a bulk account remove, or download operation, click **Cancel.**

List administrators and proctors in VMware Lab Platform

To view the list of administrators and proctors, use the Account List (User Accounts) feature.

1. Click **Users & Groups › User Accounts**. The Account List opens.
2. To list all administrators, from the **Manage Columns** menu, select **Admin Role**.
3. To list all proctors, from the **Manage Columns** menu, select **Proctor Role**.

Create a temporary user in VMware Lab Platform

As an administrator, you can create a temporary user for a “one-time” lab use, without the need to register the user in the system. This feature is useful for users who create content for quick demos or live events to demonstrate their products to users who might not use the system regularly.

1. To open the Entitlements page, navigate to **Entitlements › User Entitlements** .
2. Click **Create New**.
3. Select a lab from the lab list.
4. Select the **Send Quick launch URL to user via Email** and **Create user for entitlement** check boxes.
5. Enter all the required details in the **Create user entitlement** dialog box.
6. Click **Create Entitlement**.

An email with the lab details and lab URL is sent to the user. After the entitlement ends, the temporary user is automatically deleted from the system.

Create an anonymous user in VMware Lab Platform

Similar to a temporary user, you can create an anonymous user for a “one-time” lab use, without the need to register the user in the system. This feature is useful for administrators who create content for quick demos or live events to demonstrate their products to users who might not use the system regularly.

Enable the **Portal Settings › Configuration › General Options › Enable Anonymous User Lab access** option.

1. To open the Entitlements page, navigate to **Entitlements › User Entitlements**
2. Click **Create New**.
3. Select a lab from the lab list.
4. To display the **Create anonymous user** option, select the **Create user for entitlement** check box .
5. Select **Create anonymous user**.
6. Enter all the required details of the user in the page.
7. Click **Create Entitlement**.

   Once the entitlement is completed, the user account is deleted from the system. You can share the quick launch URL for the lab with the user.

Create bulk anonymous users in VMware Lab Platform

Account administrators can create multiple anonymous users similar to a single anonymous user for a lab or an event.

Enable the **Portal Settings › Configuration › General Options › Enable Anonymous User Lab access** option.

1. To create bulk anonymous users, use any of the following options:
   - To open the **Account Bulk Upload** form, click **Users & Groups › User Accounts › Create New> › Bulk Upload** option. To upload a CSV file for bulk users, click **Attach** and add the file.
   - **NOTE**
     To see the supported file format for anonymous bulk users, click the instructions link on the page. The supported format for anonymous user is `anonymousUser`.
   - Alternatively you can also create bulk accounts with the `Account API` call.

Bulk anonymous users are created and account administrators can send the quick launch link for the lab to anonymous users.

Working with groups in VMware Lab Platform

You can create group permissions for a user account.

You can enact the following permissions:

- All user accounts in the group you created have permissions to the catalogs associated to the group.
- When you create a catalog (Create Catalog page) you can select which groups have permission to the catalog.

Create a group in VMware Lab Platform

With groups, you can assign a group of user permissions to same catalogs in VMware Lab Platform.

1. To open the Group List page, click **Users & Groups › Groups**.
2. To open the **Create Group** form, click **Create New** .
3. Enter the following information:
   a) In the **Name** text box, enter a name for the group you are creating.
   b) In the **Accounts** text box, select the user accounts you want associated to your group. You can select from the drop-down menu, or use **Manage List View**.
   c) In the **Catalogs** text box, select the catalogs that you want associated to your group. You can select from the drop-down menu, or use **Manage List View**.
4. Click **Create Group**. The View Group pop-up appears which summarizes the group you just created.

Working with settings in VMware Lab Platform

With the **Portal Settings** option, you can customize your VMware Lab Platform environment and activate actions for per-role or per-lab basis. To work with these settings, you must have the administrator role privileges.

You edit settings from the **Portal Settings › Configuration** page.

- Details
- AWS S3 Integration
- Class and Instructor Options
- Console Interface
- Custom Messages
- Deployments
- Email
- Entitlements
- Event Mode
- Event Statistics
- Exam
- General Options
- Interface
- Lab Builder

Details

Setting	Description
Administrative Contact	Email address of the person in your organization assigned to act as your administrative contact. Manually enter the email address or you can select the email address from the drop-down of user email addresses.
Allow all users to see content creator credentials	To enable the setting, select the check box.
Cost Center	To identify your cost center, assign a value, if applicable.
Technical Contact	The email address of the person in your organization assigned to act as your technical contact. You can select an email address from the drop-down menu of user email addresses.
Display Name For Tenant	Name of the organization that VMware Lab Platform displays in the user interfaces.

AWS S3 Integration

You can add and configure their own instance of the Amazon Simple Storage Service (S3) with VMware Lab Platform.

Setting	Description
AWS API Key	Enter the AWS API key to use with VMware Lab Platform for AWS API authentication. By default, the value is empty.
AWS API Secret	Enter the AWS API secret to be used with the AWS API key. By default, the value is empty.
AWS Bucket Enabled	When activated, the AWS S3 options to import and export templates is added to the **Lab Templates** page. By default, the **awsBucketEnabled** setting is deactivated. For more information on the Export to S3 feature, see Edit a Lab Template with the Lab Builder.
AWS Bucket Name	Enter the AWS Bucket name. By default, the value is empty.
AWS Bucket Prefix	Enter the prefix value for the AWS Bucket.
AWS Bucket Role	Enter the Bucket's AWS role.
AWS Role Arn	Enter the Amazon Resource Name (ARN) of a role with access to the Bucket.
Send all the mails for tenant via Amazon SES	

Support Chat

Setting	Description
Live Chat	Activate or deactivate the Live Chat setting. If activated, an end user taking a lab can select **Help › Live Chat** to open a window to chat with the support staff.
Live Chat URL	Specifies a default URL to use for live chat. You can override this URL for an individual lab from **Labs › Update Settings**.

Console Interface

Setting	Description
Allow Console Resizing	If activated, a lab user can manually resize the lab console.

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Setting	Description
Compact Panel Tabs	Activate or deactivate. When this option is activated, the manual toolbar option and VM console tabs are moved on the top of the console. To view the console panel tabs, click the  icon . To view the manual panel tabs, click the  icon.
Console Connection Monitoring	Activate or deactivates console connection monitoring.
Default Console Location	Select from **right**, **left**, **center**, or **maximized** from the drop-down menu. When you select the **maximized** option, and enable the **Maximize Console on Fullscreen** option, the console size along with the browser is maximized.
Show 'Shortcut Keys' button on Lab Console	Activate or deactivate. When activated, the **SHORTCUT KEYS** button is displayed on the user console that prompts keyboard quick access keys. This option replaces the CTRL+ALT+DEL buttons.
Compact Panel Tabs when Maximized	Activate or deactivate. When this option is activated along with the **Maximize Console on Fullscreen** option, the manual toolbar option and VM Console tabs are moved on the top of the console. To view the console panel tabs, click the  icon . To view the manual panel tabs, click the  icon.
Maximize Console on Fullscreen	Activate or deactivate. When activated, maximizes the console screen to full screen of the device. **NOTE** Ensure that the Default Console Location is selected as maximized.
Delay for redirecting to ScreenSaver after it's turned on	Sets the time to wait before redirecting a console screen to a screen saver.
Delay before starting ScreenSaver on station	Sets the time to wait before starting a screen saver on a station in a lab.
Show Lab's console header buttons	Activate or deactivate. When activated, console buttons are visible on the console header.
Show Console Connection Strength	When activated, the console connection strength indicator is displayed in the end-user interface. If this option is deactivated, the console connection latency data is recorded and can be viewed from the administrator interface.
Show 'Notes' button on Lab Console	Activate or deactivate. When activated, the Notes option is available in the lab console.
Show network information in VM notes tab in Lab Console	When activated, shows network information in VM notes.

Custom Messages

You can replace the default content of lab messages with your own content. For example, for **Lab Submitted for Grading**, the default is `Thank you for taking this lab. An instructor will grade your lab.` You might change the default to `Your lab will soon be graded.` You can use the Custom Messages section to customize the messages sent by the labs of the current organization. To do so, highlight the message content, and click the edit symbol that appears.

NOTE
For **Error Message For Maximum Allowed Entitlement Reached**, the system replaces the token strings *{maxAllowed}* and *{current}* with the number of active and current entitlements.

Example: the system might return the message `Only four concurrent enrollments are allowed for this lab and four enrollments already exist`. You can set the values for *{maxAllowed}* and *{current}* within the User Hard Cap section.

Message Type	Message Content
Class no more seats available message	Cannot enroll in a class because all the available seats are filled.
Lab Submitted For Grading Message	Thank you for taking this lab. Your lab has been submitted for grading by an instructor.
Lab Completed Message	Thank you for taking this lab. We hope you enjoyed your experience. Please go back to the home page to select another course or lab to take.
Lab Completed Title	Lab successfully completed!
Lab Submitted For Grading Title	Lab submitted for grading!
Lab Completion Criteria Not Met	
Lab Completion Criteria Not Met (Graded Labs)	
Lab End Message When Exit Allowed	Caution! Once you end the lab, you will not be able to return. If you would like to exit so you can return to the lab at a later time, please choose to Exit the lab instead. Are you sure you want to end this lab?
Lab End Message When Exit Not Allowed	Caution! Once you end the lab, you will not be able to return. Are you sure you want to end this lab ?
Lab End Title	Are you sure you want to end this Lab?
Lab Exit Message When VMs will be powered down	Caution! Your servers will be powered down when you exit the lab. Any unsaved work will be lost. You can return to finish the lab at a later time with any saved work. Are you sure you want to exit this lab?
Lab Exit Message When VMs will not be powered down	You are about to leave this lab. You can return to finish at a later time, your VMs will remain in a running state, but you should save any unsaved work. Are you sure you want to exit this lab?
Lab Exit Title	Are you sure you want to exit this lab?
Lab Reset Confirmation	Are you sure you want to reset this lab?
Lab Reset Message	If you reset this lab, all VMs will be restored to their original state. You will not be able to recover any work you have done, so you will need repeat it. Only reset this lab if you are having unrecoverable issues.
Lab Reset Title	Are you sure you want to reset?
Lab Reset Warning	Warning!
Error message for maximum allowed entitlement reached	Only {maxAllowed} concurrent enrollments are allowed for this lab and {current} enrollments already exist.
Show/Hide VMware Lab Platform in beta message	Activate or deactivate.

Deployments

Setting	Description
Allow assignment while deploying	Activate or deactivate the assignment of cloud resources before those resources are fully enabled. When enabled, the system assigns a still deploying prepop rather than waiting for an on-demand prepop. The result is a faster start time for labs in cases where fully provisioned cloud resources are not available.
Name of default prepop assignment algorithm	Select the algorithm the system uses to assign prepops after an end user starts a lab. This assignment algorithm selection is the first step of a two-step process during which the system assigns available prepops from a specific vDC. The second step is the **Name of Default Prepop Assignment Preference**. You can override this selected customer account-wide algorithm for a particular lab template using the **Edit Template** setting. The prepop assignment algorithms are as follows: • **VMware Lab Platform Assignment Algorithm**: This is the default algorithm. It selects a prepop from any available vDC based solely upon the preference setting. • **Most Available Prepops** Sets a preference for the vDC having the most available prepops. • **Most Available Prepops for the Template** Sets a preference for the vCD having the most prepops deployed from the template being used in the assignment. • **Most Deployments** Sets a preference for the vDC having the most deployments. This deployment preference includes both prepop and assigned vApps. • **Most Deployments of the Given Template** Sets a preference for the vDC having the most deployments. This deployment preference includes both prepops and assigned vApps of the lab templates being used.
Name of default prepop assignment preference	Selects a prepop preference if there are multiple prepops available in the assigned vDC. This prepop preference is the second step of a two-step process. The first step is the prepop assignment algorithm selection. You can override this selected customer account-wide preference for a particular lab template using the **Edit Template** setting. The following preferences are supported: • **Prefer Older**: Sets a preference for the older prepops (the older the prepop, the stronger the preference). • **Prefer Newer**: Sets a preference for the newer prepops (the newer the prepop, the stronger the preference). • **No Preference**: Sets a preference randomly.
Name of default prepop placement algorithm	Select the algorithm the system uses to place new prepop deployments in cases where a template is available in multiple vDCs. The system does not apply the selected algorithm to on-demand deployments. The algorithm you select is applied on a customer account-wide basis. You can override the selected customer account-wide algorithm for a particular lab template using the **Edit Template** setting. Changing the default placement algorithm is not a best practice. The VMware Lab Platform Default Placement Algorithm is designed to offer a balanced use of available resources. The default value is VMware Lab Platform Placement Algorithm.

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Setting	Description
	NOTE All algorithms ignore vDCs that have exceeded their configured hard-limits for availability and avoid vDCs that have exceeded their soft-limits.The system can still select these vDCs with a reduced frequency. • **VMware Lab Platform Default Placement Algorithm**: This is the default. It attempts to balance distributions across all available vDCs. This algorithm presumes that if a vDC is available and can be used, then the system uses it. This algorithm does not use a “balancing step” to ensure equal load upon all vDCs in the near term. Instead, this algorithm leaves balancing as a side-effect of the way the system assigns prepops, which leads to a balanced use of available resources. • **Equal By Deployed Template Count Algorithm**: Attempts to have an equal number of vApps for the given template deployed in each vDC in which the template is available. This algorithm does not consider differences in resource availability in the vDCs (other than hard and soft limits) and aims only for equal numbers of deployments. This algorithm considers only the number of deployments that use the templates being used in the placement. • **Equal By Deployment Count Algorithm**: Attempts to have an equal number of vApps deployed in each vDC based solely on the count of deployments regardless of the actual resource usage of different types of vApp. This algorithm does not consider differences in resource availability in the vDCs (other than hard and soft limits) and aims only for equal numbers of deployments. This algorithm considers the total number of deployments in the vDC. • **Equal By Resource Utilization Algorithm** Attempts to distribute deployments across vDCs so that the relative resource utilization of vDCs is equal. While the system assigns more deployments to vDCs with more available resources than clouds with fewer resources available, the proportional resource utilization is similar.
Max amount of queued VDC tasks before ignoring location preference	The maximum number of queued tasks for a virtual data center. If the virtual data center surpasses this option value, labs are deployed from a different virtual data center that might be in a different location.
Enable Optimized Deployment Pool Usage	By default this option is activated. When activated, you cannot access the **Deployment Pool** menu option from the main navigation panel, and you need to manage your prepops on the template level in the Lab Templates page. In the Lab Builder, you can directly choose the lab template instead of deployment pools.
Daily Report for Prepop Hits and Misses	If activated, a daily report of number of prepopulated lab templates (prepop) successfully deployed is sent to you.
Enforce network wiring from metadata only	If activated the setting at account level, the deployment of prepop will not connect the cloud org virtual data center network to a vApp. If deactivated, the deployment of prepop will connect through the network settings defined in the metadata.

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Setting	Description
Explicitly ready prepops	If activated, provides the **Explicitly Ready Prepops** option in the **Deployment Pool**page. When activated for a specific deployment pool, prepopulated lab templates must be explicitly marked as available after they are powered on.
Prepop Hit/Miss Rate Period (In Hours)	The **View Deployment Pool** page lists details of the selected deployment pool. One of the details is “Hit”. A “hit” means that a user got a prepop when they requested. A “miss” means that the user got a full deployment of a lab and probably had to wait for a long time. Hit Rate Period specifies the number of hours for the hit rate calculation displayed in the Deployment Pool details. Changing the hit rate period only updates the values shown on the Deployment Pool details but the system does not actually make any substantive changes.
Maximum prepops per template	Maximum prepops that can be deployed for a template.
Prefer own cloud (BYOC) for Labs with hybrid setup	Activate or deactivate. When activated, if a user is using two clouds; their own and VMware Lab Platform cloud, then the own cloud is used instead of the VMware Lab Platform cloud.
Query size for prepop assignment	VMware Lab Platform assigns the oldest available prepop first. Once the system has found all the prepops it might assign when a user starts a lab, it can then order them by their creation date and assigns the oldest one first. The **Query size for prepop assignment** setting adds some randomness to this process by finding the available prepops, ordering them by their creation date, and then assigning one by selecting one at random from the oldest range of prepops. This oldest range is the query size for the prepop assignment value you specify.
Select a cloud by user's location	If activated, the system picks a prepopulated lab, or creates a lab deployment, that is closer to the user's location. **NOTE** The **Select a cloud by user's location** setting works only when: • Your VMware Lab Platform organization account has multiple cloud regions. • Location information for the user account is configured. In a multi-cloud environment, when a deployment pool has templates in clouds located in different regions, users will receive a deployed lab that is in the same region or that is closer to the user's location. For example; if an organization account has two clouds, in two different regions (one in the USA and another one in EU). Lab (L1) that points to deployment pool 1 (DP1). DP1 has two lab templates —T1 (located in the USA cloud) and T2 (located in the EU cloud). Now when you activate the portal setting **Select a cloud by user's location**, and a user starts a lab, VMware Lab Platform takes the account’s location field and looks for a prepop in that location, or deploys a new lab to the cloud that is closer to the user. If the account’s location is set to EU, and user starts lab L1 and the system checks if the template T2 has prepops, and assigns one prepop for the entitlement. For information on how to add user location and address, see Create a user account in VMware Lab Platform.

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Setting	Description
Explicitly Ready Prepops	Enable to turn on the assignment of VM scripts to a VM in deployment pools.

Entitlements

Entitlements refer to lab time limits and lab valid periods. A Time limit is the amount of time, in hours, users get to work on a lab. A valid period is the amount of time a lab stays active.

Setting	Description
Enforce Extension Limit For Proctor	If activated, then the specified **Maximum Extension Times** also applies to proctors. Activated proctor users cannot extend entitlements beyond the number specified in the **Maximum Extension Times** setting. When this setting is deactivated, proctors can extend entitlements any number of times.
Max number of concurrent entitlements per user	Limits each user to a maximum about how many entitlements (labs) can be taken concurrently. This setting lets you to manage large online events where concurrent users can often increase quickly and greatly.
Max number of concurrent entitlements per user per lab	Limits each user to a maximum concerning multiple entitlements to the same lab. This setting lets you to manage a student during certification. For example, a student can take a lab, end it and, while the lab is in “awaiting grade status,” enroll and take the lab again. **NOTE** The values you enter for the **Max Number of Concurrent Entitlements per User** setting and the **Max Number of Concurrent Entitlements per User per Lab** setting can affect a lab message. The Custom Message section includes a lab message setting called **Error message for maximum allowed entitlement reached**. The message includes the strings {maxAllowed} and {current}. The system replaces those strings with the values set in the **Entitlements** section. For example, the system can return the message `Only 4 concurrent enrollments are allowed for this lab and 4 enrollments already exist`.
Maximum Number of Extensions	Sets a maximum number of times lab users can extend the time remaining for a lab. For example, a value of five means that lab users cannot extend a lab’s “time remaining” more than five times.
Time Extended Per Request	Sets a value, in minutes, by which a lab’s time remaining is extended each time a user clicks **Extend**. For example, a value of two means the lab’s “time remaining” is extend by two minutes each time a user clicks **Extend**.
Time Extension Enabled	If activated, then the lab console displays the **Extend** setting to the lab user.
Time Remaining to Show Danger Prompt	Sets a time value (for example, one minute) for the lab console to display a lab “time remaining” danger prompt.
Time Remaining to Show Warning Prompt	Sets a time value (for example, three minutes) for the lab console to display a lab “time remaining” warning prompt.

Event Mode

Includes settings to manage credits, event statistics, and other event parameters.

Setting	Description
Allow users to log in without providing a password	Activate or deactivate a password-free login for end-user interface. If this option is activated, end users can login the user interface without a password.
Limit concurrent Entitlements	Limits the number of concurrent entitlements a user can have.
Event Credit Check Enabled	Activate or deactivate credit check.
Event Dashboard Content Cycle time (in milliseconds)	At events, you can provide a dashboard that lists statistics. The dashboard is somewhat like a slide show on a continuous loop. Cycle time refers to much time each slide stays up on the dashboard before going to the next slide.
Enable Slack Integration	Activate or deactivate the Slack integration with the VMware Proctor application. When the **Enable Slack Integration** option is activated, and a client Slack channel URL is provided in the **Slack Integration URL** setting, student help requests are relayed in the client Slack channel.
Event Mode Enabled	When activated, the Event Management menu is displayed in the side navigation panel on the administrator UI. Allows multiple settings, including seating assignments, dashboard updates for statistics and proctor information, and checks for assigning course credit.
Event Client Content Cycle Time (in milliseconds)	Similar to dashboard cycle time but refers to a slideshow that acts as a screensaver on station desktops.
• **Lanyon API Token** • **Lanyon HOL Token** • **Lanyon API URL (include http/https)**	The Lanyon settings are for badge scanning at events where registration is provided by the Lanyon events registration company. If you want to use the badge-scanning feature, contact the VMware Lab Platform staff.
Time to Live for Credits (in seconds)	Credits issued to a user are valid for a limited time before they expire.
Max Station Token Time (in minutes)	At events, VMware Lab Platform staff provides several desktops attendees can sit down and use. These desktops are known as stations. When the VMware Lab Platform staff registers a station, it gets a unique seat number associated with it and a cookie to identify which station it is. The token time specifies the length of time the cookie is valid before it times out and prevents anyone from potentially stealing the cookie and using it on their own machine.
Number of credits granted to a user from single scan	Specifies the number of credits VMware Lab Platform grants when an end user's badge is scanned. At an event, customers who want to take a lab have their badges scanned. The scanner application sends a request to VMware Lab Platform to add the specified number of credits to the user's account.
One Time Password Length	Number of characters for a one-time password. When an administrator first creates a user in VMware Lab Platform, the account must be verified. Normally, this is done by sending an email to the user with a verification link. At events, this practice can become problematic because many users might not access their email. To address this problem, VMware Lab Platform creates a random one-time password for the user (for example, 98fds788). The VMware Lab Platform staff then prints the one-time password on the badges for the event so users have a known, good password, which one can use to log in the first time. After the first login, VMware Lab Platform prompts the user to set security questions and change the user's password.
Treat Proctors as Normal Users	Activate or deactivate allowing an event proctor to be treated by the system as a normal user.

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Setting	Description
• **Rainfocus API Profile ID** • **Rainfocus API Widget ID** • **Rainfocus API URL (include http/https)**	Fields let you integrate VMware Lab Platform with the Rainfocus application. Rainfocus is an event performance management platform designed to help improve the event experience for exhibitors, vendors, and attendees.
Enable Roost Notification	Activate or deactivate the Roost notification application. Roost is used by the Event Proctor application. The Event Proctor application is a VMware Lab Platform application that runs on mobile devices. The application provides a convenient way for a proctor to move about the event floor and efficiently respond to help requests from students taking labs.
Slack Integration URL	Enter the client Slack URL to use in the VMware Proctor application.

Event Statistics

These settings are used to collect event statistics that can be displayed during an event.

Setting	Description
Show All Historical Labs (Stats Display)	Activate or deactivate the historical labs display.
Event Statistics Display Start Date	Day/Month/Year/Time.
Max Active Lab (Stats Display)	Maximum number of active labs to display.
Max Completed Lab (Stats Display)	Maximum number of completed labs to display.
Max Popular Labs Shown (Stats Display)	Maximum number of popular labs to display in the **Show Popular Labs** widget dashboard.
Max Active VM (Stats Display)	Maximum number of active virtual machines to display.
Max Completed VM (Stats Display)	Maximum number of completed virtual machines to display.
Show Popular Labs	Enable this setting to add the **Show Popular Labs**widget in the dashboard.
Started Labs Only (Stats Display)	Activate or deactivate started labs.

Exams

Setting	Description
"Exam Result" email body	The text content of an email that is sent the trainee or student who is taking an exam.
"Exam Result" email title	The title of the email that is sent to the trainee or student who is taking an exam.
Time (ms) to keep the vAPP of an exam that failed grading	Specifies the time to keep the vAPP of an exam that failed the automatic grading.
Time (ms) to keep the vApp of a failed exam	Specifies the time to keep the vAPP of an exam that failed.

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Setting	Description
Time (ms) to keep the vApp of a successful exam	Specifies the time to keep the vAPP of a successful exam.

General Options

Setting	Description
Acquire Console Failure Notification	This option is activated by default. A notification is sent when a VM cannot connect to its console view. After 3 unsuccessful attempts, another attempt is made after 10 minutes - meaning that if a notification for a specific VM has already been sent, another notification will be sent after 10 minutes of the previous notification.
Allow uploading screenshot in lab	Lets a user to upload screenshot in a lab.
Self Registration	Lets a user to self register for a class. When enabled, a self registration option is available for users in the end-user interface to register for a class. Students can use the self registration link and sign up or login to the class with the configured SSO authentication method.
Allow Student SSO Login	Lets administrators to provide a non-local authentication provider to users to login to the end-user interface. If the **Allow Student SSO Login** option is activated, students/end users can log in to the VMware Lab Platform user interface using a SSO authentication provider configured by the account administrator. **NOTE** • To use only the SSO authentication provider option for a user login, activate both the **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options. • If both the **Allow Student SSO Login** and **Disable Student Local Provider Login** settings are deactivated, the default VMware Lab Platform login option is provided in the user interface login page. • To deactivate the direct login option, deactivate the **Allow Student SSO Login** option, and activate the Disable Student Local Provider Login setting option. For more information on how to add an authentication provider for a user, see Working with an Authentication Provider for Single Sign-on.
Default tenant group for anonymous users	Add/remove groups for an anonymous user.
Duration of inactivity after which an anonymous an account would be deleted	Specifies the inactivity time duration for an anonymous user after which the user is deleted from the system.
Enable audit history	Activate or deactivate the audit history feature.
Auto run compatibility checker when user starts the lab	If activated, when a user starts a lab, the system prompts the user to select whether to run a compatibility check. When run, the compatibility check shows and saves a report.
Compatibility Check Bandwidth/Latency	If activated, the Compatibility Check page shows student's bandwidth and latency for the student interface.

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Setting	Description
Compatibility Check Pop-ups	If activated, check Compatibility Check page to check if the student's browser allows pop-ups for the student interface. This check is useful when a manual or other content require a popup to function.
Compatibility Check Proxy	Activate or deactivate the Compatibility check proxy feature. On the Compatibility Check page, check if the student is connecting to the student interface behind a web proxy.
Compatibility Check Resolution	On the Compatibility check page, check student's current browser size and settings. This check is useful if your lab requires a certain amount of screen space to be used.
Compatibility Check Resolution Minimum Height	Set the minimum browser window size height that is required for passing the resolution compatibility check.
Compatibility Check Resolution Minimum Width	Set the minimum browser window size width that is required for passing the resolution compatibility check.
defaultAuthenticationProvider	Add your default authentication provider for user logins. By default, the **`LOCAL`** provider is selected.
Default Wizard Input Style	Sets the wizard page layout. You can choose the option from the drop-down menu.
Seconds to wait before deleting an expired vApp	Sets the amount of time the account keeps expired entitlements and vApps.
Delete prepops in error state after specified hours	Deletes a prepop in an error state based on the value set in the **Delete prepops in error state after specified** option. 24 hours is the default value. To deactivate this option, set the value as 0.
Delete non-prepop vApps in error state after specified hours	Deletes a non-prepop vAPP that is in error state based on the value set in the **Delete non-prepop vapps in error state after specified hours** option. 24 hours is the default value. To deactivate this option, set the value as 0.
Disable Student Local Provider Login	Lets administrators to deactivate an authentication provider option from the end-user interface login page. If the **Disable Student Local Provider Login** option is selected, the SSO authentication provider option in the end-user interface is deactivated (unavailable). A remote SSO authentication provider must be configured to allow students/end users to login to the end-user interface. **NOTE** • To use only the SSO authentication provider option for a user login, activate both the **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options. • If both the **Allow Student SSO Login** and **Disable Student Local Provider Login** settings are deactivated, the default VMware Lab Platform login option is provided in the end-user interface login page. • To deactivate the direct login option, deactivate the **Allow Student SSO Login** option, and activate the Disable Student Local Provider Login setting option. For more information on how to add an authentication provider for a user, see Working with an Authentication Provider for Single Sign-on.
Enable Anonymous User Lab access	Activate, or deactivate the create anonymous user feature in Create Entitlement page. Default is false.

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Setting	Description
Enable CSP SaaS Labs (Orgs and user creation in CSP)	Allows for creation of lab environments in VMware Cloud Services Provider (org and user). After the environment is ready, the lab loads a SaaS console of the VMware Cloud Services Provider environment.
Enable Granular Permission	Activates granular permission setting for roles and permissions.
Create cloud org VDCs and cloud OVDC catalogs with preshadow enabled	Preshadow is enabled by default when creating org vDCs or org vDC catalogs.
Enable Test Drive Integration	
Enable Tracking Code for Classes	When activated, the tracking ID feature option is added in the Classes wizard. You can track entitlements for classes using the tracking ID feature in the Classes wizard. **NOTE** You cannot modify the tracking code once a class is started.
End Entitlements when Class ends	If activated, ends the entitlement after the class ends.
Minimum time between notifications about slow callbacks	Sets the minimum time between VMware Lab Platform notifications about slow callbacks.
Intercom Identity Verification Secret	
Max classes result on "Add Entitlement to Class" modal	Specifies the maximum number of loaded classes in the “Add Entitlement to Class” modal on the Entitlements list page.
Send networking configuration error notifications on vApp network setup	Sends dashboard notification if there are any network configuration issues for a vApp in the VMware Cloud Director.
Days after which a user can re-rate a lab	Sets the rating period per account. During this period, an end user cannot rate a lab again. For example, if set to 10 days and an end user rates a lab, then the end user cannot rate the same lab again during the next 10 days.
Send notifications for unused running vApps after specified hours of inactivity	Sets the time value after which notifications are sent for unused running vApps. The default value is 0 and the option is deactivated. To enable the notification setting, set any value higher than 0.
Length of verification token and URLs	A numerical value (minimum is 10 and maximum is 100) that specifies the length of the short strings that the system uses. A short string is a randomly generated string the system uses in place of longer strings. The system uses a shortened string value for verification tokens and URLs when the long string is difficult to type or copy. A short string is much easier for the system to handle but must be unique within the application. The **Length of Shortened Strings** setting specifies the length of the generated short string and has a default value of 12. For accounts with a large user-base, you might have to increase the length of this string to ensure the uniqueness of the generated strings. The system does not strictly enforce the minimum and maximum values but values outside the valid range are moved to the closest limit when the value is used.

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Setting	Description
Minutes before verification tokens and URLs become invalid	Specifies the amount of time, in minutes, that the short token string is valid for. After this time, the system expires the shortened string and the long token string also cannot be used. The default value represents one day. You can increase the default value to give users more time to validate their accounts. The verification tokens and URLs are sent to users in account verification emails.
Enable New Lab Wizard	If activated, provides a simple and easy to use lab creation form that lists only the required fields for a lab. This option is useful when administrators want to create a quick lab. For advanced lab creation options, deactivate the **Enable New Lab Wizard** option.
User Idle Count Down	Time in hours, minutes, and seconds after which VMware Lab Platform pauses the end user's lab due to exceeding the countdown.
User Idle Delay	Time in hours, minutes, and seconds to delay pausing a lab due to exceeding the user idle countdown.
useSkuInQuickLaunchUrl	

Class and Instructor Options

Setting	Description
Allow instructors to clone vApps	Lets instructors to the clone vApps from the **Update class › Manage Entitlements** page.
Allow instructors to create and edit class	Activated by default. If deactivated, instructors are restricted from creating new classes.
Allow instructors to extend class end time	Lets instructors to extend the class end time. **NOTE** If the **Allow instructors to extend class end time** option is deactivated, the date range option in the Classes wizard is dimmed out for instructors in the instructor user interface. Instructors cannot modify or extend the class time once the class starts. However, administrators can modify the class time from the administrator user interface.
Allow instructors to manage class	Activated by default. Lets instructors to manage a class. Instructors manage the class start time, enrollment and, so on.
Allow instructors to reset lab from class	Activate the **Reset** control in the Instructor-Led Lab console. The **Reset** control resets the vApp.
Allow instructors access to all classes	If activated, an instructor can go to the Class page (click **Class Management › Classes**) and view all the classes. If deactivated, then only classes to which the instructor is assigned are listed.
Enable broadcast display	If activated, hides the Broadcast button from the Instructor URL for a class.
Send student class assigned email from email template	To send students an email when enrolled for a class, enable this option. You can create custom email templates using the **Email Templates** menu option.

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Setting	Description
Add support email in cc in class emails	By default this field is empty. To send custom emails for a class, add the email address, such as support team email. You can add multiple email IDs. Ensure that the email IDs are separated by using commas. For example, username1@test.com, username2@test.com.
Send instructor class assigned email from email template	To send instructors an email when assigned for a class, enable this option. You can create custom email templates using the **Email Templates** menu option.
Send student class started email from email template	If activated, instructors can send class started emails to students using the email template in the **Classes** menu.
Create entitlements of class with quick launch link	If activated, when instructors create an entitlement for a class, it is created with the quick launch link to the lab.
Make time allotted for entitlement equal to class duration and hide remaining lab time	If activated, the entitlement duration is equal to the class duration, and the remaining time and extend time buttons are hidden for class labs.
Allow instructors to see and bulk upload users to tenant	Lets instructors to access the Account page **Security › User Controls**. The page includes **Account Bulk Upload**, **Account Bulk Remove**, and **Account Bulk Download** settings. To add users to the administrator with a CSV file, instructors can use the **Class Bulk Upload** function. **NOTE** Instructors can use **Class Bulk Upload** to specify only user name, first name, last name, and company. The system ignores all other columns in the CSV file. When an instructor does a bulk upload users action, the system creates those users with the administrator user role and adds them to the All Users group.
Instructor can connect to inactive console	Activates or deactivates an instructor's ability to connect to an inactive console.
Send an email to instructor when assigned to class	When an instructor is assigned to a class, email is sent to the instructor.
Send an email to instructor on class creation	Sends email to the instructor when the class is created. The contents of the mail can be customized by editing the **Instructor Class Created Email** template in the Email section of the Settings page.
Send notifications for unused resources in classes after a specified hours after class start	If activated, sends notifications to the instructor on unused resources like powered on vApps. Select a numeric value to enable the notification. By default, 0 is selected. 0 means no notification is sent. 2 means, a notification is sent after two hours of a class start.
Send an email to student when assigned to class	If activated, sends email notification to students when assigned to a class. You can customize the email using the **Student assigned to class Email** custom email templates.
Send an email to student email on class start	If activated, sends email notification to students when a class is about to start. You can customize the email using the **Student's class start Email** custom email templates.
Show all VMs in instructor interface	If activated, displays all the student VMs to the instructor.
Show preferred cloud location dropdown in the classes wizard	If activated, displays the cloud location option in the Classes page.

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Setting	Description
Specify the maximum refresh frequency for student console thumbnails in the instructor view (in milliseconds)	
Specify the minimum refresh frequency for student console thumbnails in the instructor view (in milliseconds)	
Add a warning if there are enough resources in cloud on class creation	If activated, the system displays a warning on the class creation page if enough resources are available for the class. This feature helps administrators to make informed decision on resource allocation.

Interface

Setting	Description
Allow Anonymous Catalog/Lab Browsing	If activated, allows anonymous browsing. Anonymous browsing means that users can browse the catalog even if they are not logged in to VMware Lab Platform. As a result, end users can view what labs are available before registering with a site (otherwise, only registered users can view what labs are available).
Allow raw CSS styles in the manual content	Activate or deactivate allowing raw CSS styles in a lab manual.
Announcements	Activate or deactivate the display of lab console announcements.
Anonymous Language	Select a language for anonymous users from the pull-down menu. When an anonymous user opens the lab console, the default language is the language you selected.
Auto Logout from Student Interface Timeout (in milliseconds)	Automatically logs out a student when the student interface is idle for the time limit set in this option.
Enroll and Start lab with one click	Immediately starts a lab after enrollment, redirecting the user to the console screen.
Default Active Tab	Selects the default active tab in the user interface, such as **Labs**, **My Enrollments**, **Completed Labs**, and **Favorites**.
Disable End button until Complete	Deactivates the End button on the user console until after the time specified by the **Minimum Time Spent for lab to be completed** and **Percentage of steps visited for the lab to be completed** settings at **Lab Management › Labs**.
Enable Cloud Media upload in the student interface	When activated, it allows lab admins to insert and eject media files to the VM console from the student user interface.
Show an overlay displaying the console dimensions when resizing it	
Average Deployment Time History in days	Specifies the time frame to consider while calculating the average time for deploying a lab. Enter the value as number of days to calculate the average deployment time.
Show EULA (and customize)	Activates or deactivates the display of Service Terms. Service Terms are rules to which a person taking a lab must consent. When activated, Service Terms are displayed when a user first takes a lab. If the lab

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Setting	Description
	user consents, then the Service Terms do not appear again for that user. Click the **Customize** button to open the Service Terms List page, which lists existing Service Terms for the account. You can use the page to: • Activate or deactivate a particular Service Terms. • Edit an existing Service Terms. • Create Service Terms (click **Create Service Terms**).
Show full lab details by default in the catalog	Activate or deactivate the display of full lab details through Catalog List.
Display VM Thumbnails	Display of the thumbnail associated on the end user lab console in the middle tab.
Default Help Request Location	Sets location of the student help request, either **Nested in Help Menu** or **Top Level**.
Hide forgot password button	When activated, the **Forgot Password** option is not available (hidden) for users from the VMware Lab Platform login page.
Hide Instructions panel when no manual	If activated, hides instruction panel when no manual is added for a class.
Instructions Panel Location	Sets default location (left or right side) of the Manual button on the instructor-led lab console.
User Interface Customization	
Lab End Page Timeout (in milliseconds)	Time in hours, minutes, and seconds. After a user completes a lab, VMware Lab Platform logs out a user having zero credits within the timeout specified. (Typically, a user with zero credits is at an event)
Allow users to unenroll themselves from a lab	Lets users enrolled in a lab to cancel their enrollment.
Offers	Activate or deactivate the display of lab console offers.
Open 'Notes' Pop-up On Lab Start	If activated, then once an end user starts any lab within this account, the lab console displays metadata (notes, user name, and password for each VM) for the lab.
Save the anonymous compatibility tests results	Save compatibility tests results for users who are logged as anonymous users. Results are located under **Reports › Compatibility Tests**.
Lab Rating Enabled	Activate or deactivate the lab rating feature on the lab console.
Show Catalog	Activate or deactivate the display of lab catalogs on the Show Catalog page.
Allow users to see a transcript certificate	Activate or deactivate users to view lab transcripts by clicking an entitlement key in the Transcripts List.
Show loading tips at lab start	
Show the Dark Theme toggle in Student UI	
Show Default Welcome Tips	If activated, then the lab console displays the account's default tips and displays any tips created by the admin using the **Content › Tips › Crea te Tip** setting. If deactivated, then the lab console displays only the tips created by the admin but not the default tips.

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Setting	Description
Show estimated deploy time at lab start	If activated, then when an end user begins a lab, the system displays an estimated time required for the lab startup. If not activated, then the system does not display an estimated time required to start the lab. **NOTE** The **Lab Templates** list includes the **Reset Average Deploy Time Statistics** setting. The setting clears the statistics for the selected template to restart the calculation for the average deploy time.
Show invite a friend links	Activate or deactivate display of the link on the lab console. VMware Lab Platform displays the link the right tab of the lab console within each listed lab.
Show a badge on catalog lab entries for "Available Now" or "Deployed on Demand" labs	If activated, this a badge, which shows if the lab has prepops or not. "Available Now" means, the lab will start much quicker that the deploy on demand.
Show the warning prompt when enrolling in a lab if prepops are not available	If activated, a warning is displayed that there are no prepops for the lab and it will take longer time to deploy.
Show "Lab is Provisioning" loading screen	If activated, and there are no prepopulated lab templates provisioned for a lab, then the Lab is Provisioning message is displayed on the user screen.
Show Split Screen/Remote Control Link	Activate or deactivate display of link. VMware Lab Platform supports mobile devices for the split screen functionality such that the lab's instructional manual appears on the mobile device while the lab console appears on a desktop or laptop.
Show UI Walkthrough	Activate or deactivate the lab console walk-through tour. When an end user starts a lab for the first time, the system automatically starts a guided console walk-through tour.
Show Welcome Tips	Activate or deactivate the Welcome Tips message for labs in lab console.
Student UI Custom Script	
Table of Contents Location	Customizes the location ("Nested in More Options Menu" or "Top Level") at which the lab console displays a lab manual Table of Contents option.
User UI URL	URL for the user interface (for example, `https://www.labplatform.vmware.com/<org_name>`). Here the org name is the name of your organization registered with VMware Lab Platform.
Video Conference Panel Location	Customizes the video conference panel location.
VM Listing Panel Location	Sets default location (left or right side) of the Console button on the instructor-led lab console.

LTI Settings

Setting	Description
LTI Consumer Key	Consumer key that you can use for LTI setup in your LMS tool.

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Setting	Description
LTI Shared Secret	Shared secret that you can use for LTI setup in your LMS tool.

Lab Builder

Setting	Description
Automatic Deployment pool creation	If activated, default pools are created for new templates created in the **Lab Builder**. Enable this setting if you want to create labs directly from the **Lab Builder**.
Auto Update Lab Builder Content	Automatically synchronize content from the cloud (vCD org) and keep the content up-to-date in VMware Lab Platform.
Default catalog	Specifies which catalog is selected initially when you open the **Lab Builder** page
Deactivate default network creation	Activate or deactivate this setting.
Enable file upload in lab builder editor	If activated, lets you to upload resource files in the **Lab Builder**.
Notification for powered on lab builder items inactive for more than specified hours	If activated, sends notifications when a powered on lab Builder items are inactive for more that the specified hours. Select a numeric value to enable the notification. By default, 0 is selected. 0 means no notification is sent.
Suspend lab builder items after specified hours of inactivity	Suspends Lab Builder items if the items remain inactive for more than the hours specified in the **Notification for powered on lab builder items inactive for more than specified hours** option.
Max number of CPUs per VM	Controls the maximum number of allowed CPUs per VM for VMs created through Lab Builder.
Max memory (MB) per VM	Controls the maximum allowed memory per VM for VMs created through Lab Builder.
Max storage (MB) per VM	Controls the maximum allowed storage per virtual machine for virtual machines created through Lab Builder.
Enable Lab Builder network settings autocorrect	
Comma separated list of public catalogs that should be visible in Lab Builder	A comma-separated list of public catalogs that must be visible in Lab Builder.
Display Lab builder resources in GB	Enable to display resources in Lab Builder in GBs. Deactivate to display resources in MBs.
Resources in GB significant figures	Sets how many decimal places are displayed when resource information is displayed in gigabytes in the Lab Builder. The default is one decimal place.
Use tasks for lab builder template operations	Enable the **Use tasks for lab builder template operations** setting to view Lab Builder tasks such as uploading files, or VMs in the Dashboard.
Remove Org network connections on replication	
Show a WiFi icon in Lab Builder to indicate VM internet Status	By default this option is enabled. The **Show a WiFi icon in Lab Builder to indicate VM internet Status** setting adds the WiFi visual icon to lab templates when you edit a lab template.

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Setting	Description
Use edge gateway for NSX-T backed vCD	When enabled, VMware Lab Platform creates additional VMs serving as a gateway to handle NSX-T firewall, NAT, and static routing feature.
Enable Vm script run	When enabled, VMware Lab Platform turns on the VM script run in Lab Builder.

Maintenance

Setting	Description
Maintenance Mode Message	Text message that VMware Lab Platformdisplays when an account is in maintenance mode.
Maintenance Mode Enabled	Activate or deactivate to put an account in maintenance mode.

Side Menu

Control the menu options that are displayed in the left-navigation pane and in the **Labs** using the **Portal Settings › Confi guration › Side Menu** options. When a **Side Menu** option is activated, it is hidden from the navigation pane and also from the **Create Lab** and **Update Lab** options in the **Labs** window.

Setting	Description
Events Menu Enabled	When activated, the **Events** menu is activated and can be accessed from the navigation panel.
Exams Menu Enabled	Activate or deactivate. When activated, activates the Exam Management menu option. This option is deactivated by default.
Show Analytics page	By default this setting is deactivated. Activate or deactivate. When activated, the **Analytics** menu can be accessed from the navigation pane.
Show API Domain Allowlist page	By default this setting is deactivated. Activate or deactivate. When activated, the **Allowed API Domains** menu is displayed in the side-navigation pane.
Show Badges page	Activate or deactivate. When activated, hides the **Badges** menu from the navigation pane.
Show Entitlement Callbacks page	Activate or deactivate. When activated, the **Entitlement Callbacks** menu is displayed in the side-navigation pane.
Show Help Desk Integrations page	By default this is deactivated. Activate or deactivate. When activated, shows the **Help Desk Integrations** menu in the navigation pane.
Show Lab Ratings page	Activate or deactivate. When activated, the Show Lab Ratings option is displayed for a lab once a user completes a lab in the user interface.
Show LTI Integrations page	Activate or deactivate. When activated, shows the **LTI Integrations** menu in the navigation pane.

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Setting	Description
	NOTE The **LTI Integrations** menu is not hidden in the **Labs** window.
Show Network topologies page	Activate or deactivate. When activated, shows the **Network Topologies** menu in the navigation pane.
Show Page Contents page	Activate or deactivate. When activated, shows the **Page Content** menu in the navigation pane.
Show Products page	Activate or deactivate. When activated, shows the **Products** menu in the navigation pane.
Show Service Terms page	Activate or deactivate. When activated, shows the **Service Terms** menu in the navigation pane.
Show Surveys page	Activate or deactivate. When activated, shows the **Survey Configuration** menu in the navigation pane.
Show Tags page	Activate or deactivate. When activated, the **Tags** menu is displayed in the navigation pane.
Show Thumbnails page	Activate or deactivate. When activated, the **Thumbnails** menu is displayed in the navigation pane.
Show Tips page	Activate or deactivate. When activated, the **Tips** menu is displayed in the navigation pane.
Get Support (Instructor)	Add a site or email address in the Get Support (Instructor) field to get personalized support site. If left empty, all support requests are redirected to the default VMware support site. You can also modify the support link for your VMware Lab Platform organization administrators by contacting the VMware Lab Platform support team.

User Hard Cap

Setting	Description
Limit the number of concurrent users	Activate or deactivate a cap on the number of concurrent users per account. **NOTE** The value limits the number of concurrent users per account who can take labs. While many users can be logged into an account, the value does limit the number of the users actually taking a lab.
Alert threshold for active/committed	Sets the threshold for VMware Lab Platform to send a warning to alert users when the maximum setting is being approached.
Committed users per month	Maximum concurrent users per month (for example, a commitment of a maximum 10 concurrent users per month). This maximum does not actually limit the number of users per month. To limit the concurrent

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Setting	Description
	users (in the example, preventing the 11th user from taking a lab), you have to enable the “Limit the number of Concurrent Users” setting. **NOTE** **Committed users per month** are not related to specific labs. Instead, the maximum regards concurrent users in your account. Users can take any lab more than the maximum if they do not take the labs at the same time.

Email

Administrators can send custom emails to users, such as “welcome” emails or “password change” emails and notify the users. To view the templates for custom messages, click **Portal Settings › Configurations › Email**.

To modify the default formats, click **Click Here to Customize Your Email** on the templates form. You can include any of the predefined tokens that are displayed. The Email page also includes the **Return Address** text box. Use this text box to update the return address for all emails sent on behalf of your account.

The following email templates are included:

Emails	Description
Email template for Add Users	Email template when a user from an existing VMware Lab Platform account is added to a new organization account. Ensure that you select the Send Welcome/Verify Email option in the Create Account page to send this email to the new users. For more information on how to create a user account and send a welcome email message, see Create a user account.
Change Password Email	Email template to notify users when the account password is changed.
Reset Password Email	Email template to notify users with URL link to reset their password.
Email template for security question reset	Email template to notify users when the security questions associated with the account authorization is reset.
Welcome Email	Email template to welcome new users on board to VMware Lab Platform. For more information on how to create a user account and send a welcome email message, see Create a user account.
Instructor Class Created Email	Email template to notify instructors of a class when a class is created. Ensure that the “Send an Email to Instructor when assigned to class” Setting option is activated for this email type.
Entitlement Quick Link Email	Email template to provide a quick launch URL for an entitlement that a user is enrolled for.
Student assigned to class Email	Email template to notify students when they are assigned to a class.
Student's class start Email	Email template to notify students when a class that they are enrolled for is started.
Exam result Email	Email template to notify a user of an exam result for a lab.
Class notification Email	Email template to notify a user when a class is about to start.

Edit portal settings in VMware Lab Platform

You can modify settings from the **Portal Settings** page on the administrator user interface.

For more information about portal settings, see Working with settings in VMware Lab Platform.

1. Select **Portal Settings › Configurations**. Administrative settings are listed by the category.
2. If needed, use the **Search** text box to locate a particular setting.
3. Highlight the setting you want to edit. Depending on the setting, you can toggle the X and check mark to activate or deactivate a setting, edit the value for the setting inline, or open a pop-up editor for further work.

   Your changes are saved automatically.

Monitoring console connections in VMware Lab Platform

Administrators, instructors, and lab users can monitor console connections to track the health of the connection for an active lab.

Lab users and instructors can view the following information in the user console:

- **Connection strength**. The student user console has a graph icon that displays connection strength.
- **Console latency**. Hover over the **Connection Strength** icon to view information about console latency.
- Saved console benchmarks and transcripts. Saved console benchmarks for active entitlements and transcripts are displayed in the **Console Benchmarks** column.
- Average, high, and low latency values per session. You can display this information as a timeline when you use the menu opened with the item's horizontal dots icon (**…**).
- The reason for leaving each lab session.

Enable console connection monitoring in VMware Lab Platform

Enabling console connection monitoring for a lab is a two-step process. First, you must activate the feature for the organization. After the option is activated for the organization, you activate console connection monitoring for organizations.

You must have administrator permissions.

Every lab that runs in a cloud org where console monitoring is activated inherently monitors console connections to lab templates. This is true for all labs that run in that cloud.

1. Enable console monitoring for your account.
   a) From the navigation panel, select **Portal Settings › Configuration**.
   b) To locate the **Console Connection Monitoring** option in the Console Interface section, use the search bar on the Configuration page.
   c) Verify that a green check mark icon is shown in the Value column. You can toggle the icon by clicking on it.
2. Configure the cloud organizations whose labs you want to use console connection monitoring.
   a) To display a list of cloud organizations, from the navigation panel, select **Cloud Management › Clouds**.
   b) From the list, click the cloud organization where the labs that you want to use console connection monitoring are deployed.
   The Update cloud page opens.
   c) Select **Console Connection Monitoring**, and click **Update cloud**.

View console connection in VMware Lab Platform

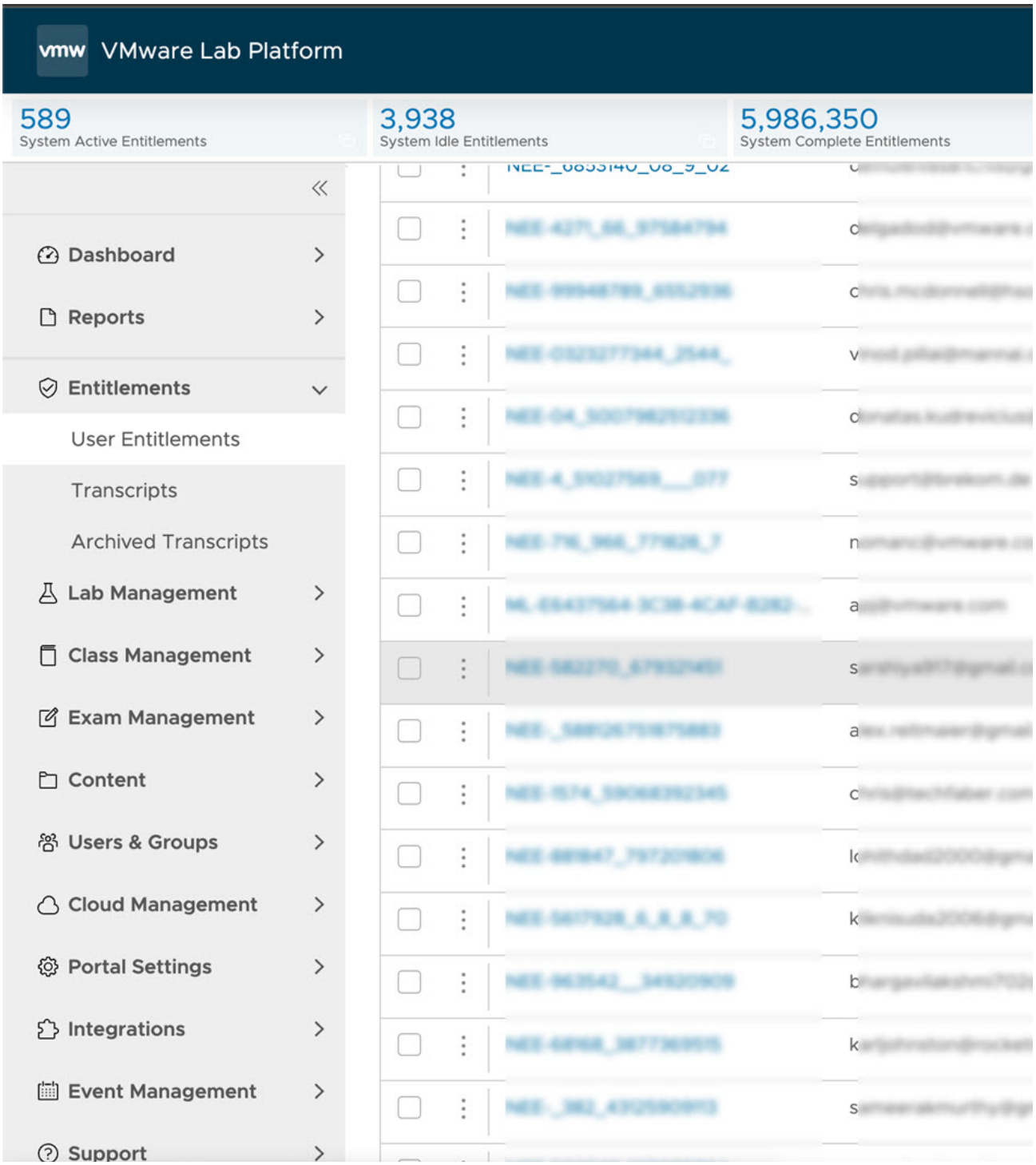
Administrators, instructors, and end users can view the console connection strength of an active lab from their respective interfaces. Administrators must enable the console connection settings for an active lab in the administrator user interface.

For more information on how to enable the console connection setting, see Enable Console Connection Monitoring.

Administrator user interface

To view the console connection strength of an active lab:

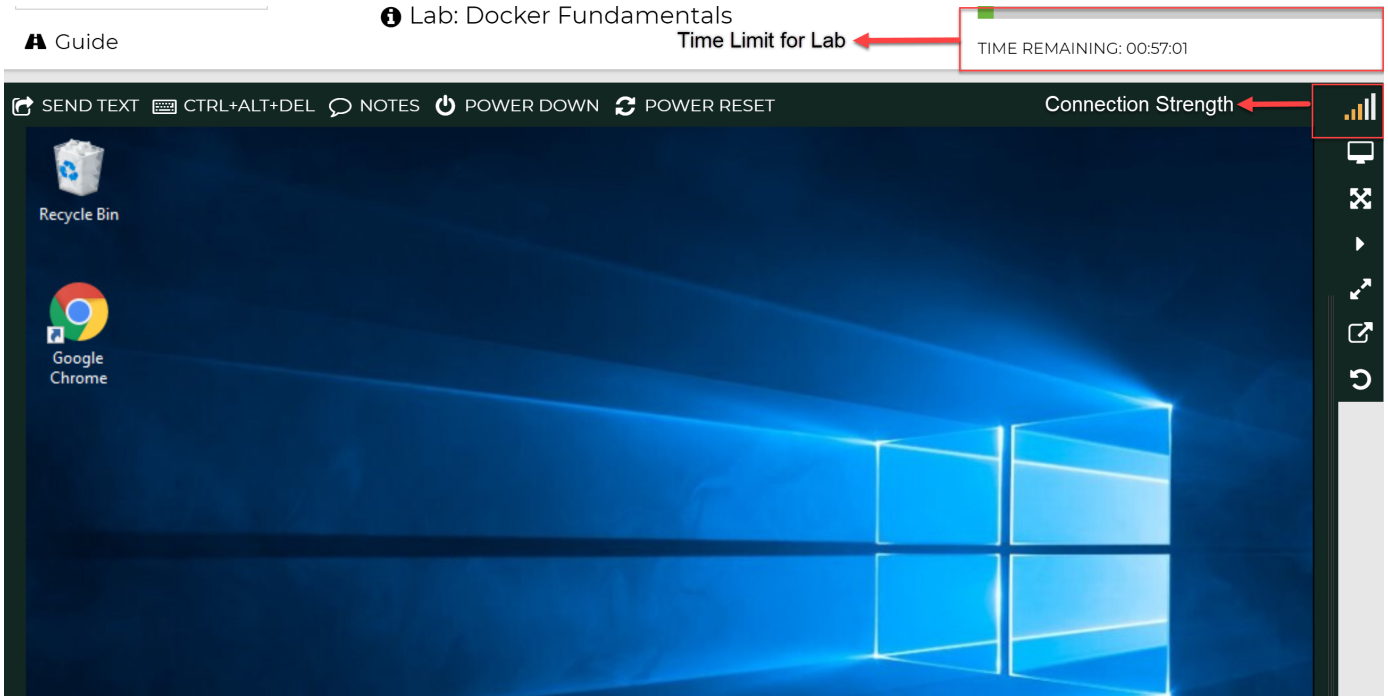
1. Enable Console Connection Monitoring.
2. Navigate to **Entitlements › User Entitlements**.
3. On the Entitlements page, view the console strength under the Console Benchmarks column for an entitlement.



End-user interface

The end-user interface goes through a few levels of connection checks and warning messages:

- A connection degraded message is shown if calculated latency is consistently high.
- A connection unstable message is shown if there is a drop in connection.
- A connection disconnected message is shown and the console connection attempts to be refreshed if there is no communication.



Instructor user interface

The instructor user interface has a console connection strength indicator showing student's latency values and other console connection indicators in addition to the phase changes such as when a student starts a lab, ends a lab, or exits. This connection strength indicator is displayed when the instructor is viewing the student's console and in the Class Overview.

Working with VMware Lab Platform speed test

You can run a speed test to measure the performance of connections from your browser to VMware Lab Platform.

The VMware Lab Platform speed test measures available network bandwidth and network latency for labs.

Any user with a VMware Lab Platform account can run a speed test. Students and other end users can run speed tests to determine whether there is an issue with their network connection for a lab. Administrators and instructors might use the speed test to evaluate potential performance issues reported by students and others taking a lab.

Speed tests measure the network bandwidth and network latency for VMware Lab Platform and your associated cloud.

Run a speed test

To run a speed test, open a browser with a URL of the form https://admin.labplatform.vmware.com/http://labplatform.vmware.com/*tenant_name*/speed-test/ where *tenant_name* is the unique name given to your account and click **Start Test**.



View speed test results

After you run the test, the speed test screen displays results for VMware Lab Platform and cloud speed for all the clouds, and for VMware Lab Platform and cloud network latency. Network latency results are color coded as follows:

- Results less than 300 ms are shown in green
- Results greater than 300 and less than 450 ms are shown in amber
- Results greater than 450 ms are shown in red

A good speed score is 1.5 MB/s or greater, and a good latency score is 300 ms or less.

Styling accounts in VMware Lab Platform

As an administrator, you can modify the design elements of VMware Lab Platform user interface with branding or styling options that specify elements such as colors, fonts, or logos. You can also create custom CSS files.

You can add or change style elements for the end-user interface, administrator UI, or event seating tickets (Event Management UI).

NOTE

- Use the **Portal Settings › Branding** styling options to make branding changes for the old user interface.
- Use the **Style** option in the new user interface to make any branding changes in the new user interface. For more information, see Add custom style for end-user interface.

Add custom style for old end-user interface

To open the Styling page, select **Portal Settings › Branding**.

View styling changes in the Preview tab

Click on **Preview** to see the styling changes as you make them.

Add color options

Click one of the color options to open a palette from which you can select a color:

- **Top Header Color**: Styles the header area color.
- **Top Header Accent Color**: Styles the header accent color.
- **Side Navigation Background Color**: Styles the side navigation background.
- **Primary Button Color**: Styles the primary button color.

Add a logo

To upload a logo, click **Upload a File** within the Logo field. You can also drag a logo into the **Load Logo File** box. The logo you load must conform to an aspect ratio of 188 width/44 height.

Add a seating ticket logo

To upload a seating ticket logo, click **Upload a File** within the Ticket Logo field. A ticket logo refers to the logo on a seating ticket. The Event Manager UI lets an administrator to print seating tickets.

Save your styling changes

Even though you see your changes as you make them on the **Preview** tab, you must click the **Save** button to actually save your styling changes. If you do not save your changes, then your changes are lost when you log out.

When you click **Save**, current users see the impact of your styling changes when they next refresh their screen.

Access advanced interface styling options

To access the advanced styling options for the user UI, click **Advanced Options**:

- **Body Background Color**: Styles the user UI background color.
- **Section Header Color**: Styles the user UI section header color.
- **Catalog Monitor Color**: Styles the user UI section monitor color.

Customize a CSS file

Cascading Style Sheets (CSS) files lets you customize the look and formatting of any document written in a markup language.

Customization made to a CSS file are added after any of the color options you use.

To access CSS options, click **Custom CSS**. You can use the following options to customize the CSS file:

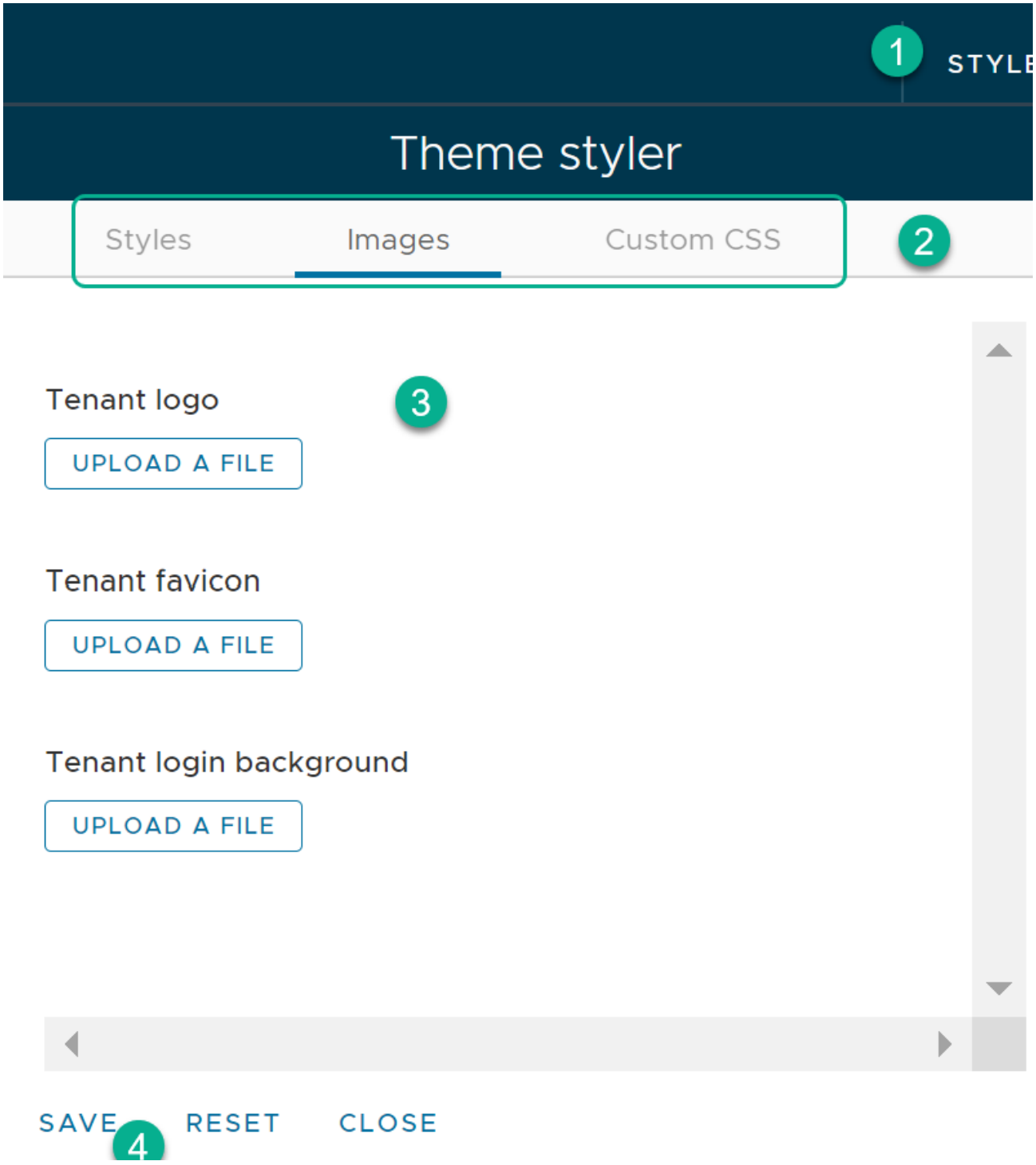
- **Edit UserUI CSS**: Opens and editing tool for the user UI CSS file.
- **Load a CSS File**: Lets you to load a CSS file for the user UI. (You can also click/drag a CSS file into the Load box.)
- **Edit AdminUI CSS**: Opens an edition tool for the administrator UI CSS file.
- **Load CSS File**: Lets you to load a CSS for the administrator UI (You can also click or drag a CSS file into the Load box.)

Add custom style for new end-user interface

As an administrator, you can personalize and tailor the VMware Lab Platform user interface login page according to your brand or organization's identity for your users. You can change the header color, page banner, add company logo, company icon, add page background, and interface buttons, using the **Theme** styler option. You also use the custom CSS script to personalize the UI page.

To customize your user login page:

1. Enter the end-user interface URL (https://www.labplatform.vmware.com/org_name) in a browser.
2. Log in as an administrator.
3. Click **Style** and use the options under the **Theme** styler.



4. To save your changes, click **Save**.

You can view your changes in the end-user interface.

Working with entitlement callbacks in VMware Lab Platform

An entitlement callback sends data about a just-completed lab to a contact or an administrator. You can select the type of event data that you want to send back.

For example, you can send data about lab start times, end times, or enrollment times of a completed entitlement. The Entitlement Callback list provides options for more targeted follow-up calls to end users from the organization conducting the lab. Once you have configured a callback, you must associate it with a lab using the **Update Settings** tab on the Create Lab form.

Understanding entitlement callback triggers

To better understand the life cycle of events and their respective boundaries, note the following guidelines:

- Every entitlement starts with an enroll event.
- Every entitlement that an end user actually uses, has at least one start event.
- Every entitlement ends with at least one of the following events: end, expired, or canceled.
- If an entitlement is being graded, the graded event occurs after the end event.

Create an entitlement callback in VMware Lab Platform

You can create an entitlement callback to send data about a completed lab.

1. Click **Integrations › Entitlement Callbacks**. The Entitlement Callback list opens.
2. To open the Create Entitlement Callback form, on the Entitlement Callback page, click **Create New** .
3. Enter the following information on the Create form:
   a) To activate or deactivate the callback you are creating, select or clear the **Enabled** check box.
   b) To apply the callback for all labs in the account, select the **Global** check box.
   c) **Name**: Enter a name for the callback (for example, TestEntitlementCallback1).
   d) **Type**: Select a value from the drop-down menu to determine how the system sends the lab data. Note that the callback type cannot be changed after the entitlement callback is created.

URL	If you select **URL**, then you must enter a web address, authentication method, username/header, and a password/token. Selecting **URL** results in the lab data being posted to the URL you have provided. A service can use that data in some manner (for example, to create statistics). **NOTE** Token authentication depends upon your server and its requirements. What you enter in the Username/Header field becomes the header KEY, and what you enter in the Password/Token value becomes the VALUE. The result is customized according to your web endpoint. For example, if you were posting back to VMware Lab Platform, then the header would be neeToken and the token would be a long string key with the encoded user name with other such data.
Email	If you select **Email**, then you must enter an email address and email subject in the **Subject** field.
Saba	Saba is a Learning Management System (LMS) that you can configure for VMware Lab Platform. If you select **Saba**, then you must enter values for the Client ID and Client Secret text boxes. You can find both those values while configuring Saba.

4. To receive the entitlement callbacks, enter an email address for it.
5. Click **Next**. The system creates your entitlement callback and opens the Update Data form.
6. On the Update Data form, enter the following information:
   a) For **Data Type**, select JSON or CSV (comma-separated values) as the format for data returned by the callback.

b) For **Date Format**, select the date format to use for dates included in the data returned by the callback.
c) For **Fields**, select the date format and the fields that the return data must include:

firstName	First name of the student who took the lab.
lastName	Last name of the student who took the lab.
displayName	Name as it appears in the lab console.
language	Language with which the student took the lab.
location	Location at which the lab was launched.
lastStart	Date on which the student started the lab.
timeAlloted	Amount of time a student is allotted to finish the lab.
timeElapsed	Amount of time that has elapsed since the student started the lab.
expirationDate	Date on which `timeAllotted` expires.
completionDateTime	Date on which the student completed the lab.
completionStatus	Status regarding whether the student completed the lab.
ipAddress	IP address used by the student that took the lab.
returnUrl	Return method URL.
extensionCount	Number of times the time limit has been extended. This value refers to the time allotted for the lab. **NOTE** The account's configuration determines whether users can extend the time limit themselves, and how many times the users can extend the time.
Primary Bookmark	Value for the primary bookmark for an entitlement. This bookmark is for the current page, the page to which the system takes the user upon starting the lab - so users do not have to remember where they last were in the manual. **NOTE** VMware La Platform automatically adds and manages bookmarks having a name starting with the characters "vlp". The bookmarks are not always present.
maxStepNo	Maximum step number that the user viewed in the lab manual. Example: If a manual has 50 steps and the highest step number the user viewed was 40, then the number 40 is returned.
entitlementMetadata	Metadata that results from adding custom metadata to the entitlement. You can add custom metadata by including it as a text string within a POST Entitlement API call, or by entering text in the **Metadata** text box of the Update Entitlement form (click **Entitlement>User Entitlements** to open the Entitlements List). For example, you can include the following metadata text string to an API call: `{'metadata': This is an example of metadata'}`. In this example, the system adds the metadata. This is an example of metadata to the entitlement.

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labMetadata	Metadata entered in the Callback Metadata text box of the Lab form. This metadata is included in the returned email or URL. The following is an example of a formatted JSON: `{ "entitlementKey": "NEE-34505_7095222167", "account": "jsmith@vmwarelearningplatform.com", "sku": "EDU-ICM-1400-HOL", "tenant": "HOL", "eventTime": "06-05-2015 12:46:38 UTC", "event": "end", "firstName": "John", "lastName": "Smith", "displayName": "John Smith", "language": "en", "location": "NA-WEST", "lastStart": "01-07-2016 12:36:34 UTC", "timeAllotted": 120, "timeElapsed": 15, "expirationDate": "01-07-2016 12:40:55 UTC", "completionDateTime": "01-07-2016 12:46:36 UTC", "completionStatus": "expired", "ipAddress": "Unknown", "returnUrl": "/", "extensionCount": 0, "laststepcompleted": 1, "metadata": "Test Callback Metadata", }`
trackingCode	Used to track the link by which an end user found a lab.
vAppinfo	Includes the information on vApp's deployed for the lab.
seatLocation	Includes the student's seat location in a class, when applicable.
requiredStepsVisitedPercentage	Includes the percentage of steps that must be completed for the lab.
requiredMinimumTimeSpent	Includes the time the student spent on the lab.
manualStepCount	Includes the number of steps completed by the student.
manualStepsVisitedCount	Includes the number of steps visited by the student.
completionStatusForLabCriteria	States the criteria for lab completion.
Entitlement Location	Includes the geographical location for the entitlement.
saasConsole	The attached SaaS console(s) to the lab. There are a couple of lab types: 1. Normal labs that have a deployment pool and are deployed as a lab template in vCD (vCloud Director). 2. SaaS labs that have SaaS consoles attached to them instead of deployment pools. For example, in VMware Lab Platform, you can create a new org account and a test vCD org, all of this through some APIs, therefore it is a SaaS lab with a SaaS console attached.
labName	Displays the name of the lab that is being/was taken.
companyName	Displays user's company name. All account are associated with a company name.

7. After you have completed the Update Data form, click **Next**. The Update Event Triggers form opens.
8. In the Update Event Triggers tab, select the events for which the system must send data back.

enrollment	System sends the data when the end user enrolls into a lab. Example:
	`entitlementKey, account, eventTime, event`
	`NEE-929289_82627133_, jdoe@xyz.com, 01-13-2015 22:15:50 UTC, enroll`

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	NOTE The enrollment event is not available for Saba Entitlement callbacks.
ready	System sends the data when a lab has been made ready, which means that a deployed vApp is running in the cloud and is associated with the entitlement even though the entitlement has not yet started. Typically, a vApp is not associated with an entitlement until the entitlement has started. An entitlement can be made ready by an administrator or by an instructor, but an end user cannot make an entitlement ready.
start	System sends the data when the end user resumes a lab. Example: `entitlementKey,account,eventTime,event NEE-929289_82627133_,jdoe@xyz.com,01-13-2015 23:13:30 UTC,resume`
resume	System sends data when an end user resumes a lab. Example: `entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 01-13-2015 23:13:30 UTC, resume`
gone	System sends the data when the end user's lab session expires due to inactivity. This event occurs when the end user did not explicitly end/exit the lab. Instead, the lab was inactive for a prolonged period and the system marked the inactivity as a gone entitlement. Example: `entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 01-13-2015 23:50:30 UTC, gone`
incomplete	System sends the data when the lab is not currently active but has not been ended. Typically, this occurs when an end user has exited the lab or has closed the browser.
expired	System sends the data when the end user reaches lab's expiration time. Example:`entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 01-13-2015 22:40:54 UTC, expire`
complete	System sends the data when the end user ends a lab. This event does not trigger a callback if the lab was canceled. Example: `entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 01-11-2015 22:39:38 UTC, canceled`
canceled	System sends the data when the end user cancels a lab. Example: `entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 01-13-2015 23:19:35 UTC, canceled`
awaitingGrade	System sends data if the lab is in the awaiting grading status. The awaiting grading status is an interim status between an entitlement being started and ended. Example: `entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 02-16-2016 13:44:27 UTC, awaiting grade`
graded	System sends the data if the lab has the **Graded** option activated (Update Settings on the Create lab form). Example: entitlementKey, account, eventTime, event NEE-929289_82627133_, jdoe@xyz.com, 02-16-2016 13:45:28 UTC, graded

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consoleReady	System sends the data when a console is ready for use.

9. After you are done selecting callback event types, click **Done**.
10. Click **Finish**. The system creates your entitlement callback.

After you configure a callback, you can preview the callback data, or test it to see if your callback successfully processes. You can find the **Preview** and **Test** command on the menu you see when you click the three dots next to a callback name.

When you are satisfied with your work, associate the entitlement callback to a lab using **Lab Configuration** on the Create lab form.

Working with auto-disabled callbacks in VMware Lab Platform

To detect the calls that exceed a completion time limit, VMware Lab Platform monitors callbacks. Once a callback exceeds the time limit, the system deactivates it and sends a notification to the dashboard to warn you (the administrator).

Once a callback is deactivated, you can:

- To resend the callback, re-enable the callback using the Entitlement Callback form .
- To retrieve the callback data, contact the VMware Lab Platform support team. Even though the system never sent the callback, the data is retained in the VMware Lab Platform database for a period. The support team can retrieve it for you.

Working with the API domain list in VMware Lab Platform

Open API call returns can potentially be susceptible to cross-site request forgery (CSRF) attacks. To manage this security risk, the administrator UI includes the API Domain List.

You can add a domain to the list for an account integration that points to VMware Lab Platform and that runs within a browser. If your account uses a hosted tool on a web server that makes API calls to VMware Lab Platform, then you must add the domain to the list. Also, if your account includes a vanity domain for a user interface, then you must add that domain into the list.

In the following example, VMware Hands-On-Labs uses the following dashboard:

```
http://web.hol.vmware.com/navigator/web.hol.vmware.com
```

This domain makes API calls to VMware Lab Platform and must be added to the domain list.

Add a domain to the Allowed API Domain List

You can add a domain to a list of allowed domains so that the VMware Lab PlatformAPI Domain circumvents susceptibility to Cross-site request forgery (CSFR) attacks.

You must have administrator permissions for VMware Lab Platform.

1. Click **Integrations › Allowed API Domains**.
   The **API Domain List** opens.
2. To open the **Create Allowed API Domain** form, click **Create New**.
3. Enter a domain in the **Domain** text box, and select **Enabled**.
4. Click **Create**.

Managing lab entitlements in VMware Lab Platform

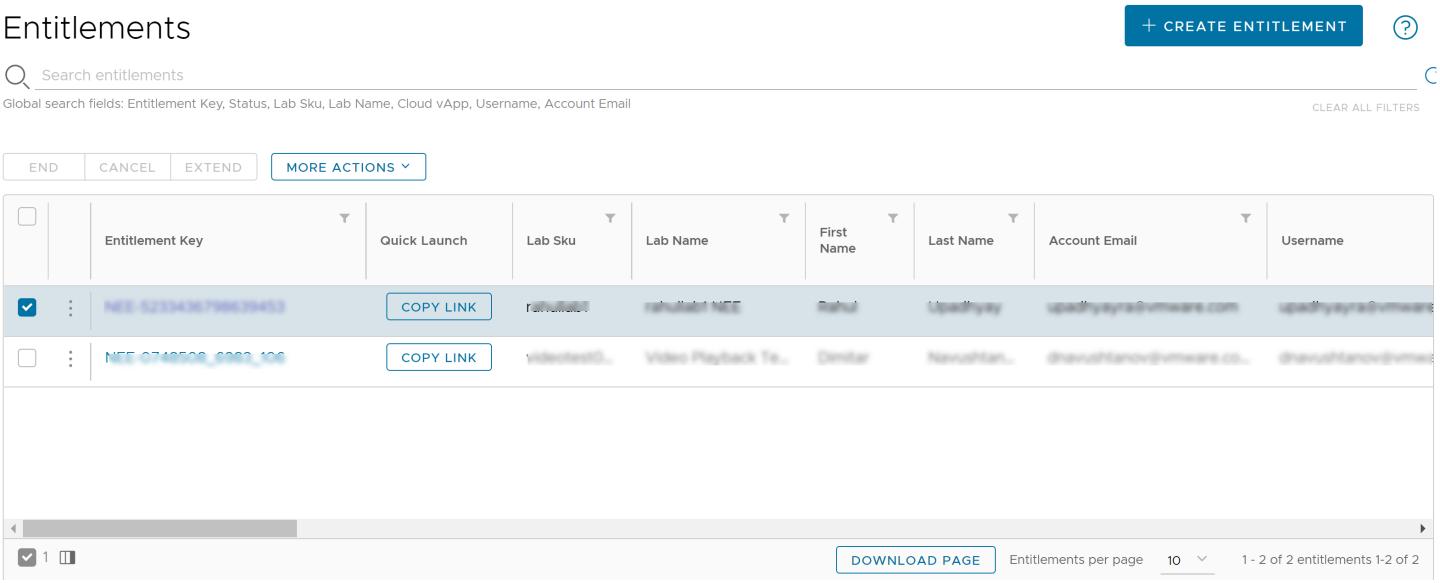
You can manage entitlements from the **User Entitlements** page on the administrator user interface.

Entitlements refer to a lab's time limit; the time a student can work on a lab. The valid period for a lab during which a lab is active and available for sign-up and use. VMware Lab Platform copies values for an entitlement from the lab record when an entitlement is created.

To open the **Entitlements** page, click **Entitlements › User Entitlements**. To see all the options for columns, use the



icon.



Action options for single entitlement

On the **Entitlements** page, select an entitlement from the entitlement list and click the vertical ellipses. The following action options are available:

- **Update**: Displays a form from which you can update entitlement properties.
- **End**: Ends an entitlement that has already started.
- **Cancel**: Cancels an entitlement that has not yet been started.
- **Extend**: Adds more elapsed time to the entitlement. When you select Extend, an entry box opens that allows you add to the remaining time.
- **Expire**: Expires an entitlement for the time being, but you can renew the entitlement.
- **Reset**: Forces the vApp in the cloud to be deleted. The next time the user starts the entitlement, the user gets a new vApp (either a prepop or an on-demand vApp).
- **Make Ready**: Assigns cloud resources (vApp) to an entitlement before the entitlement initially starts. Once the entitlement is ready, the system either takes an existing prepop from the available prepops pool or triggers an on-demand deployment. So, a vApp exists in the cloud assigned to this entitlement. The **Make Ready** option, allows you to assign cloud resources to the entitlement but does not trigger the entitlement's time allotted time to start counting down.
- **Delete**: Deletes the entitlement.
- **Console Benchmarks**: If you have activated the console benchmark setting, then this will display the console strength information.
- **Entitlement Event History**: Displays a detailed event history for each stage of an entitlement, including time stamps. This provides administrators a more comprehensive view when troubleshooting issues that affect multiple students, or when administrators need specific information for a support request.
- **Entitlement Cloud Task History**: Displays the cloud task history of an entitlement.

- **Power Reset**: Resets the power cycles for all the VMs in a vApp, except for entitlements that have **Awaiting Grade** status.
- **Add to Class**: Adds the selected entitlements to the class you select. Once you select the **Add to Class** action, the Class text box appears. To display the Class text box, click **Add to Class**, and then select a class from the Class text box.
- **Reassign Entitlement**: Reassigns an existing entitlement to another user.
- **Clone vApp**: The **Clone vApp** tab allows you to take one vApp in a cloud and create another version of it in the same virtual data center (vDC). You can select one or more entitlements and then click **Clone vApp**.
- **Capture Template**: Creates a new lab template in the Lab Builder page. This feature is useful in scenarios where administrators and content creators want their own version of the lab templates for the same lab.
- **Send Email**: Sends an email to the user with the entitlement's quick launch URL.
- **Change Step**: Changes the current manual step for the entitlement.
- **Shut down and Power off cloud vApp**: Powers off the vApp for the current entitlement. This feature is helpful in saving cost as only storage costs are charged if a vApp is powered off. When a vApp is in powered off state, vCD organization admin cannot modify the configuration of cloud vApp, but can modify the cloud vApp configuration when the vApp is in Stop state.
- **Power on cloud vApp**: Powers on the vApp for the current entitlement.
- **Usage and Cost for Entitlement**: Get resource usage and cost estimates of an entitlement in real-time.
- **More Details**: Displays a list of properties for the entitlement key.
- **Grade**: This option is activated only if the entitlement is in the “Awaiting Grading” status. The awaiting grading status is an interim status between an entitlement being started and ended.

  If a lab is not marked as requiring grading, then when it ends the system moves the entitlement to a transcript and deletes the vApp.

  If grading is required, then the system transitions the entitlement into the awaiting grading status and retains the vApp. The entitlement is no longer active and so a lab user cannot take or re-enroll into the lab.

  If the **Graded** option is activated and the admin clicks it, then the system removes the entitlement from the awaiting grade status, copies it to a transcript and deletes the vApp.
- **Rollback Awaiting Grade**: If an entitlement has the Awaiting Grade status, then this option lets you to change the entitlement to normal entitlement again. If it is awaiting grade you can just grade it.

Action options for bulk entitlements

On the **Entitlements** page, you can select one or more entitlements, click the three ellipses on the top-of the page, the **Batch Actions** drop-down menu is displayed.

- **End**: Ends an entitlement that has already started.
- **Cancel**: Cancels an entitlement that has not yet been started.
- **Extend**: Adds more elapsed time to the entitlement. When you select **Extend**, an entry box opens that allows you add to the remaining time.

  CAUTION

  If the class end date or time is changed, then the Entitlement expiration date is updated for the same. For more information, see Edit an Existing Class.
- **Reset**: Forces the vApp in the cloud to be deleted. The next time the user starts the entitlement, the user gets a new vApp (either a prepop or an on-demand vApp).
- **Power Reset**: Resets the power cycles for all the VMs in a vApp, except for entitlements that have **Awaiting Grade** status.
- **Make Ready**: Assigns cloud resources (vApp) to an entitlement before the entitlement initially starts. Once the entitlement is ready, the system either takes an existing prepop from the available prepops pool or triggers an on-demand deployment. So, a vApp exists in the cloud assigned to this entitlement. The **Make Ready** option, allows you

to assign cloud resources to the entitlement but does not trigger the entitlement's time allotted time to start counting down.

- **Send Email**: Sends an email to the users with the entitlement's quick launch URL.
- **Add to Class**: Adds the selected entitlements to the class you select. Once you select the **Add to Class** action, the Class text box appears. So, the steps are first click **Add to Class** to display the Class text box, and then select a class from the Class text box.
- **Change Step**: Changes the current manual step for the entitlements.
- **Shut down and Power off cloud vApp**: Powers off the vApp for the current entitlement. This feature is helpful in saving cost as only storage costs are charged if a vApp is powered off. When a vApp is in powered off state, vCD org admin cannot modify the configuration of cloud vApp, but can modify the cloud vApp configuration when the vApp is in Stop state.
- **Power on cloud vApp**: Powers on the vApp for the selected entitlements.
- **Download page**: Exports the entitlements data in a .csv or Excel file.

Access mirror display of a user console

You and support teams can use the **Mirror Display** link to view the entitlement console to troubleshoot any issues from the administrator user interface. The **Mirror Display** link allows you to clone, or mirror the display of the user interface of an entitlement.

To access the **Mirror Display** link:

1. Navigate to **Entitlements › User Entitlements**.
2. Use the scroll bar to scroll the table and you can view the **Mirror Display** column.

NOTE

If the **Mirror Display** column is not available in the table, use the  icon to see all the options for the table, and select the **Mirror Display** option from the list.

Edit an entitlement

To open the **Update Entitlement** form, click an entitlement on the **Entitlements** page.

The **Update Entitlement** form lets you to edit the following options:

- **Time Allotted**: Allots an amount of time for the selected entitlement.
- **Expiration Date**: Sets an expiration date for the selected entitlement.
- **Score**: Assigns a numerical score to the entitlement. For more information about scores, see “Grading Options”.
- **Grade**: Assigns a text grade (for example, “A”) to the entitlement. For more information about grade strings, see “Grading Options”.
- **Metadata**: Adds the metadata you enter. For example, you might include the following metadata “This is an example of metadata”. In this example, the system adds the metadata “This is an example of metadata” to the entitlement.
- **Tracking Code**: A tracking code is a useful way to track the link by which an end user found a lab. Consider the following examples:
  - A promotion campaign posts a link to a lab that includes the tracking code promoA:`https://www.vmwarelearningplatform.com/hol/catalogs?ti d=promoA` this example, the system adds the tracking ID `promoA` to the entitlements of anyone who clicked the link.
  - Another promotion campaign posts a link to the same lab but instead includes the tracking code promoB:`https://www.vmwarelearningplatform.com/hol/catalogs?tid=promoB` this example, the system adds the tracking ID `promo` to the entitlements of anyone who clicked the link.
- **Always Run vApp**: If activated, the system prevents the vApp from shutting down and powering off.

The Clone vApp tab allows you to take one vApp in a cloud and create another version of it in the same virtual data center (vDC). You can select one or more entitlements and then click **Clone vApp**.

View entitlement status

On the Entitlement page, you can select one or more entitlements, then click **Status** to filter the list by entitlement status. The life cycle of an entitlement in VMware Lab Platform includes the following statuses:

- **Complete**: Legacy status which the system no longer uses. When an entitlement has been completed, the system moves it to a transcript.
- **Incomplete**: Initial status of an entitlement. An incomplete status means that the entitlement is ready to be started. The entitlement still has time left and has not reached its expiration date, but is not currently active (end user is not using the lab console). An incomplete entitlement can be in either of the following states: Just created (not started yet) or Started but paused.
- **Expired**: Every entitlement has an expiration date and a time allotted. If the expiration date is reached or the time allotted is over, then the entitlement expires. This temporary status means that a couple of hours after expiration, the system ends the entitlement and moves it to a transcript.
- **Suspended**: Not yet been implemented so no entitlement must be in this status.
- **Gone**: Entitlement was active but one of the following occurred:
  - End user closed the browser without first exiting or ending the entitlement.
  - End user navigated away from the lab console without first exiting or ending the entitlement.
  - Internet connection on the client dropped.
  - No activity on the browser console has occurred for a while and as a result the system navigated the end user to an Idle page.
- **Active**: End user is using the entitlement. During an Active status, the browser pings the server to verify a connection. If the browser drops the connection or if the connection closes, then the ping does not occur, and the system sets the entitlement to the next status.
- **Canceled**: Similar to the Complete status. Canceled is a status of a transcript rather than an entitlement. If an entitlement has never been started, the administrator can cancel it. The transcript marks the entitlement as having been canceled.
- **Awaiting Grade**: Only for entitlements that require grading. If an entitlement requires grading, then, after it has been ended, the system does not move it to a transcript. Instead, the system keeps the entitlement in an Awaiting Grading status meaning that it is waiting for an administrator to grade it.
- **Ready**: Much like the Incomplete status. However, the Ready status means the system assigned a prepopulated vApp to the entitlement. When an end user starts an entitlement in a Ready status, then it starts without delay as the vApp is already assigned.

Add grading for an entitlement

You can use the **Score** and **Grade String** text boxes on the Entitlement page to add grading data to an entitlement. You can subsequently edit the grading data. To grade an entitlement:

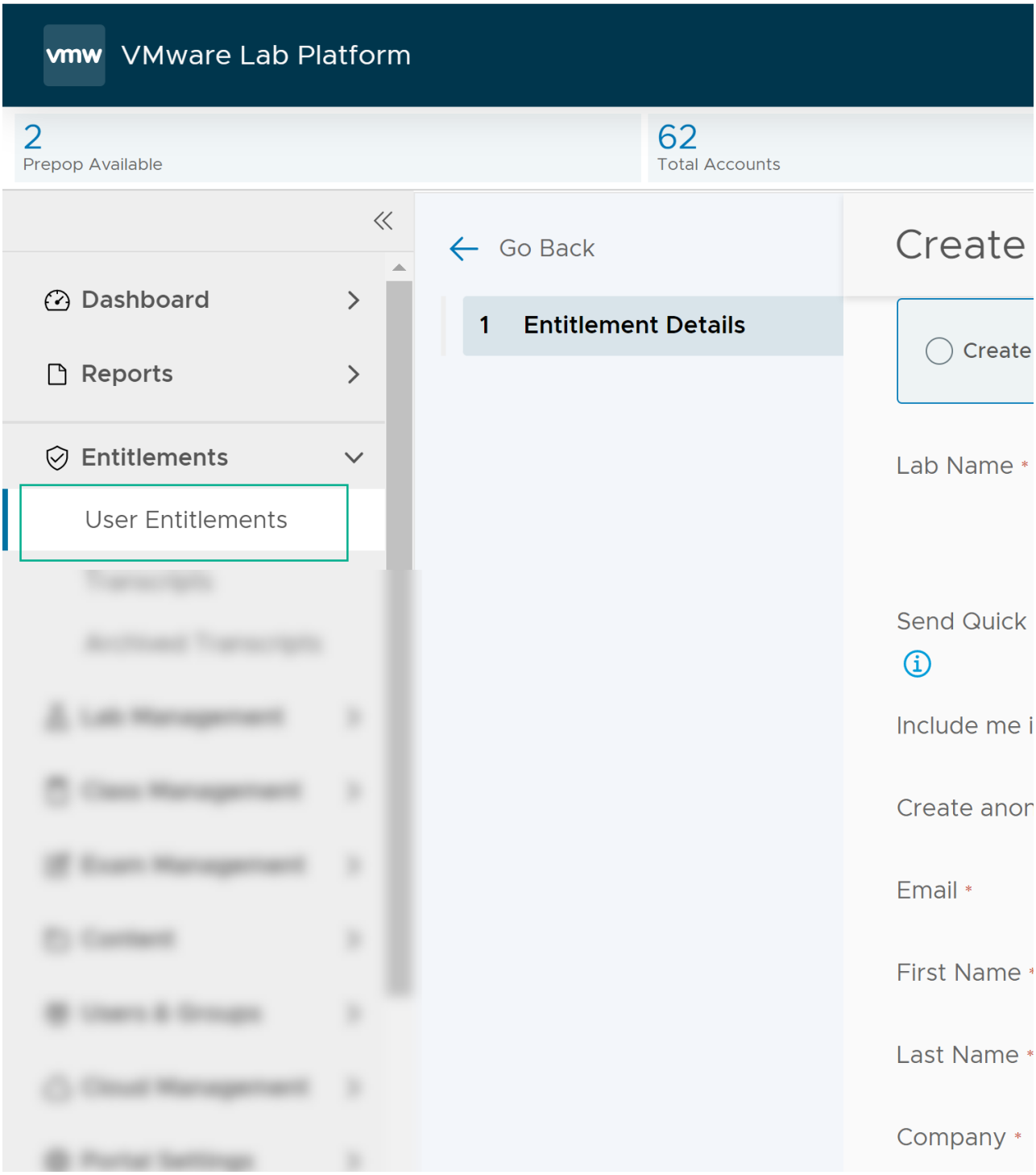
1. Enter a **Score** or **Grade String**.
2. When you use scores or grade strings, the system does not actually grade the entitlement. Grading an entitlement requires transferring its grading data to a transcript. To transfer the grading data to a transcript, you must click the **Graded** option from the drop-down menu.

Create an entitlement for a new user in VMware Lab Platform

The Create User and Entitlement feature lets you create a new user and add entitlements for the user. Entitlements refer to the time or period that a user can use a lab.

To create a user and an entitlement using the **Entitlements** wizard:

1. Open the **Entitlements** page, click **Entitlement › User Entitlements** .



2. Click **Create New**.
3. Select the **Create User and Entitlement** option.
4. Select a lab from the **Lab Name** drop-down list.
5. (Optional) To send a lab URL directly to the user, select the **“Send Quick launch URL to user via Email”** option.

a) To add the administrator in the user email as a CC recipient, select the **Include me in the user email as CC** option.

6. (Optional) To create an anonymous user for one-time lab access, select the **Create anonymous user** option, and fill the user details in the form.

   For anonymous users, once the entitlement is completed, the user details are deleted from the system. For more information on anonymous user, see Create an Anonymous User.

7. To create a user for the entitlement, fill the mandatory user details such as; email, name, and company.
8. Click **Create Entitlement**.

An email with the lab details and lab URL is sent to the user.

Create an entitlement for an existing user in VMware Lab Platform

To create an entitlement for an existing user for a lab:

1. To open the **Entitlements** page, from the navigation menu, click **Entitlements › User Entitlements** .
2. Click **Create New**.
3. Select the **Create Entitlement for existing user** option.
4. Add an account name from the account name list.

   User accounts can be created from **Users & Groups › User Accounts › Create New** option.

5. Assign a lab to the user from the **Lab Name** drop-down list.
6. To send a lab URL directly to the user, select the "**Send Quick launch URL to user via Email**".
7. To add the administrator in the user email as a CC recipient, select the **Include me in the user email as CC** option.
8. Click **Create Entitlement**.

An email with the lab details and lab URL is sent to the user.

Working with transcripts in VMware Lab Platform

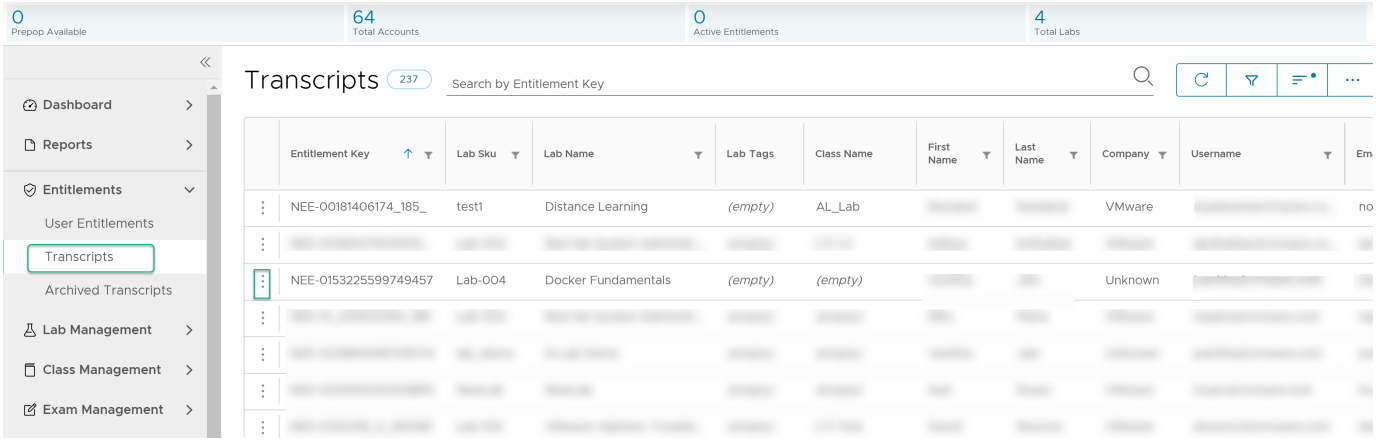
In VMware Lab Platform, transcripts provide detailed information on a user's report of using a lab. A transcript is similar to a report card that lists all the information an of entitlement. Like the percentage of a lab completed by a user, lab completion details, and so on.

As an administrator, you can view the lab transcripts from the administrator user interface using the **Entitlements › Trans cripts** page. The Transcripts page lists transcripts by lab and user.

View a transcript report for an entitlement

To view the transcript information for an entitlement key:

1. Click **Entitlements › Transcripts**.



2. To view the details of an entitlement transcript, click the three ellipses next to an entitlement key, and click **More Details**.

   a. To view all the options for a transcript, click the **Manage Columns** button and the select the options that you want to view for an entitlement in the Transcripts page.

      You can view the transcript report for the following parameters for an entitlement.

Options	Description
Entitlement Key	Name of the entitlement key. The entitlement key is created using the **Entitlements › User Entitlement** wizard.
Lab SKU	Lab Stock Keeping Unit (SKU) is the short internal name for the lab.
Lab Name	Name of the lab.
Lab Tags	Lists the tags associated with the lab.
Class Name	Name of the class.
First Name	First name of the user.
Last Name	Last name of the user.
Company	Company name of the user.
Username	Username or email Id of the user.
Email	Email Id of the user.
End Type	Status of the entitlement key.
Short Description	Description details of the entitlement.
Time Allotted	Displays the amount of time, in hours, a user gets to use the active console or the entitlement active time.
Time Elapsed	Displays the time a student has spent in the lab.
Required Minimum Time Spent	Labs have this property. Available in both Entitlements and Transcripts. Example, if the minimum time spent is set to 5 minutes for a lab, a student cannot end the lab until 5 minutes have passed.
Start Date	Start date details of the entitlement.
Completion Date	Completion date details of the entitlement.
Score	Displays score and grading information of the entitlement.
Grade	Displays score and grading information of the entitlement.

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Options	Description
	This is an Exam feature for grading. This is a score number (example: 100).
Language	Language in which the lab was taken by a user.
Location	Location details of the user at the time of taking the lab.
IP Address	IP address of the last person to start the entitlement. Most likely will be the student's IP address, but could also be instructors, class owners, automation as well.
Extension Count	Number of times an entitlement time was extended.
Return URL	If a return URL for LTI integration is configured, then this data is displayed here.
Metadata	Additional data for the entitlement.
Tracking Code	Class tracking code. Tracking code can be specified when creating a class. When the entitlements are created for the class, they will have the class tracking code.
Station ID	Used in the Events feature. Events can have Seatmaps with stations.
Last Start	When the entitlement was last started.
Expiration Date	When the entitlement expired.
Manual Step Number	Student's current manual step number.
Manual Steps Visited	Number of steps that a student/user has completed from the list of steps for a given lab.
Manual Step Count	Lists the remaining steps that a student needs to complete for a lab.
Percentage Steps Visited	Percentage of the manual steps visited by a student. For example, Manual Steps Visited divided by Manual Step Count.
Required Percentage Steps Visited	Labs have this property. Available in both Entitlements and Transcripts. Example, if the minimum percentage steps is 5%, a student cannot end the lab before they have visited 5% of the lab manual.
Completion Status	Dsiplays the status of entitlement when the lab was completed. For Transcripts this may be Completed or maybe Expired. For Entitlements this can be many things: Incomplete, Gone, Canceled, Suspended, Ready, Active, Awaiting Grade.
Cloud Org Vdc	Most labs usually run in vCD (vCloud Director). The vDC (virtual data center) provides resources for the lab to run (example: 4GHz CPU, 16 GB RAM, 100 GB Storage, etc.) and the **Cloud Org Vdc** option specifies which vDC the lab was running from.
Cloud Lab Template	Most labs in VLP are represented by a template in vCD (vCloud Director) and this is the backing template of the lab.
Console Benchmarks	View the average, high, and low latency values of the console connection. An option called "**Console Connection Monitoring**" must be enabled in the Cloud for this.
Creation Type	Describes how the entitlement was created and can either be on-demand or Prepop. A Prepop means the student was given a prepop when they first started the entitlement. On-demand means there were no prepop, so a new deployment had to be done for the student to get a lab.
Instructors Name	Instructor names is displayed if the entitlement was part of a class.

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Options	Description
Cu Usage	Displays the CU (Compute Units) the entitlement has used.
Su Usage	Displays the SU (Storage Units) the entitlement has used.
Cost (Dollar Value)	Calculates the cost for the entitlement.

View the End Type option

The **End Type** option lists the completed, canceled, or expired transcripts. To view the expired transcripts, click **Entitlements › Archived Transcripts**.

Export transcripts

The **Download Page** option allows you to download all transcripts to your local system and export it. To view the **Download Page** option, click the three ellipses on top of the table.

Archive transcripts

When a user account is deleted from the system, the transcripts of the user are archived and you can access the transcripts from the **Entitlements › Archived Transcripts** page. You can access the user account details from the Archived Transcripts page. The Archived Transcripts feature is useful to capture statistical data about users, or events.

Working with manuals in VMware Lab Platform

The **Manuals** page lists the lab manuals associated with the current account.

You create manuals with the tools of your choice and can import manuals that are in the following formats: Word docx, PDF, or ScreenSteps. Alternatively, you can use the VMware Lab Platform editing tool called the Content Editor to develop a manual or edit a manual after it is imported.

To ensure that the manual you select on the **Manuals** page has the same content in both the administrator dashboard and the Content Engine interface, synchronizing (Sync) is necessary. Creating and editing the content of an instructional manual is done separately from the VMware Lab Platform administrator dashboard.

> **NOTE**
> PDF resource files used as lab manuals are not displayed here.

To access the Content Engine user interface, click **Content › Content Engine**. The Content Engine allows an administrator to manage lab manuals. The *Content Editor Reference Guide* describes how to use the Content Editor.

Add a PDF manual to a lab in VMware Lab Platform

You can add PDF manuals for a lab.

- To use a PDF document as a lab manual, deactivate the **Portal Settings › Configuration › Hide Instructions panel when no manual** option.
- Ensure that the **Labs › Update lab › Manual** option is left empty.

1. To open the Labs Resources page, from the navigation panel, select **Labs › Lab Resources** .
2. Click **Create New**.
   The Create Lab Resource page is displayed.
3. In the **Name** field, enter a name for the lab manual.
4. Click **Upload a File**, and select a PDF manual from the local system.
5. Click **Create Lab Resource**.

The PDF document is listed on the **Lab Resources** page.

6. To add the PDF manual to a lab, on the **Labs › Labs** page, click the lab for which you want to add the PDF manual.
7. In the **Update a lab** page, select **Update Content**, and scroll to the **Resources** option. Select the PDF manual from the list view that you added in the **Lab Resources** page.
8. Click **Done**.
9. Click **Update Lab**.

The PDF manual is added to the lab. Access the lab with the PDF resource in the user interface. If you select multiple PDFs, only the first selected file is displayed as a manual in the user interface lab. However, the other PDFs are available for downloading along with other lab resources in the running lab, from the **More Options › Resources** menu.

Creating page content for labs in VMware Lab Platform

Page content refers to offers, announcements and promotions that appear in the Announcements section of the lab console. As an administrator, you can create and modify page content.

To create a page content:

1. Navigate to **Portal Settings › Page Content › Create New**.
2. Fill the details in the **Create Page Content** form.
3. Click **Create Page Content**.

You can view, edit, or delete the page content from the **Portal Settings › Page Content** page.

Create content for the Announcements page in VMware Lab Platform

You add content for the announcement page from the Page Content form.

1. From the navigation panel, select **Portal Settings › Page Content**.
2. To open the **Create Page Content** form, click **Create New**.
   a) In the **Type** text box, select **Offer**, **Announcement**, **Dashboard Promo**, or **Client Promo** from the drop-down menu.

      NOTE
      Promotions display as individual slides either as dashboard promotions on a large, central screen during an event, or as a client promotion in the form of a screensaver, on a user's screen. An example of a promotion is: `This lab sponsored by our Company`.

   b) To activate or deactivate the display of page content, clear or select the **Enabled** check box.

      You can enable content for only for a certain period each day by using this enable-disable toggle.

   c) In the **Title** text box, enter a title for your page content.
   d) In the **Content** text box, enter text that appears under your title.
   e) If anonymous browsing is activated, you can control what content is viewed in anonymous browsers and by which logged-in users by selecting or clearing the **Public Visible** check box.
3. Click **Create Page Content**.

Change the order of page content in VMware Lab Platform

You can optionally change the order of the page content as it appears in the **Page Content** list and in the lab console.

The Page Content Order feature allows you to order individual content within the **Announcements**, **Offers**, and **Promos** tab. You cannot order the tabs themselves.

1. From the navigation panel, click **Portal Settings › Page Content**.
2. Click **Change Display Order**.
3. On the **Change Display Order** page, select the icon next to the content you want to move and drag the row to move it to a new place in the display order.
4. Continue moving rows until you have the order you want, then click **OK**.

The lab console reflects the new page content order.

Working with lab groups in VMware Lab Platform

You can assign labs to groups and associate characteristics, such as a color, to the group.

At events, proctors often wear a color tag associated to a category of labs. For example, a red tag can be associated to labs about the network security. Students having questions about a security lab will know which proctor to contact.

Create a lab group in VMware Lab Platform

You can group labs and assign a color to the group.

The following procedure shows you how to create a lab group and associate it with a color.

1. Click **Events › Lab Groups**. The **Lab Group List** opens.
2. On the **Lab Group List**, to open the **Create Lab Group** form, click **Create Lab Group**.
3. Enter a name for the lab group, for example Security.
4. Open the color chart from which you can select a color.
5. Select a lab from the Labs drop-down menu.
6. Click **Create**. The View Lab Group page opens that shows the grouping of a name, color and lab.

Working with dashboards in VMware Lab Platform

The dashboard is a great solution to keep administrators fully informed about the system and its performance. You can even create your own dashboard to customize the information it displays.

VMware Lab Platform provides you with great flexibility in selecting the type of data your dashboards display. You can:

- Select the dashboard you want to display
- Put a dashboard in Edit or Full screen mode
- Create your own dashboard
- Select a dashboard layout
- Move (click/drag) individual widgets within dashboard
- Reset dashboards

Selecting a dashboard

When you click **Dashboard › Default Dashboard**, the system displays the default dashboard, but you can create your own dashboard. In fact, you can create multiple dashboards and select the dashboard you want to display. Click the **Default Dashboard** drop-down menu and select the dashboard you want to view.

Putting a dashboard in full screen or edit mode

Dashboards include the full screen option and edit mode option:

- Use the  edit icon to put the dashboard in Edit mode.
- Use the  maximum icon to expand the dashboard to the full screen view.

In edit mode, you can:

- Add a widget
- Change the layout
- Delete the dashboard
- Save the dashboard
- Closes Edit Mode

Create a dashboard in VMware Lab Platform

You can create and customize new dashboards in Key definition for "VLM_name_long" not found in the DITA map..

1. Select **Default Dashboard> › New Dashboard**.
2. Click **New Dashboard**. The New Dashboard form opens.
3. In New Dashboard form, enter a name for your dashboard and select a layout. The layout you select determines the number and size of panes for your dashboard.
4. Click **Save**. The system saves and opens your dashboard.

Working with widgets in VMware Lab Platform

The dashboard includes predefined widgets that you can add to your dashboard.

On the current dashboard, to put your dashboard into the Edit mode, click the  gear icon. In Edit mode, the dashboard displays a list of widgets that you can add to your dashboard. To add widgets to your dashboard, click **Add Widgets** on the **Add Dashboard Widgets** form.

The following widgets are available for a dashboard.

Widget	Description
Aggregated Statistics	Displays aggregated statistics data on labs deployed lab templates, VMs, labs delivered, active lab sessions, and total users for a defined period for an organization account. Select the date range to populate the data.
Bar Chart Display	Displays total count of various system values for VMs, vApps, labs, entitlements, prepops, and seating in a bar chart format.
Broadcast Messages	Lists event broadcast messages. These message are sent to proctors from students about any issues with the labs during an event.
Cloud OVDC Capacity Usage	Displays overall organization vDC resource allocations for the organization account.
Concurrent Users	Displays a timeline graph of concurrent users.
Content Migration Tasks	Displays a table of Content Migration tasks.
Daily Entitlement Statistics	Displays a bar chart of entitlements per day.

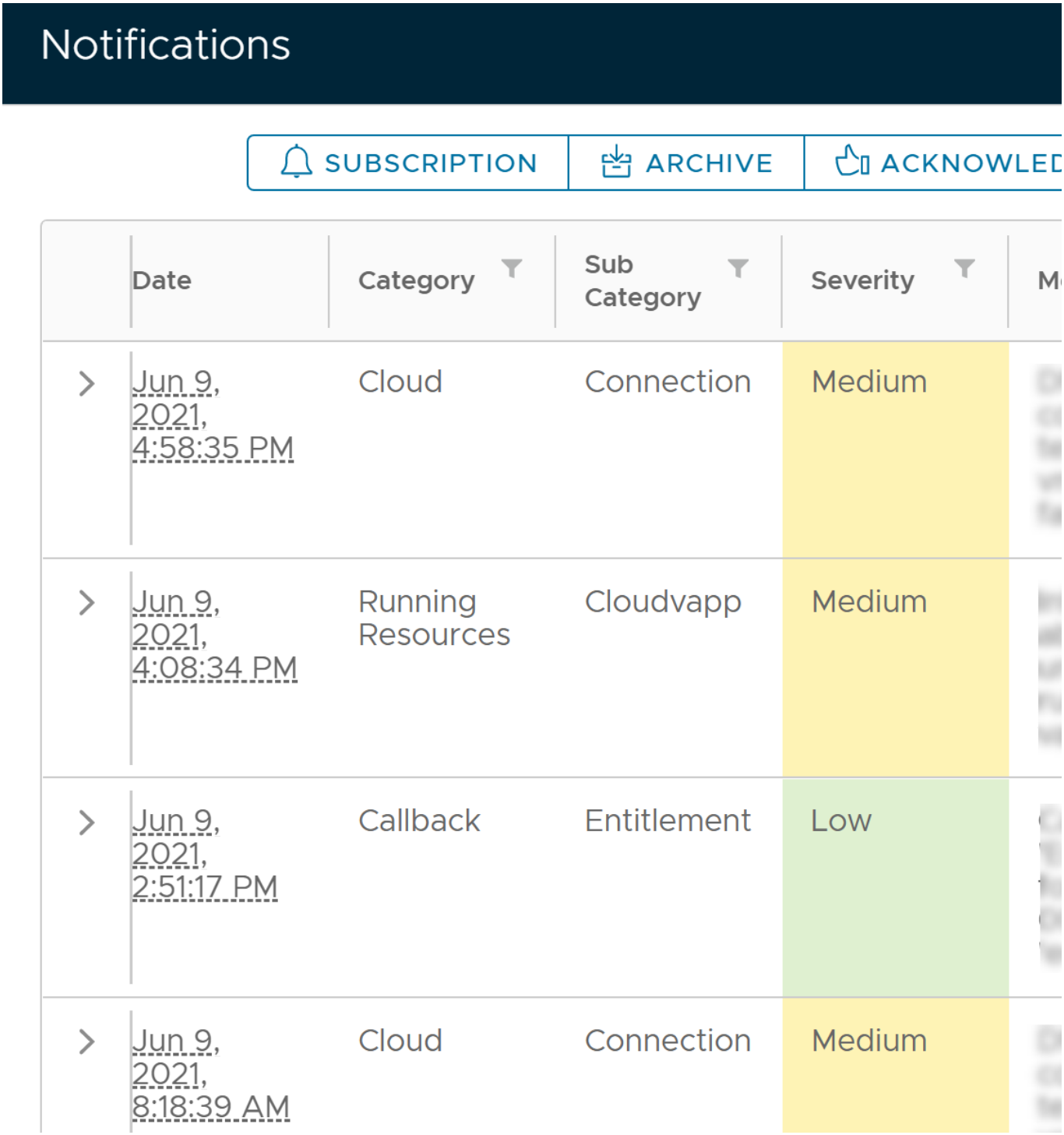
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Widget	Description
Download Entitlement Statistics	Generates a CSV report of the entitlement statistics.
Entitlement Duration Statistics	Time spent on a unique completed entitlement relative to previous 7 days.
Get Started! Set up your first lab!	Displays the Get Started! Set up your first lab! panel.
Help Requests	Lists active event Help Requests.
Hourly Entitlement Statistics	Displays a stacked column graph of entitlements by hour of the day.
Labs	Lists total and current lab's count.
Manage Unresolved Templates/Media	Lists all unresolved templates,and media.
Manage Uploads	View all uploads.
Notifications	Displays a table of important notifications.
Numeric Display	Lists a configurable container for numeric display of various system values.
Pie Chart	A configurable pie chart for display of various system values.
Popular Labs	An ordered column graph of popular labs.
Upcoming Classes	A table that shows the next 10 upcoming classes.
Upcoming Classes Resource Usage	A timeline graph of estimated resource usage for upcoming classes.
vAppTemplate Placement Statistics	Pie graphs of vApp template deployment placement statistics.
vDC Placement Statistics	Pie graphs of Org vDC deployment placement statistics.
VMs	Total and current VMs count.
Weekly Entitlement Statistics	A stacked column graph of entitlements per day of the week over a configurable period.
Weekly Resource Usage	A stacked timeline graph of concurrent resource usage by the week.

About the Notifications Widget

The **Notifications** widget is the most informative and configurable widget.



It includes the following features:

- Severity Levels: Green (low), yellow (medium), or pink (high).
  - High severity notifications are related to issues impacting multiple users and often across multiple accounts. Typically, a high severity notification stems from a cloud being unavailable. The pink color highlights the high severity notifications.

  - The medium severity notifications issue signifies that someone is unable to use the system (examples include failure to the start labs or the errors deploying vApps). The yellow color highlights the medium severity notifications.
  - Low severity notifications are related to information such as daily summaries. The green color highlights the low severity notifications.

    NOTE
    You can sort notifications based on the severity of the notification.

- To expand and view the details of a notification, click the **Expand** arrow.
- Notification Subscription: While notifications appear on a dashboard, you can also enable the system to email you the notifications. To open the **Notifications Subscription List**, click **Subscription**. From there, you can add an email address and set criteria for which the system must send out notifications. Once you define a subscription, each time a notification that meets your criteria occurs, the system emails it to everyone who has subscribed, and showing it on the dashboard.

    NOTE
    In the Notifications Subscription list, severity levels are represented as 1, 2, 3 (1= Pink, 2=Yellow, and 3=Green).

- Archive: Displays the Notification Archive List which lists all archived notifications.

    NOTE
    The Notification Archive List includes an Export option. To export all notifications to a CSV, Excel, or PDF file, use the **Export** option.

- Acknowledge All: Archives all the notifications. Once you archive a notification, it is not displayed on the dashboard. To view acknowledged notifications, follow the archive link at the top of the page.

View notification categories

Notifications categories are:

- **Default**
- **Cloud**
- **Administrative**
- **cloud vApp**
- **System**
- **generic task**
- **prepop**
- **callback**
- **billing**
- **cloud template**
- **management**
- **egw**
- **running resources**

View notification subcategories

The notifications are also sub categorized to form the following category/subcategory pairings.

- **administrative, myLearn**
- **cloud, connection**
- **cloud, sync**
- **cloud, task**

- **cloudvApp, ""**
- **genericTask, massUpload**
- **genericTask, task**
- **genericTask, userDownload**
- **genericTask, userMove**
- **prePop, "Daily Summary"**
- **prePop, "provision failure"**
- **system, monitoring**
- **administrative, myLearn**
- **cloud, connection**
- **cloud, deployment**
- **cloud, sync**
- **cloud, task**
- **cloudvApp, ""**
- **genericTask, emailTranscipt**
- **genericTask, massUpload**
- **genericTask, statsDownload**
- **genericTask, task**
- **genericTask, userDownload**
- **genericTask, userMove)**
- **prePop, "Daily Summary"**
- **prePop, "provision failure"**
- **system, cloud**
- **system, monitoring**

Download entitlement statistics widget

The **Download Entitlement Statistics** widget generates a CSV file report of entitlement statistics. The report includes the following information: `username`, `firstName`, `lastName`, `company`, `labId`, `entitlementId`, `entitlementStart`, `entitlementEnd`, `entitlementComplete`, `exitCount`, `totalTime`, `stepNo`, and `language`.

Select widgets for your dashboard in VMware Lab Platform

A new dashboard does not contain any widgets. You must select the widgets and the type of data each widget must contains.

1. On the dashboard you created, click the gear icon to put your dashboard into Edit mode.
2. To open the Add Dashboard Widgets form, click the plus sign icon. Select widgets to include in your dashboard.

   NOTE
   The widgets available in the Add Dashboard Widgets form depends upon your user role. For example, an instructor role accesses fewer widgets than an admin role.

3. Click **Add Widgets**. The system displays the dashboard with the widgets you selected.

Configure dashboard widgets in VMware Lab Platform

A widget you add may need configuration.

1. To open a widget configuration form, click the gear icon on a widget.
2. Specify a configuration .
3. Click **Add**, then click, **Save**. The system displays your dashboard with the newly configured widget.

Change the layout of a dashboard in VMware Lab Platform

You can change the layout of the dashboard in following ways.

- Click or drag a widget.
- Click the cube icon to open the Update Layout form. Use it to select a different layout.

Working with reports in VMware Lab Platform

The reports functionality allows you to access detailed information about lab entitlement, billing, compatibility results, and audit history about the system based on your data. Use this insight to make informed decisions on what actions to take.

You can view the following reports:

Report Category	Description
Billing	Provides compute and storage report for your account.
vDC Resource Usage	Provides a snap shot of vDC resources, such as memory, CPU, storage, and capacity used for your account.
Compatibility Tests	Provides the compatibility test report for various compatibility parameters for users accounts.
Audit History	Provides audit history for various domains such as Exam, Lab, Entitlement and so on.

View the audit history report in VMware Lab Platform

The Audit History page provides a view into the actions performed by all the VMware Lab Platform users. You can view all the changes made to your infrastructure and the lab environment in the Audit History page. The log information in the Audit History page is useful to analyze events and failures in the system, such as accidentally ending an entitlement and so on.

To view the audit history details:

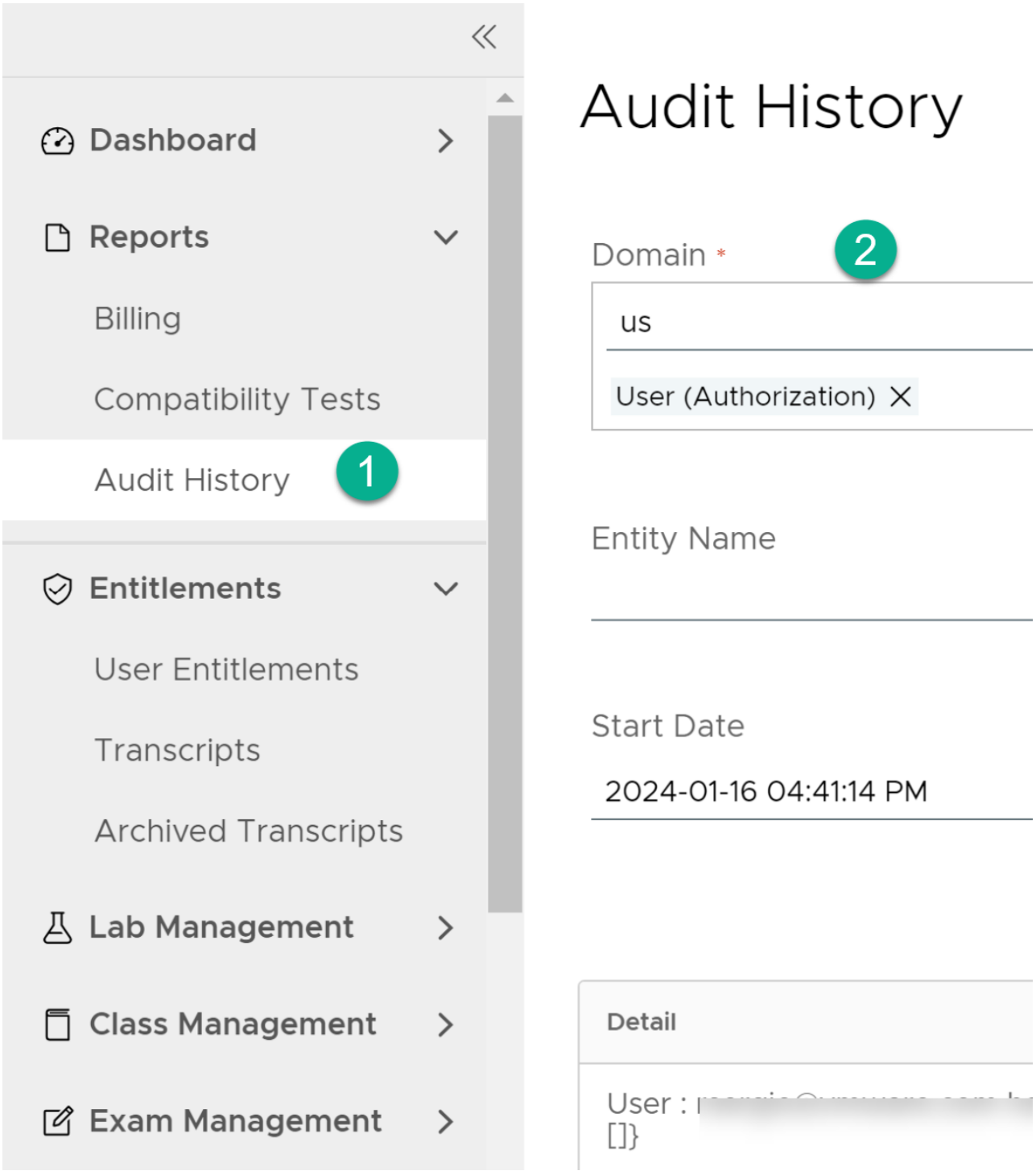
1. From the navigation panel, select **Reports › Audit History**.
2. Select a domain from the **Domain** drop-down menu. A domain refers to a key category in VMware Lab Platform.

   You can select only one domain type per report.

3. Select an action for the domain from the **Action** drop-down menu.

   The action options correspond to the domain type that you select.

4. Enter an entity name for the domain type. For example, if you run an audit report for the Exam domain, the list of entities is listed on the **Exam Management › Exams** page. Ensure that the name and the case for the entity is same.
5. Select a date range to run the report. Enter the start date and end date.
6. Click **Submit**.



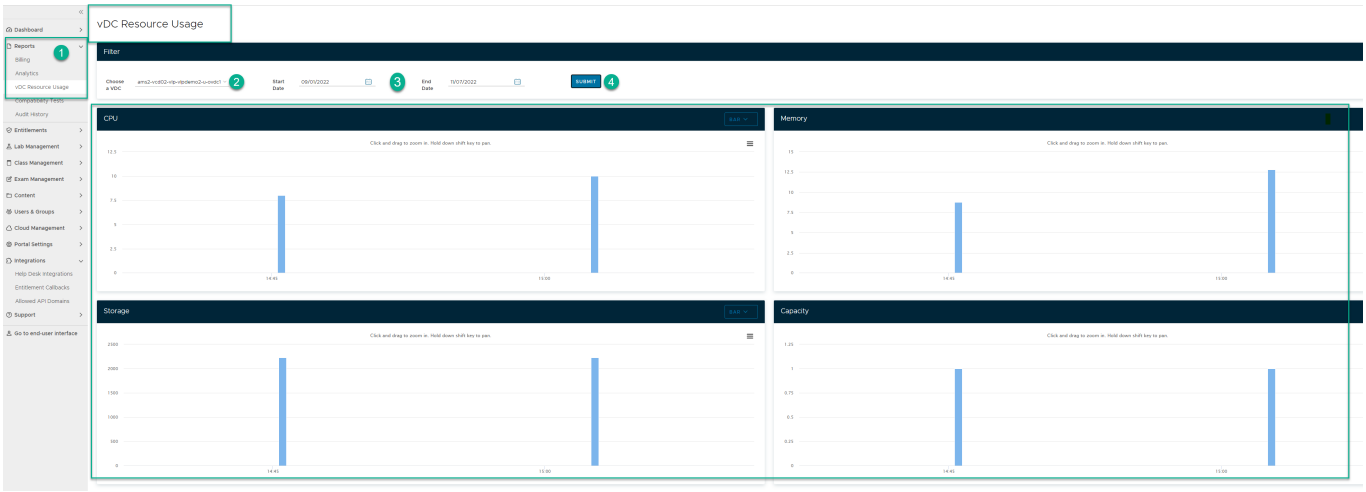
The audit report for the given domain type, action, and entity is displayed. Use the XLS, or CSV tabs to download the report to your local system.

View the vDC resource usage report in VMware Lab Platform

The vDC Resource Usage report displays the resource consumption details per virtual data center for your VMware Lab Platform account, or organization.

To view the resource consumption report for your data center:

1. From the navigation panel, select **Reports › vDC Resource Usage**.
2. Select your virtual data center from the **Choose a VDC** drop-down menu.
3. Select a start date and end date range for the report generation.
4. Click **Submit**.



The vDC Resource Usage page displays four different graphs for the CPU, memory, storage, and capacity consumption for the vDC. The graphs provide an easy and clear view of the consumption units and you can compare the consumption units over a range of period. By default, the graphs are displayed in bar graph format. To change the graph view, select an option from the graph drop-down menu. To download the report, use the view icon and select a download option.

Working with notifications in VMware Lab Platform

You can view and receive real-time notification alerts from the system. The alerts are categorized based on the severity and priority of the issue such as High, Medium, and Low.

To view all notification alerts for an account, click the  icon on the top of the page header. You can view the list of subscribed notifications from the **Notifications** menu option.

Figure 12: Notification quick view



Severity of alerts and the color codes associated with a notification alert are:

- Green (low): Low severity notifications refer to information such as daily summaries.
- Orange (medium): Medium severity notifications mean that a user is unable to use the system (for examples include failure to start labs or errors deploying vApps).
- Red (high): High severity notifications refer to issues that impact multiple users. Typically, a high severity notification stems when a cloud is unavailable.

You can perform the following operations in the **Quick Notifications** pane:

- **All Notifications**: To view all the notifications.
- **Unread**: To view all the unread notification.
- **High Priority**: To view all high priority notifications.
- **Medium Priority**: To view medium priority notifications
- **Low Priority**: To view low priority notifications.
- **Flagged**: Select notifications that you want to flag and take an appropriate action
- **More Details**: Click the arrow to expand and view the details of a notification alert.
- **Settings**: Click the  icon to open the **Settings** option. You can select the alert categories, and specify the number of notification alerts, display time of the alerts and also, subscribe to alerts and receive email notifications.
- **Do Not Disturb**: Move the slider to stop notification alerts.
- **Advance Settings**: To view the **Notification Subscription** page and create notification subscriptions, use the **Advance Settings** option.
- **Notification Subscription**: While notifications appear on the page, you can also enable the system to email the notifications. To open the **Notifications Subscription List**, click **Subscription**. From there, you can add an email address and set criteria for which the system must send out notifications. Once you define a subscription, each time a notification that meets your criteria occurs, the system emails it to everyone who has subscribed, and showing it on the dashboard.
- **Delete**: Deletes a notification from the notification list.
- **Activate**: Activates the subscription for the notification.
- **Deactivate**: Deactivates the subscription for the notification.
- **View All Notifications**: Opens the Notifications page. You can perform single and bulk functions for notification alerts such as:

  - **Acknowledge All**: All notifications, including those marked as read and flagged, are moved to the Notification Archive page.
  - **Read All**: All notifications are marked as read and the notification count under the bell icon is reset.
  - **Flag All**: All notifications are marked as flagged and you can take action for them later. However, the flagged notifications are not marked as read.
- **Archive**: Displays the **Notification Archive** page that lists all archived notifications.

To get the notifications within the app, select the notifications listed under the **In-App Notification** menu. To get the notifications in an email, select the notifications listed under the **Email Notification** option. Use the Advanced Settings option to customize the maximum notification counts and duration for each notification.

> **NOTE**
> The **Notification Archive** page includes an export option. The **Download** option lets you to export all notifications to a CSV, Excel, or PDF.

Notification categories

Notifications are categorized as follows:

- Administrative
- Billing
- Cloud Template
- Cloud vApp
- Cloud
- Generic Task
- Prepop
- System

Create a notification subscription in VMware Lab Platform

You can create subscriptions for notifications and customize them as per their requirements.

1. Click the  icon and open the **Notification** page. To open the **Notification Subscriptions** page, from the **Advanced Settings**, click **Advanced notification subscriptions**.
2. To create a new subscription for a notification, click **Create New**.
3. Update the **Create Notification Subscription** form:

Notification Email	Add email to which you want the notifications to be sent.
Notification Category	Select the notification category.
Severity	Assign a severity for the notification category.
Enabled	Select **Enabled** to enable the notification.

4. Click **Create**.

Create custom notification messages in VMware Lab Platform

You can push custom notifications to all or selected logged in users in the end-user interface.

To create a custom message:

1. Click **Portal Settings › User Messages › Create New**.
2. Enter the details in the Create user message form.

Title	Name for the custom message.
Content	Custom message content.
Type	You can send the message immediately or schedule a date and time to send the custom message to lab users. The immediate message type sends an instant message to the user only if the user is active on the user console. To send the message immediately, select the **Immediate** option from the drop-down menu. To send the message on a specific date and time, select the **Scheduled** option from the drop-down menu. The option to select the date and time is populated in the form. Select the date and time to send the message. You can also specify if you want to send the message now or later in the **Send When** option. For schedule messages, user has to acknowledge the message notification on the system.
Enabled	When selected, custom message notification is activated.
Severity	Severity level for the message. You can categorize the severity of the message as: • Low • Medium • High
Send When	Define the time and date to send the custom message. You can choose to send the message now or at a later time. Use the drop-down menu option to make the selection.
Start Date	If you select the **Later** option in the **Send When** field, you need to enter the start and end date to send the scheduled messages.
End Date	Define the time to stop sending the custom message.
Send to	Select the accounts to whom you want to send the custom messages from the drop-down menu.

3. Click **Create User Message**.

Custom user message is created and listed in the User Messages pages. To update and modify the custom message, click the three ellipses next to the custom message.

Log in to the user interface to verify the custom message pop ups.

Working with instructor-led labs in VMware Lab Platform

You must assign the instructor role to a VMware Lab Platform user and associate the instructor with a lab.

The general workflow is as follows:

1. Using the Edit Account form, assign an existing user the role of instructor. Users assigned this role appear in a list accessible from the Create Class form. An instructor can monitor the progress of selected students (a class) taking a specified lab (instructor-led lab).
2. Using the InstructorLabMapping List to associate instructors with specific labs. By creating such instructor-lab associations, you can restrict instructors to access only specific labs. If you do not create any instructor-lab associations, then no instructor restrictions exist, and instructors are able to access any lab in the account.
3. Using the Create Class form, create a class. A class is composed of selected users (students) to take a specified lab, at a specified time and to be monitored by a specified instructor.

4. Instructor, by means of the Instructor Console, monitors the progress of each student in a class who has started the lab specified in the class.

Associate an instructor with a lab

Assign an instructor for a lab to the give the instructor full access to the lab.

1. To open the **InstructorLabMapping List**, click **Classes › Instructor-Lab Map**.
2. Click **Create InstructorLabMapping**. The **Create InstructorLabMapping** form opens.
3. Complete the fields of the form.

Account	Allows you to select a user from an account list. Select a user having an instructor role.
Labs	Allows you to select labs from a labs list. Select labs that you want to associate with the selected instructor.

4. Click **Create**. The system creates an association between the instructor you selected and the lab(s) you selected. This association restricts the instructor's access only to the labs you selected.

Create a class in VMware Lab Platform

Create a class for instructor-led labs with the **Create Class** form on the administrator user interface.

A class, consists of a lab and class attributes such as class time, instructor, and so on. When a class is created, the class listing provides an **Instructor Launch URL** and a **Registration URL**. Lab instructors or administrators can share the URL with students, which students can use to self register for the class. The **Instructor Launch URL** option links to the instructor console for the class. You can create a class with for a single lab or multiple labs.

1. From the navigation panel, click **Class Management › Classes** .
2. To open the Create class form, click **Create New**, and from the drop-down menu select **Single class**. To create bulk class upload, select the **Bulk Upload** option.
3. Complete the form. Required values are marked with an asterisk.

Basic Details	
Class Name	Specify a class name for your class. Text string associated to the class you are creating. An example is "Introduction to Network Security."
Unique Class ID	Specify an unique identifier for the class that you are creating. For example, Network Security101.
Class Instructor(s)	To display a list of users who are assigned the role of instructor, click the **Manage List View** icon. Select one or more instructors for the class.
Class Owner(s)	To display a list of users who are assigned the role of class owner, click the **Manage List View** icon. The class owner is the person who creates and maintains the class.
Student class started email	Select an email template to send to students when a class has started. Ensure that the **Send student class assigned email from email template** Portal Settings is activated. Create custom email templates using the **Email Templates** option.

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Student assigned to class email	Select an email template to send to students when a class is assigned to them. Ensure that the **Send student class started email from email template** Portal Settings is activated. Create custom email templates using the **Email Templates** option.
Instructor assigned to class email	Select an email template to send to an instructor when a class is assigned to an instructor. Ensure that the **Send an email to Instructor when assigned to Class** Portal Settings is activated. Create custom email templates using the **Email Templates** option.
Describe your Class	A brief description of the class.
Tracking Code	This option is visible only if the **Portal SettingsConfigurationEnable Tracking Code for Classes** setting is activated. Add a tracking code to track entitlements for classes. For large events, or multiple classes with same entitlements, you can track the entitlements based on the tracking ID associated with the class. **NOTE** You cannot modify the tracking code once a class is started.
Lab Details	
Lab	• **Single lab class**: Click the **Manage List View** icon on the right side of the text box and select a lab from the list. This is the lab that the students will take in the class. You can also select a lab from the **Recently created labs** section. • **Multiple labs for a class**: Click on all the labs that you want to add for the class. Or select labs from the **Recently created labs** section. Use the Actions on the Classes wizard to create entitlements for lab users. For more information on entitlements in the Classes wizard, see Single and Batch Action Options.
Create New Lab	To create a new lab for the class, click **Create New Lab**. For more information on creating a lab, see Create a Lab. After a lab is successfully created, it is added in the **Recently created labs** section.
Schedule	
Timezone	Specify a time zone for the location where the class is being held. Select a value from the drop-down menu.

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Date	Specify a start date and end date for the class given in local time for the class location. The date format is in YYYY/MM/DD. **NOTE** If the **Portal Settings › Configuration › Allow instructors to extend class end time** setting option is deactivated, the date range option is dimmed out for instructors in the instructor user interface. Instructors cannot modify or extend the class time once the class starts. However, administrators can modify the class time from the administrator user interface.
Time	Specify a start time and end time for the class given in local time for the class location. The time format is in HH/MM/SS.
Preferred cloud location for class resources	This option is available only if the **Portal Settings › Configurations › Show preferred cloud location dropdown in the classes wizard** tenant setting is activated. Use case scenario: • When the **Preferred cloud location for class resources** setting is activated; – If there is a prepopulated lab in a user's location, the lab is assigned to the user. – If there is no prepopulated lab in a user's location, then on-demand lab is deployed in the correct virtual cloud director. – In a class, if a user account has multiple clouds associated for the account and the same lab is deployed across all the cloud regions, you can choose the preferred location of the cloud to deploy the lab from. This option overrides the user's location set in the user account page. • When the **Preferred cloud location for class resources** setting is not activated; – In a class, if there is no preferred cloud location set, then all deployments are based on the user's location.
Students	
Maximum Limit for the class	Specify the maximum number of students allowed to register for a class.
Upload CSV	Create bulk student accounts and add them directly to the class.

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Add Student	To view the list of VMware Lab Platform users from which you can add students, click **Add Student**. A student must have an account in the system to be eligible to be added to a class.
Cloud Org VDC	Displays the name of the cloud org VDC for the user name.
Send New Token	To send a verification email to the user when a student is assigned to a class, click **Send New Token** from the ellipses drop-down menu. The purpose of the email is to validate that the user's email address is authentic.
Send Notification Email	To send notification email with information about the class to students, click **Send Notification Email**. The email can be customized in **Portal Settings › Configuration › Email › Student assigned to class email**. You can also send the quick launch URL for the class in the email. To auto-send the quick launch URL, activate the **Portal Settings › Configuration › Email › Send an email to Student when assigned to Class**.
Clone vApp	To clone a vApp for the class.
Remove	To remove a student from a class, click **Remove**.
Allow self-enrollment	Expand the **Opening your class to more students form** to view this option. Students can enroll themselves for a class.
Allow self-registration	Lets students without VMware Lab Platform accounts to create an account and self-register for the class. Students can use the **Register** option to create their VMware Lab Platform account. A registration URL is generated and sent to students to log in and enroll themselves in the class.
Video Conferencing	
Type	If a video conferencing application is supported for your organization, select the video conference option from the drop-down menu. Add the conference URL or meeting ID and password for the conference. **NOTE** Currently, VMware Lab Platform supports only Zoom video conferencing. To enable video conferencing for VMware Lab Platform, contact the support team.
Resource Management	
Pre-Provision Labs	Specify a point in time before the class start date when the system must start making prepops for the lab. A pre-provision period saves student's time to wait for a vApp to power on when the lab is launched. Move the slider button to activate this feature and specify the time to pre-deploy a lab and keep additional lab resources in reserve over and above the data

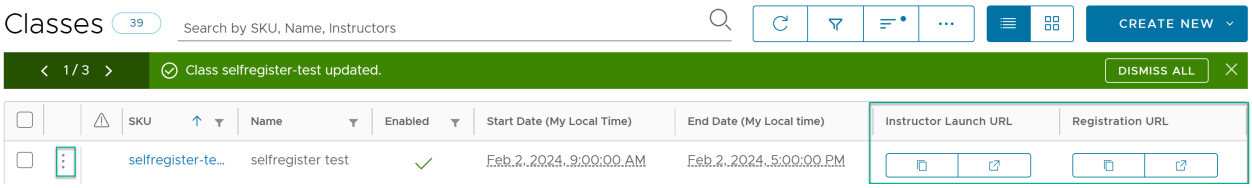
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	mentioned in the deployment pool. Resources for the entitlements are assigned from the deployment pool or the special class reserved resources. **NOTE** The **Pre-Provision Labs count** option is grayed out, or unavailable if multiple labs are selected for the class. You can pre-provision labs only for a single lab instance.
Skip provisioning of entitlements	Select the **Skip provisioning of entitlements** option, to ensure that the resources are not assigned to students unless the lab is launched. This feature is useful in saving resources and cost, as it doesn't allow students to launch a lab before the scheduled lab date and time. If a student tries to launch a lab before the scheduled date and time, a warning message is displayed on the user interface. If the **Skip provisioning of entitlements** option is selected, you cannot provision an entitlement from the **Classes** page.
Resource Saving Options	Specify resource management settings. Slide the **Resource Saving Options** button to enable saving of resources for a lab. You can suspend or power off a lab after the class ends each day to save the resources.
Publish	Provides a snapshot of approximate resource usage for the class based on labs, number of students, and duration of the class.

4. Click **Create**.
   The class is created and added to the Classes page. In the Classes listing page, the **Instructor Launch URL** and **Registration URL** columns have the options to “copy to clipboard” and “open in new tab”.

Figure 13: Classes listing view



A class instructor can now go to the Class page, select the class you created and start the instructor console. From the console, an instructor can monitor students of the class as they take the lab and even take over a given student’s lab console. For more information about the instructor console, see the Using and Managing the VMware Lab Platform Instructor Interface guide.

Optionally, you can continue and add students to the class roster, or manage entitlements for the class. To create entitlements, on the Classes page, click the three ellipses icon next to the class, and click **Create Entitlements**. If a user is assigned multiple labs, then entitlements for each of the labs is created.

NOTE
If a class does not have a lab template or deployment pool with a lab template, a warning message is displayed for the class on the **Classes** page.

View single and batch action options in VMware Lab Platform

The **Classes** page lists all the classes. You can view the classes in list view or in tile view. List includes a drop-down menu that lets you apply an action to a single class and a drop-down menu for actions you want to apply to multiple classes.

Single action options for a class

To modify a single class instance, select the class and click the ellipses next to the class.

Option	Description
More Details	Shows details for the selected class.
Update	Provides actions to edit or modify an existing class.
View Usage and Cost for Class	Provides cost details and usage cost.
Start	Creates, enables, and provisions the selected classes. Also sets the statuses of the selected classes to “Active”. Administrators can also edit the start time of a started class. The new start time of the class will be effective from the following day.
End	Sets the statuses of the selected classes to “Complete” which will archive them and prevent further updating.
Create Entitlements	Creates lab entitlements for students in the selected classes but does not enable them. Until activated, a student’s lab console will not list the entitlements in the Enrollments tab.
Provision Entitlements	Powers-up the vApps for each entitlement (manually triggered prepop) for the selected classes.
Enable Entitlements	Activate the created entitlements for the selected classes.
Recreate Completed Entitlements	Recreates a completed entitlement.
Send Notification Email	Sends notification email to all students. You can modify the content of the email from **Portal Settings › Configuration › Email**.
Duplicate	Creates a replica of a class along with the details such as students, instructors, lab and so on. When you duplicate an existing class, a new SKU Id is added to the duplicated class. You need to set the start and end date, and time for the duplicated class.
Open Instructor Interface	Opens the instructor interface for the class.
Delete	Deletes the selected classes form the Class List.

Batch action options for classes

The Class List includes Batch Actions that lets you to update/modify multiple classes. First, select the classes you want to take a batch action upon, then click the ellipsis to open the drop-down menu.

Actions	Description
Start	Creates, enables, and provisions the selected classes. Also sets the status of the selected classes to “Active”.
End	Sets the status of the selected classes to “Complete” which will archive them and prevent further updating.
Create Entitlements	Creates lab entitlements for students in the selected classes but does not enable them. Until activated, a student’s lab console will not list the entitlements in the **My Enrollments** tab.
Provision Entitlements	Powers-up the vApps for each entitlement (manually triggered prepop) for the selected classes.
Enable Entitlements	Activates the created entitlements for the selected classes.
Send Notification Email	Sends notification emails to the students enrolled for the selected classes.
Delete	Deletes the selected classes.
Download page	Exports the Classes page information in an excel sheet format to your local system.

Modify a class in VMware Lab Platform

As an instructor, you can modify a class from the **Class Management › Classes** page.

- The following settings must be activated, if you want to modify or update a class.

  Allow Instructors to Manage Lab and Entitlements of a Class

  Allow Instructors to Clone vApps

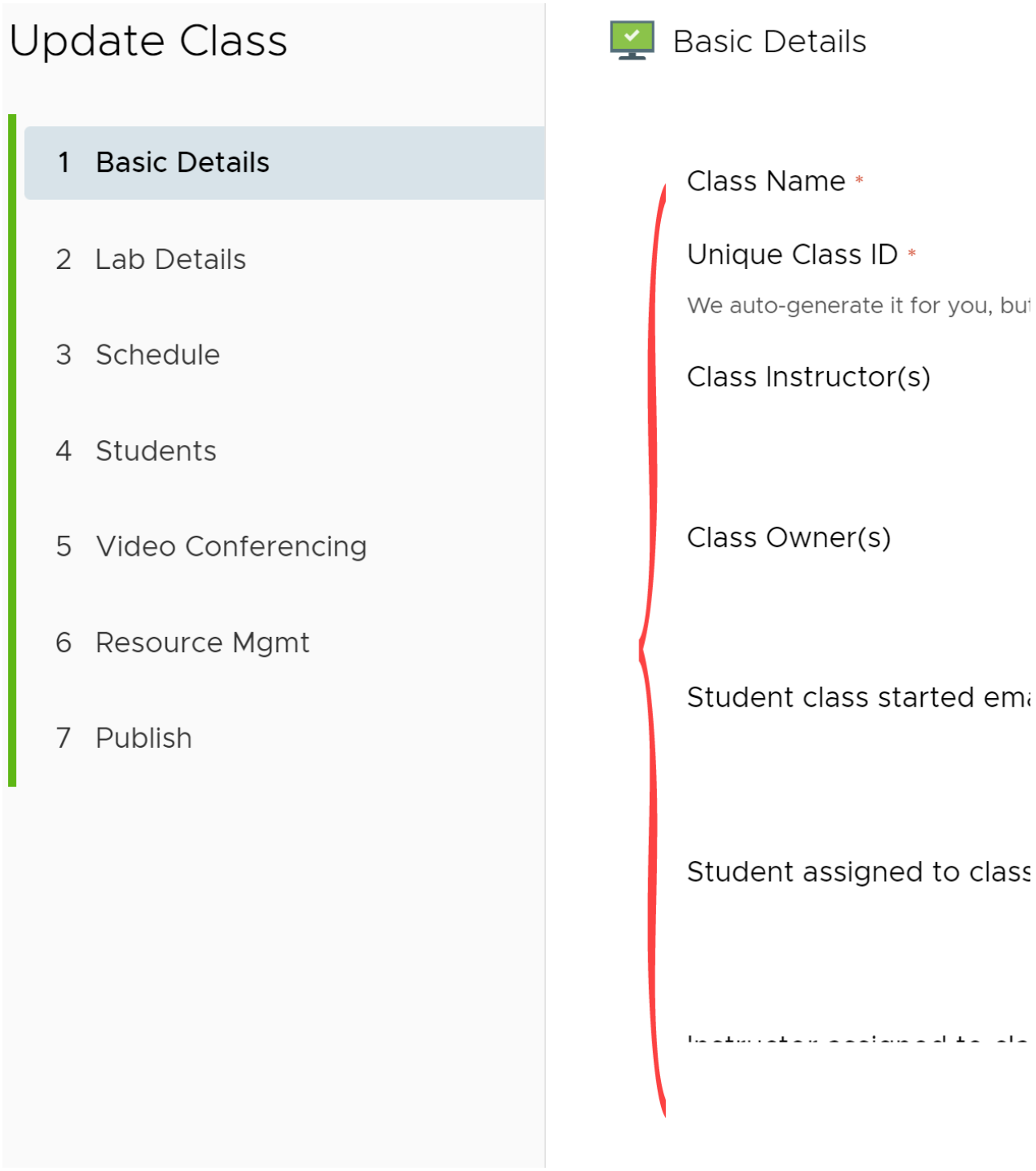
For information about other instructor-related settings, see Working with settings in VMware Lab Platform

You can modify settings for a class including adding or removing students, managing labs or entitlements, modify the start time of a started class, or cloning a vApp.

NOTE
Each class can have one lab, and the lab cannot be changed after the class is started.

IMPORTANT
If the class end date or time is changed, then the Entitlement expiration date is automatically updated to the same. For more information, see Managing Lab Entitlements.

1. To display a list of available classes, from the navigation panel, click **Classes**. If you have activated Advanced Options, select **Class Management › Classes**.
2. To update a class form, click a class name, and click **Update**.



3. In the **Update Class** form, edit information for your class, and click **Update Class**. If you modify the time and date values of an already started class, the modified values will be effective from the next day.
   The updated class is listed in the **Classes** section.

Working with the exam module in VMware Lab Platform

To grade the entitlements (users) on completion of labs, administrators, instructors, proctors can use the exam feature. You can define the exam in VMware Lab Platform and add various grading options in the system. Administrators can customize email templates to send the exam report to users. To enable the exam menu option, activate the **Portal Settings › Configurations › Exam Menu Enabled** option.

The grading statuses for an exam are:

- Not graded: Exam is not evaluated and there is no grade.
- Pass: Exam is evaluated and the criteria for passing the exam are met.
- Fail: Exam is evaluated and the criteria for passing the exam are not met.
- Error: An error occurred during the automated exam evaluation process.

Create an exam definition in VMware Lab Platform

To grade an entitlement for an exam, you need to create an exam definition. Administrators create exam definition, which contains a grading script, an exam code, a lab, and email templates for different grading statuses. You can customize these email templates.

To associate an exam with a lab, select the **Lab Management › Labs › Update lab › Require grading** option.

To create an exam definition:

1. To open the Exam Definitions page, from the navigation panel, select **Exams › Exam Definitions**.
2. Click **Create New**. The Create Exam definition form opens.
3. Enter details about the exam definition that you want to create.

Name	A descriptive name for the exam in human readable format. This field has a character limit of 255.
Lab	Associates an existing lab. When an user takes the exam, an instance of this lab is deployed .
Exam Code	An inventory code for the exam. A free form string. You can also cross-reference exams between VMware Lab Platform and external systems.
Enabled	If selected, an exam can be created from the definition.
Suspend Exam vApps after grading	If selected, the VMware Lab Platform immediately suspends the exam vApps after grading the exam.
Grading Script	A grading script that is fetched from the lab during an exam grading phase and based on the script, evaluates whether the exam's conditions have been met or not. **NOTE** VMware Lab Platform does not send or execute this script in the lab automatically. Administrators need to automate a system to fetch and execute this script. For information on the API calls for the grading script, click **Support › REST API Guide**.
Duration (minutes)	Default duration of an exam. Can be overridden in individual Exams.
Send email on grading	If activated, when the grade of an exam is changed, an email is sent to the user taking the exam.
Success Email Template	Email template to be used when an exam is graded as PASS.

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Override Success Email Template	Overrides the success email template and lets administrators to create a new email template and use it.
Failure Email Template	Email template to be used when an exam is graded as FAIL.
Override Failure Email Template	Overrides the failure email template and lets administrators to create a new email template and use it.
Error Email Template	Email template to be used when an exam is graded as ERROR.
Override Error Email Template	Overrides the error email template and lets administrators to create a new email template and use it.
Persist Period Success	Define the time period for how long the lab instance must be kept running after an exam has been graded as PASS.
Persist Period Failure	Define the time period for how long the lab instance must be kept running after an exam has been graded as FAIL.
Persist Period Error	Define the time period for how long the lab instance must be kept running after an exam has been graded as ERROR.

4. Click **Create**.

The new exam definition is created and listed on the Exam Definitions page.

Create an exam in VMware Lab Platform

To create an exam:

1. To open the Exams page, from the navigation panel, select **Exams › Exams**.
2. Click **Create New**.
3. In the **Create Exam** form, enter the following details:

Account	Add a user account to associate with the exam.
Exam Definition	Add the exam definition for the exam. For more information on exam definition, see Create an Exam Definition.

4. Click **Create Exam**.
   The new exam is listed on the Exams page.

Update an exam definition in VMware Lab Platform

As an administrator, you can edit or modify the exam definitation for an exam in VMware Lab Platform.

To update an exam definition:

1. To open the Exam Definitions page, from the navigation panel, select **Exams › Exam Definitions**.
2. Select the exam definition from the list that you want to update. Click the three ellipses, and click **Update** from the list of options.

More Details	Provides a snapshot view of the exam definition.
Update	Lets you to update the exam definition form.
Activate	Activates the exam definition form for a user.
Deactivate	Deactivates the exam definition form for a user.

3. Update the form and, click **Update**.

Batch actions for exam definitions in VMware Lab Platform

The Exam Definitions list includes batch actions that lets you to update or modify multiple exam definitions.

Select the exam definitions that you want to perform a batch action, and select any of the following options:

Option	Description
Activate:	Activates the exam definitions for an exam.
Deactivate:	Deactivates the exam definitions for an exam.
Download page	Exports the page information in a Excel format to your local system.

Single and batch actions for exams in VMware Lab Platform

The Exam page lists all the exams. You can view the exams in a list view and perform an action for a single exam or multiple exams.

Single actions

Select an exam, and click the three ellipses to view the action options.

Option	Description
More Details	Provides detail view of the exam.
Entitle	Creates an entitlement for the user from the lab without deploying a vApp. Sets the exam status to Entitled.
Make Ready	Creates an entitlement if it does not exist and makes it ready be deploying a vApp. Sets the Exam status to Ready.
Start	Ensures that there is an entitlement with running vApp and starts the exam. Starts the countdown for the duration of the exam. Sets the Exam status to Running.
Archive	Archives graded exam entries. Discarded exams are deleted. Non graded exams are left untouched.
End	Ends an exam. The grading script for the exam can be retrieved. Sets the exam status to Grading.
Grade	Lets administrators to grade an exam. Administrators or instructors can add notes for the exam in the Notes section. You can manually add keys for questions and answers for a manually graded exam. **NOTE** You can manually grade exams that are not graded by the exam script.
Change Account	Lets you to change the account of the user entitled to take the exam.
Send email report	Sends an email report to the linked account when the exam is graded.
View grading report	Displays the grading report the selected exam.
Export grading report to PDF	Exports the grading report in PDF format. Administrators can download the file from the local system.

Bulk actions

Archive	Archives selected exams. **NOTE** Only graded exams can be archived.
Send email report	Sends an email report about the exam to the users. **NOTE** Only graded exams report can be emailed.
Download page	Exports the exam page information to the user's local system.

NOTE
You can upload a list of exams with the bulk upload operation.

Create exam email templates in VMware Lab Platform

You can customize the email templates for an exam.

To customize emails for success, fail, or error reports:

1. Open the **Exam Email Templates** page, from the navigation panel, select **Exams › Exam Email Templates**.
2. Click **Create Exam Email Template**.
3. Fill the **Create** form.

From	Add sender's email address to be added in the email message.
Subject	Subject of the email message. You can substitute entries—[firstName], [lastName], [username], [examGrade], [examScore], [examName] along with any of the question entries.
Body	Body of the email message. You can substitute entries—[firstName], [lastName], [username], [examGrade], [examScore], [examName] along with any of the question entries.

Working with Content Engine in VMware Lab Platform

You use the VMware Lab Platform Content Engine interface to import and manage manuals used with hands-on labs.

Instructional content

Typically, a technical course instructor or a subject matter expert (SME) create instructional content for a lab using an authoring tool such as ScreenSteps or Microsoft Word. After the content is created and in some scenarios content is translated, you use the Content Engine interface to import the technical content into VMware Lab Platform.

After you import your manual and manual translations, you can make modifications with the Content Engine editor.

Language support

You can use the Content Engine to make multiple, translated drafts of a manual and add them to your customer account. When an end user starts a lab, the lab console defaults to the language version according to the user's language preferences. Language preferences are set using the My Profile option on the lab console. For more information on language preferences, see the *VMware Lab Platform User Guide*.

Content Engine workflow

To add a lab manual:

1. Create a lab manual project. Think of a lab manual project as a kind of container for lab manuals.
2. Upload or import instructional content. The result is an instance with an imported status. You can upload instructional content from your local machine or import it from the ScreenSteps authoring tool.
3. Optionally, upload or import additional content in different languages. This language support is one of the most useful features of the Content Engine.
4. Promote an imported instance to be a draft. A draft is an intermediate status that allows you to check instructional content before publishing.
5. Publish drafts to make them available to your administrator. Once available, an administrator can then use the Admin UI to associate a manual to a lab.

For more information on how to use the Content Engine interface, see the VMware Lab Platform Content Engine Guide.

Working with survey configuration in VMware Lab Platform

The Survey Configuration feature allows you to roll out custom surveys to end users in the VMware Lab Platform user interface. You create surveys using third-party survey tools, and add the survey link in the VMware Lab Platform and roll out the survey to the lab users.

Enable the **Portal Settings › Configuration › Show Surveys page** setting.

To create a survey configuration for a lab:

1. Navigate to **Content › Survey Configuration**.
2. Click **Create New**.
3. Add the required details in the Survey Details form.

Provider	Enter a unique name for the survey.
URL	Add the survey link.
Image	(Optional) You can upload an image file and use it as a backdrop to add your survey link.
Enabled	Select the **Enabled** check box to make the survey link available to lab users.
Auto Display	Select the **Auto Display** option, to auto display the survey link when a user ends or completes a lab.
Show Survey Promotion Page	Select this option to display promotion pages associated with the survey.
Survey Promotion Page Content	Add content to display in the Survey Promotion Page on the user interface.

4. Click **Create Survey**.

After the survey configuration is created, the survey details are listed on the Surveys page. To edit, or delete the survey, click the three ellipses next to the survey name and perform the required action.

To associate a survey with a lab, use the **Lab Management › Labs › Lab Content** page. For more information on how to add a survey to a lab, see Create a Lab.

Administering events in VMware Lab Platform

VMware Lab Platform provides an event management user interface for administering and managing large-scale events.

The event management user interface allows user roles such as, administrators, proctors, and help desk staff to manage lab station registration and seating maps.

For more information, see the VMware Lab Platform Event Management Guide.

Working with Account Lab Mapping in VMware Lab Platform

The Account Lab Map option allows you to associate a proctor to a lab and promote that proctor to a captain. A captain is typically the person who wrote, or co-wrote the lab, and has an extensive knowledge of the content.

Create an account lab map in VMware Lab Platform

To associate a proctor to a lab or promote a lab proctor to lab captain, administrators can use the **Account Lab Map** option.

1. Select **Events › Account Lab-Map**. The Account Mapping List opens.
2. To open a form, click **Create Account Lab Mapping** .
3. Complete the form to associate a proctor to a lab and promote that proctor to a captain. Select **Alert** to send an alert to the user you promoted.
4. Click **Create**.

Working with Lab Stats in VMware Lab Platform

You can view and reset the ratings for a single lab or a collection of labs from the **Lab Stats** page.

Managing existing Lab Stats

To open the **Lab Stats** page, click **Events>Lab Ratings**. You can sync, view details, or delete any lab on the list of the **Lab Stats** page.

1. Click the menu icon next to a lab shown on the **Lab Stat** list. A drop-down menu opens.
2. To sync, view details, or delete the lab rating information, click an option.

Clearing or syncing all Lab Stats

You can clear or sync all lab ratings by clicking **Clear All Ratings** or **Sync All Ratings** on the **Lab Stats** page.

Working with the Event ID list in VMware Lab Platform

An event ID is a unique key on a badge for use at an event. You can use the Event ID List to view or delete Event IDs.

1. To create a new event Id, click **Create EventId**.
2. Enter the details in the Create EventId form.
3. Click **Create**.

Modifying an event ID

To modify an event Id:

1. To open the Event ID List, click **Events › Event ID**.
2. Click the • • • icon along an existing event ID on the Event ID List. A drop-down menu with options to view details or delete the event ID opens.
3. To delete all event IDs, click **Delete All**.

Working with an authentication provider for single sign-on in VMware Lab Platform

As an administrator, you can set up various third-party authentication providers to access the VMware Lab Platform administrator user interface and the end-user interface. The authentication integration works with Facebook, Google, VMware Workspace One, and every certified OpenID provider. You can configure multiple authentication providers, and users can use any one of the authentication provider listed on the login page to log in to the VMware Lab Platform interfaces. You can add an authentication provider from **Users & Groups › Authentication Providers › Create New**, and associate the provider with a user account from **Users & Groups › User Accounts › Account Details** page.

The **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options in the administrator user interface lets you to add different authentication providers for users.

- **Allow Student SSO Login**: This option lets administrators to provide a non-local authentication provider to users to login to the end-user interface.
- **Disable Student Local Provider Login**: This option lets administrators to deactivate a non-local authentication provider option from the end-user interface login page.

With the **Allow Student SSO Login** and the **Disable Student Local Provider Login** setting options, you can have the following four scenarios for the authentication provider setting in the end-user interface:

- VMware Lab Platform login only: Deactivate both the Allow Student SSO Login and the Disable Student Local Provider Login setting options.
- SSO only login: Activate both the Allow Student SSO Login and the Disable Student Local Provider Login setting options.
- SSO and VMware Lab Platform login: Activate the Allow Student SSO Login option, and deactivate the Disable Student Local Provider Login setting option.
- No direct login: Deactivate the Allow Student SSO Login option, and activate the Disable Student Local Provider Login setting option.

Add an authentication provider in VMware Lab Platform

Ensure that the default authentication provider that you want to use for your logins is added in the **Portal Settings › Configurations › General Options › defaultAuthenticationProvider** settings.

To add an authentication provider in VMware Lab Platform:

1. From the navigation panel, click **Users & Groups › Authentication Providers › Create New**.
   The Create Authentication Provider form is displayed.
2. Enter the required details and the OpenID configuration URL for the authentication provider in the Create Provider form.

Name	Enter a name for your authentication provider. This same name will be displayed when logging in the Administrator user interface.
Visible	If you activate this option, this authentication provider option will be visible to users on the login page.
Client ID	Enter the authentication provider client ID.
Client secret	Enter the client secret of your authentication provider.
Open ID Configuration URL	Enter the authentication provider's open ID connection URL.
Scopes	Add the scope values supported by the authentication provider. For example, profile, email, address, and Open ID.
First Name Claim	Enter a value for the 'First Name' claim. The value of this claim is mapped to the user accounts' first name.
Last Name Claim	Enter a value for the 'Last Name' claim. The value of this claim is mapped to the user accounts' last name.

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Company Claim	(Optional) Add company's name. If left blank, the default value, 'Unknown' is used.

3. Click **Create Provider**.

All the authentication providers associated with your account are listed in the Authentication Providers page. For more information on the settings for authentication providers, see the Working with Settings topic.

Modify the password policy for a local user

You can efficiently manage the password policy settings such as password expiry, password length, password for users using the local authentication provider.

Administrator user role access and privileges.

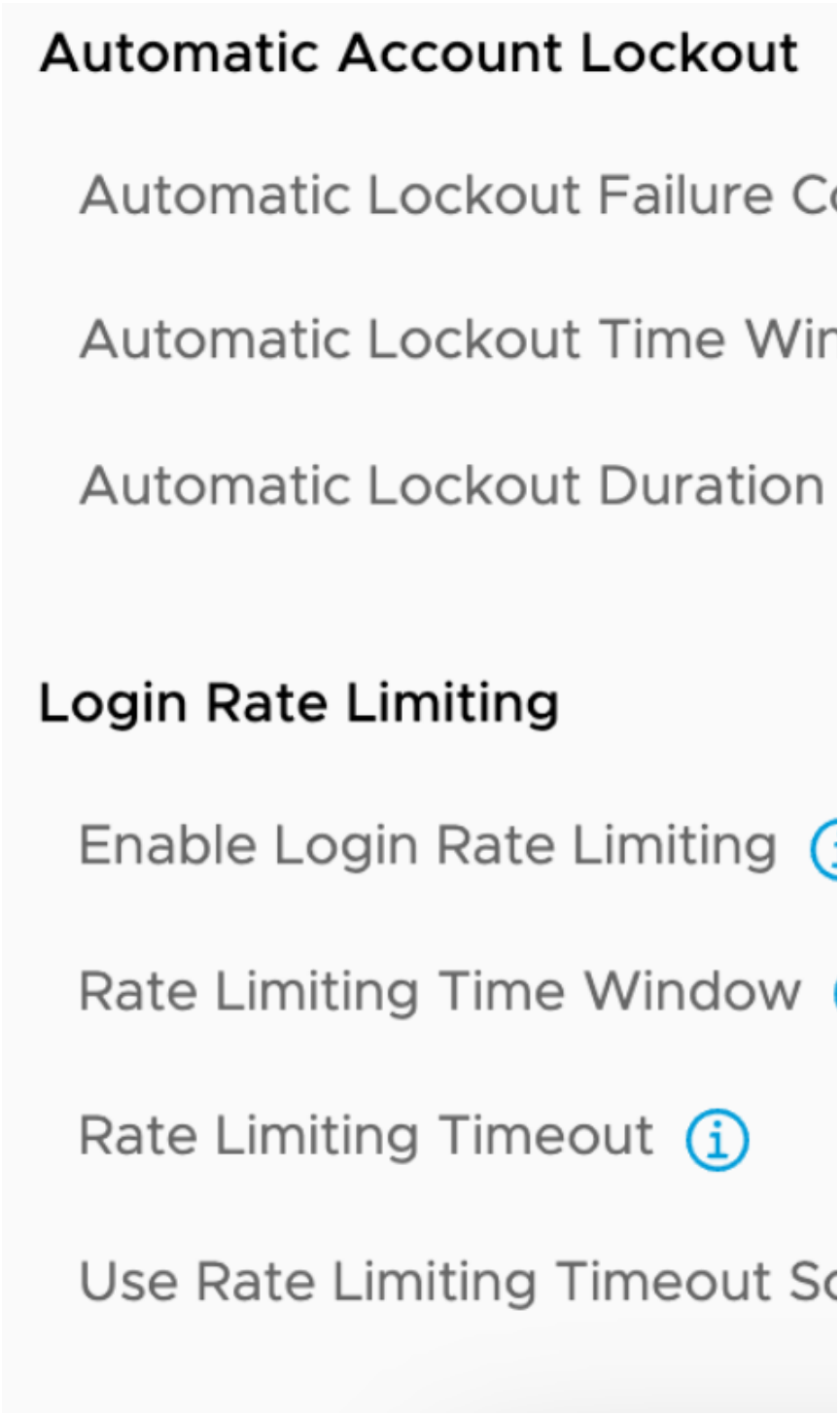
To set or modify password policies for local authentication provider (that is **VMware Lab Platform** authentication provider):

1. From the navigation panel, click **Users & Groups › Authentication Providers**.
2. Click the three ellipses icon next to the **Local** authentication provider, and click **Update**.
3. You can set or modify the following password policy information:

Strong Passwords	If you select the option, users need to set a password that must have at least one uppercase character, one digit, and one symbol character in the password. By default, the Strong Passwords option is deselected.
Password Length	Set the length for password. The default value is 8 characters.
Previous Passwords	The number of times that a user cannot use a previous password as a new password. The default value is set as 0.
Password Expiration	Number of days before a password expires and a password change is required. **CAUTION** Setting the value to 0 disables the password expiration check and allows users with expired passwords to login.
Automatic Account Lockout Failure Count	The number of failed authentication that is allowed before the account is locked. By default this feature is deactivated. Set a value more than 0 to activate the feature. As a best practice, we recommend setting the value to 5. Setting a value for **Automatic Account Lockout Failure Count** prevents a malicious user from trying many different passwords to login to an account. It locks an account if too many failed login attempts occur. The lockout can be temporary or permanent.

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When activated, the following options are displayed:

Automatic Lockout Failure Count

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Specifies the number of failed login attempts before the system locks out the user.

- The number of failed authentication attempts allowed before an account is locked out. After a successful login, the failed login counter for the account is reset.
- The default value is 0 (deactivated). The recommended value is 5.
- A value of 0 deactivates this feature altogether.
- If this feature is enabled (value greater than 0) and then is deactivated (value is 0), any accounts temporarily locked out is unlocked. However, any permanently locked accounts will stay locked.

Automatic Lockout Time Window

Specifies the time period in seconds for failed login attempts specified in the **Automatic Lockout Failure Count**option.

- The time window during which the authentication attempt failure-count triggers a lockout. The time window starts from the date/time of the first failed login attempt for an account.
- Recommended value is 900 seconds (15 minutes)
- A value of 0 sets the time window to unlimited.
- If the **Automatic Lockout Time Window** is set to unlimited (a value of 0), and the user hits the **Automatic Lockout Failure Count** number of failed logins over any time period (without a successful login), the account is locked.
- After the **Automatic Lockout Time Window** period passes, the failed login attempt counter on the account is reset to 0.

Automatic Lockout Duration

Specifies the temproray lockout period of an account if a user has crossed the number of failed login attempts specified in the **Automatic Lockout Failure Count**option.

- The duration for which an account is locked out.
- The recommended value is 900 seconds (15 minutes)
- A value of 0 locks the account permanently.
- If not permanent, after the specified duration has elapsed, the account is unlocked and a login can be attempted again.

To unlock a failed account:

- If the lockout was temporary, wait until the **Automatic Lockout Duration** is over.
- User can use the **Forgot Password** option to regenerate a new password in the user interface.
- You can send new token in a new verification email from the User Accounts page. User can click the verification link from that email.
- You can use the **Unlock Account** option in the **User Accounts** page for the user and unlock the account.

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	• If the lockout is permanent (**Automatic Lockout Duration** is set to 0), toggle the **Account Locked** option on the **User Accounts** page table, or from the **User Access** page. Click **Update Account** to save the changes.
Enable Login Rate Limiting	When enabled, it prevents repeated attempts to login the account with invalid credentials. The **Login Rate Limiting** option is similar to the **Automatic Account Lockout** option and can be used in along with **Automatic Account Lockout**. The **Login Rate Limiting** option is used to throttle too many login attempts that occur with in a specified time window. When enabled, the following options are displayed: **Enable Login Rate Limiting** • **Login Rate Limiting** prevents repeated attempts at logging in to an account with invalid credentials. • Default value: deactivated. **Rate Limiting Time Window** • The time window during which the authentication attempt failure-count triggers rate limiting. • The default value is 3600 seconds (one hour). • A value is 0 is unlimited (really 720 days behind the scenes). • The time window starts from the most recent failed login attempt, meaning that it keeps getting updated after every failed login attempt, which is different than how it works in **Automatic Lockout Time Window**. **Rate Limiting Timeout** • Specifies how long a user must wait before attempting another authentication attempt after a failed authentication. • The default value is 10 seconds. **Use Rate Limiting Timeout Scaling** • The **Rate Limiting Timeout Scaling** uses the **Rate Limiting Timeout** to increase the wait time for each failed authentication attempt, up to 24 hours. The initial 3 attempts are not rate-limited. • The default value deactivates the service. • This option is provided, if a non-malicious user attempts to login and mistyped their password will not be throttled in any way. It allows the first 3 failed attempts to happen without any throttling at all and then the time out starts to grow for every group of 3 failed attempts after that. It uses the **Rate Limiting Timeout** value for how much the timeout grows by.

4. To save your changes, click **Update Authentication Provider**.

You can view the details of the authentication provider using the **Users & Groups › Authentication Providers** page. Click the three ellipses icon, and click **More Details**.

Working with email templates

To create a custom email template for exams and classes to send to students and instructors, administrators can use the Email Templates feature.

Create an email template

1. Navigate to **Portal Settings › Email Templates**, click **Create New**.
2. On the Create Email Template page, enter the email template details.
3. Select the category type from the Category drop-down menu.
4. Enter the contents for the custom email in the body text field.
5. Click **Create Email Template**.

The email template is added to the list on the Email Templates page.

Monitoring tasks

Administration includes monitoring tasks that are running, queued, on-hold, or have failed for your cloud org vDC.

To open a page that displays all long-running tasks, click **Task Management**. A long-running task is any task that does not return data instantly. The Task Management page allows you to ensure that accounts are not over-burdening the cloud. Task counts and task lists are dynamically updated.

Downloading grid entries

In VMware Lab Platform you have the option to download the grid data on most of the pages. You can either download the grid data on the page that is currently visible or download all data.

Download grid entries

1. Navigate to any grid page.
   For example navigate to **Lab Management › Badges**.
2. Click on the **vertical ellipsis** drop-down in the header row and select **Download current page** or **Download all pages**.
   Generate CSV file modal opens.
3. Select **Only properties in the view** or **All properties on record** option and click **Generate CVS**.

   The option **Only properties in the view** generates a CSV that contains only the columns that are currently visible on the grid. If you want to include all columns you must select **All properties on record**.

The generated CSV file is downloaded in your browser.

Advanced administrator topics

Working with cloud organizations

If you have opted out from the VMware Lab Platform cloud and opted for Bring Your Own Cloud (BYOC) subscription, you must create a cloud organization and connect your virtual cloud director instance to VMware Lab Platform.

Port and proxy requirements

VMware Lab Platform requires API access to the virtual cloud director URL running your HTTP service. This is configured to run on port 443. Console access is directed through the console proxy, which runs on port 8443 on the same IP as the HTTP service.

The IP address configured as the console proxy endpoint must not be located behind an SSL-terminating load balancer or reverse proxy. All console proxy requests must be relayed directly to the console proxy IP address. For an installation with a single IP address, you can customize the console proxy address from the Service Provider Admin Portal. For example, for the VMware Cloud Director service, you must customize the console proxy address to `vcloud.example.com:8443`.

Understanding the definitions of VMware Lab Platform actions

When setting up and managing cloud organizations, it is important to understand the definitions of VMware Lab Platform actions and how these actions interact with the cloud.

Deployment

In VMware Lab Platform, deployment means creating a new vApp in the cloud, either as a prepopulated vApp or as an on-demand creation.

Assignment

Assignment is the action of assigning an existing prepopulated vApp to an entitlement that is starting for the first time or that has its vApp reset. These settings specify a baseline for determining whether VMware Lab Platform can use the cloud for the action. If an action is deactivated for the cloud org, then the organization is not used for this action regardless of any other settings. If the action is activated for this cloud, then the settings on the VMware Cloud Director and/or template are used to determine if the action can proceed.

Action Enablement

VMware Lab Platform determines whether a cloud is activated for a specific action by following a series of steps, in a specific order, where each step checks one setting. If the setting for the cloud is deactivated for the action, then the cloud is deactivated for that action. Only if all the steps are completed, with no step indicating that the action is deactivated, does VMware Lab Platform consider the action activated. So, if an early step in the process indicates that an action is deactivated, then no subsequent step in the process can make the action activated.

Add a cloud to VMware Lab Platform

If you opt out from the VMware Lab Platform Cloud and choose a BYOC subscription, you must first connect your organization's virtual cloud director instance to VMware Lab Platform before you can create labs.

> **NOTE**
> The VMware Cloud Director term cloud organization, or cloud org, is equivalent to the VMware Lab Platform term cloud.

1. To display the Clouds page, from the navigation panel, click **Cloud Management › Clouds**.
2. On the Clouds page, click **Create New**. The **Create cloud** tab of the Cloud form is displayed. Enter details for the cloud organization that you are creating.

Tenant Assignment and **Tenant Deployment**	Both apply only to the current customer account and does not affect other accounts using this cloud org. Their settings only have meaning if **Enabled** has been set to true and the corresponding activity setting is also set to true (for example, Tenant Assignment only has meaning if Assignment has been set to true). Both can be used to deactivate deployments or assignments for a

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	specific account, without affecting the settings for other accounts.
Tenant Maintenance Mode	Activates or deactivates maintenance mode for the cloud organization. This value is specific to the customer account, that is, it affects only your account. This option is useful when you do not want activity in the selected cloud organization (for example, due to problems or other issues or an upgrade).
Use Edge Gateway	Activates the NSX-T organization network connections and service settings (DHCP, firewall, and routing).
Edge Gateway DNS	Name of the edge gateway domain name server.
Org Name	Name of your cloud organization. Organization names are case sensitive.
Host Name	Host name for your organization's VMware Cloud Director (Example: `abc-vclass1.xyzedu.com`). **NOTE** VMware Lab Platform confirms that the host you specified has a certificate, but does not validate the certificate.
Org Admin	Administrator user name.
Password	Password for the VMware Cloud Director cloud org administrator.
Display Name (optional)	Name to display in the list of cloud organizations. Enable display name by clicking the gear icon on the Cloud Org page and selecting **Display Name**.
Location	Select a location from the drop-down menu.
Console Connection Monitoring	Select the check box to enable console connection monitoring for this organization's labs. With console connection monitoring, students and instructors can monitor connection health when a lab is in use.

3. Enter the details in the form, click **Create Cloud**.
4. On the **Cloud Org Vdcs** tab, you can view the Cloud Org VDC details.
5. On the **Cloud Catalogs** tab, enable at least one catalog for the cloud org.

   A catalog is a collection of lab templates. Activated means that the templates contained in a catalog are available to VMware Lab Platform. VMware Lab Platform requires access to templates to build a lab.

6. Click **Update Cloud**.

The **Clouds** page displays the new cloud added in the list.

Cloud login for VMware Lab Platform

VMware Lab Platform enforces limits on cloud login failures and notifications of failure.

When a cloud login attempt fails, VMware Lab Platform does not attempt another login for one minute. Such cloud login failures can occur in many cases, for example when an administrator performs administrative tasks or when an end user tries to take a lab.

Most cloud login failures trigger a notification to the customer. VMware Lab Platform throttles such notifications, for example, it only sends one notification within any ten-minute period. Multiple login failures within this ten-minute period do not trigger multiple notifications.

VMware Lab Platform enforces both these limits on a per-node instance and typically runs multiple nodes. As a result, it is possible that it will send more than one notification in a ten-minute period if, for example, different nodes experience a failure during that time.

Manage cloud organization actions

You manage the cloud organizations you create from the **Clouds** page.

1. From the navigation panel, click **Cloud Management › Clouds**.
2. Select the check box for the cloud organization that you want to update.
3. Click the three vertical dots icon to open a drop-down menu and choose an item from the menu.

Menu Item	Description
More details	Displays a list of properties for the cloud organization.
Update	Displays a form from which you can update cloud organization properties, view and edit cloud org virtual data center properties, and assign or create a cloud catalog.
Remove from Tenant	Removes the selected cloud org from the customer account.
Sync Cloud vApps	Syncs the selected cloud vApps. For more information, see vApp Sync from VMware Cloud Director.
Sync Lab Templates	For all of the cloud's templates, the sync operation checks the cloud catalog version and syncs only those catalogs that have changed.
Full Resync Lab Templates	For all cloud templates, syncs all templates without checking the cloud catalog first.
Update Cloud Org VDC capacity	Updates the record of cloud usage (cloud capacity) for each virtual data center in the selected cloud org.
Delete Prepops without templates	Deletes and removes the prepops that do not have a lab template in a cloud. This action allows the system to properly maintain the cloud.
Invalidate Connection	Removes cached connection in cloud engine and force login on next connection.

Perform batch cloud organization actions

You can manage and update multiple cloud organizations from the **Clouds** page.

1. From the navigation panel, click **Cloud Management › Clouds**.
2. Select the check boxes for all the cloud organizations that you want to update.
3. Click the three vertical dots icon on the top of the table to open a drop-down menu and choose an action item from the menu.

Menu Item	Description
Sync Lab Templates	For all of the cloud's templates, the sync operation checks the cloud catalog version and syncs only those catalogs that have changed.
Full Resync Lab Templates	For all of the cloud's templates, syncs all templates without checking the cloud catalog first.
Update Cloud Org VDC capacity	Updates the record of cloud usage (cloud capacity) for each virtual data center in the selected cloud org.

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Menu Item	Description
Delete Prepops without templates	Deletes and removes prepops that do not have a lab template in a cloud org. This action allows the system to properly maintain the cloud.

Using cloud organization virtual data center settings

You can manage settings for a cloud organization virtual data centers (VDCs) from the VMware Lab Platform administrator user interface. As an administrator, you can edit settings that affect only your account. Only VMware Lab Platform technical staff can modify settings that affect all customer accounts.

To view the Cloud Org VDC page, click **Cloud Management › Cloud oVDCs**.

The **Enabled**, **Deployment**, **Assignment**, **Tenant Deployment**, and **Tenant Assignment** options control the deployment and assignment for the selected cloud organization VDC.

Deployment means deploying a new vApp either as a prepopulated vApp or as an on-demand deployment. An assignment means assigning an existing prepopulated vApp to an entitlement once a user starts the lab.

To display the information option, use the **Manage Columns** button in the lower-left corner of the page. The following settings are displayed by default.

Setting	Description
Name	Name of the cloud.
Cloud Org	Name of the cloud org VDC.
Description	Short description provided for the cloud organization VDC.
Tenant Deployment Enabled	This option applies only to the customer account and does not affect other accounts using this cloud organization VDC. The settings have meaning only if **Enabled** is set to True and the corresponding activity is set to True. For example, Tenant Assignment only has meaning if **Assignment** is set to True. You can use both to deactivate deployments or assignments for a specific account without affecting the settings for other accounts.
Tenant Assignment Enabled	Similar to **Tenant Deployment Enabled** option. The settings have meaning only if **Enabled** is set to True and the corresponding activity is set to True.
Tenant Maintenance Mode	Activates or deactivates maintenance mode for the selected cloud org VDC(s). This field is specific to an account, that is, it affects only your account. This field is useful when you do not want activity in the selected cloud org VDC (for example, due to problems or other issues or an upgrade).
Display name	Enter and save a display name for the cloud org VDC.
Enabled	Indicates whether a cloud org VDC can be used. If set to True, then the system does not use the cloud org VDC for deployments or assignments regardless of any other settings. VMware Lab Platform technical staff activates or deactivates this setting.
Deployment Enabled and **Assignment Enabled**	Both are active only if **Enabled** has been set to True. These fields activate or deactivate an assignment or a deployment for all customer accounts. If both are set to False, the result is the same as **Enabled** being set to False. VMware Lab Platform technical staff activates or deactivates these settings.

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Setting	Description
Maintenance Mode	Activates or deactivates maintenance mode for the selected cloud org VDCs and affects all accounts. If set to True, the system does not deploy any new vApps in the cloud org VDC and does not use any existing vApps from the cloud org VDC. Only VMware Lab Platform technical staff activates or deactivates this field. This field is useful to deactivate a cloud org VDC for maintenance.
Preshadow Enabled	If activated, it automatically creates a shadow of a template at VCD.
Date Created	Lists the date when a cloud org VDC was created.
Last Updated	Lists the last updated date of a cloud org VDC.

You can also modify the fields for Urn, usage threshold, CPU allocated and used, memory allocated and used, storage threshold, and vCPU values.

You use the three vertical dots on the right for each cloud org VDCs to drain all prepops, see full details, or delete a cloud org VDCs.

Examples

Example 1: To enable assignment for a cloud org VDC, **Enabled**, **Assignment**, and **Tenant Assignment** must be set to True. If any one of these is set to False, then the system cannot use the cloud org VDC for assignment.

Example 2: If a cloud org VDC is to be taken out of service, then the cloud org VDC can be activated only for assignments but not for deployments. Thus, the system cannot make new deployments to the cloud org VDC. However, the system still allows the use of existing prepops on that cloud org VDC.

Example 3: If a cloud org VDC is to be brought online, then it can be activated for deployments but not assignments. Thus, starting to deploy prepops to the cloud org VDC is valid. However, the system cannot use the prepops for labs until the cloud org VDC has officially been made available.

Additional settings for cloud org vDC

Select a cloud organization from the list and click the three ellipses next to it.

Option	Description
Drain All Prepops	Drains all prepops from the selected cloud org VDC but retains prepops for the template in other VDCs.

Create a cloud system credential

If you want a cloud to use cloud shadow information, you must create a cloud system credential for the cloud.

You can create a credential or view existing ones. You cannot edit or delete the existing credentials.
For more information about cloud shadows, see Monitor shadow copies

1. To open the Cloud System Credential page, click **Cloud › Cloud System Credentials**.
2. To open the Create cloud system credential form, click the **Create Cloud System Credential** button.
3. Enter the following required information for your new credential.

Hostname	Select a host name from the drop-down menu.
Username	Enter your user name.

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Password	Enter your password.

4. Click **Create Cloud System Credential**.

VMware Lab Platform cloud placement algorithm

In VMware Lab Platform, you can configure and use algorithms to define and deploy a vApp from any data center. To determine which data center and vApp to use for deployments, the system provides many configurable options. For example, for on-demand deployments, based on the geographical preference, or the speed of the deployment, the choice of a virtual data center is made. However, for prepopulated vApps, or other advanced deployments, VMware Lab Platform uses the VMware Cloud Director placement algorithm to determine which VDC or VDCs to use for a vApp deployment.

About the Placement Algorithm

The purpose of a placement algorithm is to determine which VDC or VDCs is used for a new prepopulated lab template and reserved deployments. You can overwrite the default placement algorithm in a lab template. To determine the deployments, the algorithm overrides the default settings for per-lab template, and the uses the current state of the system.

Each deployment algorithm shares a common first phase, and determines which VDC can deploy the new vApps. For a VDC to be available, it must have access to at least one of the template that is used for the deployment (either directly or as a shadow). The template must be available, enabled, and have sufficient storage for the deployment (storage must not be over-provisioned, so when the available storage is used, the VDC will no longer be used for deployments). If this phase makes more than one VDC available, the deployment algorithm determines how to make the deployment – either by selecting a single VDC for a single deployment, or by specifying how multiple deployments must be made across VDCs if multiple deployments are being made at once.

VMware Lab Platform supports six deployment algorithms. As described earlier, each algorithm has a standard first phase and only if there are multiple VDCs available, the algorithm is used to determine the placement.

Algorithm options

The supported algorithms are:

- VMware Lab Platform Placement Algorithm
- Equal by Resource Utilization Algorithm
- Equal by Total Deployments for Template Count Algorithm
- Equal by Total Deployment Count Algorithm
- Equal by Deployment Count Algorithm
- Equal by Deployed Template Count Algorithm

VMware Lab Platform Placement Algorithm

The VMware Lab PlatformPlacement Algorithm is the default algorithm, and we recommend that you do not change this algorithm or the configuration of this algorithm without the support of the VMware Lab Platform support team.

The aim of this algorithm is to use all the available VDCs equally. The equal placement is done by randomly selecting VDCs for deployments so that over the time, the usage of the available VDCs is roughly equal. The algorithm assigns a resource usage value to each VDC and lowers the probability of that VDC being selected once the usage value passes a configured value, thus favoring the fewer utilized VDCs.

The usage value is a weighted average of the currently used number of the CPU divided by the total number of CPUs, and the currently used amount of the memory divided by the total memory. The weight relies more on the memory value, so the utilization score grows higher and quicker (thus making a less likely to be selected) as memory utilization increases when compared to just the CPU utilization increase.

Equal by Resource Utilization Algorithm

The Equal by Resource Utilization Algorithm balances deployments so that vDCs are utilized in equal proportions. That is, a VDC with more total resources is likely to be assigned more placements than a vDC with fewer available total resources.

Like the VMware Lab Platform Placement Algorithm, the Equal by Resource Utilization Algorithm uses the utilization score of a VDC to determine what proportion of the available vDC resources are currently in use. The algorithm then assigns deployments to lesser used VDCs. The VMware Lab Platform Placement Algorithm treats all clouds equally until they reach a certain threshold of usage, where as Equal by Resource Utilization Algorithm always favors less used clouds.

Equal By Total Deployments for Template Count Algorithm

The Equal By Total Deployments for Template Count Algorithm balances the number of deployed vApps that use the same template(s) in current deployment across all available VDCs so that the same number of vApps using these templates are deployed to each cloud.

Equal By Total Deployment Count Algorithm

The Equal By Total Deployment Count Algorithm balances the total number of deployments across all the available VDCs regardless of the vApp state. This algorithm does not consider what the deployed vApps are, but aims to make the placement decisions so that the total number of deployed vApps is equal in all VDCs.

Equal By Deployment Count Algorithm

The Equal By Deployment Count Algorithm balances the number of running and currently deployed vApps across available VDCs. The size of the vApp is not considered for this deployment algorithm.

Equal By Deployed Template Count Algorithm

The Equal By Deployed Template Count Algorithm balances the number of running and currently deployed vApps derived from the templates that are used across available VDCs in this placement decision. The size of the vApp is not considered for this deployment algorithm.

Managing cloud shadows

The administrator UI includes a shadow management feature that allows an administrator to see where shadow copies reside in the various storage pools.

In VMware Cloud Director (vCD), if you deploy a vApp that resides in a different storage pool other than where it actually had been deployed, then vCD first deploys a shadow copy in the new storage pool. VMware Cloud Director then deploys a vApp from that shadow copy, which is essentially a double-deployment that can take a lengthy amount of time to deploy and leaves the end user waiting.

The administrator can predeploy a shadow to pools that do not already have a copy of that lab to reduce the wait time for the end user. For more information, see Pre-deployed shadow copies of VMware Lab Platform labs .

You can monitor shadow activity in VMware Cloud Director from the Cloud Shadow page in VMware Lab Platform. Information about the cloud org name, availability, and replica status per lab template is shown. More detailed information is displayed when you click the arrow next to the template name.

You can create replicas from this page by clicking on the blue replica oval and following the prompts.

Monitor shadow copies

You can use the VMware Lab Platform administrator interface to see where shadow copies of vApps reside in storage pools in VMware Cloud Director.

You can monitor shadow copies in VMware Cloud Director from the Cloud Shadow page.

1. Click **Cloud Management › Cloud Shadows**.
2. Optionally, you can click **Refresh Shadow Information** to run a shadow update.

3. View the information you are interested in. Detailed information is available for a template when you click the arrow next to the template name.
4. If needed you can create a replica by clicking on a value in the Replica column and following the prompts.

Predeploy a shadow to pools that do not already have a copy of that lab, as a result reducing the wait time for the end user.

Pre-deployed shadow copies of VMware Lab Platform labs

You can pre-deploy a shadow to pools that do not have a copy of the lab, and thus reduce the waiting time for the end user.

- Login as an administrator.
- Enable **Advanced Options** on the administrator UI control panel.

1. To open the Cloud Shadow page, from the navigation panel, click **Cloud Management › Cloud Shadows** .
2. Select the Cloud orgs you want to display. The system displays where shadow copies reside in the various storage pools.
3. Optionally, you can click **Refresh Shadow Information** to run a shadow update for the clouds you selected.

Working with support options

The **Support** menu provides links to user support options.

In the administrator UI, click the  icon next to the **Support** menu option to view the support options.

The following support options are available:

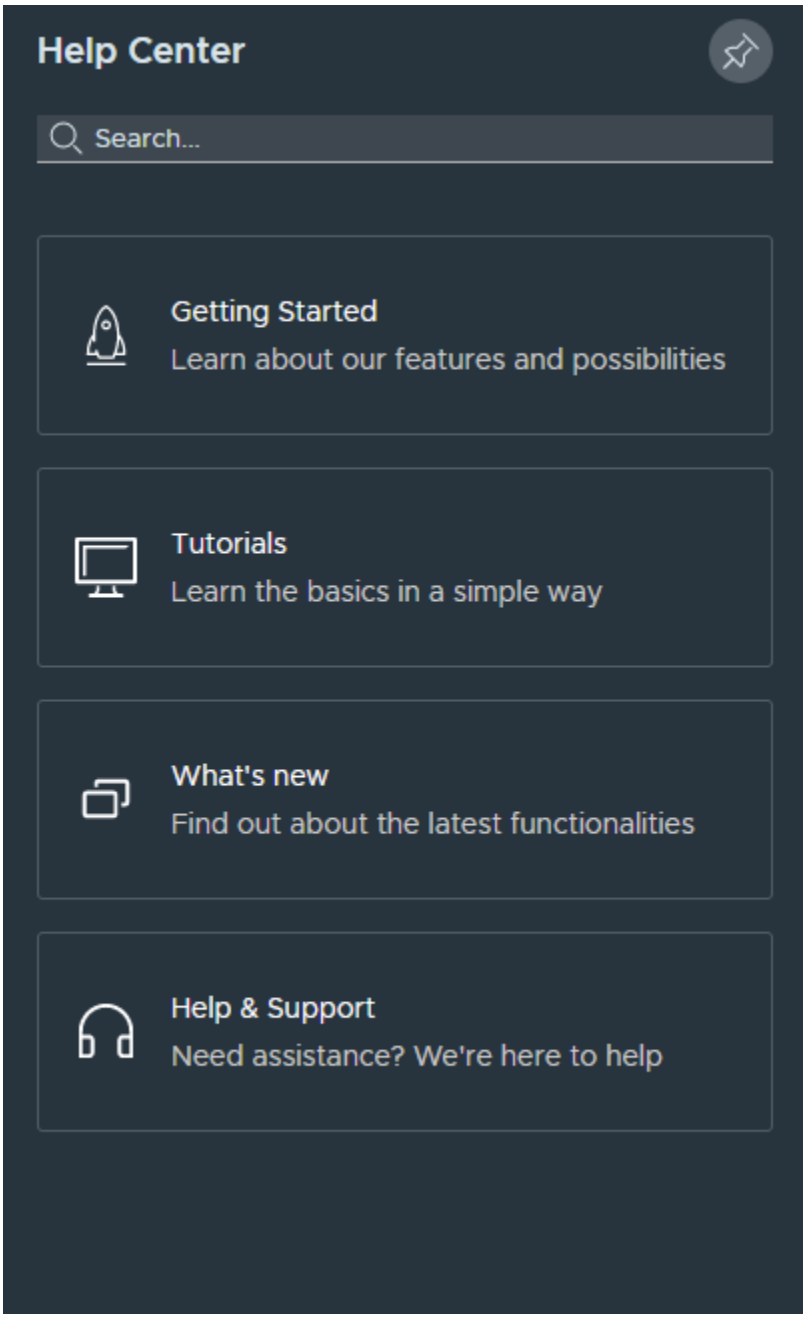
Support Site	Link to the VMware Cloud services support page.
Documentation	Link to the VMware Lab Platform documentation page.
REST API Guide	Link to download the VMware Lab Platform REST API guide.
YouTube	Link to the VMware Lab Platform training videos on YouTube.

Working with Help Center

With the VMware Lab Platform in-product Help Center, you can view help content, tutorial guides, use the powerful search for relevant VMware Lab Platform knowledge-base articles, product documentation such as getting started guide, release notes, and other technical guides. By embedding support channels and capabilities within VMware Lab Platform, administrators can access personalized support from the application instead of navigating to external support portals, phone queues or multiple search engines for the help they need to.

To view in-product Help Center support options, click the  help icon. The Help Center panel is a floating window and you can move, resize the panel as per your requirements.

Figure 14: Help Center Support Channels



The Help Center page displays the following options:

Getting Started

Provides link to the *VMware Lab Platform Getting Started guide*.

Tutorials

Displays a list of in-product tutorials and videos.

What's New

Provides link to the *VMware Lab Platform Release Notes*.

Help & Support

Provides an option to raise support tickets.

Provide feedback with the VMware Lab Platform forms

You can use the improved feedback tool in the administrator user interface to provide timely feedback to our developers.

To provide feedback for VMware Lab Platform:

1. Click the  feedback icon on the top of the page. The feedback form opens.
2. Enter the details in the feedback form and click **Submit**.

Using and Managing the VMware Lab Platform User Interface

VMware® Lab Platform™ (formerly known as VMware® Learning Platform™) is a complete learning environment that integrates instructional materials, rich virtual machine (VM) labs, videos, social media, and more as a lab. VMware Lab Platform is a cloud-based SaaS service, and its content management tool includes features to run and track training labs and courses. The VMware Lab Platform user interface lets you take on-demand labs or instructor-led labs.

Intended audience

The primary audience for VMware® Lab Platform™ user interface is end users and students who intend to take an on-demand lab or an instructor-led lab.

Where to begin when using the VMware Lab Platform user interface

As a user with a student or an end-user role of VMware Lab Platform, you use the VMware Lab Platform user interface to take an on-demand lab. Your lab administrator creates the labs in the VMware Lab Platform administrator interface. If you are new to VMware Lab Platform user interface, you will need to know where to start.

Day in the life of a lab user

As a lab user, you use the VMware Lab Platform user interface to perform the following tasks:

- Register for a lab.
- Enroll and start a lab.
- Use the lab options.
- Customize your lab console.
- Use a lab manual to complete a lab.
- View and email a lab transcript that shows training labs and courses that you have completed.
- Share and recommend on-demand labs with friends and team members.
- View a catalog of all the available VMware training labs and courses.

Access the VMware Lab Platform user interface

You access the VMware Lab Platform user interface through a web browser.

- User interface URL: https://www.labplatform.vmware.com/<org_name>

NOTE
You must have valid credentials from your organization administrator to log in the VMware Lab Platform user interface. Your organization administrator defines the authentication provider for the login page in the VMware Lab Platform administrator user interface.

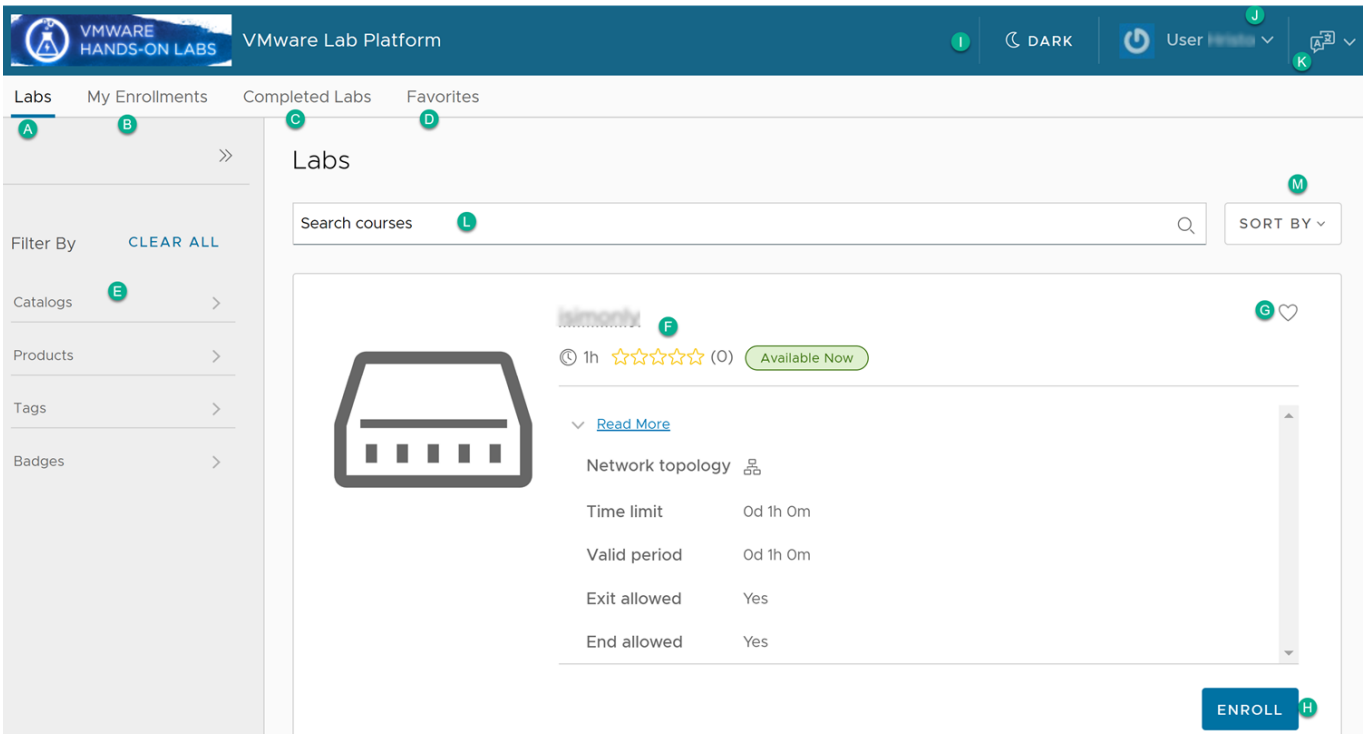
Or you can use the **Register** option to self register and create an account to use a lab.

For more information on how to self register for a class, see How to self register for a class.

NOTE
Passwords are case-sensitive. Contact your VMware Lab Platform administrator, if you have trouble logging in to the application, or locate a lab.

After you login, VMware Lab Platform displays your lab portal.

Figure 15: VMware Lab Platform user interface view



The following table describes the main areas of the **VMware Lab Platform** user interface.

Menu	What the menu option does
Labs—A	Displays all the labs that are available to you. You can enroll for a lab from the available lab list.
My Enrollments—B	Displays all your lab enrollments. For more information on how to enroll for a lab, see How to enroll and start a lab.
Completed Labs—C	Displays a list of labs that you have completed.
Favorites—D	Displays all your favorite labs. To add a lab to the favorite lab list, click the heart icon in the lab.
Classes	Use the **Classes** tab to see your active and upcoming classes. **NOTE** The **Classes** tab is available only when your part of an upcoming class or an active class.
Filter By—E	Use the filter option to search for a lab. You can search a lab by a product name, tag name, badge, or a catalog type.
Lab details—F	Displays all the details about the lab you have selected.
Favorites icon—G	Use the heart icon to tag all your favorite labs. You can use the **Favorites** tab to list all your favorite labs.
Enroll—H	Use the **Enroll** tab to enroll for a lab. After you enroll for the lab, you get the **Start a Lab** option.

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Menu	What the menu option does
Style selector—I	Use the style selector to set the theme of your interface window. You can choose from light and dark mode user interface setting.
User name—J	Displays the user name of the user logged in the system. If you have logged in as an anonymous user, then the user name field is blank and does not display any name.
Language selector—K	You can choose the language in which you want to see the user interface information. VMware Lab Platform supports more than 12 languages.
Search option—L	Use the search option to search for any labs assigned to you or available to you. You can conduct a search for a lab by entering the title of a lab in the search box or by entering a keyword. For example, you might enter `Introduction` to list all labs in the current catalog that include the word “Introduction” in their titles.
Sort By—M	Similar to the search option, you can use the sort option to filter your search results for labs.
Announcements/Offers tab	Displays announcements, offers, and social media posts added by your account administrator.

Supported browsers and environments

The following browsers are supported. As a best practice, use the latest available version of the browser.

- Internet Explorer version10 and above
- Firefox version 13 or above
- Chrome version 18 or above
- Safari version 6.0 or above

> **NOTE**
> Your browser settings must accept cookies, run JavaScript, and allow web sockets.

We recommend that you run the compatibility test to identify if there are any system compatibility issues to take a VMware Lab Platform lab. For example; https://www.labplatform.vmware.com/<org_name>/compatibility.

Support for mobile devices

VMware Lab Platform supports mobile devices for split-screen functionality such that the lab’s instructional manual appears on the mobile device and the lab console appears on a desktop or a laptop. With the split-screen feature, you can use a full screen for the console of the lab and have the manual for the lab display on another desktop screen or other device. For more information on how to use the split screen feature, see Using the lab manual.

> **NOTE**
> VMware Lab Platform does not support the console of the virtual machine on a mobile device.

VMware Lab Platform lab workflow

As a VMware Lab Platform lab user, you typically follow a standard workflow.

1. Access the lab through your browser and use the credentials sent by your administrator to authenticate yourself.

NOTE
User interface URLs to access the lab.
- User interface URL: https://www.labplatform.vmware.com/<`organization_name`>

  Note: The <*`organization_name`* is a variable in the URL. Contact your lab administrator to get the lab URL for your organization account.

2. Select a lab for which you want to enroll, and click **Enroll**, or resume a lab that you left without completing (click **Enrolled**).

   VMware Lab Platform loads the lab you selected. After VMware Lab Platform finishes loading the lab, it displays a lab console that shows your enrollments. The length of time to connect to a lab varies depending on the lab environment's complexity and configuration.
3. (Optional) If the lab you started includes multiple consoles, click **Consoles** and select the one you want to work with. For example, your lab might include a Windows console icon and a Linux (RHEL) console icon. Most labs only have a single visible virtual machine that you interact with for the exercises, so the console tab is not shown.
4. Test your browser compatibility with VMware Lab Platform, click **Run Test**.
5. To start the lab, click **Start this Lab**. If you start a lab and then leave a lab, then VMware Lab Platform displays the **Resume the lab** control instead of the **Start this Lab** control. When you start a lab for the first time, you get the option to start a guided tour of the console controls.
6. To display the manual for your lab, click **Manual**. Manuals are typically made up of modules that you complete sequentially.
7. Progress through the modules of your lab and end or exit the lab as you choose.

Using the lab action options in VMware Lab Platform

Each lab listed on the VMware Lab Platform lab portal includes the lab status and the lab action options. As a lab user, you can use the lab action options to optimize your lab experience. While you are working on a lab, you can use lab options to extend the time remaining in your lab, to end or temporarily exit a lab, or to submit your lab results for grading. You can also use lab options to set your lab keyboard and console attributes.

Table 2: VMware Lab Platform lab action options

Lab status	Description
Enroll	If you are not enrolled in a lab, then VMware Lab Platform displays the **Enroll** option. To enroll for a lab, click the **Enroll** option. After you enroll in the lab, you see the **Start This Lab** option. If you are already enrolled in a lab, then the VMware Lab Platform system displays the **Enrolled** option. To display the **Start This Lab** option, click the **Enrolled** button. To view the all labs that you have enrolled for, use the **My Enrollments** option.
Start the Lab	After you enroll for a lab, you see the **Start the Lab** option that let's you to start the lab.
Log out	To log out of the VMware Lab Platform interface, open the user name drop-down menu options, and click **Log out**.
Resume This Lab	If you exited a lab (but not ended it) before completing it, then the system displays the **Resume This Lab** option. If you want to resume the lab where you left, click **Resume This Lab**.

Lab status options

Lab status options are located on the ribbon above the console pane. They include the following settings:

Table 3: Lab status options and descriptions

Status Options	Descriptions
Time Remaining	Displays the amount of time remaining for you to complete your lab. **NOTE** Labs have two time limits: Time Remaining and Expiration Date. Time remaining is the total amount of time that the lab can be active. The counter for Time Remaining only changes when the lab is in use. When the Expiration Date arrives, the lab ends even if there is allotted time still available. For example, a lab might have been configured with a one-hour limit to do the lab, but expire in three days. As a result, you can use the lab for up to one hour of total time and break up that time into smaller intervals spread out over three days. In this case, the lab would end when the first of the time limits is reached.
End	Displays a screen that allows you to confirm or cancel the ending of your lab. If you confirm that you want to “end” your lab, then VMware Lab Platform considers you have completed the lab and returns you to the VMware Lab Platform lab console login page. As a result, you are unable to resume the lab. You must click **Exit** to come back and resume your lab.
Exit	Displays a screen that allows you to confirm or cancel the exiting from your lab. If you confirm that you want to “exit” your lab, VMware Lab Platform returns you to the lab console login page. VMware Lab Platform also provides you with an option to resume the lab. The **Exit** option is displayed only if your admin has enabled it.
Extend	Allows you to extend the time remaining in your lab. The **Extend** option is displayed only if your admin has enabled it.
Finish	If your admin has configured your lab for grading, then the system displays the **Finish** option instead of the **End** option. When you click **Finish**, the lab is ended and your work is sent for grading. Note that once you click **Finish**, you are not able to return to the lab.
Reset	If you reset a lab, all its virtual machines (VMs) are restored to their original state. You are not able to recover any work you have done. The **Reset** option is displayed only if your admin has enabled it.

Keyboard and Console Options

To view the console and keyboard options, click the  gear icon. Select the checkbox for the options you want to use.

Console Options	Descriptions
Keyboard Layout	Select your keyboard layout to adjust any incorrect keys being sent to the console.
Intercept Paste	Intercept the paste shortcut keys and send the text copied from your browser (or elsewhere) to the console. This option could interfere with the console's own copy and paste commands.

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Console Options	Descriptions
Remap iOS Key to CTRL Key	If you are using iOS operating system, the VMware Lab Platform automatically remaps the iOS keyboard shortcuts equivalent to the Windows shortcut keys. You can remap the system keyboard shortcuts from the Settings menu.
Manual Shortcuts	If enabled, you can change the page in the manual with `Ctrl` key combinations even when the console has focus. See the gear menu for key combinations for your environment.
Fix non-ANSI US keyboard layouts	Remap the keys pressed on your keyboard to send US ANSI key code equivalents to the console. If you are using a non-US keyboard or have the keyboard set a non-US language in your operating system, enabling this option could help send the correct keys to the console.
Reverse Vertical Scroll Direction	Change the mouse scrolling direction sent to the console. This setting can help with mice or track pads that have the scroll direction reversed (sometimes called "natural").

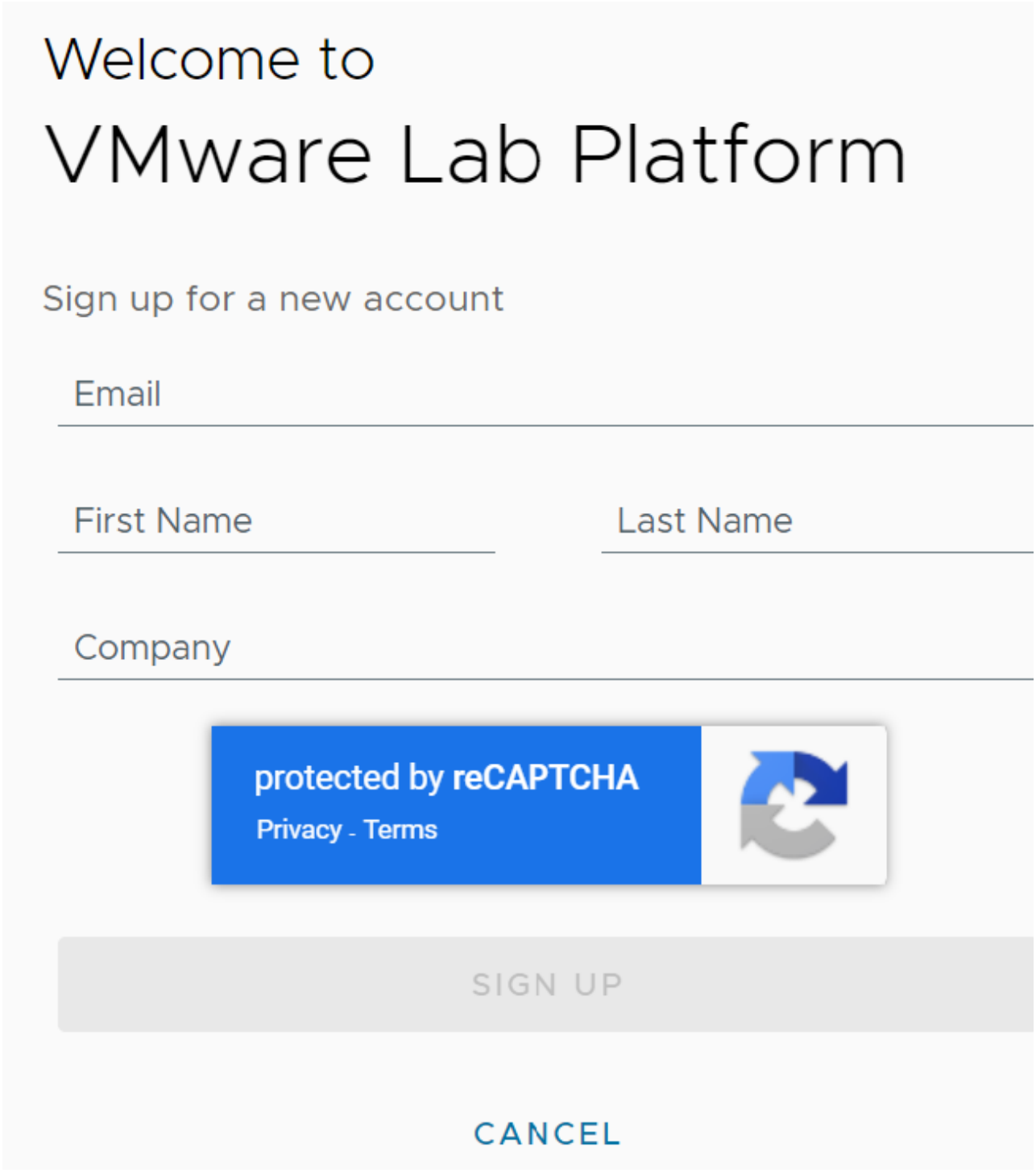
How to register and verify your account in VMware Lab Platform

Register and verify your VMware Lab Platform account so that you can use the VMware Lab Platform user interface to take a lab.

- Lab URL for the lab assigned to you.
- **NOTE**
  If your lab administrator has shared the quick launch URL for a lab, then you don't need to register or create a user account with VMware Lab Platform. You can directly access the lab.

To self-register for a lab:

1. Open your self-register lab URL in a web browser.
2. On the **VMware Lab Platform** login page, click **Register**.
3. Enter the user details.



4. Click **Sign Up**.

You receive the welcome email with steps to verify your account in your registered email address. Follow the steps in the email to verify your account.

NOTE
After the succesful login, the system redirects you to the page or URL based on your user role. For example, as an administrator or an instructor, you self-registered for a lab from the administrator or the instructor user interface, after the registration the system redirects you to the user login page based on your user role.

To get familiar with the user interface, see Where to begin when using the VMware Lab Platform user interface.

How to enroll and start a lab in VMware Lab Platform

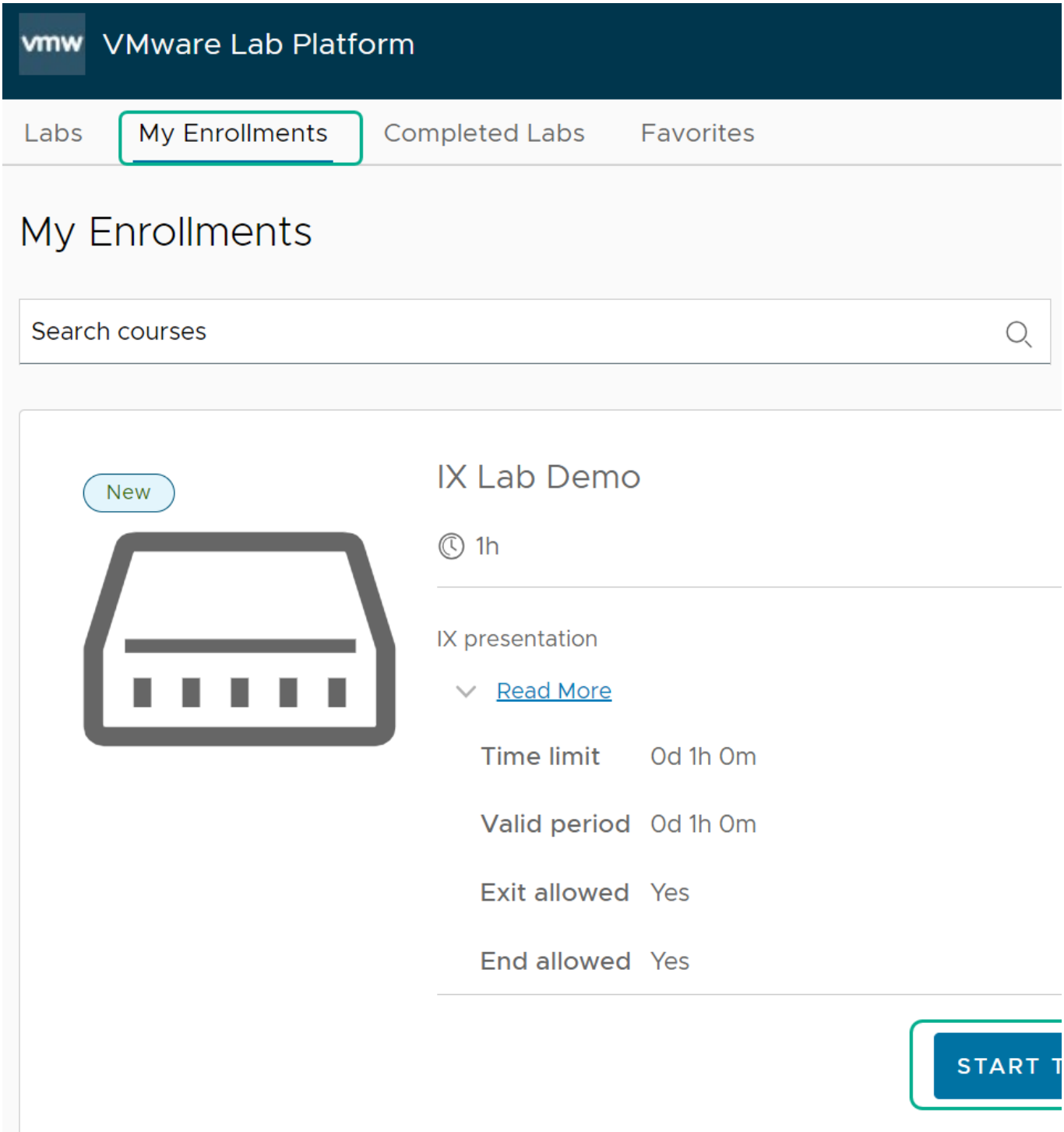
In a VMware Lab Platform lab, if you are not enrolled in a lab, then VMware Lab Platform displays the **Enroll** option. You need to be enrolled for a lab to start the lab.

1. Launch the lab using the lab URL provided by your account administrator.
2. To enroll for a lab, click the **Enroll** option. If you are already enrolled in a lab, then the VMware Lab Platform system displays the **Enrolled** option.
3. To view all your enrollments, click **My Enrollments**.

After you enroll in the lab, you see the **Start This Lab** option.

My Enrollments

The following sample enrollment page shows lab details and the **Start the Lab** button. This example also displays the **Announcements** section.



After you enroll for a lab, to start the lab and complete the lab course, click **Start This Lab**. For more information on how to start a lab, see

How to enroll as a temporary user in VMware Lab Platform

If your account administrator has given you permissions to access a lab for one-time use, you can log in as a temporary user using the lab URL (provided by your account administrators) and enroll for a lab in the user interface.

1. Use the lab URL and access the lab through your browser.
2. On the lab login page, enter your first name, last name, company name, and email ID, and you can enroll for the lab.
   You can use the lab until the entitlement is complete, or until the valid period of the lab is reached.

   NOTE
   Once you complete the lab, you cannot take the lab again. You cannot access non-anonymous labs from the lab catalog page . A warning message is displayed for non-anonymous labs. After the entitlement is completed, all user details are deleted from the system.

How to search for a lab in VMware Lab Platform

A quick search checks all types of labs that are available for your to enroll.

- Labs assigned to you.

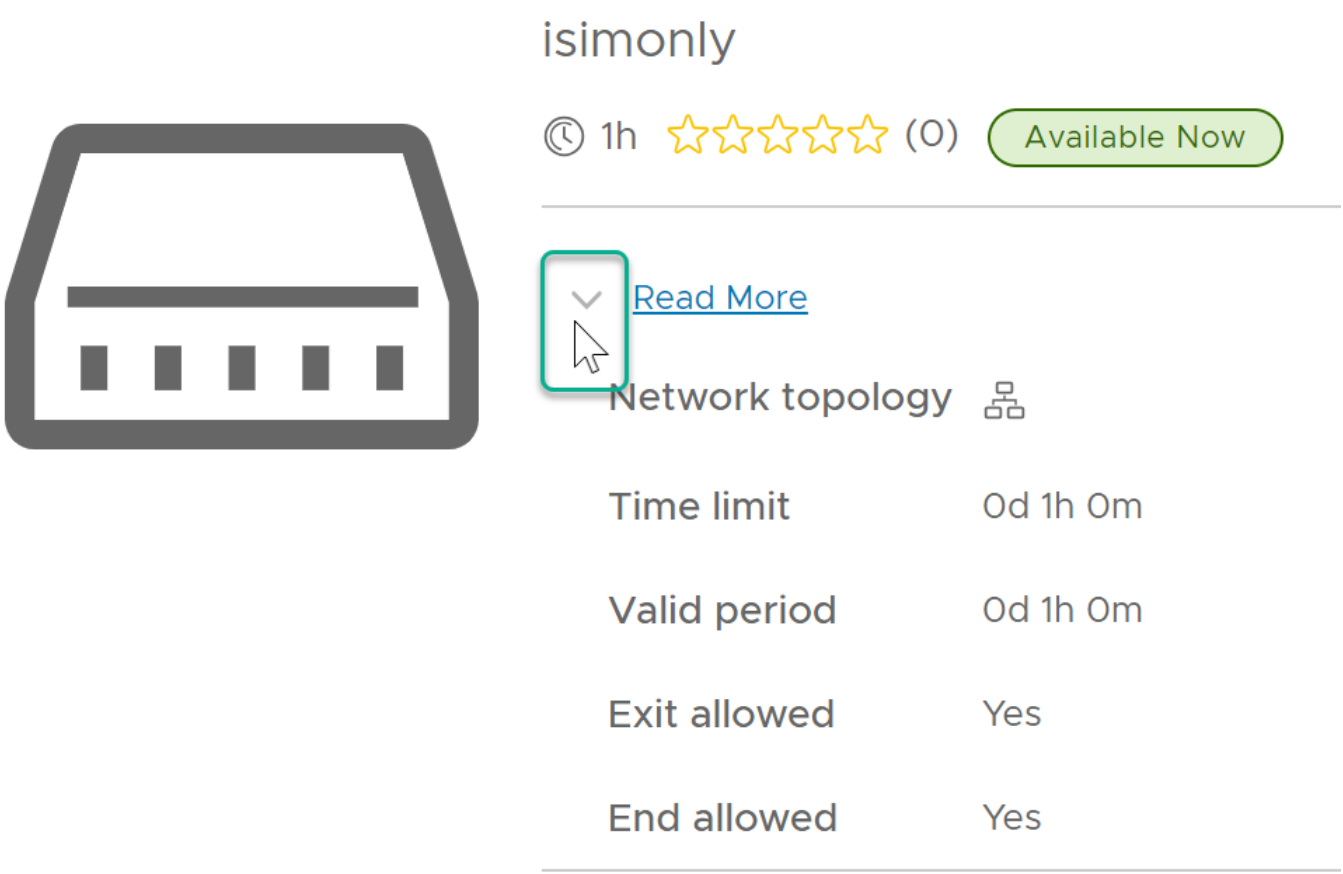
1. Enter the lab name in the search box at the top of the user interface window. For example, you might enter `Introduction` to list all labs in the current catalog that include the word “Introduction” in their titles.
2. (Optional) You can also use the sort and filter option to narrow down your search results based on product type, catalog, or tag type.

How to view details of a lab in VMware Lab Platform

In the labs page, you can view the lab details such as brief description of the lab, including running time, network topology, and lab costs.

- Lab assigned to you.
- User credentials to access the lab.

1. Access the lab through your browser and use the credentials sent by your administrator to authenticate yourself.
2. To view the lab details, click the expand arrow for a lab on the lab page.



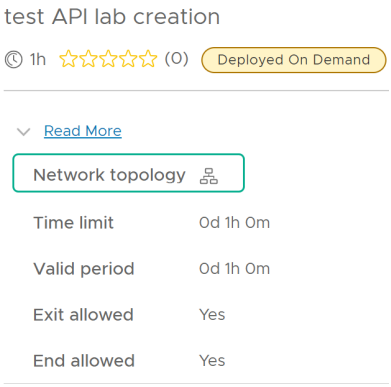
If you have not enrolled for the lab, then enroll for the lab and start the lab.

How to view the network topology of a lab in VMware Lab Platform

Your lab comprises virtual machines. If your administrator has included a network topology image in your account, then your lab console includes the **Network Topology** option.

- Lab assigned to you.
- User credentials to access the lab.

1. Access the lab through your browser and use the credentials sent by your administrator to authenticate yourself.
2. To display an image that shows the network topology of the virtual machines that are used in your lab, click the Network Topology icon.



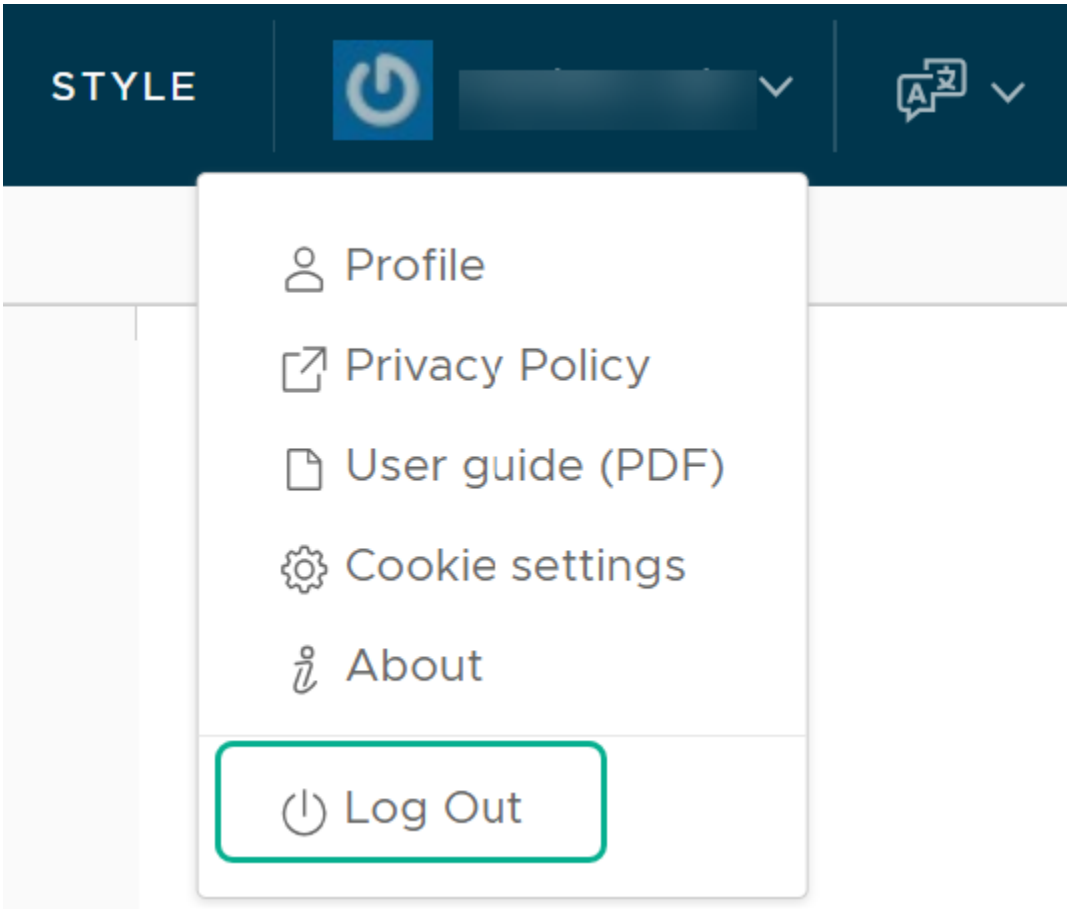
Start the lab.

How to log out of a lab in VMware Lab Platform

On every page of the lab, the **Log Out** command is nested under the user name menu in the top-right corner of the banner of the lab web page. Logging out ends your current session.

- Lab assigned to you.
- User credentials to access the lab.

1. To log out of a VMware Lab Platform lab console and return to the Login page, open the drop-down menu options from the user name. Click **Log Out**.



If you accidentally logged out of a lab or want to resume your lab where you left, select the lab and click **My Enrollments › Resume This Lab**.

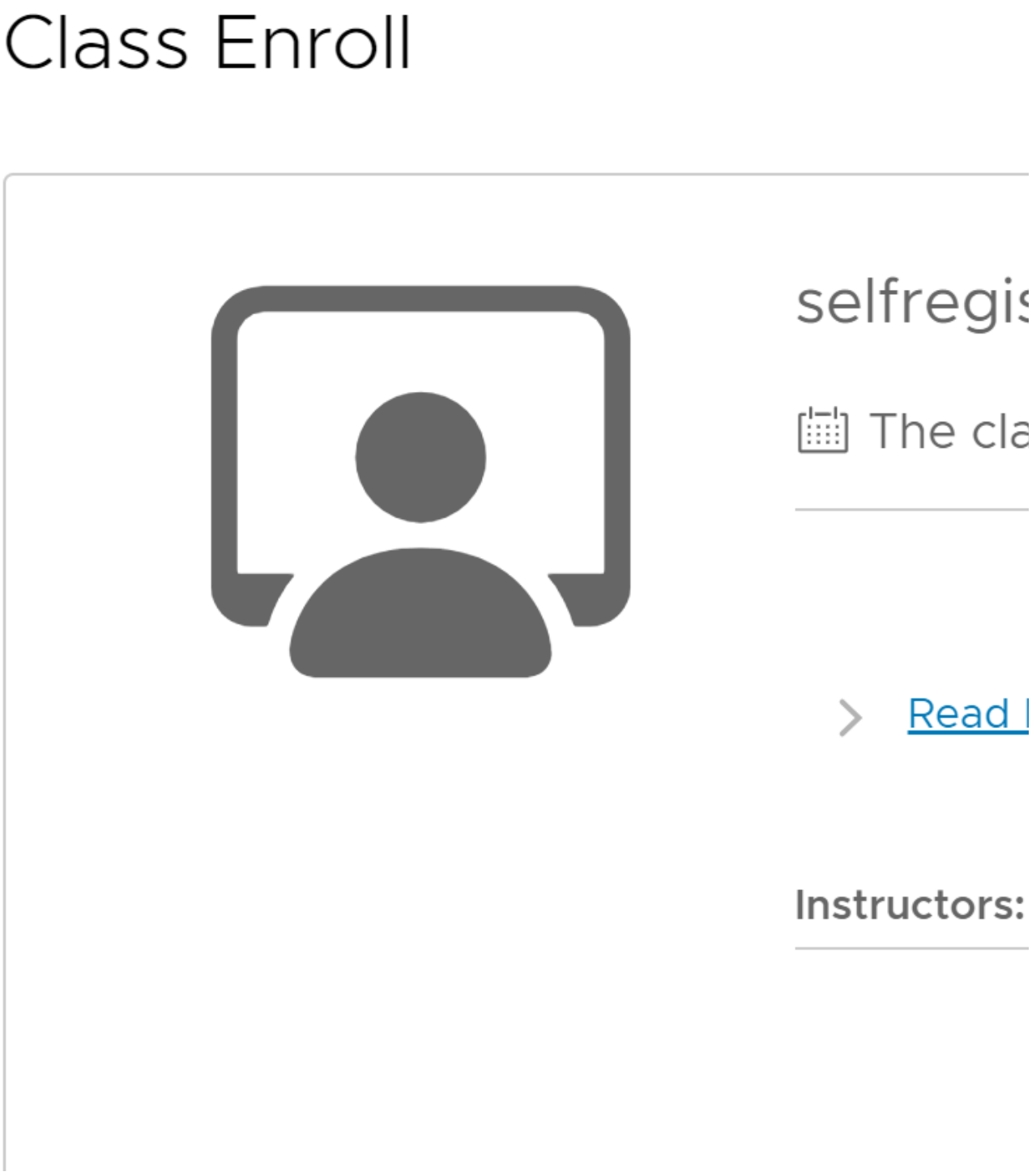
How to self register for a class

One of the easiest way to enroll for a lab is to self register for a class.

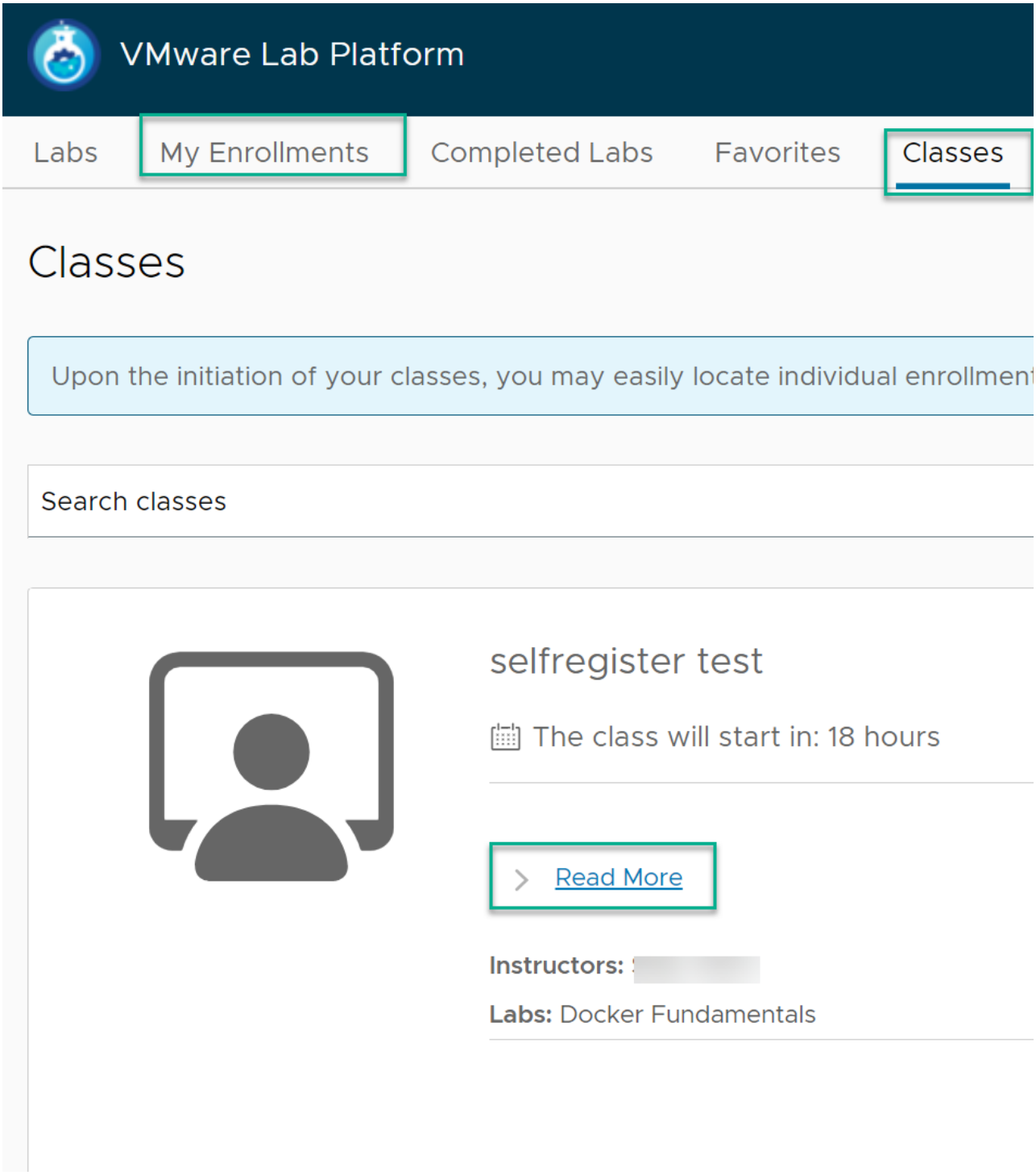
You need to be have the registration URL for the class that you want to self register for. Your account administrator provides the registration URL.

To self-register for a class:

1. Enter the class registration URL in a web browser. For example; https://www.labplatform.vmware.com/vlpdemo2/catalogs/studentToken
2. Enter your email and register your account, if you don't have an account in VMware Lab Platform. For information on how to register your account, see How to register your account.
3. In the Class Enroll window, click **Enroll**.



4. If there are seats available in the class, you can enroll for it.
5. Go to the **Classes** tab.



The **Classes** tab will be visible only when:

- A student has an upcoming class or has active classes.
- Note: You cannot see completed classes that are not yet deleted from the system.

- The class tile in the tab displays the name of the class, a label indicating remaining time for the class, status of the class if the class is active, description, start and end date details, instructor name, and other relevant information.
- The Classes tab has an option that will redirect you to the **My Enrollments** page with filtered results. This means that if the class has more than one lab in it, the filtered result will list all the enrollments related to the class for you. The filter will be visible and present in the search bar.
- The title in the **Classes tab** will indicate only the class information, you cannot manage the individual entitlements from the page.

Click the **See Enrollments** tab to see all your enrollments, and start your lab.

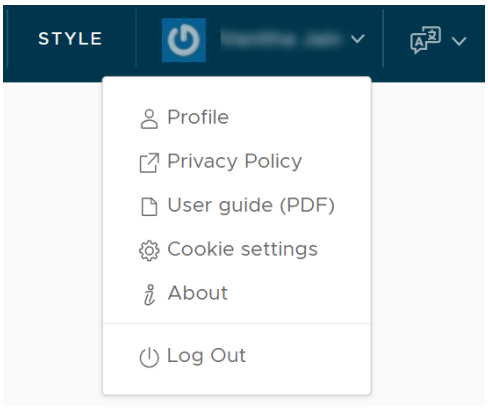
Managing my profile in VMware Lab Platform

All users can view and modify their own information on the My Profile page in the **VMware Lab Platform** user interface. You can view privacy policies, modify user preferences, help options, and the log out option.

To access the **My Profile** page:

1. On the username menu in the upper-right corner of the page, click the username drop-down menu.

Figure 16: User profile options



The Profile page for your user account displays the following user options:

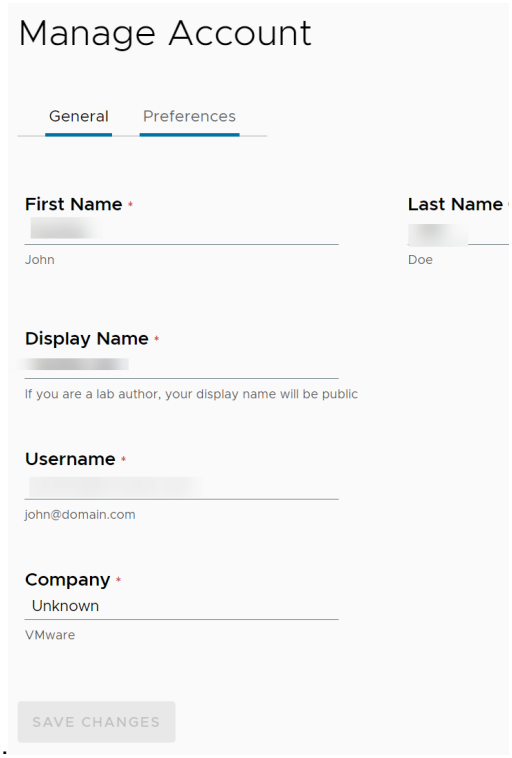
- Profile
- Privacy Policy
- User Guide
- Cookie Settings
- Log Out

How to view my profile details in VMware Lab Platform

You can use the **Profile** option to view and manage your account information associated with your VMware Lab Platform user account.

1. To view your profile information, open the user name options, and click **Profile**.

2. The **Profile** option, when clicked, displays the Profile page. The Profile page comprises of several sub pages beginning with the General page. To close the Profile page, click outside the page. Make sure that you have saved



your changes before you close the page.

> **NOTE**
> When you switch to the language version you selected, you must end the lab and then restart the lab.

General

Add your profile details such as name, display name, user name, and company name.

Language and Time Zone preferences

To select the language and the time zone to use, click **Preferences**. Select a language sets the language of the lab console to the selected language. For example, if you select German, the system switches the user interface to German. Also, selecting a language determines the language version of the instructional manual. For example, if you select German, the system uses the German version of the instructional manual associated with the lab.

> **NOTE**
> When you switch to the language version you selected, you must end the lab and then restart the lab.

About (Bio)

To add a short biography about yourself, click **About**.

Password

To change your password, click **Password**.

Security Questions

To update your security questions, click **Security Questions**. Security questions help validate who you are in case you forgot your password.

Social

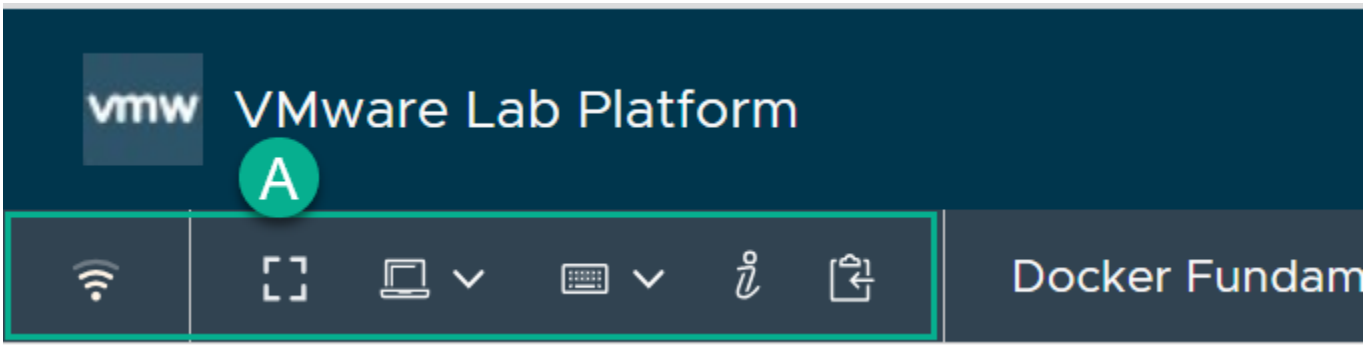
To log into one or more social media services (Facebook, Twitter, or LinkedIn), click **Social**. VMware Lab Platform never shares social media password information.

To view the privacy policy for a user, open the user profile drop-down menu, and click **Privacy**.

Using the lab console toolbar in VMware Lab Platform

From the lab console, you can view the lab console walk-through or use the console toolbar to get help, set up your profile, select a language, or log out.

Lab menus and toolbar settings control how the menus and toolbar option appear when a lab is in full screen and windowed mode. Hover on each of the menu option to see the menu name and its details.



Console toolbar—A

Use the options in the console toolbar to manage your lab console screen appearance. Hover over each icon to view the action option name. You can:

- **Check connection strength**: To check the console connection strength.
- **Go Fullscreen**: To change the console screen size.
- **Console actions**: Use the Console actions to reset or power off a VM. Click the drop-down menu to view all the available options.
- **Keyboard shortcuts**: Use the keyboard shortcuts to perform keyboard actions to help you work quickly and efficiently.
- **Check the console notes**: Copy and send notes to the console.
- **Send text to console**: Copy and send notes to the console.

For more information on using the console toolbar, see Using the screen display options.

Help and Setting options—B

Use the help option to see all the possible support options for the lab.

For more information on the Help options, see How to view the help options in VMware Lab Platform.

Lab status—C

You can extend your lab, end your lab, or exit from a lab.

Language setting—D

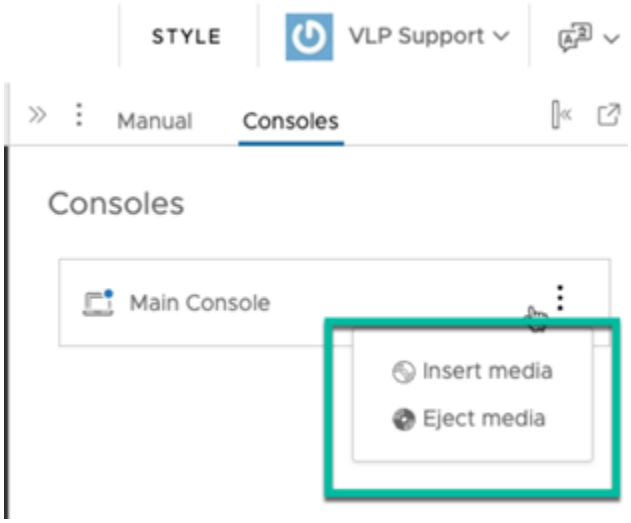
Use the language icon to change the language setting for your lab. For more information on language setting, see How to change the language settings in VMware Lab Platform.

Manual and Consoles— E

Lab resources such as lab manuals are useful and assists you to complete the lab modules.

Click the **Consoles** tab to view the consoles and console order for your lab. To insert a media file, click the vertical ellipses next to the console, and click **Insert Media**. To eject a media file or CD, click **Eject Media**.

NOTE
The **Insert Media** and **Eject Media** options are available only if your account admin has enabled the **Enable Cloud Media upload in the student interface** option in the administrator interface.



This lets you to host labs or classes that require a walk through of an OS installation and other use cases for mounting a CD on a VM.

How to view the help options in VMware Lab Platform

VMware Lab Platform provides various support and help options to help you navigate through the user interface or troubleshoot any issue with the lab or the user interface.

1. To display the Help options, click the  icon:

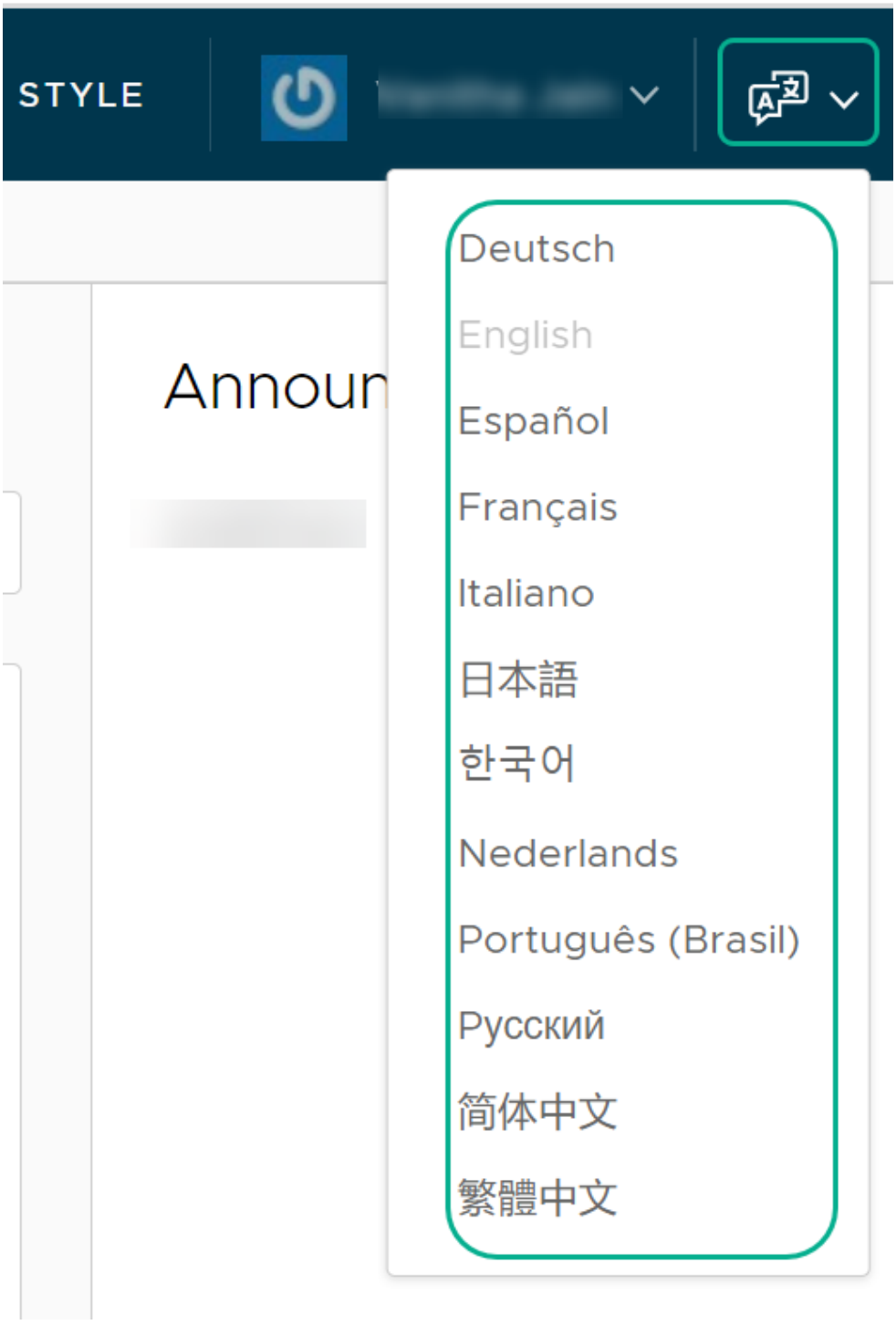
Help options	Descriptions
Live Chat	If enabled by your organization administrator, Live Chat opens a window that you can use to chat with the support staff.
Raise Hand	To activate a help request for you and alerts the class instructor that you need help, click **Raise Hand**. For more information about requesting help in instructor-led labs, see the Taking an instructor-led lab in VMware Lab Platform.
Support Options	Displays an email address for support requests and a support phone number.
Tips	Displays tips about using the lab console.
User Guide	Opens a PDF version of this document.
About	Displays VMware Lab Platform product information.

How to change the language settings in VMware Lab Platform

VMware Lab Platform provides labs in more than 12 languages. You can select a preferred language from the available language options.

The current language is displayed followed by the **language icon** icon.

1. To change the language, click the language icon, drop-down menu and select a language option.



The lab console language changes to the language that you select.

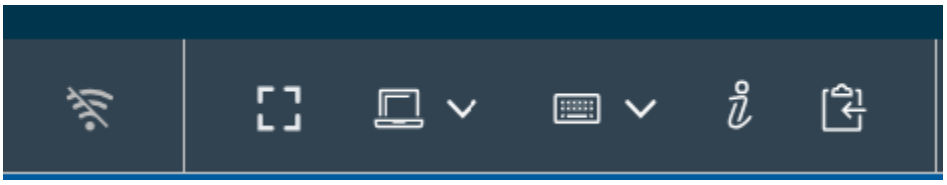
NOTE
For anonymous users (users who are not logged in and therefore do not have a profile), the administrator sets the default anonymous user language. Logged in users do have a profile. If a logged in user changes language, the user's profile is updated accordingly.

Using the console screen display options in VMware Lab Platform

You can use the screen display options to manage your console screen appearance and behavior.

Display options are located in the upper-left corner of the console.

Figure 17: Screen display options



Connection Strength	The **Connection Strength** control displays information about console latency. Your instructor or administrator can use this information to diagnose any connection problems. This feature must be enabled by an administrator.
Toggle Full-Screen	Use the **Toggle Full-Screen** control to resize the console. Console resizing includes the following actions: • When the end-user console is docked in the center or maximized, then VMware Lab Platform resizes it to fit into the space available in the browser. • If the console is floated, docked-left, or docked-right, then you can resize the console manually by dragging the handles near the edge of the console frame. • If the end-user console is supported with VMware tools, then the guest operating system resolution is updated to the new console size. • In unsupported consoles, the guest operating system resolution remains unchanged, but the console is scaled to the browser.
Console Actions	Use the console action options to change the console size and the console location. Expand the **Console Actions** menu to see all the options. The following console actions helps you manage the console: • **Undock**: Use the **Floating window** control to toggle between a docked or floating screen. • **Refresh**: Use the **Refresh** control to re-establish your console connection and reload your console. This option does not perform a screen refresh. Use this option only if your console locks up or your connection drops. • **Reset VM**: Hard-reboots the machine as if the reset button on the physical hardware had been selected. It is independent of the power options. • **Reset VM And Enter BIOS**: Resets a virtual machine (VM) of a lab, and starts the BIOS and make changes to the boot order. • **Shutdown VM**: Powers down the VM for the lab that you are running

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Keyboard Shortcuts	• **CTRL+ALT+DEL**: Sends the `CTRL+ALT+DEL` key combination to the virtual machine (VM). The result of sending the key combination is operation-system dependent.
Console Notes	Displays metadata about the lab (notes, user name, and password for each VM) as configured by your administrator.
Send Text	Displays the Send Text to Console window. You can copy and paste text from your lab manual or enter new text in the window. When you click **Send Text**, VMware Lab Platform sends the text you entered to your lab console. The text appears at the pointer location inside your console. You can also highlight text in the lab manual then drag the selected text onto your console.
Maximize	Use the **Maximize** control to magnify (full screen) your screen.
Arrow	Use the **Arrow** controls to dock your screen display controls to the left or right side of your screen.
Place in Center Screen	Use the **Place in Center Screen** control to center your screen display.

How to change the console screen size in VMware Lab Platform

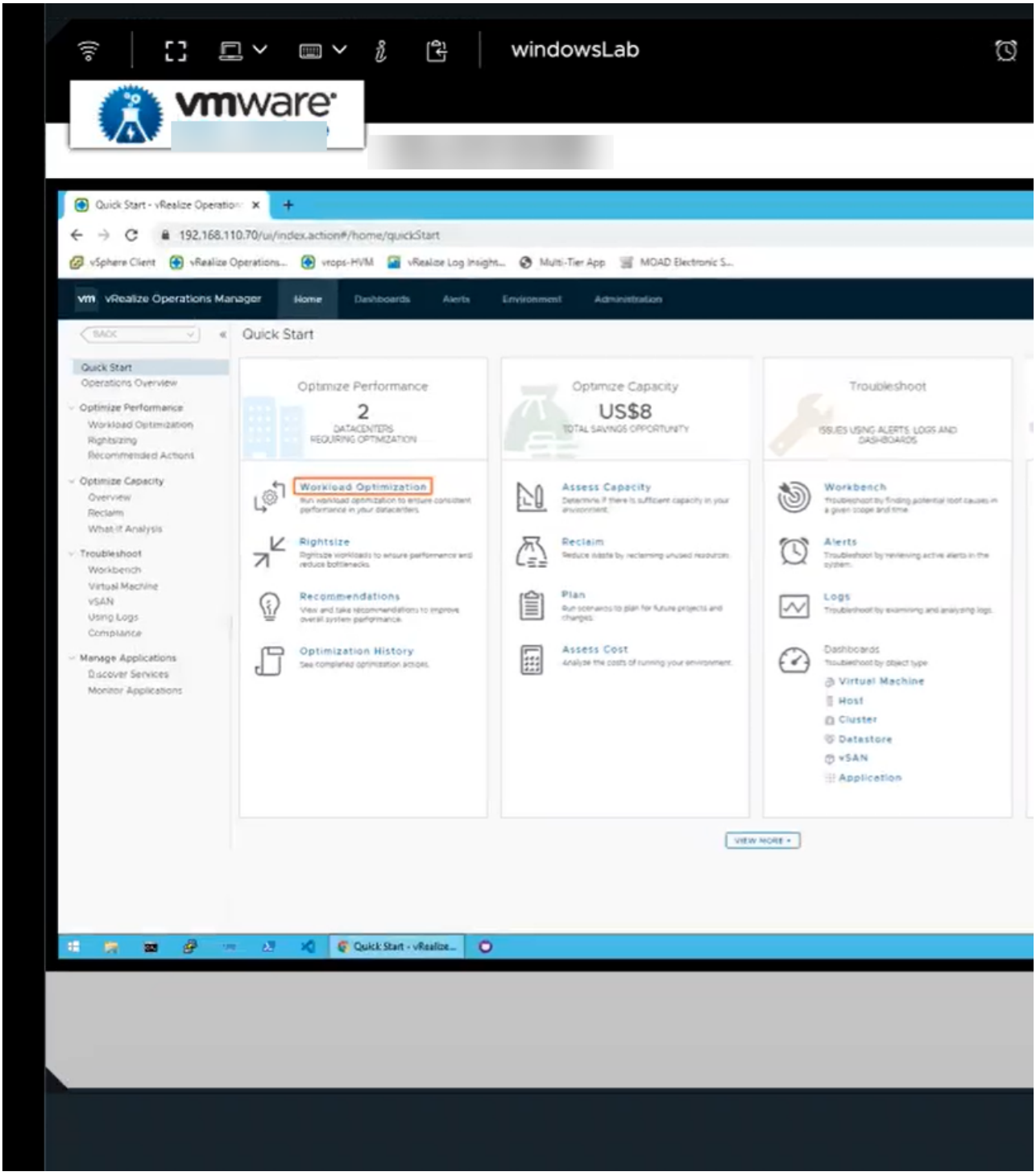
As a lab user, you can easily change the VMware Lab Platform console size to suit your device preferences.

You can use any of the following methods to resize your console screen size:

1. Click the right arrow next to the Manual panel to expand the console size.



2. Or, you can custom resize the windows. Move the cursor to any edge or corner of the window until a double-headed arrow appears. When this arrow appears, click-and-drag to make the window larger or smaller.
3. Select the specific console from the **Consoles** menu on the right side of the page. To resize the VM console of a lab, click **Console Actions › Undock**. Use the double-headed arrow to click-and-drag the console window according to your preference.



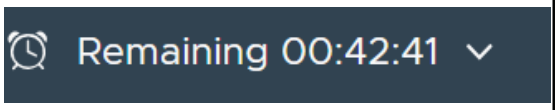
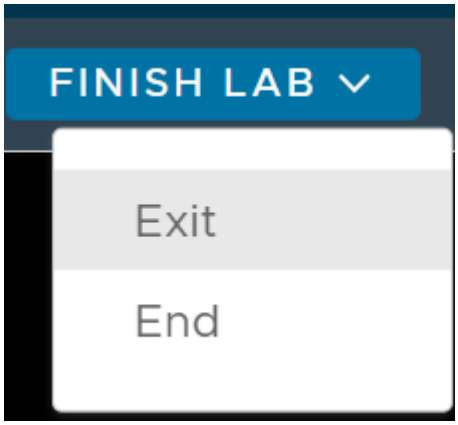
To see different key board options supported for your lab, see Using the lab action options in VMware Lab Platform.

Using the lab options in VMware Lab Platform

While you are working on a lab, you can use lab options to extend the time remaining in your lab, to end or temporarily exit a lab, or to submit your lab results for grading. You can also use lab options to set keyboard and console attributes.

Status options

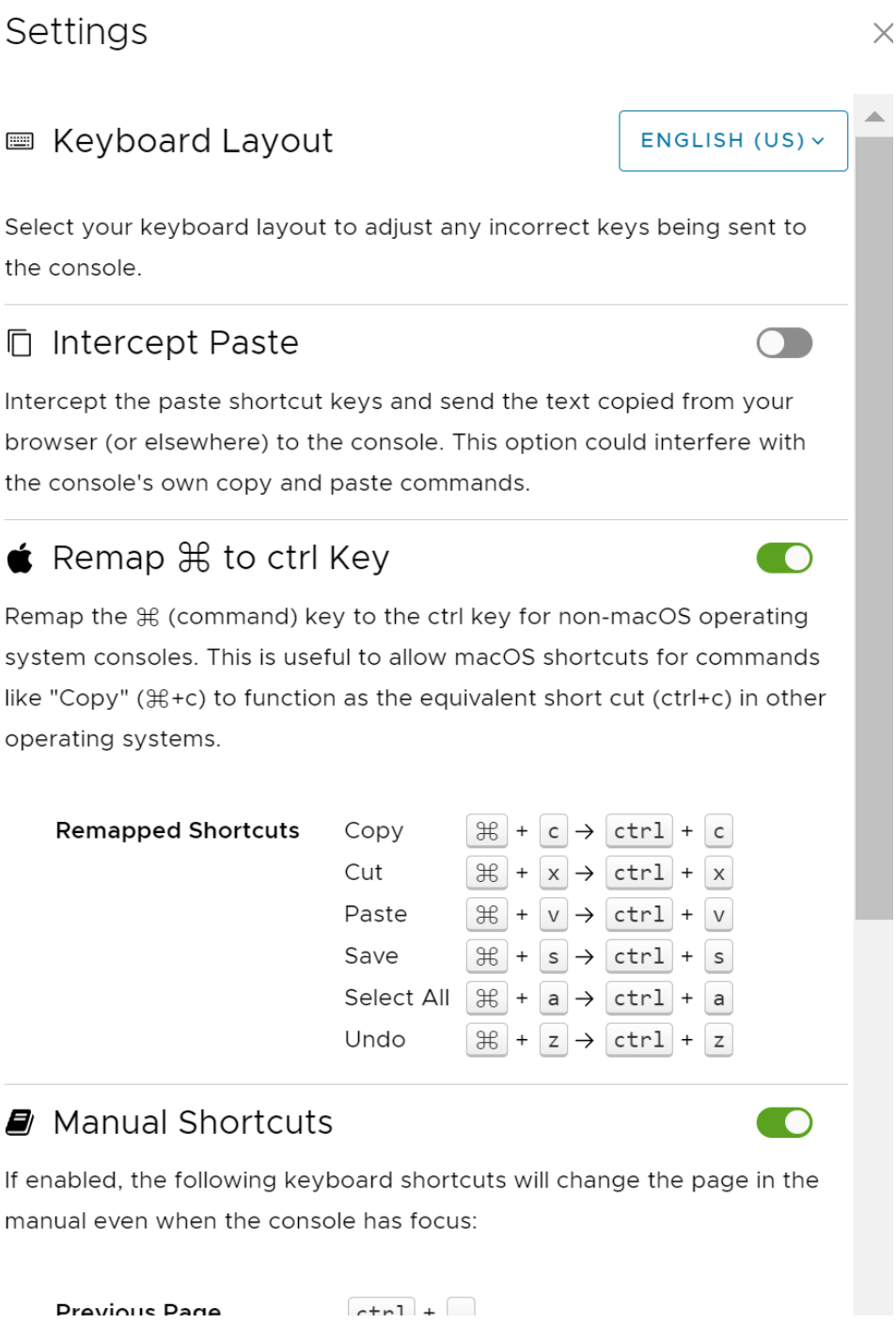
Status options are located on the ribbon above the console pane. They include the following settings:

Status Options	Descriptions
 **Time Remaining**	Displays the amount of time remaining for you to complete your lab. **NOTE** Labs have two time limits: Time Remaining and Expiration Date. Time remaining is the total amount of time that the lab can be active. The counter for Time Remaining only changes when the lab is in use. When the Expiration Date arrives, the lab ends even if there is allotted time still available. For example, a lab might have been configured with a one-hour limit to do the lab, but expire in three days. As a result, you can use the lab for up to one hour of total time and break up that time into smaller intervals spread out over three days. In this case, the lab would end when the first of the time limits is reached.
Extend	Allows you to extend the time remaining in your lab. The **Extend** option is displayed only if your admin has enabled it.
 **Finish Lab**	If your admin has configured your lab for grading, then the system displays the **Finish** option instead of the **End** option. When you click **Finish**, the lab is ended and your work is sent for grading. Note that once you click **Finish**, you are not able to return to the lab. • **Exit**: Displays a screen that allows you to confirm or cancel the exiting from your lab. If you confirm that you want to “exit” your lab, VMware Lab Platform returns you to the lab console login page. VMware Lab Platform also provides you with an option to resume the lab. The **Exit** option is displayed only if your admin has enabled it. • **End**: Displays a screen that allows you to confirm or cancel the ending of your lab. If you confirm that you want to “end” your lab, then VMware Lab Platform considers you have completed the lab and returns you to the VMware Lab Platform lab console login page. As a result, you are unable to resume the lab. You must click **Exit** to come back and resume your lab.
Reset	If you reset a lab, all its virtual machines (VMs) are restored to their original state. You are not able to recover any work you have done. The **Reset** option is displayed only if your admin has enabled it.

Keyboard options for console

To view the keyboard options for the lab console, click the  icon. Use the slider button to activate the options you want to use. After you select the keyboard options, you can view the options under the  keyboard icon in the console toolbar.

Figure 18: Keyboard settings for the lab console



Console Options	Descriptions
Keyboard Layout	Select your keyboard layout to adjust any incorrect keys being sent to the console.

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Console Options	Descriptions
Intercept Paste	Intercept the paste shortcut keys and send the text copied from your browser (or elsewhere) to the console. This option could interfere with the console's own copy and paste commands.
Remap iOS Key to CTRL Key	If you are using iOS operating system, the VMware Lab Platform automatically remaps the iOS keyboard shortcuts equivalent to the Windows shortcut keys. You can remap the system keyboard shortcuts from the Settings menu.
Manual Shortcuts	If enabled, you can change the page in the manual with Ctrl key combinations even when the console has focus. See the gear menu for key combinations for your environment.
Fix non-ANSI US keyboard layouts	Remap the keys pressed on your keyboard to send US ANSI key code equivalents to the console. If you are using a non-US keyboard or have the keyboard set a non-US language in your operating system, enabling this option could help send the correct keys to the console.
Reverse Vertical Scroll Direction	Change the mouse scrolling direction sent to the console. This setting can help with mice or track pads that have the scroll direction reversed (sometimes called "natural").

Using the lab manual in VMware Lab Platform

You can choose the lab manual's display, either docking or floating the manual, or displaying the manual on a second device using the lab manual options in the interface.

General options

To display the instructional manual panel, click **Manual** in the top-right corner..

Manual Options	Descriptions
X	The **X** icon closes the panel.
Arrow-on-Screen	The **Arrow-on-Screen** icon makes the manual panel a floating window. Once you make the panel a floating window, you can click the **X** to dock the panel on the previous side and close it.
Arrow	Use the **Arrow** icon to toggle between a left dock or right dock of the manual panel.
Table of Contents	To display the links to each module of the manual, click **Table of Contents**. Manuals are typically comprised of modules which you complete sequentially.
Network Topology	Your lab comprises virtual machines. If your administrator has included a network topology image in your account, then your lab console includes a **Network Topology** option. To display an image that shows the network topology of the virtual machines that are used in your lab, click **Network Topology**.
Split Screen	Displays the Use a Second Screen or Device screen that includes links to open the manual in a new window and email a link to open the manual with another device such as an iPad or a second window on your desktop. Thus, you can go full screen on your console and have the lab’s manual on another desktop screen or on your iPad. VMware Lab Platform keeps the manual

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Manual Options	Descriptions
	steps on the second screen in sync with the main screen. When you change to the next step on one of your screens, it changes on your other screen. In order to get to that second screen, you can email yourself a link, scan a QR code or copy and paste a special key.

Drag-and-drop option

The **Drag and Drop** feature lets you to drag text from the lab's instructional manual, and drop it into a field in the lab console. To use the **Drag and Drop** feature:

1. Select some text in the manual.
2. Drop the text into the appropriate area on the lab console.

Copy and paste option

You can copy the text from your browser or elsewhere, including a lab manual to the console. To enable the feature, go the gear icon on the ribbon on the top of the console and select **Intercept Paste**. For MAC OS users, the system automatically remaps the user's system OS shortcut keys for the copy and the paste operation to the labs OS shortcut keys.

Taking an instructor-led lab in VMware Lab Platform

An instructor-led lab is a specified lab taken at a specified time with an instructor present. There are a few differences between what you see on your lab console in instructor-led labs and what you see in on-demand labs.

Classes

- You can check for whether you have been added to a class.
- Your instructor can use the Instructor Console to monitor your progress while the class is in session.

Impact on the lab console

When you start the lab, your lab console does the following:

- Alert you if your lab is being monitored.
- Provide a button to send a message to the instructor console.

Additional user features for VMware Lab Platform

VMware Lab Platform includes additional features to help users as they complete labs.

Detection of closed browsers

VMware Lab Platform can detect when you have walked away from a lab. For example, if you quit the browser without clicking **End Lab**, then VMware Lab Platform pauses the lab and bookmarks where you left in the manual. When you return to the lab, VMware Lab Platformreturns you back to where you left off.

Detection of inactive browsers

VMware Lab Platform can detect if you have not moved the mouse or pressed a key on your keyboard for a period of time (the default is 10 minutes). If you have left the browser inactive, then VMware Lab Platform pauses the your lab and bookmarks where you left in the manual. When you return to the lab, VMware Lab Platform returns you back to where you left off.

Lab time running out

When the time is nearly out for a lab, VMware Lab Platform flashes the timer in yellow. When the remaining lab time is low, VMware Lab Platform flashes the whole screen in red.

To add more time to the lab, you can click **Extend** in the warning box or on the console frame.

Email transcript

Transcript pages include the Email Transcript option. If you want VMware Lab Platform to send your lab transcript to the email address you have provided, Click **Email Full Transcript**.

How to check connection speed and strength

A speed test helps diagnose a slow or intermittent connection from your browser to VMware Lab Platform.

To run a speed test, open a browser with the lab URL,http://www.labplatform.vmware.com/*<account_name>*/speed-test/ where *account_name* is the unique name given to your organization or account and click **Run Test**.



The speed test screen displays information about VMware Lab Platform and Cloud speed and VMware Lab Platform and Cloud network latency.

- A good speed score is 1.5 MB/s or greater.
- A good latency score is 300 ms or less.

Using and Managing the VMware Lab Platform Instructor Interface

The VMware® Lab Platform™ (formerly known as VMware® Learning Platform™) instructor interface allows you to monitor and manage a class and provides you a class roster with a list of students assigned for each of the class.

As a user with the instructor role, you use the VMware Lab Platform instructor interface to view the instructor console, monitor student's progress of a lab, assist students when required, and perform actions related to a class.

A user with the administrator role assigns an existing user the role of an "instructor". For information on the administrator tasks for setting up classes and instructor accounts, see the *VMware Lab Platform Administrator Guide*.

At VMware, we value inclusion. To foster this principle within our customer, partner, and internal community, we have updated this guide to remove instances of non-inclusive language.

Intended audience

The primary audience for this guide is VMware Lab Platform lab instructors and administrators.

Where to begin when using the VMware Lab Platform instructor interface

Before you begin your instructor tasks in VMware Lab Platform, it is important to the understand the workflow of creating a class, adding users to a class, starting a class, and ending a class.

Day 1 for administrator:

- Activate the **Portal Settings › Configuration › Enable the New Instructor Interface** setting.

  NOTE
  Only super users or VMware Lab Platform support team can activate the **Portal Settings › Configuration › Enable the New Instructor Interface** setting for an account.

- Activate the **Portal Settings › Configuration › User UI URL** setting to point to the new user interface (https://labplatform.vmware.com/<org_name>). For more information, see Working with settings in VMware Lab Platform.
- Create a user account with an instructor role. For more information, see Working with roles and permissions in VMware Lab Platform.

Day 1 for the instructor:

- Ensure that you have the instructor role assigned to you. When the instructor role or user account with an instructor role is created, an email is sent to your email id with the login credentials for the VMware Lab Platform instructor interface. For more information, see Working with roles and permissions in VMware Lab Platform.
- You can upload multiple users to the VMware Lab Platform using the **Account List's Mass User Controls** option. As an instructor, you can use the **Mass User Controls** option to add users using a CSV file.
- Create a class. A class is made up of selected users (students) who take a specified lab, at a specified time and are monitored by a specified instructor. For instructions on how to create a class, see the Create a class topic.
- Start a class and monitor the progress of students in the class. For information about how to start the instructor interface, see Open the VMware Lab Platform instructor console.
- End a class. For more information, see Explore the VMware Lab Platform instructor interface.

Account activation and verification for an instructor role

Verify that you can authenticate to VMware Lab Platform with the instructor role account.

When an account administrator creates your user account as an instructor, you receive the `Welcome to VMware Lab Platform` email to your email address. The `Welcome to VMware Lab Platform` email includes a verification link.

You use the link to verify the account and configure a password for your account. Once the account is activated by you, the account is verified in the system.

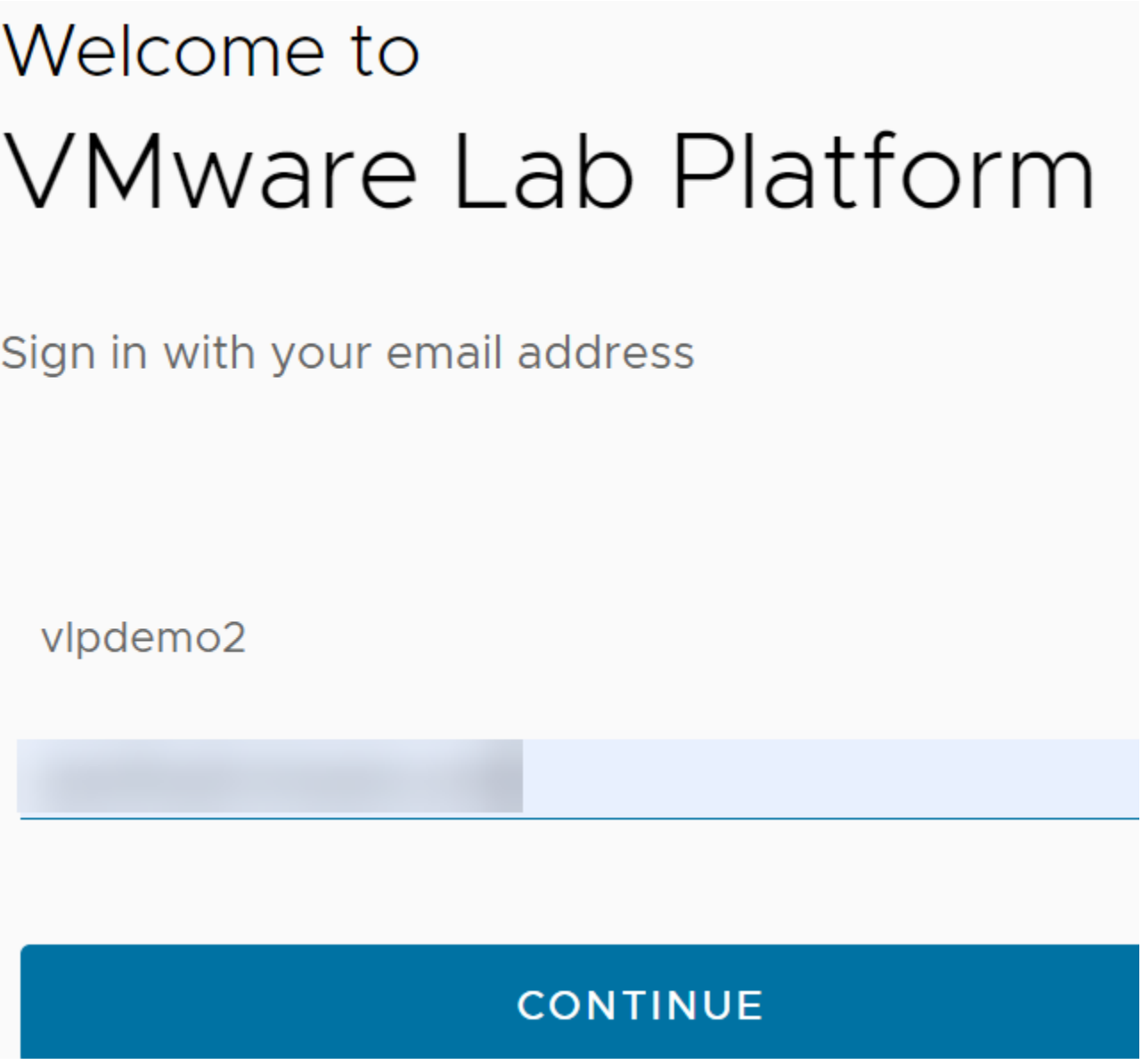
Log in to the VMware Lab Platform instructor interface

VMware Lab Platform provides you an instructor interface from where you manage your account and classes assigned to you.

- VMware Lab Platform instructor role and account URL to log in.
- Class assigned to you.

To log in to the VMware Lab Platform instructor interface:

1. Enter the https://admin.labplatform.vmware.com/login/ URL in your browser.
2. In the **Organization** field, enter your customer account name. For example, in the following image, `vlpdemo2` is the organization account name.



3. Enter your login credentials.
4. Click **Continue**.

Open the VMware Lab Platform instructor console.

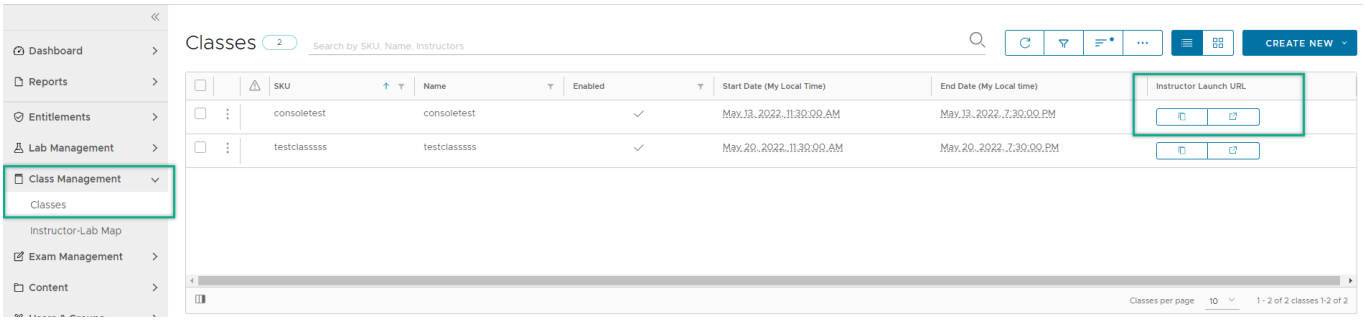
Open the VMware Lab Platform instructor console

You can start the instructor console for your lab from the administrator user interface navigation panel.

Ensure you have the VMware Lab Platform instructor role and account URL to log in.

1. Enter the https://admin.labplatform.vmware.com/login/ URL in your browser.
2. Enter your login credentials and click **Continue**.
3. To open the **Classes** page from the navigation panel, select **Class Management › Classes**.
4. Verify that the **Enabled** and **Instructor Launch URL** columns are displayed.

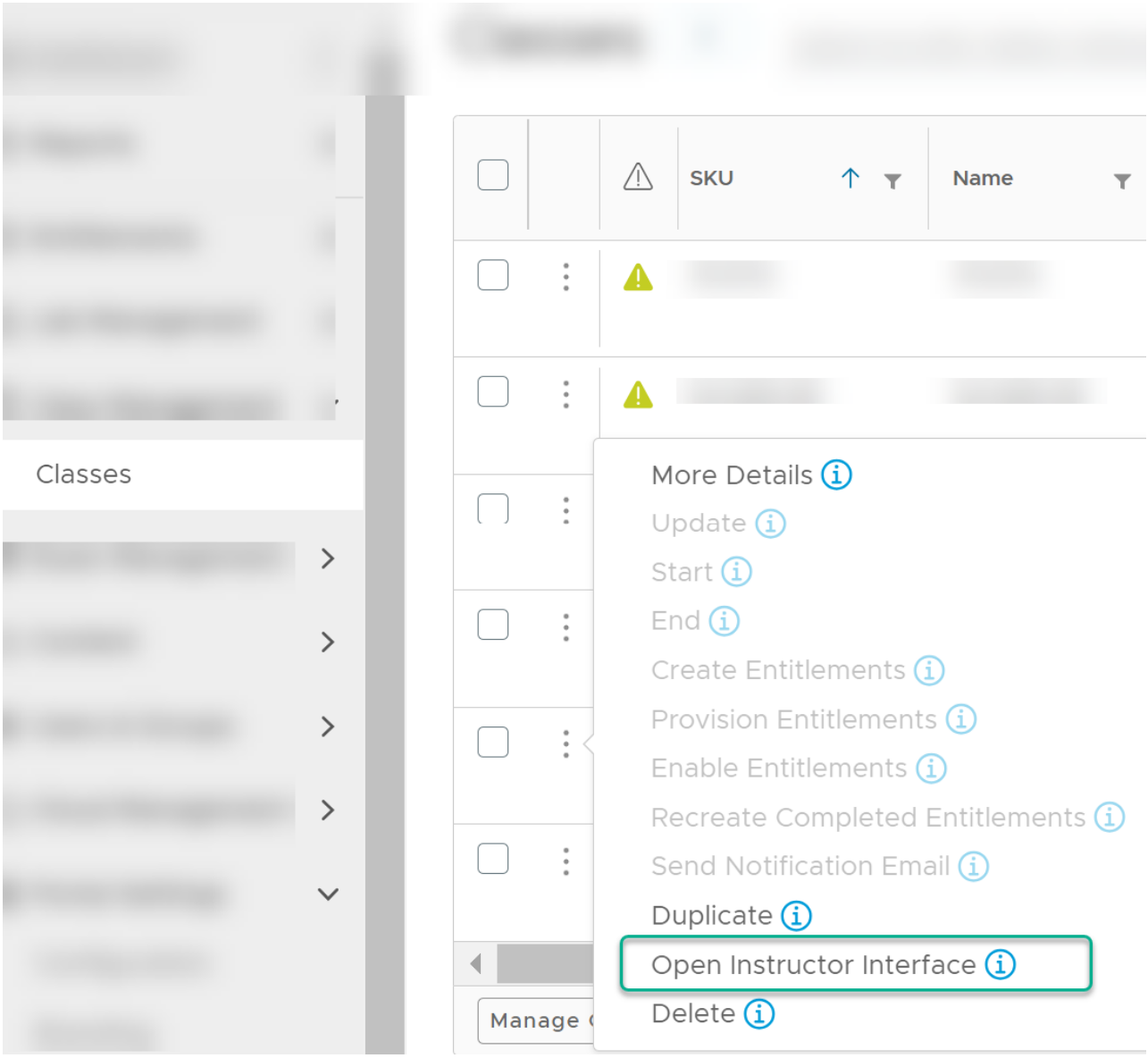
   To select what columns to display, click the view columns icon on the left, below the list of classes .



> **NOTE**
> You may have access to all the classes or only to classes that are assigned to you as an instructor. If your account administrator has activated the **Portal Settings › Configuration › Allow instructors to access all classes** setting, you can view all classes in that account. If the setting is deactivated, you can only view the classes that are assigned to you as an instructor.

5. To open the instructor interface, click the three ellipses next to the classes name and click **Open Instructor Interface**.

Figure 19: New instructor interface



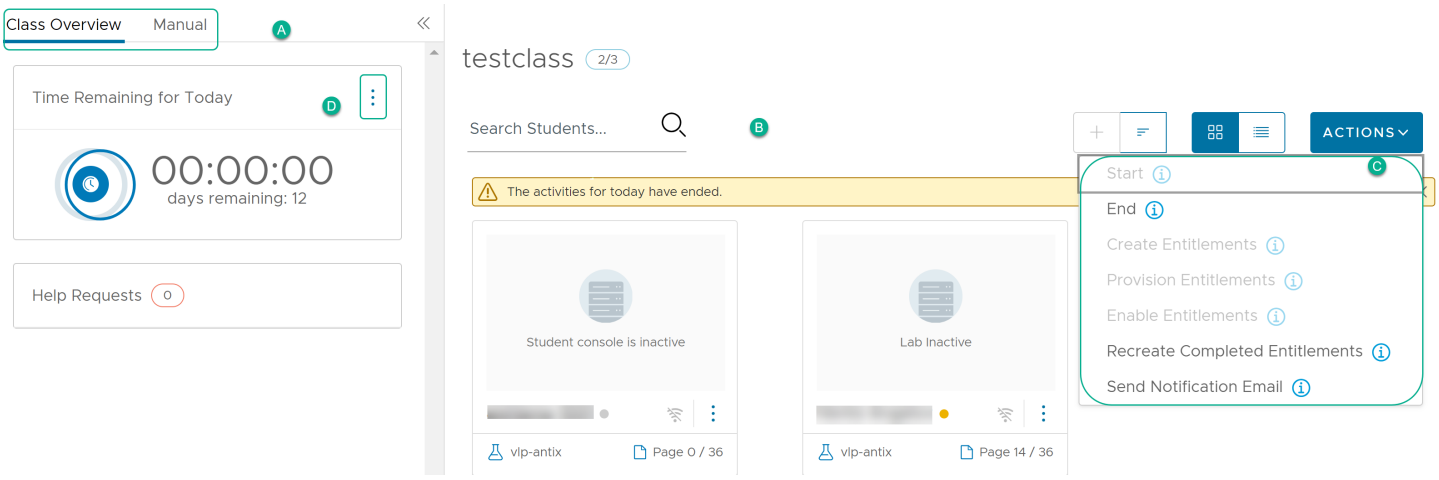
6. If the lab is not enabled, click the vertical ellipsis icon next to your class name and select **Start** from the menu. The system enables the class, provisions entitlements, and starts the class.

To learn more about the instructor interface, see the Explore the VMware Lab Platform instructor interface topic.

Explore the VMware Lab Platform instructor interface

Get familiar with the VMware Lab Platform instructor interface. Learn how to use the instructor interface, manage, and monitor a class.

1. To view the instructor interface, go to **Class Management › Classes**, click the three ellipses next to a class name, and click **Open Instructor Interface**.



Menu options	Description
A - Class Overview	The default dashboard provides an overview of the class, lab manual, timer for the class, help requests, and a list of action options to manage the class. When a Raised Hand help request occurs, a counter appears with the help request.
B - Student's Lab View	Displays a tile for each student in the course. Each tile includes the following information: • Student name and photo, if available. • The stop/start status for the lab. • Tiles are outlined in green when the student starts the lab. Otherwise, the tile is dimmed and the status says "not started". • A progress bar displaying how much of the lab the student has finished. Refer the lab manual page count for each student to view the lab progress. • Click the + icon to add more lab capacity and resources for the class. Click the filter icon to view the status of active and inactive labs. Use the tile view or list view to display the details of students and student lab console in the instructor interface. • Click the vertical ellipses to see the action options for an individual student lab. You can perform the following actions: – **Explore Console**—Lets you to view the student's console view. To open each lab console as a new tab with the student name, click the three ellipses on the student tile, and then right-click **Explore console** and select Open in a new tab. This feature helps in navigation and visibility in each student's lab console. – **Take Over Machine**—Allows you to take over the student's machine so that you can resolve any issues or troubleshoot an issue that a student might be experiencing with the lab. – **End Lab**—Ends the student lab. – **Clone Lab**—Clones the existing student lab. – **Copy Quick Launch URL**—Copies the Quick launch URL of the lab.

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Menu options	Description
C- Action Menu	You can perform the following actions from the instructor interface for a class: • **Start**—Starts, enables, and provisions the selected classes. Also sets the statuses of the selected class to “Active”. You can also edit the start time of a started class. The new start time of the class will be effective from the following day. • **End**—Ends a class. Sets the status of the selected class to “Complete” which will archive them and prevent further updating. • **Create Entitlements**—Allows you to create entitlements for your students in the selected class, but does not enable them. Until activated, a student’s lab console will not list the entitlements in the Enrollments tab. • **Provision Entitlements**—Powers up the lab templates and lab resources for each entitlement (manually triggered prepop) for the selected classes. **NOTE** If a class does not have a lab template or deployment pool with a lab template, a warning message is displayed for the class. • **Enable Entitlements**—Enables entitlements for a class. • **Recreate Completed Entitlements**—Recreates a completed entitlement for a class. • **Send Notification Email**—Sends notification email to students such as class is starting, or reminder to log in for the class and so on.
D- Timer	The timer helps you to monitor and manage the time of the class. If your students need more time to complete a lab, you can also extend the class time. **NOTE** The following settings must be activated, if you want to modify or update a class. • **Allow Instructors to Manage Lab and Entitlements of a Class** • **Allow Instructors to Clone vApps** Click the vertical ellipses, and select **Extend Class**.
Zoom App	If your class has the Zoom app integrated for the lab, you can view the details of the Zoom conference in the left-navigation pane of the instructor interface.
Student list view	You can change the way the student entries are listed on the page. Use the **Tile** icon or the **Card** icon to change the page listing.

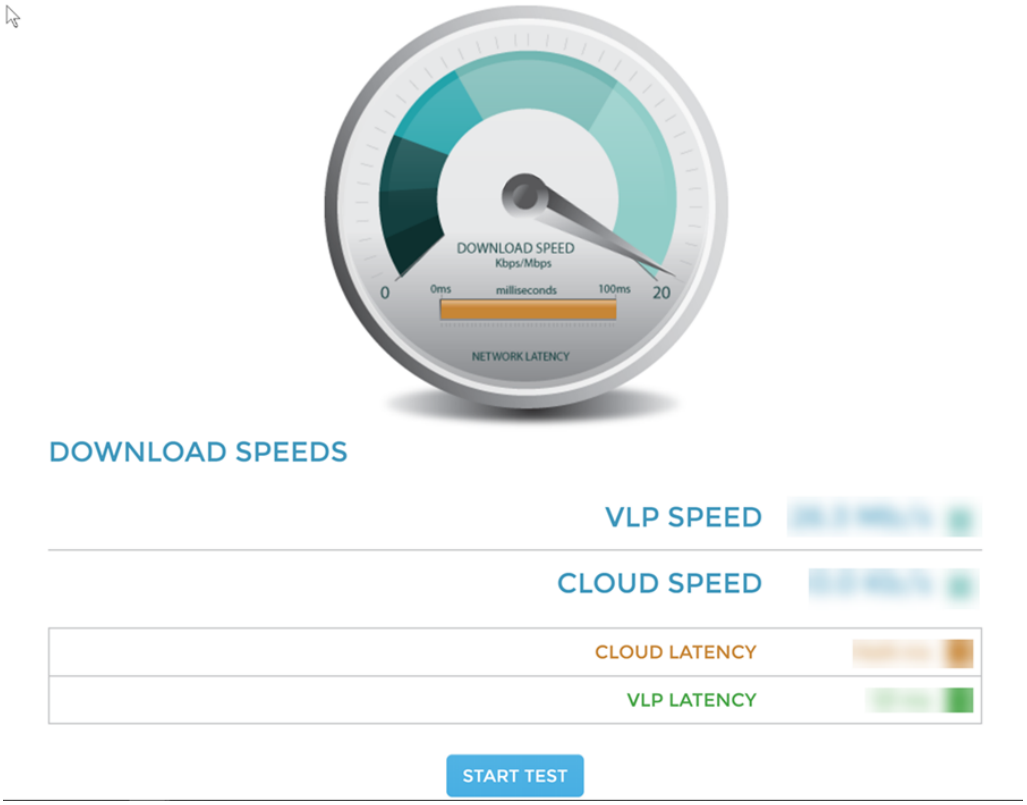
Speed Tests

You can use speed tests to help diagnose a slow or intermittent connection for a lab.

Speed tests measure network bandwidth and network latency for VMware Lab Platform and for your cloud, if you are using a hybrid configuration.

Running a Speed Test

To run a speed test, open a browser with a URL of the form http://labplatform.vmware.com/*account_name*/speed-test/ where *account_name* is the unique name given to your organization or account, and click **Start Test**.



Speed Test Results

After you run the test, the speed test screen displays results for VMware Lab Platform and cloud speed, and for VMware Lab Platform and cloud network latency. Network latency results are color coded.

- Results less than 300 ms are shown in green
- Results greater than 300 and less than 450 ms are shown in amber
- Results greater than 450 ms are shown in red
- A good speed score is 1.5 MB/s or greater
- A good latency score is 300 ms or less

Managing and monitoring a class in VMware Lab Platform

As a user with the instructor role in VMware Lab Platform, the primary task of an instructor is to manage and monitor a class, and facilitate virtual learning for your students in a class.

Modify a class in VMware Lab Platform

As an instructor, you can modify a class from the **Class Management › Classes** page.

- The following settings must be activated, if you want to modify or update a class.

  Allow Instructors to Manage Lab and Entitlements of a Class

  Allow Instructors to Clone vApps

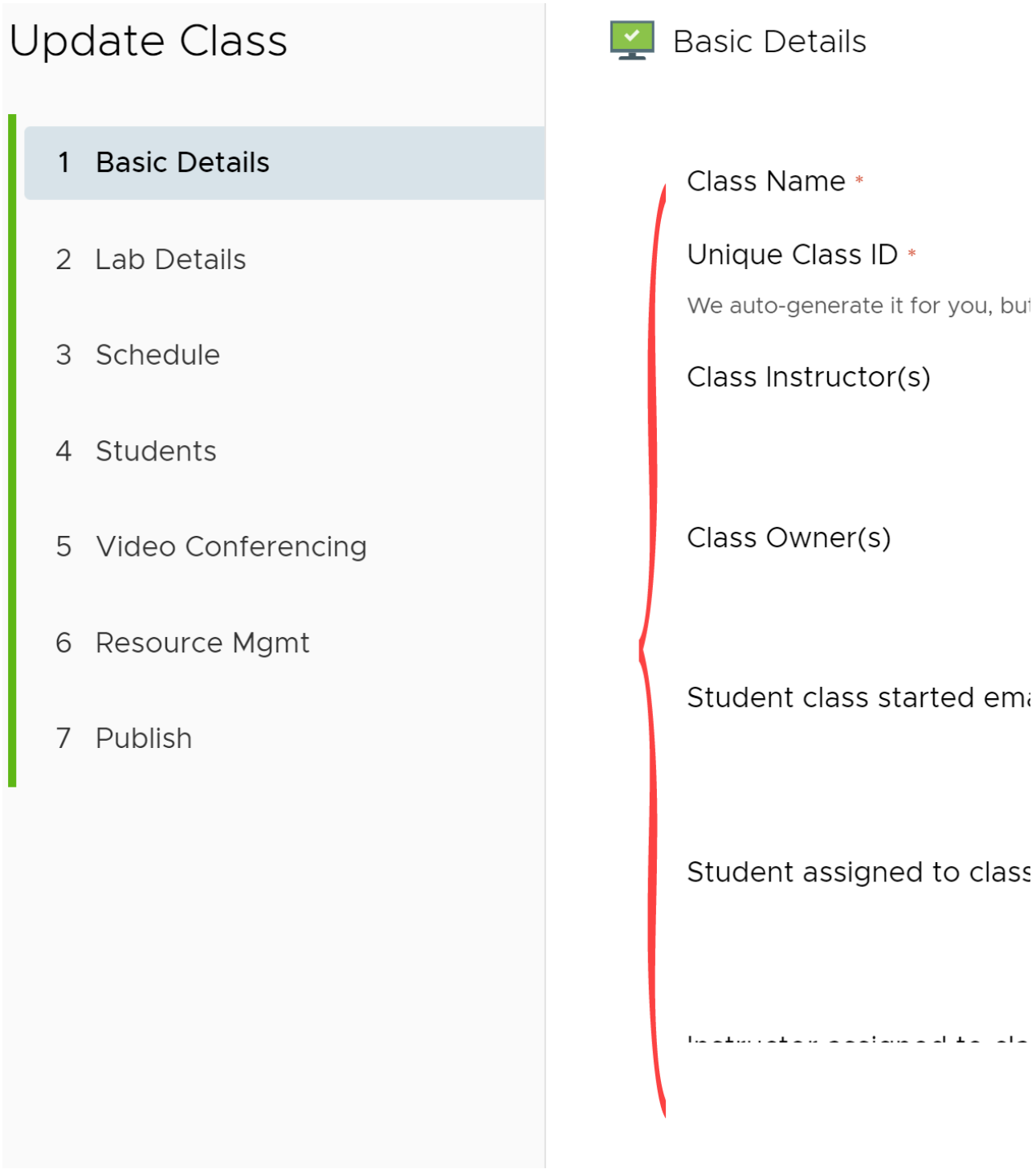
For information about other instructor-related settings, see Working with settings in VMware Lab Platform

You can modify settings for a class including adding or removing students, managing labs or entitlements, modify the start time of a started class, or cloning a vApp.

NOTE
Each class can have one lab, and the lab cannot be changed after the class is started.

IMPORTANT
If the class end date or time is changed, then the Entitlement expiration date is automatically updated to the same. For more information, see Managing Lab Entitlements.

1. To display a list of available classes, from the navigation panel, click **Classes**. If you have activated Advanced Options, select **Class Management › Classes**.
2. To update a class form, click a class name, and click **Update**.



3. In the **Update Class** form, edit information for your class, and click **Update Class**. If you modify the time and date values of an already started class, the modified values will be effective from the next day.
   The updated class is listed in the **Classes** section.

Extend a class in VMware Lab Platform

As a user with the instructor role, you can extend the class time if your students need more time to complete a lab.

- Instructor role assigned to you.
- Class assigned to you.

To extend the class time:

1. Go to **Class Management › Classes**, click the vertical ellipses for your class and select **Open Instructor Interface**.
2. On the instructor interface, click the vertical ellipses in the time section, and select **Extend Class**.
3. In the **Extend a Class Schedule** form, enter the date and time till when you want to extend the class for.
4. Click **Save**.

Assist students and raise help requests

In VMware Lab Platform, as a user with the instructor role, you facilitate a class and help your students when required. When a Raised Hand help request occurs, a counter appears with the help request.

To view if a student needs help, click the help request counter.

1. Click the **Help** button. The system opens or collapses a panel. The panel consists of all raised hand help requests. You can click the check mark to indicate that the request is resolved. The computer icon next to the resolve check mark opens the student's console.
2. To take control of a student's lab, click the vertical ellipses in the student lab view and select **Take Over Machine**. This allows you to take over the student's machine so that you can resolve any issue or troubleshoot an issue that a student might be experiencing with the lab.
3. For speed related issues, you can run a speed test. For more information, see the Explore the VMware Lab Platform instructor interface.

To learn more about VMware Lab Platform, see the VMware Lab Platform Release Notes.

Using and Managing the VMware Lab Platform Event Management Interface

The VMware® Lab Platform™ (formerly known as VMware® Learning Platform™) event management interface helps you to create and manage events, seat maps and control seating arrangements for events, and resources for one or more physical, or virtual events to deliver on-demand training labs.

Intended audience

The primary audience for this guide is administrators, proctors, and the VMware Lab Platform technical support team.

Administrators

Users with the “Admin” role can access all features of the event management user interface. Administrators can create, edit, and delete events, seat maps, and add users to events. The Admin role is the most powerful user role within an account.

Proctors

Users with the "Proctors" role are familiar with the lab content for the various labs for an organization account. When end users have a question with a lab and need help, users can use the **Help Request** option on their lab console screen. This option notifies the proctor about the help request through an application on the proctor’s mobile device. A user having an Event/Proctor role can also log in to the administrator user interface.

Technical support

A user having a support role can log in to the administrator user interface and has a subset of the administrator permissions.

Overview of the VMware Lab Platform Event Management interface

VMware® Lab Platform™ provides the event management user interface (UI) for administering and managing labs at large-scale events.

To manage lab station registration, seating maps, and help requests for virtual events and physical events, you use the VMware Lab Platform event management interface. Users with the administrator role can define and schedule multiple events per VMware Lab Platform account.

VMware Lab Platform Event Management concepts and terminology

Before you start using the VMware Lab Platform event management interface, it is good to be familiar with several concepts essential to the VMware Lab Platform event management user interface.

Event

In VMware Lab Platform, an event represents a real event such as Dell World or VMworld. An event is either only virtual, or both virtual and physical at the same time. Similar to real-time events, you can define a time period and the capacity limit for an event in the event management interface. There are many different types of formats to deliver online events.

- Self-paced labs: This is the most commonly used format as it allows users to casually browse the lab catalog and take the labs they are interested in.
- Webinar presentation and demos: This is ideal for large numbers of attendees learning about at specific product. In some cases, they are also structured as a series of webinars. A product expert presents the product or solution, and provide context and then demo the product using a lab.
- Expert-led workshops: The format is similar to the webinar demo presentation, but here the attendees take a lab with you.

For more information on how to create an event, see Create an event in VMware Lab Platform.

Event Virtual Attendees

Event attendees refers to the number of users that can attend the event or take the lab in the event. Based on an event or lab limit, attendees can start a lab. As an administrator, to view all the information of a virtual event attendee, navigate to **Event Management › Events**, click the three ellipses for an event entry, and click **View Virtual Attendees**.

Lab Queues

You can add lab queues to events. A lab queue defines the capacity limit of a lab. Each event can have multiple lab queues associated with it, depending on the labs that are in the event's group catalog. You can create a lab queue with lab capacity and add it to an event.

When users start a lab that has 'Lab Queue' associated with it, users are put in the queue based on the lab capacity limit. If all the Lab Queue's capacities exceed the capacity of the event, then queuing happens on an event level.

NOTE
Lab queues are applicable only for virtual events.

For more information on how to create a lab queue for an event, see Create a lab queue in VMware Lab Platform.

Waiting Room

Users are placed in a waiting room when an event capacity limit is reached. A waiting user can subscribe to notifications, and is notified when the users turn comes.

NOTE
Waiting room is applicable only for virtual events.

For more information on how to create a waiting room for an event, see Create a waiting room in VMware Lab Platform.

Seat Maps

A seat map is the graphical representation of a physical event’s layout. You define a seat map and add sections with seats, where attendees of physical events can be placed. This represents the real desks and stations in the physical room.

The stations of a seat map are color coded:

- Black: unregistered
- Green: occupied
- Green with lab icon: user is taking a lab
- Gray: empty (registered, but not occupied)
- Red: need help (user has raised hand)

For more information on how to create a seatmap for an event, see Create a seat map in VMware Lab Platform.

Capacity Management

To see if entitlements are not part of a class, when an event is happening, a check is performed before each entitlement can start. Enable the **Limit Concurrent Entitlements** setting option to perform the entitlement check. The event management service tracks all entitlements, and if the capacity limit is reached, users are put in a waiting list until a place is available.

There is another view that appears when a spot is freed and the user is allowed to start the requested lab. It has counter that counts down to 0 and shows the time within which the user can start the requested lab. Otherwise, the user loses its place in the queue, and in case the user requests the same lab again, user is placed at the end of the queue. After the time becomes 0, the lab expires.

NOTE
Capacity management is applicable only for virtual events.

Lab Station

A lab station is a virtual lab working seat, with chair, table, and computer. You go to each computer and register each work station as a lab station before a regular user comes to take the lab. When a lab user comes to take the

lab, you assign the user to one of the registered lab station. When the user logs in VMware Lab Platform, the live seat map changes its state. You register a station for each lab user.

For more information on how to register a lab station for an event, see Register a lab station in VMware Lab Platform.

BYOD

Bring your own device (BYOD) for a traditional lab setup. You can use your own device to take a lab.

Traditional

A traditional seating section in a seat map with a virtual lab working seat, chair, table, and a computer.

Using the VMware Lab Platform Event Management interface

Users with administrator role can create, edit, and delete virtual events, and traditional, in person events using the VMware Lab Platform**Event Management** user interface. To access the **Event Management** user interface, you must have the administrator credentials .

To use the **Event management** user interface:

1. To use the event management menu, activate the **Portal Settings › Configurations › Side Menu › Events Menu Enabled** setting option.
2. From the navigation pane, click **Event Management › Events**.

You can configure an event for a virtual event or a traditional event. Use the following workflows to configure an event:

- Workflow for Managing a Virtual Event
- Workflow for Managing a Traditional Event

Workflow for managing a virtual event in VMware Lab Platform

As an administrator, you can create, edit, delete virtual events using the **Event Management** user interface.

- Administrator user role permissions.
- Lab catalog for the event.
- Users or a user group for the lab.

Typically, you use the following set of tasks to create and run a virtual event.

1. Create a Waiting Room.
2. (Optional) Create a Lab Queue.
3. Create an Event. (Add waiting room, capacity and queue expiration properties related to virtual events). There are no seat maps for a virtual event.
4. Start a Lab.

Create a waiting room in VMware Lab Platform

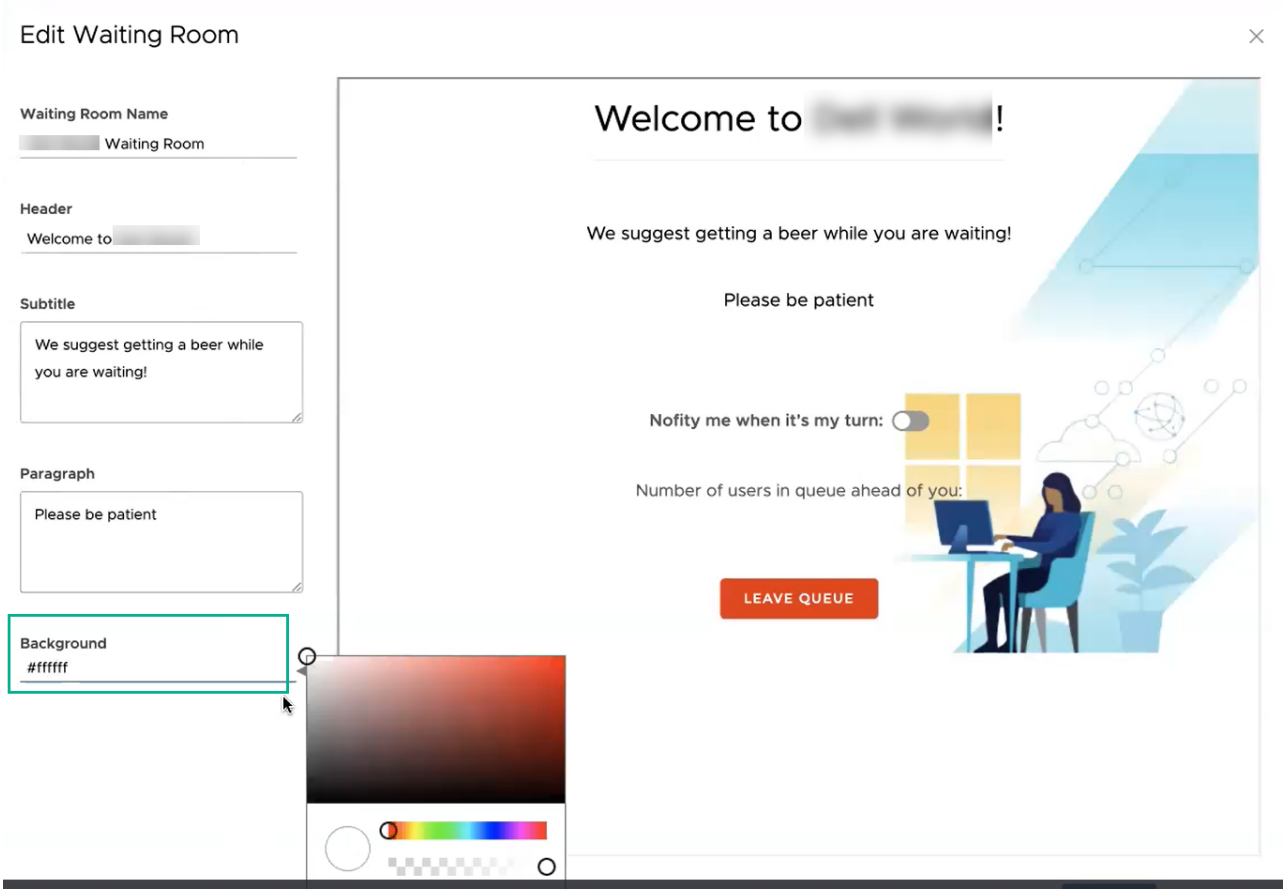
Managing large events can be challenging especially large virtual events. With the Waiting Room feature in VMware Lab Platform, you can customize the view that waiting users for an event sees. You can define a waiting room, select a background color for the waiting room page, and attach messages for your lab users. When an event reaches maximum capacity, and a user logs in, the user is placed in the waiting room. When a spot is available or free, the user can start the lab.

To create a waiting room:

1. From the navigation pane, click **Event Management › Waiting Rooms**.
2. Click **New Waiting Room**.

3. On the New Waiting Room page, enter the details and select a background color for the waiting room, and add a message if you want.

Figure 20: Sample View of the Waiting Room Page



4. Click **Create**.

Assign the waiting room to an event.

Create a lab queue in VMware Lab Platform

As an administrator, you can attach lab queues to events in VMware Lab Platform. A lab queue allows the capacity limitation at lab level.

Each event can have multiple lab queues associated with it, depending on the labs that are in the event group catalog. Labs associated with a lab queue are queued on a lab basis when a user starts such labs. If all lab queues capacities exceed the capacity of the event, queuing happens at event level.

> **NOTE**
> The lab queues parameter is applicable only for virtual events.

To create a lab queue for an event:

1. From the navigation pane, click **Event Management › Lab Queues › New Lab Queue**.
2. On the New Lab Queue page, enter the details, and click **Create**.

Option	Description
Lab SKU	To add a queue to it, select the lab SKU from the list.
Capacity	Enter the queue capacity value.
Event	To add the queue for an event, select an event from the list.

All the queues are listed on the Lab Queues page. To edit, or delete a lab queue, click the vertical ellipses next to the queue name.

Create an event in VMware Lab Platform

You can create an event for a virtual class or a traditional in-person class, using the **Create Event** wizard in VMware Lab Platform.

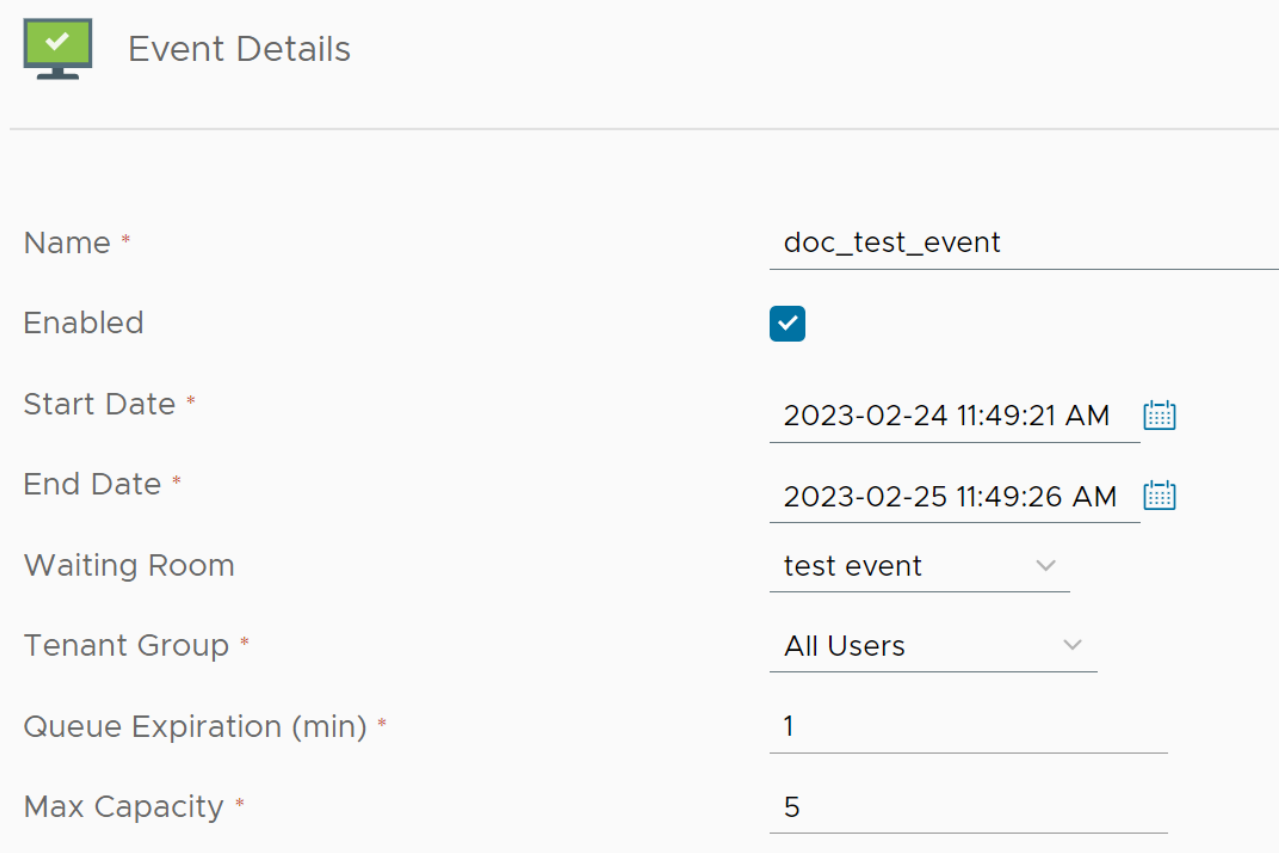
To use the event management menu, activate the **Portal Settings › Configurations › Side Menu › Events Menu Enabled** setting option.

To create an event:

1. From the navigation pane, click **Event Management › Events › Create New**.
2. On the Create Event page, enter the event details.

Option	Description
Name	Enter a name for the event.
Enabled	Ensure that the **Enabled** check box is selected.
Start Date	Select the start date and time for the event.
End Date	Select the end date and time for the event.
Waiting Room	For virtual events, you can add a waiting room page, where users in the waiting queue can be placed. For information on how to create a waiting room, see Create a Waiting Room.
Tenant Group	Select and add a group to associate for the event from the drop-down list. Only users in that group can participate in the event.
Queue Expiration	**NOTE** The Queue Expiration parameter is applicable only for virtual events. Enter a time period in minutes for users after which they are allowed to start their lab. For example, when an user gets queued and the users their turn comes, the user will have X minutes before the user can take the lab.
Max Capacity	**NOTE** The Max Capacity parameter is applicable only for virtual events. For traditional events, use the Seat Map feature. For more information on Seat Map, see Create a Seat Map. Enter the value for the maximum capacity of users that can start a lab from the event simultaneously. If the maximum capacity is reached and a user starts a lab, the user is moved to the waiting room page. You can customize the waiting room page. When a place is available in the queue, the user is moved from waiting to pending state before the user can start the lab.

Figure 21:



3. Click **Create Event**.

All the events are listed on the **Events** page. To edit, update, delete, or unregister lab stations, and view live information about virtual attendees for the event, click the vertical ellipses next to the event name.

Start the lab using the event URL. For example: https://www.learningplatform.vmware.com/event/.

To view the live status of a user, on the Events page, click the three ellipses next to the event row, and click **View Virtual Attendees**.

Edit an event in VMware Lab Platform

You can edit a room, rename a room or a section, reconfigure the class map, or renumber lab stations in a class using the Key definition for "VLM_name_long" not found in the DITA map. event management interface.

1. On the event management toolbar, click **Admin**. The Admin page opens.
2. Click **Edit** for the room you want to edit.
3. Click the monitor icon to toggle between the trad or the BYOD view.

NOTE
Options to work with a room's rows, columns, and sections vary depending upon whether you are in the Trad view or the BYOD view.

4. Work with the row/column and section options to edit your room.
5. Click **Save** to save the changes for your room.

Start a lab in VMware Lab Platform

Lab users and students use the VMware Lab Platform user interface to access and use a lab.

- To use a lab, you must have a user account and an entitlement for the lab.
- Lab URL to access the lab.

  Example: https://www.labplatform.vmware.com/event/.

  For example; Lab URLs: https://www.labplatform.vmware.com/<*account_name*>/event. In this example, account name refers to the account organization that you belong to.

To start a lab:

1. Enter the event lab URL in a browser.

   NOTE
   For example, the URL for the test account for a user is `https://www.labplatform.vmware.com/<test>/event`.

   In this example, *test* is a variable. Replace it with your organization account name that is registered with VMware Lab Platform.

2. Enter your user credentials.
3. Click **Login**.

The end user interface with the lab console is displayed. Users can start the lab. For more information on how to use the VMware Lab Platform user interface for a lab, see VMware Lab Platform User Guide.

Workflow for managing a traditional event in VMware Lab Platform

As an administrator, you can create, edit, delete traditional, in-person events using the VMware Lab Platform**Event Management** user interface.

- Administrator user role permissions.
- Lab catalog for the event.
- Users or a user group for the lab.

For a traditional, physical event, perform the following tasks:

1. Create an event.
2. Create a seat map with some sections.
3. Register a lab station.
4. Start a lab.

Create a seat map in VMware Lab Platform

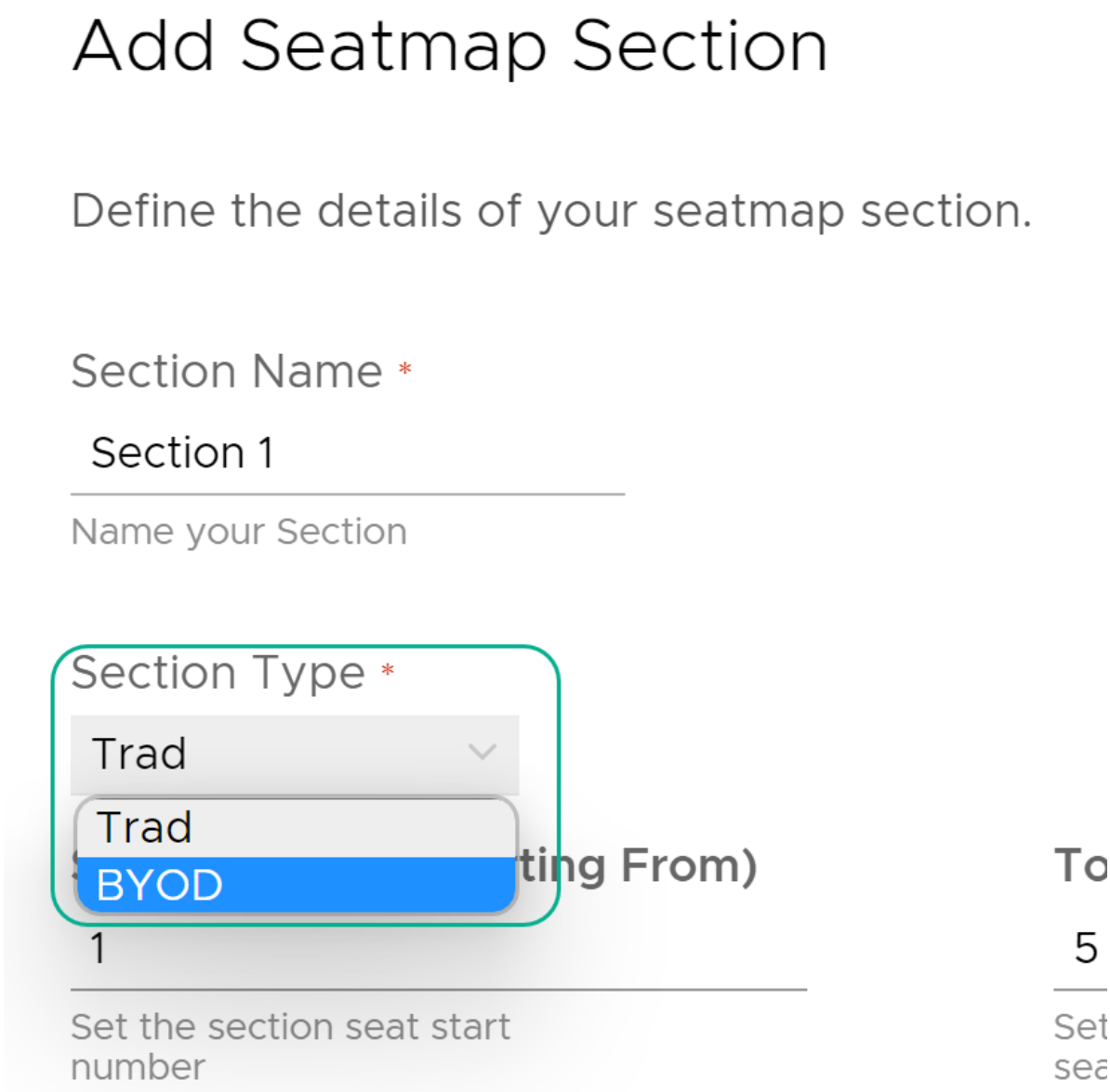
A seat map is a graphical representation of an event's layout. You can define a seat map and add sections with seats, where attendees of physical events are placed.

To create a seat map for an event:

1. From the navigation pane, click **Event Management › Seat Maps › New Seat map**.
2. On the New Seat map page, enter the event details.

Option	Description
Label	Enter a name for the label.
Event	Add an event for the seat map from the list of events. For more information on how to create an event, see Create an Event.

3. Click **Create**.
4. (Optional) To add sections for the seat map:
   a) On the Seat Maps page, click the three ellipses next to a seat map in the list, and click **Edit**.
   b) Click **+ Add Section**.



c) Enter the details in the Add Seatmap Section page.

In the Add Seatmap Section page, there are two types of section types that you can choose from:

- Trad: A traditional lab setup.
- BYOD: Allows users to use their own device and setup for a lab.

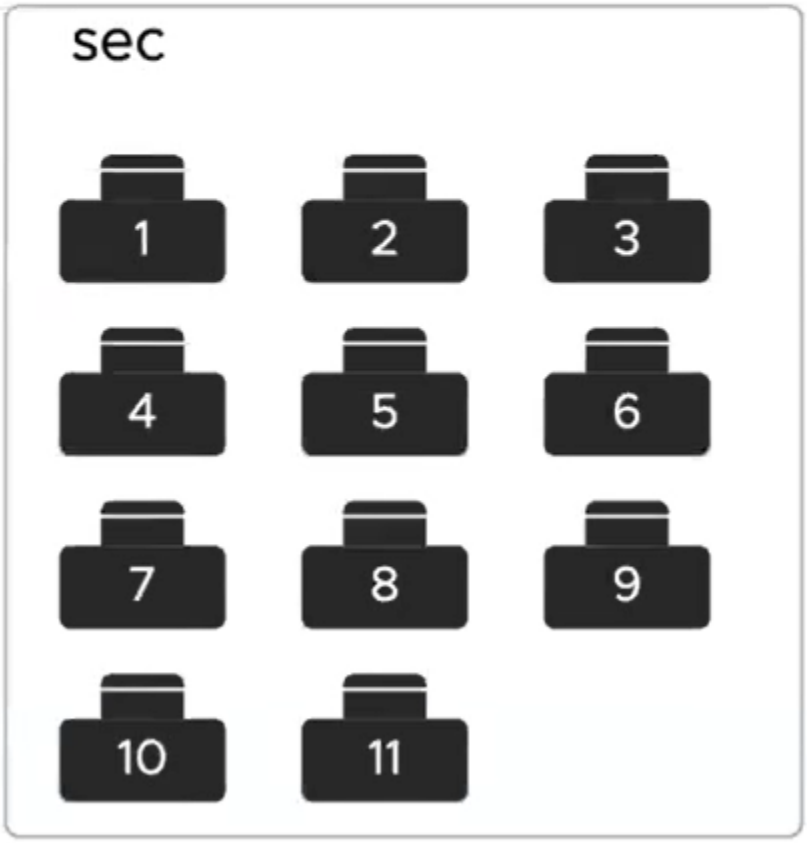
d) Click **Add Section**.

To edit a section, click the three ellipses next to the section, and click the action option. You can add more seats, delete seats and clone the section.

5. Click **Create Seat Map**.

All seat maps for events are listed on the Seat map page. To view the live seat map, on the Seat Maps page, click the three ellipses next to the seat map, and click **Live Seat Map**.

The following is a sample seat map representation for an event.



The stations of a seat map are color coded:

- Black: unregistered

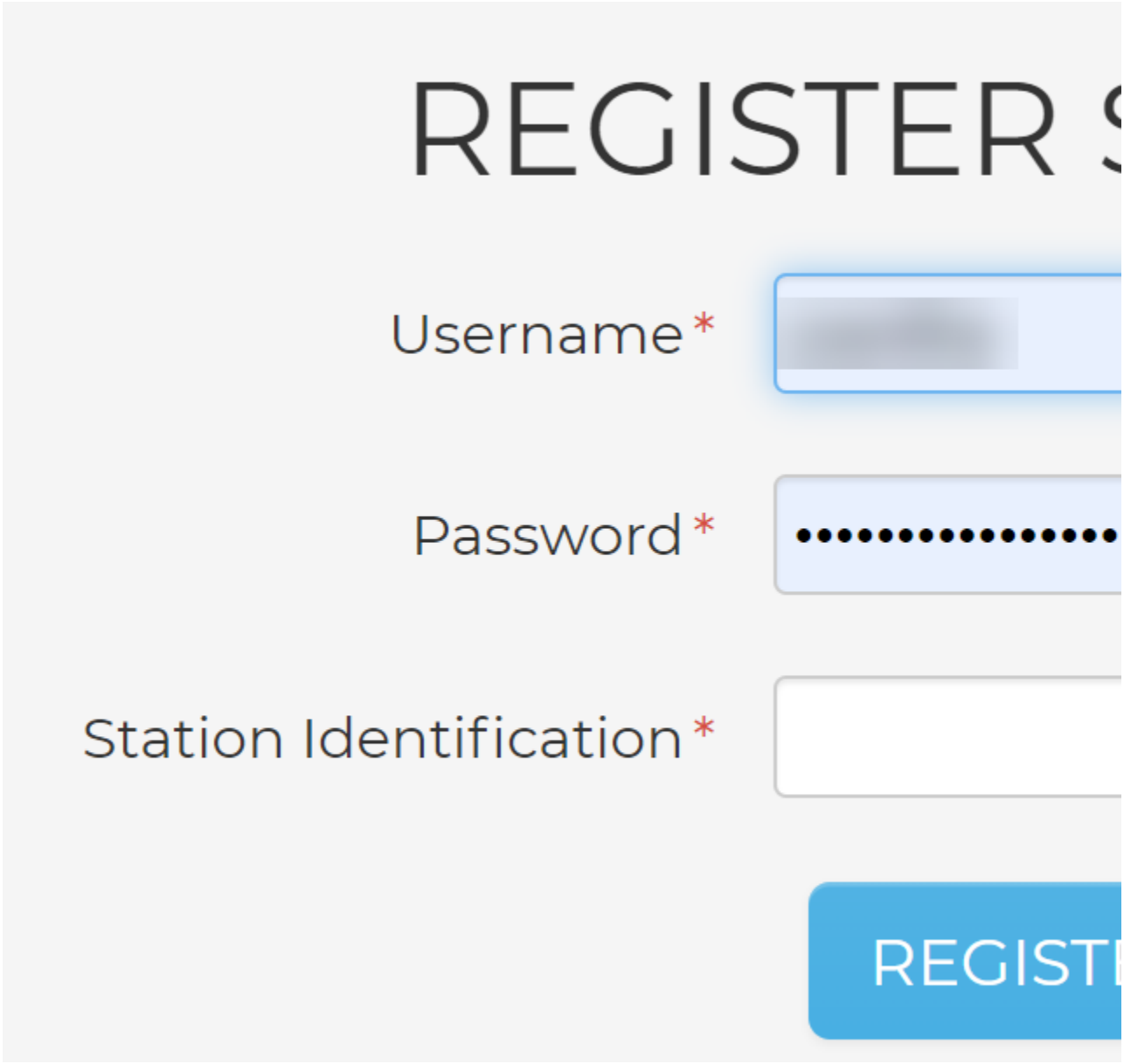
- Green: occupied
- Green with lab icon: user is taking a lab
- Gray: empty
- Red: need help

Register a lab station in VMware Lab Platform

For a traditional event, which is an in-person event, typically you have a room with computers and chairs for each event participant. A user with the proctor role, goes to each computer and registers it as a lab station before a regular user comes to take the lab. Proctors register each lab station at https://www.learningplatform.vmware.com/event/. When a user comes to take the lab, proctors assign the user to one of the registered lab station. When the user logs in VMware Lab Platform, the live seat map changes its state. You register a station for each lab user.

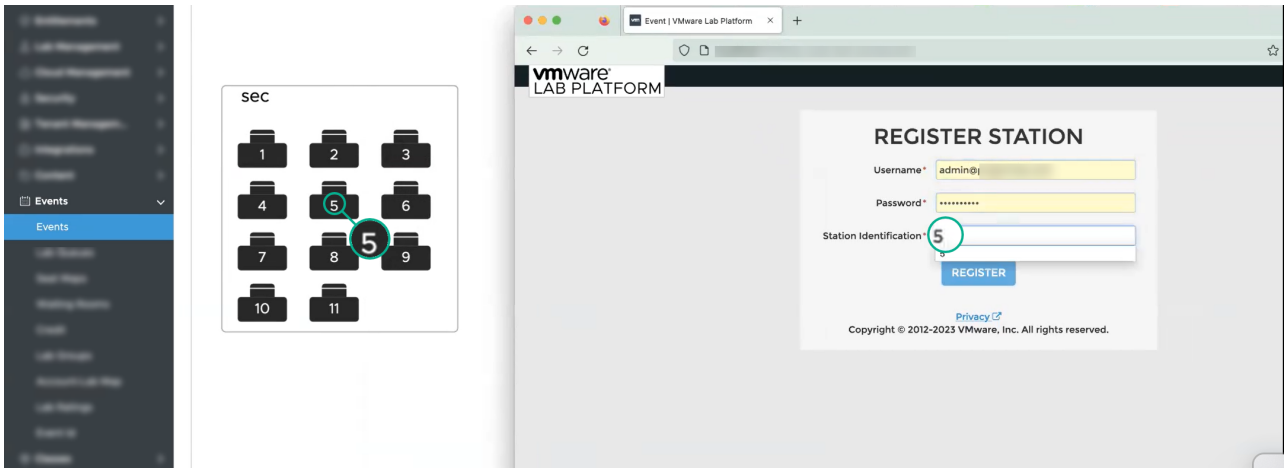
- You need to have an administrator, a proctor, or a super user privileges to register a lab station.
- A seat map with seat IDs configured in the Seat Map page.

1. To register for a lab, go to the event URL for your VMware Lab Platform account. Example: https://www.learningplatform.vmware.com/event/.



2. Enter your administrator credentials and a station ID from the seat map, then click **Register**.

Figure 22: Sample view of the Register Station



Upon a successful attempt, the system displays a message that you have registered the station. In order to succeed with that, a seat map should be created first, and then the station is available for registration.

Users can log in to the lab, and the assigned station ID appears on the user's lab page.

Lab URLs: https://www.labplatform.vmware.com/<*account_name*>/event. In this example, account name refers to the account organization that you belong to.

In the seat map view, the status of the lab station changes to green. If a user needs help, the user can request for help using the Help Request option in the lab.

Working with the VMware Event Proctor application

VMware Lab Platform enables users with account administrator and proctor role, to conduct and monitor online tests and events similar to in-person tests and events using the VMware Event Proctor mobile application. When lab users have a question or need help with a lab, users can use the Request option on their lab console screen. The option runs an application on the proctor's mobile device to notify the proctor that someone needs help. A user having an Event/Proctor role can log into the VMware Lab Platform administrator user interface.

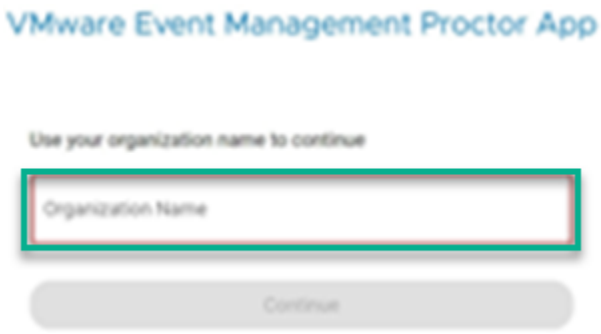
The following section gives instructions on how to install and use the VMware Event Proctorapplication for VMware Lab Platform services.

Install the VMware Proctor application on Android devices

You install the VMware Event Proctor mobile application, so that lab users can easily communicate with you using the proctor application if the users need any help or request while taking a lab.

To install the VMware Event Proctor application on an android mobile device:

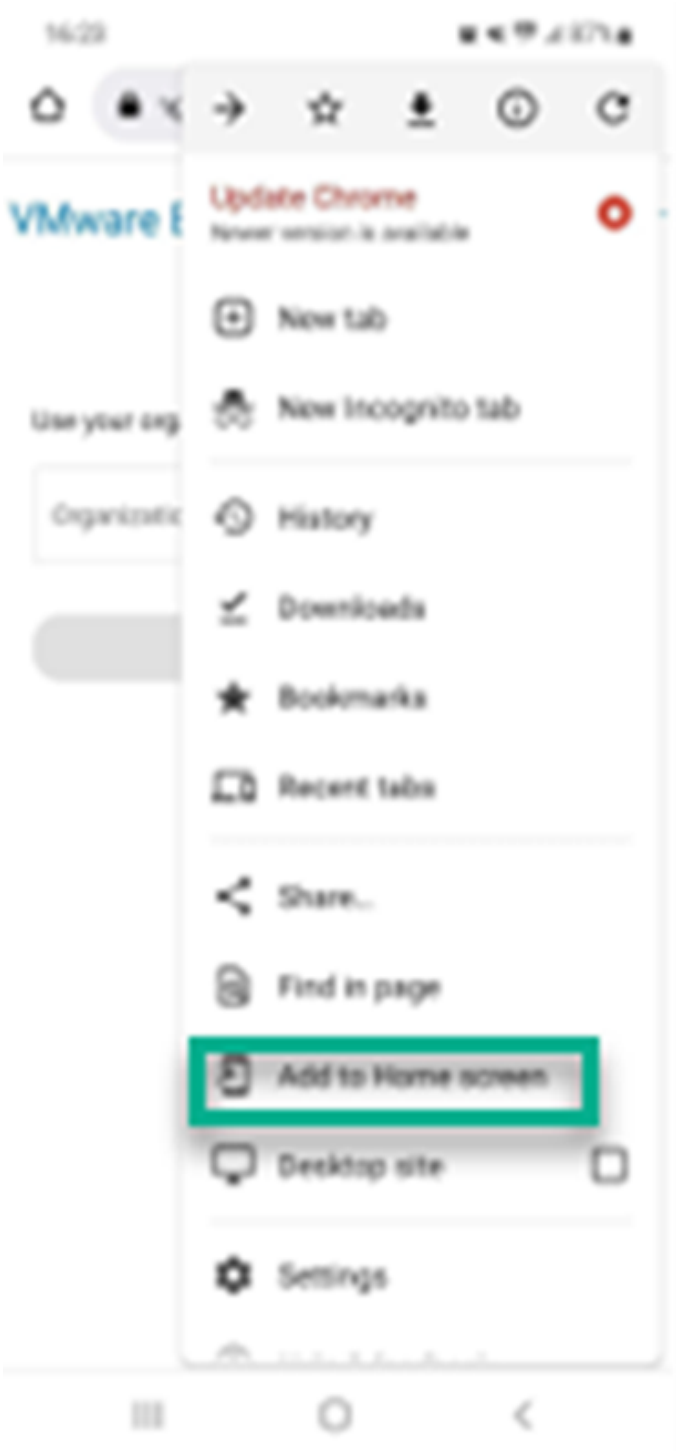
1. On an android mobile device, enter the proctor application URL in a web browser. URL:https://proctor.labplatform.broadcom.com/login
2. On the **VMware Event Management Proctor App** page, enter your organization name in the Organization field box.



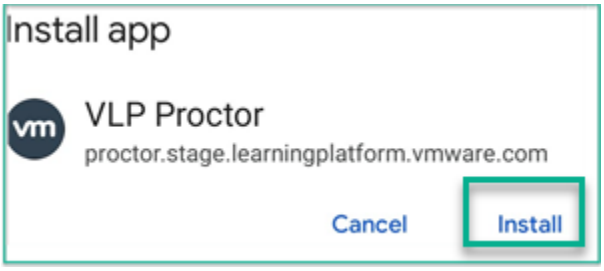
3. Click **Continue**.
4. On the **VMware Event Management Proctor App** page, enter your login credentials.
5. For better user experience, add the VMware Lab Platform Proctor application to your device home screen page. On your screen, click **Add VLP Proctor to Home Screen**.



Alternatively, you can also add the application to home screen from the browser. Click the three ellipses on top-right of the page and open the menu options. From the list, select **Add to Home screen**.



6. To install the application, click **Install**.


Install app
VLP Proctor
proctor.stage.learningplatform.vmware.com
Cancel
Install

NOTE
If you do not install the application and use the web browser version of the application, the menu items in the menu bar overlap and can impact the seamless user experience.

Figure 23: Web version of the VMware Event Proctor application

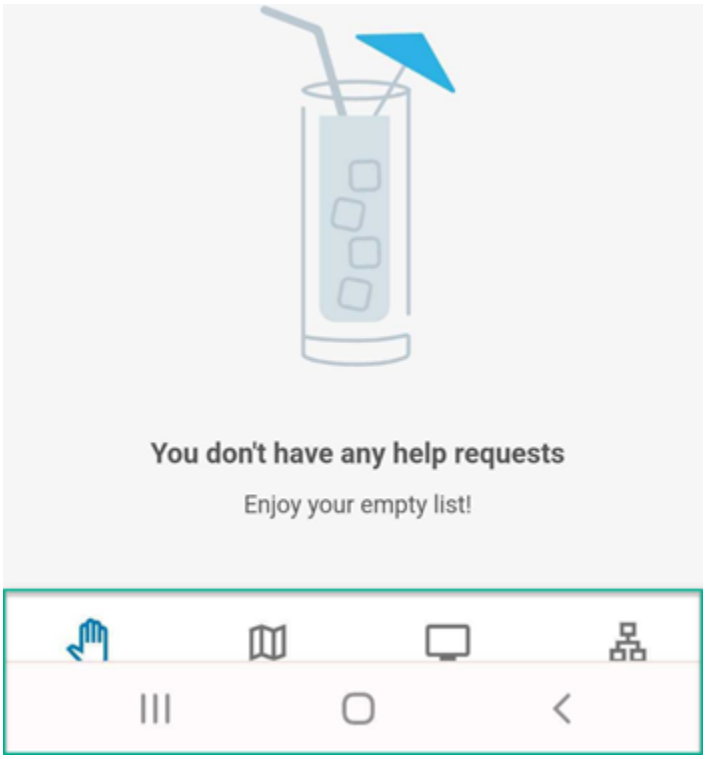
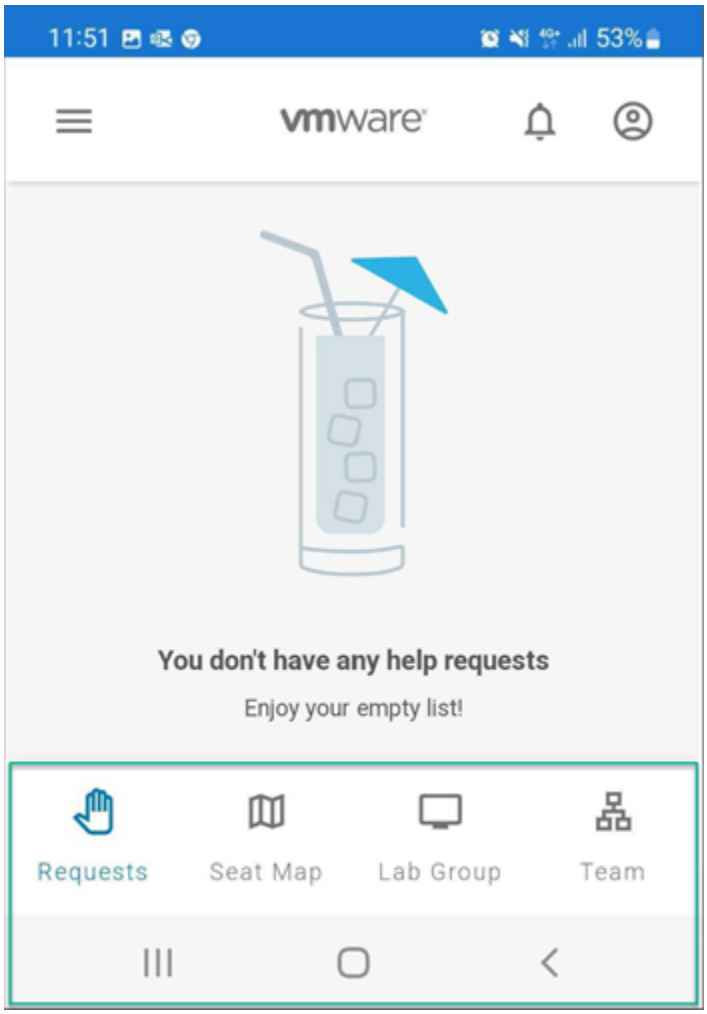
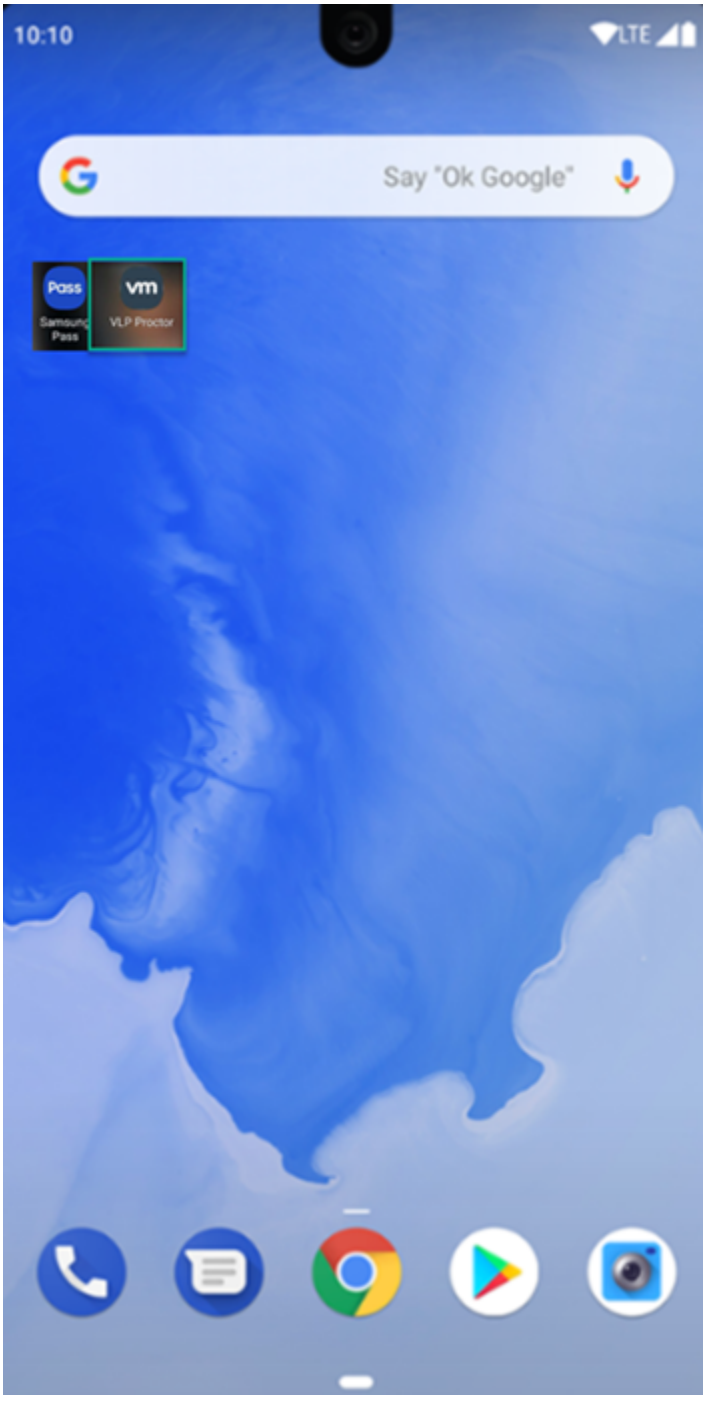


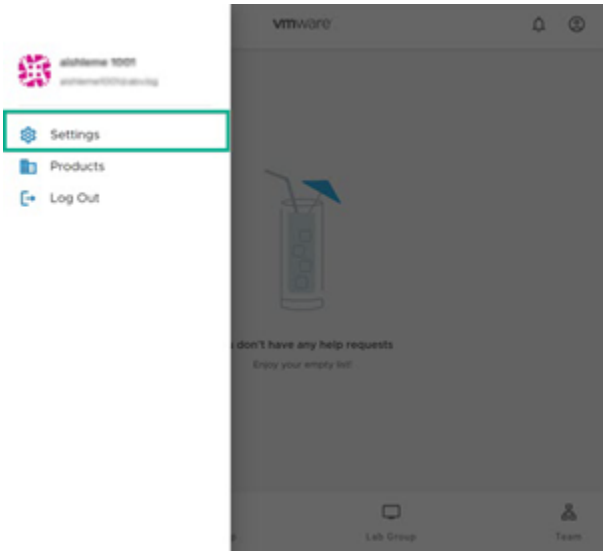
Figure 24: Mobile version of the VMware Event Proctor application



7. When the application is installed on the mobile device, you can access the application from the device dashboard.



8. To change the settings for your profile, click the  list icon and select **Settings** from the list.



9. To view the seat map of the event, on the bottom pane, click **Seat Map**.

   A seat map is the graphical representation of the event's layout. Proctors can view a seat map and get live information of seats in the event.

Figure 25: Seat map view



10. To view requests, click the **Requests** icon.
11. To view lab groups associated with the event, click the **Lab Group** icon.

    For additional support or information, contact the VMware Lab Platformsupport team or your organization administrator.

Integrate Slack Channel with the VMware Lab Platform Proctor application

You can configure the Slack messaging application with the VMware Proctor application to receive help requests from users. You must configure a webhook to post messages from the VMware Proctor application to the Slack application.

Enable the **Enable Slack Integration** and **Slack Integration URL** settings in the VMware Lab Platform administrator user interface. For more information on the Slack settings, see Working with Settings.

1. To configure a webhook, enter the URL https://api.slack.com.
2. Log in with your credentials.
3. Create an application.
4. On the Slack API page, navigate to **Features › Incoming Webhooks**.
5. Create a webhook.
6. Add your message application channel to the webhook.
7. Add the webhook in your tenant configurations.

The help requests raised in the VMware Proctor application are relayed in the client Slack channel.

For more information on Slack integration, refer to the Slack documentation.

VMware Lab Platform Content Engine Guide

The *VMware® Lab Platform™ Content Engine Guide* describes the VMware Lab PlatformContent Engine (formerly known as VMware® Learning Platform™Content Engine) interface and how you can use it to manage lab instructional manuals.

Intended audience

This information is intended for administrators and instructors who manage instructional manuals for labs developed and delivered with VMware Lab Platform.

Working with Content Engine in VMware Lab Platform

You use the VMware Lab Platform Content Engine interface to import and manage manuals used with hands-on labs.

Instructional content

Typically, a technical course instructor or a subject matter expert (SME) create instructional content for a lab using an authoring tool such as ScreenSteps or Microsoft Word. After the content is created and in some scenarios content is translated, you use the Content Engine interface to import the technical content into VMware Lab Platform.

After you import your manual and manual translations, you can make modifications with the Content Engine editor.

Language support

You can use the Content Engine to make multiple, translated drafts of a manual and add them to your customer account. When an end user starts a lab, the lab console defaults to the language version according to the user's language preferences. Language preferences are set using the My Profile option on the lab console. For more information on language preferences, see the *VMware Lab Platform User Guide*.

Content Engine workflow

To add a lab manual:

1. Create a lab manual project. Think of a lab manual project as a kind of container for lab manuals.
2. Upload or import instructional content. The result is an instance with an imported status. You can upload instructional content from your local machine or import it from the ScreenSteps authoring tool.
3. Optionally, upload or import additional content in different languages. This language support is one of the most useful features of the Content Engine.
4. Promote an imported instance to be a draft. A draft is an intermediate status that allows you to check instructional content before publishing.
5. Publish drafts to make them available to your administrator. Once available, an administrator can then use the Admin UI to associate a manual to a lab.

For more information on how to use the Content Engine interface, see the VMware Lab Platform Content Engine Guide.

About lab manual projects

Think of a lab manual project as a kind of container that consists of manual instances. An instance can have a draft, imported, or published status.

Add a lab manual project

Create a lab manual project as the first step of adding a lab manual to a lab.

A lab manual project is a kind of container that consists of manual instances. An instance can have a draft, imported, or published status.

1. On the Lab Manual Projects page click **Add New**. A form opens that allows you to create a new lab manual project.
2. On the form, click **Start a Manual Project**. The Start a Manual Project form opens.

   NOTE
   The **Upload a Manual** option allows you to upload instructional content from your local machine to a manual project you select. The result is an imported instance. The **Import from ScreenSteps** option allows you to import instructional content from the ScreenSteps authoring tool into a manual project you select. The result is an imported instance.

3. On the Start a Manual Project form, specify a name for your manual project and then click **Create**.

   NOTE
   First Numbered Step # refers to the step at which to start numbering the manual. The number 1 would be the normal behavior, while the number 2 would show the first step as a • in the step bubbles. The number 3 would show two bubbles and then the third step would be labeled 1 and so on.

The Lab Manual Projects page reopens and now lists the manual project you just created.

Lab manual project actions

For each listed manual project, you can click the vertical ellipses icon to display a list of actions.

Table 4: The VMware Lab Platform Lab Manual Project action menu

Menu Item	Description
View Project	Lists draft, published, and imported instances of the manual. The list provides options to manage each manual instance.
Edit Project	Opens the Update Manual Project form. From here, you can rename the manual project and indicate at which step to begin numbering the manual. The number 1 is the normal behavior. Number 2 shows the first step as a bulleted item (•) in the step bubbles. The number 3 display two bubbles and then the third step is labeled 1 and so on.
Add New Manual	Opens the Create a Manual (draft) form. From here, you can create a draft by entering a title and language. The draft you create does not have any content.
Upload Manual	Opens the Upload Manual form. From here, you can upload instructional content by uploading a file. Uploading a file creates an imported manual instance.
Import Manual	Opens the Import from ScreenSteps form. From here, you can import content from the ScreenSteps authoring tool. Importing from ScreenSteps creates an imported manual instance.
Delete Project	Deletes the manual project including all contents (drafts, imported, and published instances).

Working with the Lab Manual Project contents options

Each lab manual project includes a link that opens its contents.

Clicking the horizontal dot icon opens a drop-down menu of actions.

Action	Description
Edit details	Opens a form that allows you to change the title and language of an instance.
Preview	Opens a preview of the draft as it would appear within the lab console.
Export	Opens a form that allows you to export the draft to a server (typically, the same server your browser is using). Once you export a manual, it becomes available for you to upload (Upload Manual option).
View/Edit Draft	Starts an editor that allows you to edit the contents of the draft. The editor includes an options ribbon and options to insert chapters, articles, and steps.
Publish Draft	Creates a published instance of the draft. Publishing a draft makes it available to your customer account.
Delete Draft	Deletes the draft.

Adding instructional content to a Lab Manual Project

After you have created a new lab manual project, you add instructional content.

You can add instructional content in the following ways:

- By uploading a Word document (.docx file format) or a manual previously exported from the Content Engine (such exports are ZIP files).
- By uploading a PDF file from your local system to the Content Engine.
- By importing manuals residing in the ScreenSteps authoring tool.

> **NOTE**
> VMware Lab Platform supports import for Word DOCX files, but not for DOC files.

If your instructional manual is in PDF format, it does not need to be part of a lab manual project. Instead, upload the PDF file as a resource as described in Add a PDF Manual to a Lab.

About multiple language support

The Content Engine cannot translate a manual. Translations must be done by a content provider. But the Content Engine does support uploading and importing multiple language drafts of the same instructional content. This multiple language support is very useful because lab consoles include a Profile Settings option. A lab user can set the Profile Settings to associate the lab console to a version of the lab manual in a selected language.

Import word documents

You can use `MS Word DOCX` files as a source for lab manuals when you import them into VMware Lab Platform.

You import a word document by uploading it to a lab manual project. The Content Engine converts the file to a lab manual format compatible with VMware Lab Platform.

> **NOTE**
> Word `DOC` files are not eligible for import. Convert `.doc` files to the `.docx` format before you upload the file.

1. On the **Lab Manual Projects** page, click the horizontal dots icon next to your manual project to open a drop-down menu.
2. On the drop-down menu, click **Upload Manual** to open the Upload Manual form.
3. On the **Upload Manual** form, click **Choose File** and navigate to the word file you want to import.
4. Select the file, then click **Upload**. The Content Engine uploads the document to your manual project and converts it to the VMware Lab Platform lab manual format.
5. Review your result in the **Preview Document Conversion** window.
6. After reviewing the results, choose one of the following options.

- To save the converted manual to your lab manual project, click **Save**.
- To delete the converted document, click **Cancel**.

Importing instructional content from ScreenSteps

The Content Engine supports ScreenSteps integration such that you import instructional content directly from ScreenSteps into a manual project.

To use ScreenSteps Integration support, you must:

1. Create an external service which integrates ScreenSteps with the Content Engine. For more information on external service, see Create an External Service.
2. Import content from an external service into a manual project.

Create an external service

Creating an external service integrates the service with VMware Lab Platform. Once you do so, you can import content from the external service.

1. To open the **External Services** list, from the navigation panel, click **Content › External Services**.
2. To open the **Add New External Service** form, click **Create**.
3. Enter the following information in the **Add New External Service** form.

URL	Enter your organization's ScreenSteps URL. For example, vmware.screensteps.com.
Username	Enter your ScreenSteps user name.
Password	Enter your ScreenSteps password.
Language	Select a language for the instructional content from the drop-down menu.
Treat SVG file attachments as images	To add SVG files as image files in ScreenSteps, select the **Treat SVG file attachments as images** option.
Add Chapter Table of Contents	To add a manual's table of contents including chapter titles (modules) followed by the articles contained by each chapter, enable the **Add Chapter Table of Contents** option.
Add Conclusion Page	To add a conclusion page at the end of the manual, enable the **Add Conclusion Page**.

4. To add the external service you just created to the **External Services List**, click **Create**.

The external service is listed on the **Content › External Services** page. To edit, delete, or open the external services web link, see Working with the External Services List.

Working with the external services list

The Content Engine supports ScreenSteps integration such that you import instructional content directly from ScreenSteps. The External Services list provides forms that you can use to create an external service and import content from an external service into a manual.

For each listed external service, you can click its three vertical dots icon to open a drop-down menu of actions.

Action	Description
Edit External Service	Opens the Edit External Service form to update the parameters of the external services.
Delete External Service	Deletes the external service from the list.

Import content from ScreenSteps

Now that you have created an external service, you can import the instructional content from it to a manual project.

1. To open the **Lab Manual Projects**, click **Content › Manual**.
2. Click **Add New**.
3. On the form, click **Import from ScreenSteps**.
4. On the Import from ScreenSteps form, select **New Project** from the Choose Project drop-down menu.
5. Enter your project name for the manual. Click **Next**.
6. On the Choose Integration form, enter your ScreenSteps integration details. If you added an external service list, you can select the service list from the drop-down menu. To create a new login:
   a) Enter your ScreenSteps URL. For example, vmware.screensteps.com.
   b) Enter your ScreenSteps username.
   c) Enter your ScreenSteps password.
   d) Click **Next**.
7. On the Select Manual tab, select a ScreenSteps manual to open and then click **Next**.
8. Specify the following information on the Import Operations form.
   a) Select the language used by the ScreenSteps manual.
   b) Position the manual's images above or below text.
   c) Optionally, add a chapter table of contents. If you enable **Add chapter table of contents**, then the resulting table of contents includes the chapter and module titles.
   d) Optionally, you can add a conclusion page to the manual.
9. Click **Import**. The Content Engine imports the instructional content into your project.

To review your result, go to the list of lab manual projects, and click on your project link to open its contents.

Updating from ScreenSteps

The Update from ScreenSteps option reimports the instructional content from ScreenSteps.

This option is useful if after importing, you go back to ScreenSteps and make changes to the manual. The option serves as a shortcut to reimport and thus keeps your ScreenSteps manual and Content Engine import in sync.

You can refer to the sample ScreenSteps manual to see the various formatting tags for the lab manual, and its rendition in the VMware Lab Platform user interface.

Using PDF documents as lab manuals

You can use a PDF file as an instructional guide by importing it through the Admin UI as a lab resource or from the Create a Lab page.

PDF files are included in lab manual projects. For more information about using PDF files as lab manuals, see Create a Lab.

Managing lab manuals

Lab manuals can have a status of imported, draft, or published. You cannot edit VMware Lab Platform manuals after they are published.

Promote a lab manual from import to draft status

Draft is an intermediate status that you can assign to a document when you want to verify instructional content before publishing. After you have uploaded or imported a document, you can promote the resulting imported instance to a draft status.

1. On the Lab Manual Projects list, click the link to your manual project. The contents of the project are displayed.
2. Within the contents, click the ellipses • • • icon to open a drop-down menu.
3. On the drop-down menu, click **Promote to Draft**.
   The Content Engine promotes your import to a draft status.

Editing draft and imported contents

You can apply the **View/Edit** menu item to an imported instance or a draft. This starts an editor that you can use to edit the instructional content of a manual directly in the Content Engine.

Editing ribbon

When you click a text element you want to change, a window with an editing ribbon is displayed. Enter your new text here. You can modify the style and format of the text with the editing ribbon.

Add chapters, articles, and steps

If your current hierarchy is at the chapter level, then you can use the **Add New Chapter** menu item.

If your current hierarchy is at the articles level, then you can access the **Add New Article** menu item.

If your pointer is active within an article, then you can access the **Add Step** menu item to add a page within the article, specify a heading, or add content.

Send text to the lab console

You can use HTML markup to indicate text that lab users can move to their console by clicking the text. Use the **Source** button to insert HTML code. As a result, the end user can easily send text, such as a user name or password, directly from the manual to the lab console. The following example HTML code allows a lab user to send the text string "some text to send" from the manual directly to the lab console with the click of a button:

```
<!-- looks for .send-text css class -->

<span class="send-text">some text to send</span>

<button class="send-text" value="some text to send">Call to action</button>
```

Insert variables into a manual

Lab manual content creators have the ability to display some of the important information in lab manuals with help of variable tags. You can include a variable within a step of a manual. The value of the variable is assigned when the end user finishes the step. For example, you might include a step within your manual that includes the following link:

```
http://example.com/test?foo={{email}}
```

When an end user takes the lab and clicks the link, the end user's email address is assigned as the value for the variable. As a result `{{email}}` might become `jsmith@mycompany.com`. Such variables allow you to make the results of a step customized to the user taking the lab. The Content Engine supports the following variables:

- User
  - `Userusername: username`
  - `email: email@address.com`
  - `firstName: firstName`
  - `lastName: lastName`
- Lab
  - `labSku: sku`
  - `labName: lab name`
  - `ClassclassId: {{classId}}`
  - `className: {{className}}`
  - `classSku: {{classSku}}`
  - `classDescription: {{classDescription}}`
- Entitlement
  - `entitlementId: entitlement-id`
  - `entitlementKey: {{entitlementKey}}`
  - `endAllowed: {{endAllowed}}`
  - `exitAllowed: {{exitAllowed}}`
  - `resetAllowed: {{resetAllowed}}`
  - `requiresGrading: {{requiresGrading}}`
- Progress
  - `progress: {{progress}}`
  - `requiredMinimumTimeSpent:{{requiredMinimumTimeSpent}}`
  - `requiredStepsVisitedPercentage: {{requiredStepsVisitedPercentage}}`
  - `completionStatusForLabCriteria: {{completionStatusForLabCriteria}}`
- Manual
  - `stepNumber: 6`
  - `activeStepNumber: {{activeStepNumber}}`
  - `totalSteps: {{totalSteps}}`
  - `manualStepCount: {{manualStepCount}}`
  - `remoteControlMode: {{remoteControlMode}}`
- Time
  - `timeAllotted: 60`
  - `timeElapsed: 0`
  - `timeRemaining: 60`
  - `expirationDate: {{expirationDate}}`
  - `time: 1631806973255`
  - `canExtend: {{canExtend}}`
  - `extensionsRemaining: 1`
  - `extensionCount: {{extensionCount}}`

- Event Lab Groups
  - `groupName: groupName`
  - `groupColor: #FF0000`
- Miscellaneous
  - `location: {{location}}`
  - `language: {{language}}`
  - `locale: {{locale}}`
  - `tenant: NEE`
  - `tid: {{tid}}`
  - `trackingCode: {{trackingCode}}`

NOTE
For the time variable, the system assigns a unique timestamp each time a user completes the step. Example: `{{email}}&_={{time}}` becomes jsmith@mycompany.com&_12569537329.

The system does not actively update `timeElapsed`, `timeRemaining`, `extensionsRemaining`. Their values are accurate only when the step is displayed to the end user. For example, if the lab has 55 minutes remaining, then `timeRemaining` displays a value of 55 and does not count down (to 54, 53) while that step remains visible.

To get the supported list of variable tags, contact the VMware Lab Platform support team.

Insert Markup content

The Content Engine supports markup content. Consider the following example of markup content:

```
Log into the Windows console, press [vlp:ctrl-alt-del|Control-Alt-Delete].

If you need more space to work, enter your browser's [vlp:fullscreen|Toggle

Fullscreen] mode.

[vlp:maximize|Maximize] you VM's work space
```

If you use the editor to insert such markup, the resulting instructional manual, in the lab console, would include the following:



Supported Markup commands

The following markup commands are rendered as buttons.

- `[vlp:power-down|Power Down Active VM]`
- `[vlp:power-reset|Power Reset Active VM]`
- `[vlp:power-on|Power On Active VM]`
- `[vlp:refresh-vm|Refresh VM]`
- `[vlp:send-text|text to send|Send Text to Console]`
- `[vlp:next-step|Next Manual Step]`
- `[vlp:previous-step|Previous Manual Step]`
- `[vlp:goto-step|5|Goto Step 5]`
- `[vlp:network-topology|Show Network Topology]`
- `[vlp:table-of-contents|Show Table of Contents]`
- `[vlp:increase-font|Increase Manual Font]`
- `[vlp:decrease-font|Decrease Manual Font]`
- `[vlp:extend-time|Extend Time]`
- `[vlp:exit|Exit Lab]`
- `[vlp:end|End Lab]`
- `[vlp:reset|Reset Lab]`
- `[vlp:grade|Grade Lab]`
- `[vlp:ctrl-alt-del|Control-Alt-Delete]`
- `[vlp:fullscreen|Toggle Fullscreen]`
- `[vlp:maximize|Maximize VM]`

Markup commands as HTML

You can insert markup commands as HTML and with data-* properties on any tag.

- <a data-power-down="">Power Down Active VM
- <a data-power-rest="">Power Reset Active VM
- <a data-power-on="">Power On Active VM
- <a data-refresh-vm="">Refresh VM
- <a data-send-text="">Send Text to Console
- <a data-next-step="">Next Manual Step
- <a data-previous-step="">Previous Manual Step
- <a data-goto-step="5">Goto Step 5
- <a data-network-topology="">Show Network Topology
- <a data-table-of-contents="">Show Table of Contents
- <a data-increase-font="">Increase Manual Font
- <a data-decrease-font="">Decrease Manual Font
- <a data-extend-time="">Extend Time
- <a data-exit="">Exit Lab
- <a data-end="">End Lab
- <a data-reset="">Reset Lab
- <a data-grade="">Grade Lab
- <a data-ctrl-alt-del="">Control-Alt-Delete
- <a data-fullscreen="">Toggle Fullscreen
- <a data-maximize="">Maximize VM

Export a manual

When you are done with working on a draft, you can export it.

Exporting a draft sends it to a default server (typically, the same server your browser is using). Once you export a manual, it becomes available for you to upload with the **Upload Manual** option.

1. On the Lab Manual Projects list, click a lab manual project link to open its contents.
2. Click the • • • icon next to a draft name to open a drop-down menu and then click **Export**. The Opening Manual Export form opens.
3. On the form, select **Save File** and then click **OK**. The Content Engine saves the file to a default directory as a ZIP file that you can upload later.

Publish a manual

You must publish drafts for them to be available to your account.

Once available to your organization, an administrator can associate a manual to a lab. After an administrator associates a manual to a lab, the system refers to the manual name (for example, First Manual). Then when an end user starts a lab, the console takes the language version of the manual as set in the user's profile setting.

1. On the list of lab manual projects, click a lab manual project link to open its contents.
2. Click the three-dot icon next to a draft to open a drop-down menu. Click **Publish Draft**. The Content Engine publishes the draft.

> **NOTE**
> You must publish all the language versions of a manual to make them available to your customer account. For example, if you have English and German drafts, you must publish both.

VMware Event Proctor Application Guide

Working with the VMware Event Proctor application

VMware Lab Platform enables users with account administrator and proctor role, to conduct and monitor online tests and events similar to in-person tests and events using the VMware Event Proctor mobile application. When lab users have a question or need help with a lab, users can use the Request option on their lab console screen. The option runs an application on the proctor's mobile device to notify the proctor that someone needs help. A user having an Event/Proctor role can log into the VMware Lab Platform administrator user interface.

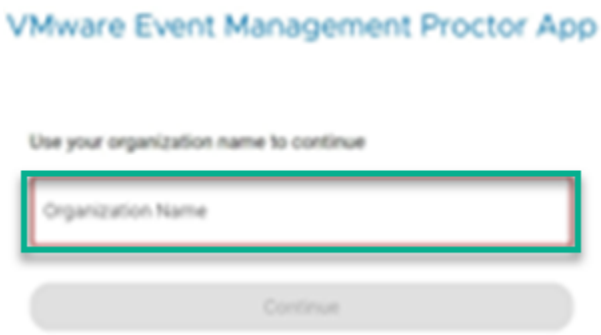
The following section gives instructions on how to install and use the VMware Event Proctorapplication for VMware Lab Platform services.

Install the VMware Proctor application on Android devices

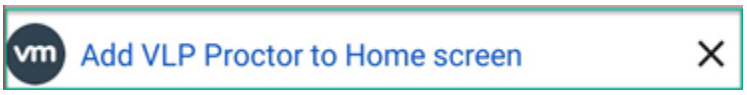
You install the VMware Event Proctor mobile application, so that lab users can easily communicate with you using the proctor application if the users need any help or request while taking a lab.

To install the VMware Event Proctor application on an android mobile device:

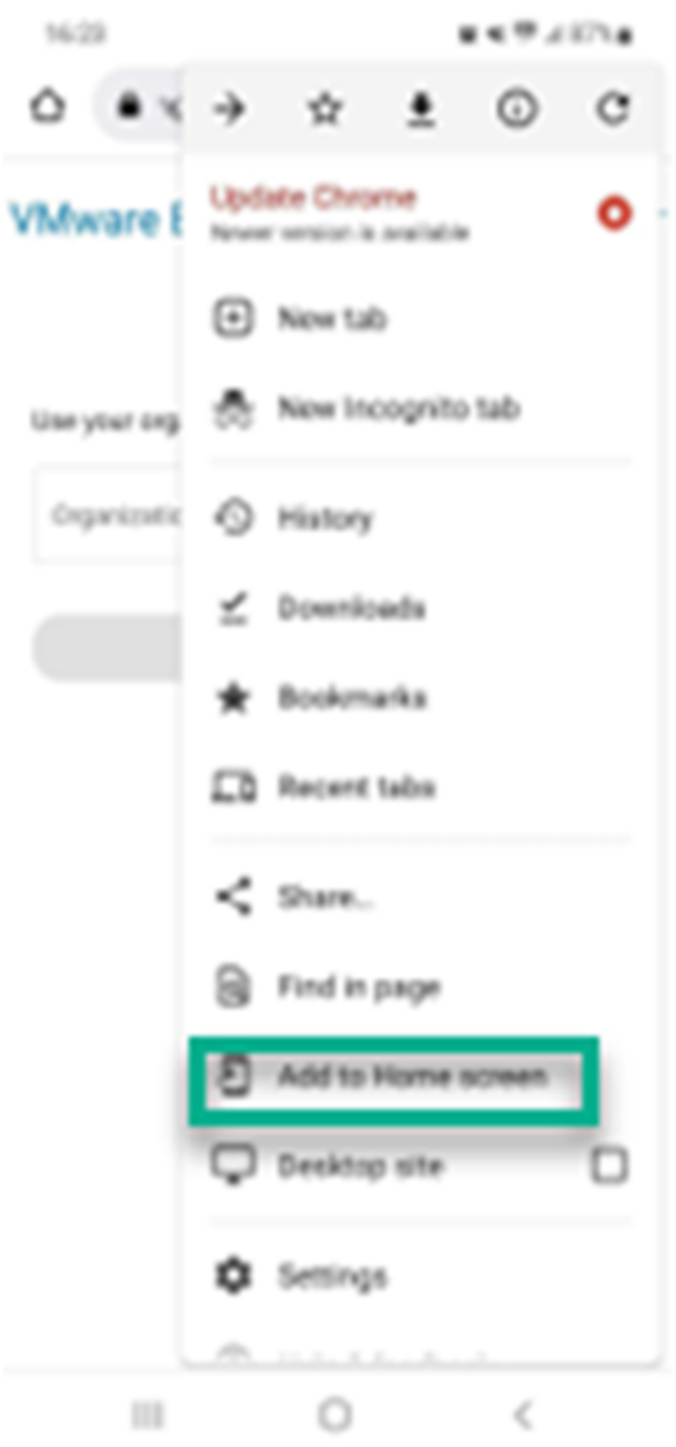
1. On an android mobile device, enter the proctor application URL in a web browser. URL:https://proctor.labplatform.broadcom.com/login
2. On the **VMware Event Management Proctor App** page, enter your organization name in the Organization field box.



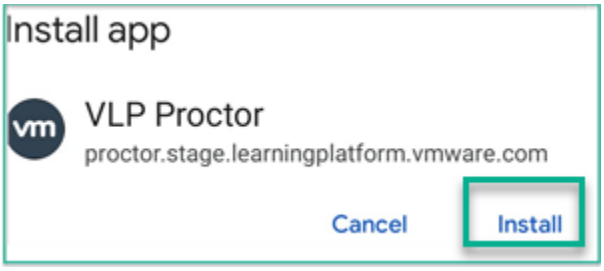
3. Click **Continue**.
4. On the **VMware Event Management Proctor App** page, enter your login credentials.
5. For better user experience, add the VMware Lab Platform Proctor application to your device home screen page. On your screen, click **Add VLP Proctor to Home Screen**.



Alternatively, you can also add the application to home screen from the browser. Click the three ellipses on top-right of the page and open the menu options. From the list, select **Add to Home screen**.



6. To install the application, click **Install**.


Install app
VLP Proctor
proctor.stage.learningplatform.vmware.com
Cancel
Install

NOTE
If you do not install the application and use the web browser version of the application, the menu items in the menu bar overlap and can impact the seamless user experience.

Figure 26: Web version of the VMware Event Proctor application

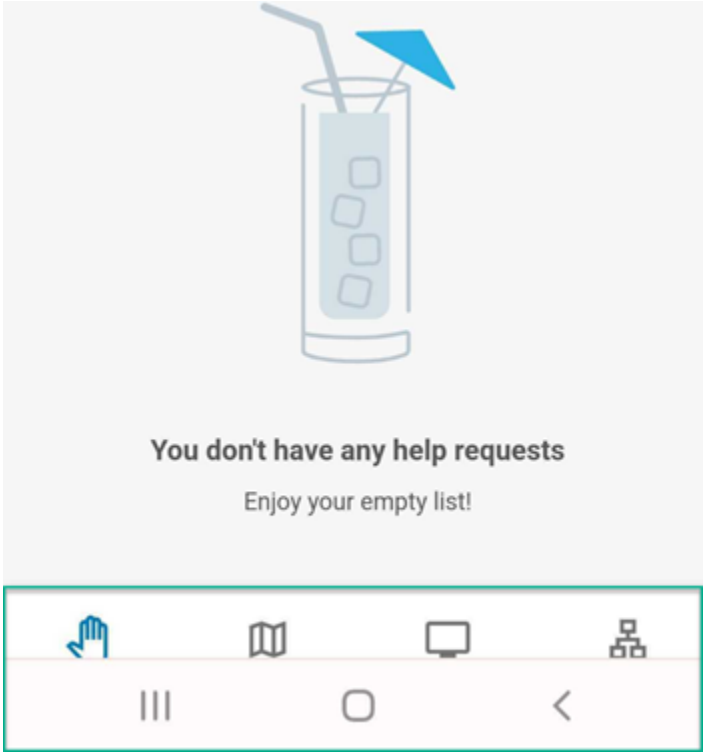
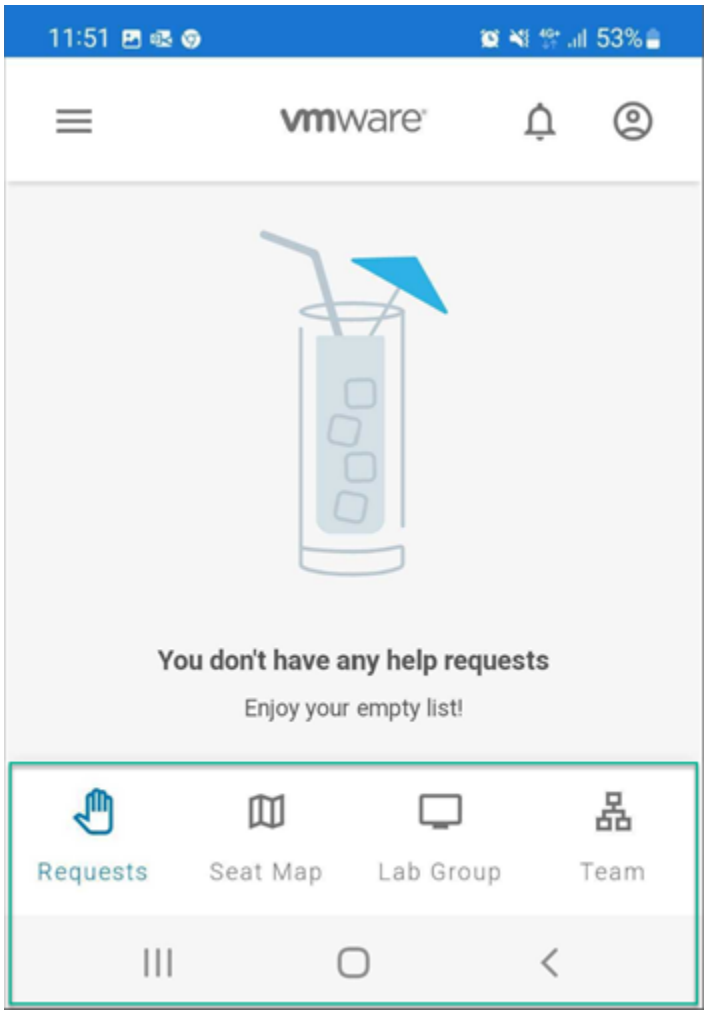
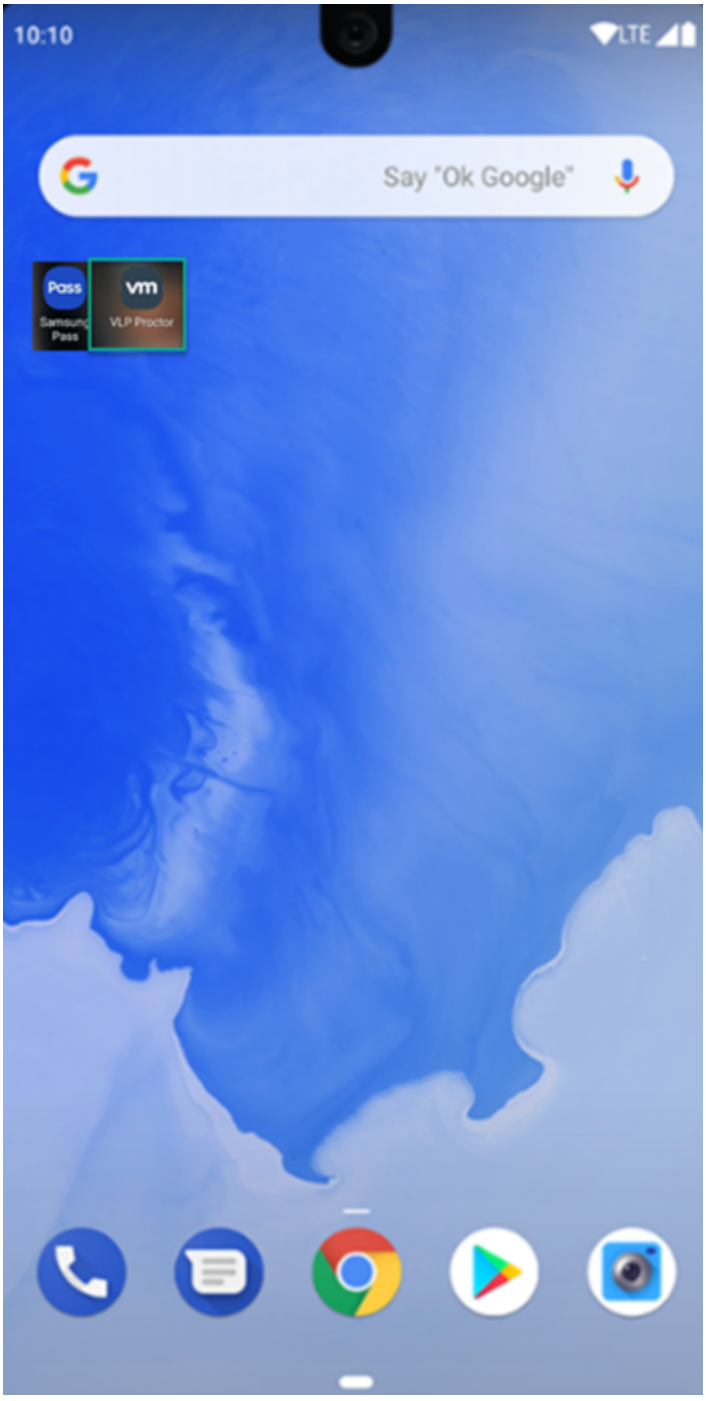


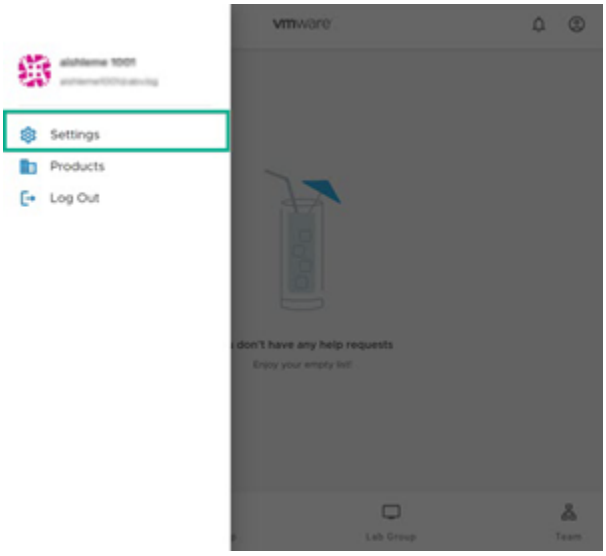
Figure 27: Mobile version of the VMware Event Proctor application



7. When the application is installed on the mobile device, you can access the application from the device dashboard.



8. To change the settings for your profile, click the  list icon and select **Settings** from the list.



9. To view the seat map of the event, on the bottom pane, click **Seat Map**.

   A seat map is the graphical representation of the event's layout. Proctors can view a seat map and get live information of seats in the event.

Figure 28: Seat map view



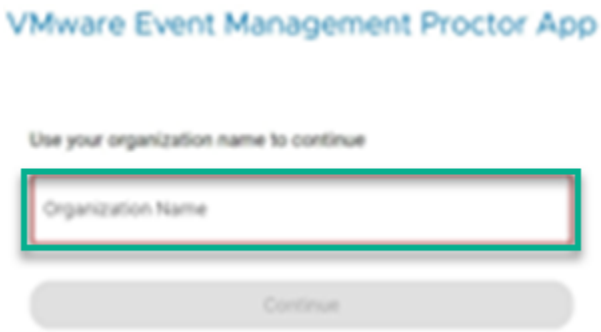
10. To view requests, click the **Requests** icon.
11. To view lab groups associated with the event, click the **Lab Group** icon.

    For additional support or information, contact the VMware Lab Platformsupport team or your organization administrator.

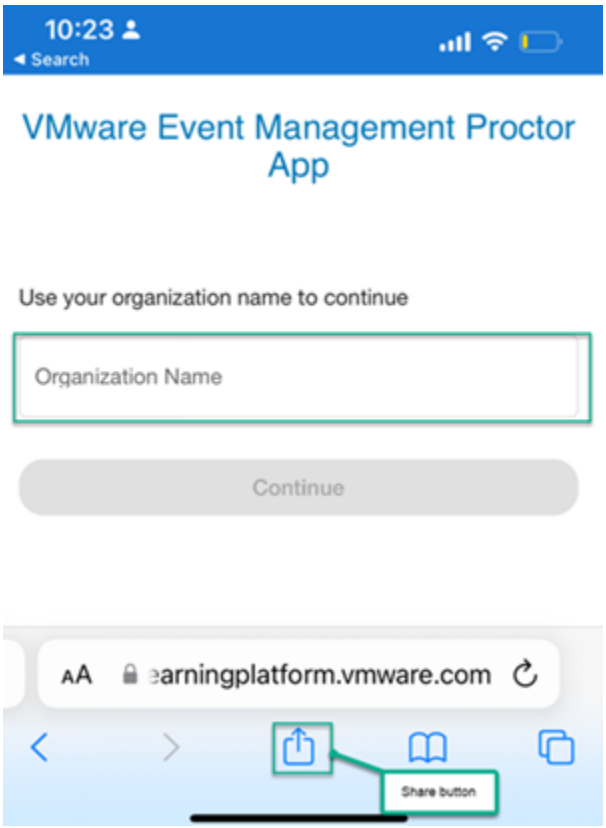
Installing VMware Event Proctor Application on iOS Devices

To install the VMware Event Proctor application on an iOS mobile device, perform the following steps:

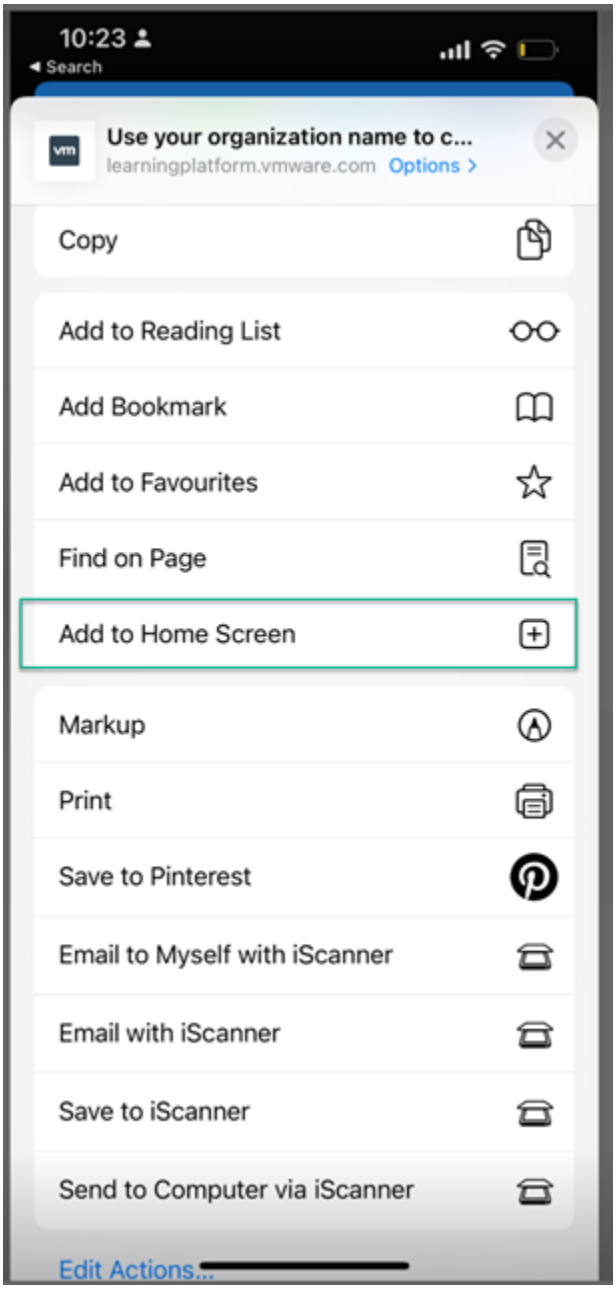
1. Open the Safari browser on your mobile device and enter the proctor application URL in the web browser. URL: https://proctor.labplatform.broadcom.com/login.
2. On the VMware Event Management Proctor App page, enter your organization name in the **Organization** field box.



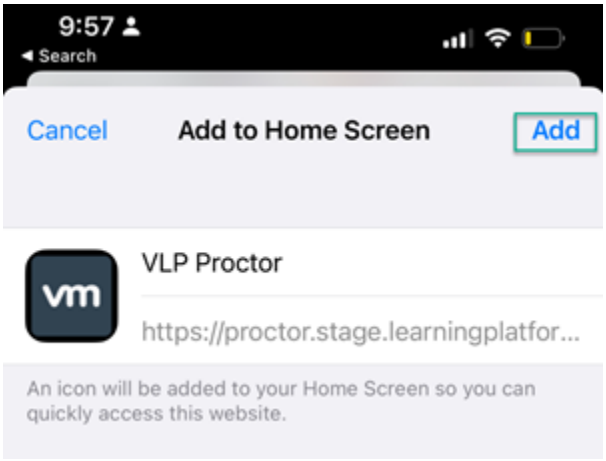
3. Click **Continue**.
4. On the **VMware Event Management Proctor** App page, enter your login credentials.
5. To add the application shortcut on your home screen, click the **Share** button located at the bottom of the screen.



6. From the list of options, select **Add to Home Screen**.



7. Add a name for the website shortcut. Click **Add**.



8. When installed on the mobile device, you can access the website shortcut for the application from the device dashboard.

Working with Third-Party Learning Applications

In addition to using VMware Lab Platform (formerly known as VMware Learning Platform) as a standalone service for delivering labs, you can integrate VMware Lab Platform with any Lab Management System that conforms to the Lab Tools Interoperability (LTI) standard, versions 1.1 or 1.3.

You can use the following definitions for reference.

Learning Management System (LMS)

A learning management system is a software that is explicitly designed to create, distribute, and manage the delivery of educational content. Think of a learning management system as a technology that can improve learning, make it faster, more productive, more cost-effective, and trackable.

Learning Tools Interoperability (LTI)

Learning Tools Interoperability is an initiative managed by the IMS Global Learning Consortium to integrate learning applications used by instructors into their courses. It includes a standard protocol for establishing a trusted relationship between the tool provider and the Learning Management System so that students and teachers can have a seamless, integrated experience of using the tool within the context of their course.

In LTI terms, VMware Lab Platform is considered an LTI Tool Provider and is configured as an external tool. An LMS is considered as the LTI Tool Consumer.

The VMware Lab Platform administrator user interface supplies VMware Lab Platform-related information required for integrations that use the LTI 1.1 protocol. Additional functionality is provided for LTI version 1.3-based integrations, including to view a list of all integrations, modify and integration, or delete an integration.

Creating and Managing LMS Integrations

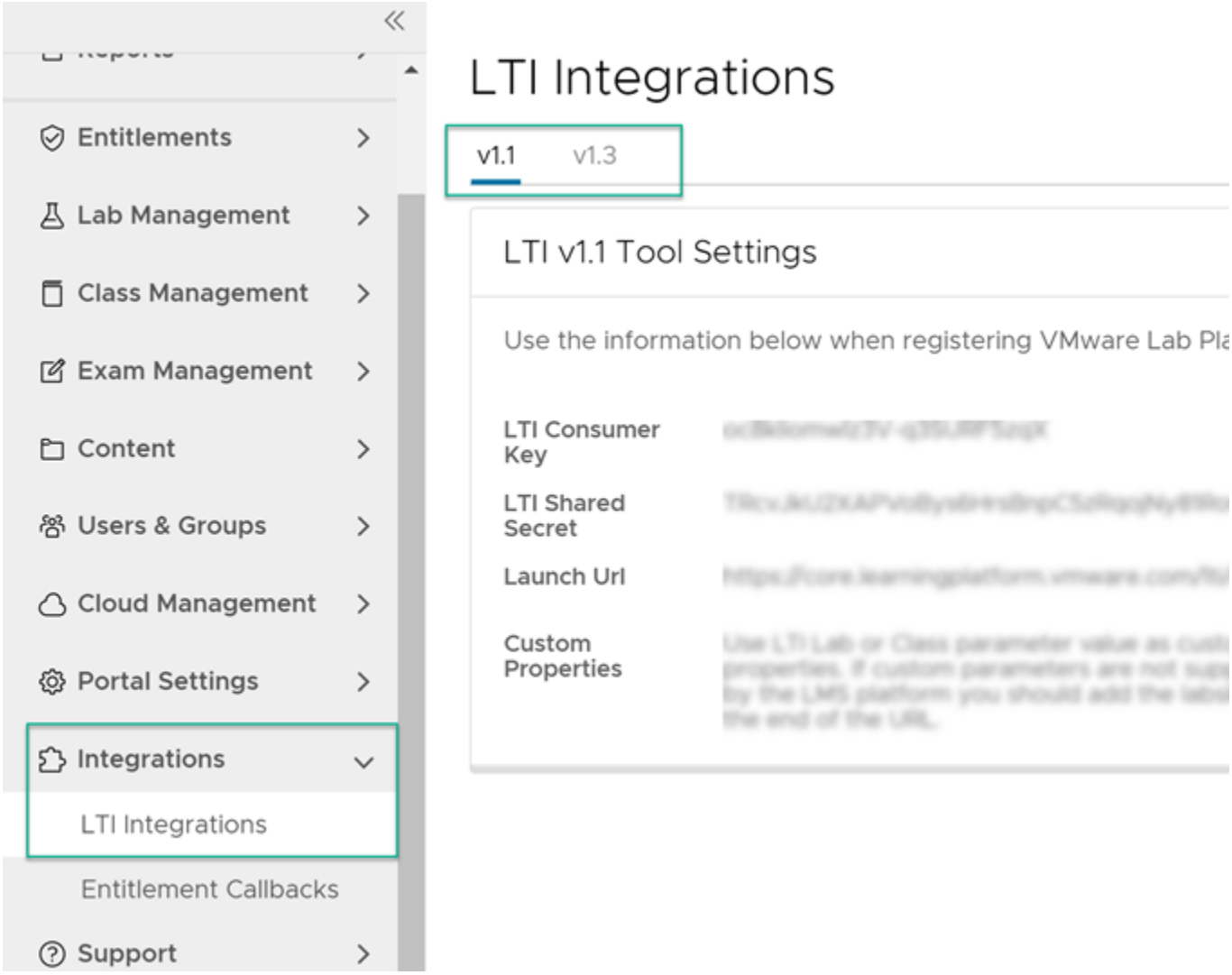
The LTI Integrations page on the administrator user interface provides information required to integrate VMware Lab Platform with external learning management systems (LMS) that use LTI version 1.1 or version1.3 protocols.

The following VMware Knowledge Base articles describe the procedure for registering VMware Lab Platform with the Blackboard, Moodle, or Docebo learning management systems:

- VMware Learning Platform LTI integration - Blackboard LMS (79111)
- VMware Learning Platform LTI integration - Moodle LMS (79113)
- VMware Learning Platform LTI integration - Docebo LMS (79418)

Learning Tools Interoperability, Version 1.1

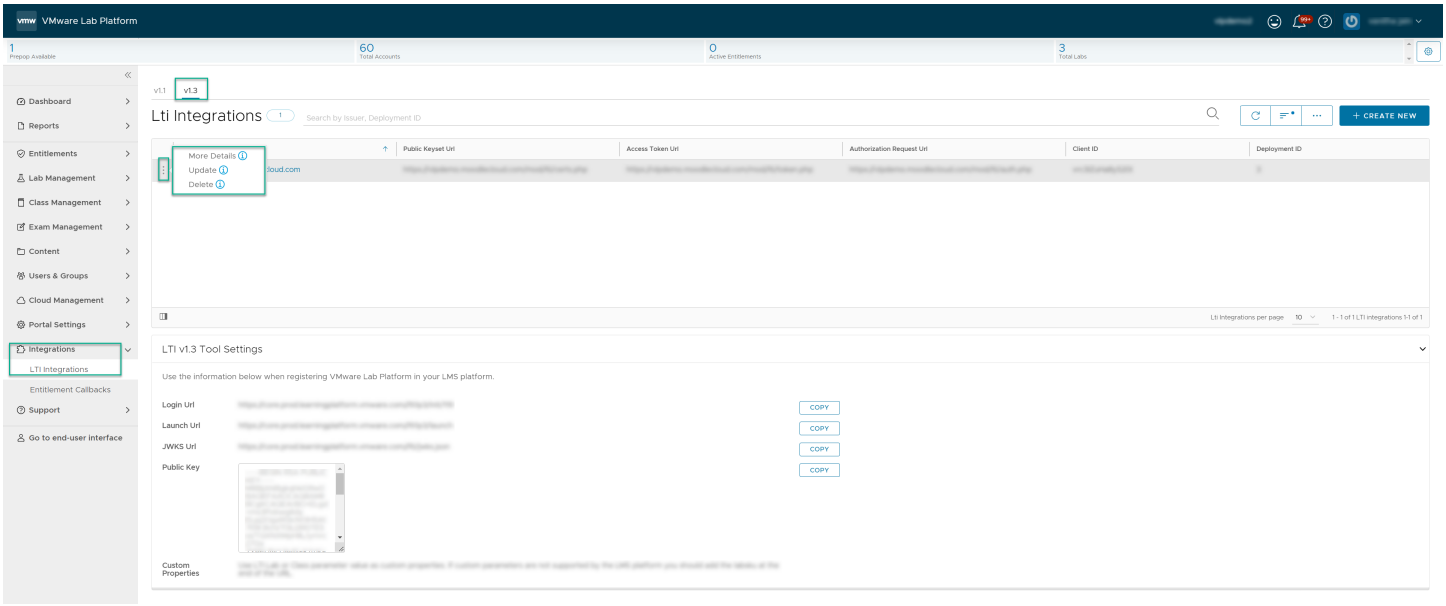
The **v1.1** tab on the LI IntegrTI Integrations page provides the VMware Lab Platform information for registering with your LMS, including the URL that starts your VMware Lab Platform lab. Go to **Integrations › LTI Integrations** and click **LTI 1.1** to open the page.



Copy or modify the information as needed and save it to use when you register VMware Lab Platform on your LMS.

Learning Tools Interoperability, Version 1.3

1. Go to **Integrations › LTI Integrations**.
2. Click the **v1.3** tab.



You can perform the following actions from the **LTI Integrations** page.

Option	Description
LTI v1.3 Tool Settings	Click the arrow to view the VMware Lab Platform information you need when registering VMware Lab Platform as an LTI tool through the LMS interface. You can copy these values and paste them into your LMS forms.
Create New	Click **Create New**. On the form, enter information from your LMS after you have registered VMware Lab Platform as described in the Knowledge Base article for your LMS.
Available actions menu	Click the three ellipses next to a listed integration to open the action menu.
Actions	The action menu includes the following options: • **More details**. Displays the parameter values associated with the integration. • **Update**. Opens the Create LTI Integration form, from which you can modify and update your integration. • **Delete**. Removes the integration.

Documentation Legal Notice

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